

SLB

Communications

Handbook

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Legal and Marketing

The SLB Name, Marks, and Copyright

Legal information

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1.0 SLB—both a company name and a trademark

The word “SLB” is both a trade name (used to identify the company and its business) and a trademark (used to identify the source of goods and services). Both have important roles in promoting the SLB brand, which is the continuously evolving corporate image discerned by our customers, our shareholders, and the industry.

The rules for correct use of the trade name and trademark are different. These rules must be carefully adhered to because improper use can reduce the value of a trademark.

1.1 Correct use of the SLB trade name and SLB trademark

The word “SLB” can be used correctly as a noun (trade name) or as an adjective (trademark) but cannot be used as a verb.

SLB trade name usage

- 1.1.1 As a trade name, use the word “SLB” only as a noun.
Correct: SLB creates technology that unlocks access to energy for the benefit of all.
- 1.1.2 As a trade name, “SLB” is used as a singular noun.
Incorrect: SLB have announced management changes.
Correct: SLB has announced management changes.
Incorrect: Worldwide SLBs provide services for the life of the reservoir.
Correct: SLB offers worldwide access to new technology.
- 1.1.3 The possessive form is correct only when “SLB” is used as a trade name. Do not form a possessive if “SLB” is used as a trademark to designate the source of a product, service, technology, or solution.
Incorrect: SLB’s new tool was launched in Indonesia last week.
Correct: The new SLB tool was launched in Indonesia last week.
Correct: SLB’s stock increased in value this year.
Correct: SLB’s business philosophy encompasses teamwork and cultural diversity.

SLB trademark usage

- 1.1.4 As a trademark, use the word “SLB” only as an adjective modifying a noun.
Incorrect: SLB’s log was acquired before the hole collapsed.
Correct: The SLB log was acquired before the hole collapsed.
Correct: SLB technology is supported by a global team of experts.
- 1.1.5 Never modify the word “SLB” or use it in coined words.
Incorrect: e-SLB, SLBlog
Correct: SLB e-commerce, SLB log

Combinations, formatting, and other common uses of the word “SLB”

- 1.1.6 “SLB” can be used as a trade name in combination with other trade names and trademarks. Standard grammar rules apply.
Incorrect: The Shell/SLB alliance was formed in 1999.
Correct: The Shell-SLB alliance was formed in 1999.

- 1.1.7 In copy, “SLB” is always written in capital letters. Lowercased “slb” is used only in the company logo and as a component in electronic addresses, including the public website slb.com. Do not use “Schlumberger” as the spelled-out version of “SLB,” as “SLB” is not an initialism.
- Incorrect:** Schlumberger today announced a merger with a leading manufacturer.
- Correct:** SLB (NYSE: SLB) today announced the acquisition of a leading manufacturer.
- Incorrect:** Schlumberger personnel are located around the world.
- Correct:** SLB personnel are located around the world.
- Correct:** For more information on stimulation, go to slb.com or contact us at stiminfo@slb.com.
- 1.1.8 SLB is always pronounced as individual letters.
- 1.1.9 Routine mentions of “Schlumberger” that require updating should be replaced with “SLB.” If necessary, when “Schlumberger” is discussed in the context of its history, the phrase “Schlumberger (now SLB)” may be used. For names of physical buildings, foundations, or other instances that use “Schlumberger” in recognition of the company’s founders or otherwise long-standing history, some exceptions may apply and should be verified with the brand department.
- Incorrect:** Schlumberger has extensive expertise across the oilfield life cycle.
- Correct:** Schlumberger (now SLB) was founded in 1926.
- Incorrect:** The global oilfield enterprise now known as Schlumberger owes its vitality to more than 100,000 people in 100+ countries.
- Correct:** SLB has extensive expertise across the oilfield life cycle.

1.2 Companies acquired by SLB

- 1.2.1 Companies acquired by SLB, including associated products and services, integrate into the SLB masterbrand. The section summarizes how companies acquired by SLB are to be classified and defined. For detailed information and processes, refer to the [SLB Masterbrand Architecture](#) (also available on the [Brand myHub page](#)). This framework organizes and unifies our brand, and all the products and services in our portfolio, inclusive of our acquired brands, so that they are easy to understand, navigate, and buy.

The equity in the names of companies acquired by SLB is considered during the integration process. Division Marcoms should be consulted to determine if key elements are to be retained to respect the acquired company’s strong heritage while aligning their product portfolios under the SLB master brand. It is important to recognize that each brand requires its own, well-thought-out approach. Individual logos (the visual identities of those legacy brands) have been fully retired to ensure that SLB goes to market as one brand identity without losing the value of those product names. Additionally, to align with this strategy, do not refer to acquired companies as “an SLB company.” Anything deemed as an exception is referred to the Brand Council.

Acquired Company	Classification	Definition	Example
For examples of companies that fit each classification, refer to the SLB Masterbrand Architecture .	Descriptive (nonmarked) names for new product families unless identified as an “exception” as defined in the “SLB Masterbrand Architecture”	New product and service families must adopt descriptive names. Assume the SLB masterbrand visual identity.	Family name: Sand consolidation fluids Product or service within the family: Water-based sand consolidation system Usage note: Both family name and products and services within the family use descriptive (nonmarked) names.
	Proprietary product family names identified as an “exception” as defined in the “SLB Masterbrand Architecture”	The current name has high value or equity as a product company name. Retain names only. Assume the SLB Masterbrand visual identity.	Family name: Cameron Wellhead Systems Product or service: Cameron Fontus configurable compact wellhead system Usage note: Cameron is the proprietary brand treated only as an adjective preceding the product or service; “Fontus” is the marked product name; and “configurable compact wellhead system” is the identifier.

2.0 SLB marks and companies’ marks

SLB marks are trademarked names of the company’s products and services. Marks are approved by Brand Checkpoint and maintained by the central Marketing Communications (MarCom) department in coordination with the Division MarComs. Collectively referred to as the “marks list,” published on myHub (<http://markslist.slb.com/>), it is the definitive source for marks used to correctly identify products and services in public-facing materials. There is also a [nonmarks list](#), which contains a list of descriptive product names. These lists are for internal publication and distribution only.

It is always necessary to reference the marks list for guidance on consistent and correct use in brand or marketing material. The marks list also includes additional information about the history of particular marks. The marks list contains active marks and retired (legacy) marks; the latter are those used to identify SLB goods or services that are no longer commercially. Only the Division that owns a retired mark can restore it to active status; other Divisions cannot repurpose retired marks that are not theirs. Whether a mark is active or retired, it must be used correctly.

Proper Use of Marks

Each mark has an identifier, which is a descriptive phrase ending in a noun. The identifier listed on the marks list is considered the “long” or “full” identifier. A mark must always be accompanied by an identifier. There may also be a usage guide for how to use a particular mark, a list of any products within a family mark, or how to shorten the identifier after its initial introduction in the copy. For mark attribution(s) (TM symbol and copyright usage), refer to the guidelines in section [2.1.2 Correct usage](#).

For questions or updates regarding the marks and nonmarks lists, contact the MarCom editorial team.

2.1 Marks of SLB and SLB companies

SLB has both registered and unregistered trademarks, which identify the source of goods, and registered and unregistered service marks, which identify the source of services. All are referred to as “marks.”

2.1.1 Naming of SLB marks and usage guidelines

Understanding how SLB product names are created, organized, and used is vital in maintaining and promoting the company’s brand. SLB Marketing Communication helps to maintain and position our products and product names. Implementing this includes how product names are developed, as well as how these names are treated in marketing collateral. This section provides the name usage guidelines most types of marketing material. a variety of material. Refer to the detailed guidance in the [SLB Naming Process and Convention Guide](#).

SLB product names are organized into the three naming categories: descriptive, retained, and proprietary names. Many of these names identify a family of products, and these product families can fall into any category. A product family is described as a collection of related products, services, applications, tools, software, or solutions that share a single name.

Name categories

Category	Definition	Example
Descriptive names	• Apply to new products without current brand equity • Include new product families • Do not require a trademark symbol and are always lowercase as indicated by the nonmarks list	Example: 3D far-field sonic service • Assumes the Masterbrand verbal identity by using a descriptive name that is representative of the actual offering • Easy to understand what the product is/does and where it fits in our offering • Does not use the trademark symbol and follows the lowercased capitalization as shown on the nonmarks list
Retained names	• Have existing equity in the product name • Retain the product (family) name identified on marks list	Example: Litho Scanner™ high-definition spectroscopy service • Assumes Masterbrand verbal identity by using a marked name that is

	Require a trademark symbol and capitalization according to the marks list	representative of the actual offering and an identifier that is descriptive of the service - Has existing equity in the product name from its commercialization in May 2012 - Uses a trademark symbol immediately following the mark name on first use and follows the capitalization as shown on the marks list
Proprietary names	Have existing equity from an acquired company name, OR - new product families that are large enough and differentiated enough to warrant a proprietary name, along with the marketing commitment to invest (\$\$) in building recognition and equity - require a naming proposal to Brand Council to discuss how the names and products from the company will be added to the marks list, nonmarks list, or dropped entirely	Example: GeothermEx™ geothermal consulting services - Assumes the Masterbrand verbal identity by becoming an SLB mark after company acquisition. These proprietary names must never be used as entities or company names themselves. By becoming a mark, they are treated as adjectives that precede a noun or noun phrase - Has existing equity in geothermal domain - Uses a trademark symbol immediately following the mark name on first use and follows the capitalization as shown on the marks list

All proposals to create a new product family name must be referred to the Brand Council. This process can be initiated through the [Comms and Branding Support](#) portal. Any exceptions to naming are reviewed and approved by Brand Checkpoint.

2.1.2 Correct use

Correct and consistent use of a mark strengthens its marketing value and enables SLB to protect its right to the mark. Use the following guidelines and also refer to special uses in [Section 3, "Marks in tables, titles and headings, figures, digital material, and slides."](#)

2.1.2.1 Marks should be used exactly as they appear on <http://markslist.slb.com/>, paying close attention to capitalization, punctuation, font, and special treatment of components. Marks can be composed of one or more letters, characters, numerals, or a combination of these; marks can also be words, designs (e.g., artwork, graphics, logos, and icons, although Divisions, services, and marks themselves cannot have a logo or icon), or slogans.

2.1.2.2 Mark attribution(s) (TM symbol and copyright usage)

Video

These rules apply for using the ™ for marks within videos (live action or animated):

On first mention of an SLB or SLB-company mark (verbally or visually), an attribution is required in the form of a superscript nonregistered trademark symbol (™). The ™ symbol should always be placed immediately following the mark on the mark's first use.

If a mark is introduced in an opening title card, the ™ symbol will suffice as its first use. If there is no opening title card, use a ™ symbol the first time the mark is shown onscreen. The ™ symbol should always be used on the ending title card if there is a mark used for the title or subtitle (even if a ™ symbol was used on the first use earlier in the video, use the ™ symbol on the ending title card if a mark is present).

The "XXX is a mark of SLB" copyright line is no longer needed if a ™ symbol is used or if there are no marks present in a video. The exception to this rule is when a mark is only mentioned verbally and never shown onscreen –or– when a mark is only present as the short URL name. For example, if a mark is part of the voiceover but is never written out onscreen, then include the copyright line, "XXX is a mark of SLB". When only one mark is present, it will be written as "[Mark] is a mark of SLB." If two marks are present, "[Mark] and [Mark] are marks of SLB. If three or more marks are present, "[Mark], [Mark], and [Mark] are marks of SLB.

Template examples and detailed instructions for video title and ending cards, including mark attributions, are available in the Media Bank Design Template in the [SLB Starter Motion Package](#) (slide 16). If necessary, refer to the [video split screen template](#).

If you have any questions or a situation that's not covered by these rules, contact Patric Perez at pperez19@slb.com.

Electronic, printed or print-ready, or PowerPoint collateral

On first mention of an SLB or SLB-company mark within text, a superscript nonregistered trademark symbol (™) is placed immediately following the mark. Even if the mark is legally registered through the US Patent and Trademark Office (USPTO), use the nonregistered trademark symbol. The symbol is not placed after the mark when the mark appears for the first time in a title or heading; rather, it is placed at the first use of the mark in subsequent text, e.g., body copy. An exception is if the mark appears in only a headline, e.g., a poster or PPT slide, in which case the symbol is placed after its first and only use in the headline.

In printed or print-ready material (e.g., PDF, product sheets), the nonregistered trademark symbol is accompanied by the following copyright or attribution note, or both.

Instructions for mentions of SLB marks, including SLB marks when used with marks of other companies:

When only SLB mark(s) are mentioned	For example:
Use only the copyright line and project docket number in the footer (or near the bottom of collateral):	The NeoSteer™ at-bit steerable system (ABSS) can drill in one run. NeoSteer ABSS enables . . .

Copyright © YYYY SLB. All rights reserved. XX-XX-XXXXXXX	Copyright © 2024 SLB. All rights reserved. 23-WC-XXXXXXX
When SLB mark(s) and third-party marks are mentioned Use the following attribution with the copyright line in the footer (or near the bottom of collateral): [Mark] is a mark of SLB. Other company, product, and service names are the properties of their respective owners. Copyright © YYYY SLB. All rights reserved. XX-XX-XXXXXXX.	For example: The Reda Thermal™ power-efficient geothermal electric submersible pumps reliably lift fluids in high-enthalpy geothermal wells. REDA Thermal pumps enable . . . Third-party marks are mentioned in text or tech specifications: INCONEL® . . . Modbus® . . . Reda Thermal is a mark of SLB. Other company, product, and service names are the properties of their respective owners. Copyright © 2024 SLB. All rights reserved. 23-WC-XXXXXXX
When one or more SLB marks and third-party marks are mentioned Use the following attributions with the copyright line in the footer (or near the bottom of collateral): [Mark] and [Mark] are marks of SLB. Other company, product, and service names are the properties of their respective owners. Copyright © YYYY SLB. All rights reserved. XX-XX-XXXXXXX. If three or more marks are present, use: “[Mark], [Mark], and [Mark] are marks of SLB. Other company, product, and service names are the properties of their respective owners. Copyright © YYYY SLB. All rights reserved. XX-XX-XXXXXXX.	For example: The PowerDrive vortex Max™ high-powered RSS is part of the PowerDrive™ RSS family of fully rotating steerable systems that enable . . . If third-party marks are mentioned in text or tech specifications: PowerDrive and PowerDrive vortex Max are marks of SLB. Other company, product, and service names are the properties of their respective owners. Copyright © 2024 SLB. All rights reserved. 23-WC-XXXXXXX
When only third-party marks are mentioned: Only use the blanket attribution in the copyright line: Company, product, and service names are the properties of their respective owners. Copyright © YYYY SLB. All rights reserved. XX-XX-XXXXXXX	For example: Bing® . . . Company, product, and service names are the properties of their respective owners. Copyright © 2024 SLB. All rights reserved. 23-PS-XXXXXXX

2.1.2.3 A mark is used only as an adjective, except where noted for special use cases.

Marks and identifiers with superscript trademark symbols or subscripts	Correct: eXpand^{BG}™ near-wellbore to reservoir-scale modeling software application for seismic interpretation (BG is a superscript trademark symbol.) Correct: InSitu CO₂™ reservoir fluid CO₂ sensor (The 2 is not subscript in the mark but is in the identifier.)
On first mention of a mark in text The mark should be followed by the identifier noun or noun phrase as listed with it in the marks list. The identifier should be in lowercase, except for any proper nouns within it. Subsequent mentions of the mark can use the last word of the full identifier as a shortened version of it (also called a short identifier.) Use of nouns or noun phrases that are not in the identifier is not permitted.	Incorrect: ABC™ delivers optimal solutions for operators. This brochure describes the growing market for ABC™ technology. Correct: ABC™ analysis behind casing services deliver optimal solutions for operators. This brochure describes the growing market for ABC services. (The noun phrase “analysis behind casing services” is the spelled-out identifier from the marks list and the last word “services” is the short identifier.) Correct: The PowerDrive™ rotary steerable system is available in different sizes. The steerable PowerDrive RSS increases the efficiency of rotary drilling applications. The various BHAs of the PowerDrive system are often used in horizontal wells. (“Rotary steerable system” is the identifier from the marks list, “RSS” is a stand-alone abbreviation for the identifier phrase, and “system” is the short identifier.) Correct: NATCO DUAL FREQUENCY™ electrostatic treaters, coalescers, and desalters use the load-responsive NATCO LRC-II™ smart interface to deliver industry-leading NATCO™ separation technology, which typically provides process improvement that more than doubles the performance of conventional approaches. (NATCO DUAL FREQUENCY, NATCO LRC-II, and NATCO are individual marks, each with its own identifier.)

When a mark includes a string of nouns In these cases, you may choose one as a short identifier, depending on the sentence’s message and context.	Correct, first mention: NATCO DUAL FREQUENCY™ electrostatic treaters, coalescers, and desalters Correct, subsequent mention: NATCO DUAL FREQUENCY electrostatic treater Correct, subsequent mention: NATCO DUAL FREQUENCY electrostatic coalescer Correct, subsequent mention: NATCO DUAL FREQUENCY electrostatic desalter
When a family mark has specific applications, such as services, tools, or solutions, etc. A family mark can substitute its identifier with a specific application of the mark, such as a distinctive service, solution, tool, or component within that family. An application must always be preceded by the family mark and cannot be used on its own. In other words, the family mark can be paired with either its identifier or one of its specific applications as the identifier.	ReSOLVE iX™ intelligent extreme wireline intervention service Applications include: active debris removal tool, anchor-linear actuator, machining tool, milling tool, setting tool, slot cutter, smart shifting tool, tubing cutter. ReSOLVE iX™ tubing cutter
When a mark includes the name of another mark The included mark is not considered the first mention and is not attributed for that mention.	The NATCO™ Jewel T™ hydrocarbon dew point control system recovers valuable hydrocarbon liquids while preparing gas to meet pipeline specifications. Jewel T systems are designed for high-pressure remote well services where centralized gas gathering facilities are not available or practical. (The first mention of the Jewel T mark must be prefaced with the NATCO mark. The NATCO mark is attributed for this as its first mention.)
When a mark is a software Marks for software are allowed to substitute their descriptively named modules (which include plug-ins and apps but not workflows) or the date-based version number for the identifier. Due to this unique relationship, a module, plug-in, or app name can never be used on its own; it must always be preceded by the mark.	For software only Correct, allowed on first use: Petrel™ Reservoir Engineering seamlessly links to INTERSECT™ 2018.1 and established numeric simulators. (The marks list identifier is Petrel <u>subsurface software</u>, and the application name is Petrel Reservoir Engineering. The marks list identifier is INTERSECT <u>high-resolution reservoir simulator</u>, and

It is recommended to avoid using the nonregistered superscript trademark symbol within a mark and module pairing. To satisfy a first mention, state the mark™ and the full identifier of the parent software before naming the module.	the date-based version is INTERSECT 2018.1.) Correct, preferred on first use: The Petrel™ subsurface software includes Petrel Reservoir Engineering. (Petrel subsurface software is the parent software that includes Petrel Reservoir Engineering, the application name.)
Special use for all software on the Delfi digital platform This includes both marked software and unmarked application or module names.	When the Delfi™ mark is used with other software marks or unmarked application names or domain profiles, the phrase “on Delfi” is appended at the end. When the Delfi mark is combined with other software marks, use of the identifier for the other marked software is preferred following the “on Delfi” phrase. Subsequent use of the software mark or unmarked application name omits the “on Delfi” phrase. Correct for marked software on the Delfi platform only (examples): Option 1, first use (preferred): DrillOps™ on Delfi™, an on-target well delivery solution Option 1, subsequent use (preferred): DrillOps solution (Superscript trademark symbol nonregistered trademark symbols are used for both marks on first mention. Subsequent use drops the “on Delfi” phrase and uses at least the marked software’s short identifier.) Option 2, first use: DrillOps™ on Delfi™ Option 2, subsequent use: DrillOps on-target well delivery solution (Subsequent use drops the “on Delfi” phrase and must use the full identifier of the marked software, because it was not previously introduced.)

	Correct for marked software on the Delfi platform only, first mention for press releases and material for print (examples): Option 1, first use (preferred): ProcessOps™ on Delfi™, an assured process optimization solution, enhances facility performance and operations uptime. Option 1, subsequent use (preferred): With ProcessOps solution, your facility's approach to maintenance no longer needs to be preventive or reactive and always playing catch-up. (Nonregistered superscript trademark symbols are used for both marks on first mention. Subsequent use drops the "on Delfi" phrase and uses at least the marked software's short identifier.) Option 2, first use: ProcessOps™ on Delfi™ provides key insights for asset management and process optimization across the facility space. Option 2, subsequent use: ProcessOps assured process optimization solution also powers exploring your future performance options and setting priorities. (Superscript trademark symbol nonregistered trademark symbols are used for both marks on first mention. Subsequent use drops the "on Delfi" phrase and must use the full identifier of the marked software, because it was not previously introduced.) Correct for unmarked applications or domain profiles on the Delfi platform only (examples): First use: Drilling Interpretation on Delfi™ Subsequent use: Drilling Interpretation (Subsequent use drops the "on Delfi" phrase.) First use: Petrophysicist on Delfi™ Subsequent use: Petrophysicist
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	(Subsequent use drops the “on Delfi” phrase.) First use: Reservoir Engineering on Delfi Subsequent use: Reservoir Engineering (Subsequent use drops the “on Delfi” phrase.)
When a mark has new generations A generational phrase can be used with a mark instead of creating a new mark. The generational phrase must precede the mark and cannot be added to the identifier.	Incorrect: TRC-II™ second-generation tubing retrievable charged safety valve Correct: second-generation TRC-II™ tubing-retrievable charged safety valve
When a mark uses models or version numbers Model or version numbers are allowed between the mark and the identifier to denote incremental changes for products or services, instead of creating a new brand identity or mark. If the model is designated by an alphanumeric, it cannot be used within the mark and identifier pair.	Incorrect: PowerJet™ model 2506 deep-penetrating shaped charge Correct: PowerJet™ 2506 deep-penetrating shaped charge Correct: The PowerJet Nova™ 4505 extradeep penetrating shaped charge significantly outperforms previous-generation shaped charges. Incorrect: LEDEEN™ Model HS actuators Incorrect: LEDEEN™ HS actuators Incorrect: LEDEEN HS™ actuators Correct: model HS LEDEEN™ actuators Correct: LEDEEN™ actuator model HS
Adjectives or other modifiers cannot interrupt the mark and identifier pair.	Incorrect: LEDEEN™ scotch-yoke actuators Correct: Scotch-yoke LEDEEN™ actuators Correct: LEDEEN™ actuators featuring a scotch-yoke design
Marks as initialisms or acronyms	Correct: ACQG™ advanced crystal quartz gauge Correct: BEST™ buildup evaluation shut-in tool

2.1.2.4 Never use a mark in possessive or plural form.

Incorrect: eFire’s advanced technology offers a significant safety advantage.

Correct: The advanced technology employed by the eFire™ electronic firing head system provides a significant safety advantage.

Incorrect: Three CT EXPRESSES™ were used in the field.

Correct: Three CT EXPRESS™ rapid-deployment coiled tubing units were used in the field.

- 2.1.2.5 Never use an engineering code name, field term, or internal mnemonic for a product or service that has a mark.

Incorrect: The sensors of the PSP production services platform are used to identify multiphase flow.

Correct: The sensors of the PS Platform™ production logging services platform are used to identify multiphase flow.

Incorrect: PEX measurements set a new benchmark for accuracy and efficiency.

Correct: The combination of wireline measurements from the Platform Express™ integrated wireline logging tool sets a new benchmark for accuracy and efficiency.

- 2.1.2.6 Never form a compound by using an SLB mark and another company's trade name, trademark, or product name. See Section 2.2 for guidelines on the use of other companies' marks.

Incorrect: The OrientXact™-Aqwanot® completion successfully controlled sand and water production.

Correct: Completion with the OrientXact™ tubing-conveyed oriented perforating system and an Aqwanot® downhole oil and water separator successfully controlled sand and water production.

- 2.1.2.7 Marks may also be company slogans. Marks in this category are the only marks that can routinely stand alone and that are not used exclusively as adjectives.

Correct: Apply everywhere what we learn anywhere™ illustrates the company philosophy of sharing best practices and lessons learned.

- 2.1.2.8 For materials submitted to other scientific or technical publications or conferences, overt commercialism is prohibited. The publication or conference organization requirements and guidelines must be followed. Items deemed to be commercial in nature will not be allowed or disseminated per SLB's MyPaper3 protocol and review process. The following guidelines apply when preparing any material for presentation or publication, including content for SPE submissions:

- Avoid any form of commercial promotion or advertising. Replace commercial trade names or trademarked material with generic descriptions whenever possible. Instead of using a SLB mark name, use its identifier.
- Refrain from using brand names, company logos, trademarks, or any branding, including in figures and tables.
- Ensure the text is purely informational and educational in tone or intent. Focus on delivering value through research findings, technical insights, and industry knowledge.

If the organization does not specify guidelines, the guidelines for SPE apply. Adhering to these guidelines maintains the integrity and professionalism of SLB's technical content.

2.2 Other companies' marks

Marketing materials may contain marks of other companies, marks held jointly by SLB and other companies through joint ventures, and marks for products distributed by or under license to SLB.

When marks of other companies are mentioned, the following copyright line can be used at the bottom of the collateral: "Other company, product, and service names are the properties of their respective owners." Because we may not always be aware of a particular company's legal status, or whether companies have changed ownership or are in the process of being acquired (either now or in the future), the copyright statement ensures our collateral remains current. Note: the copyright line is not required on SLB.com web pages, which contain a Terms of Service.

Special uses

- 2.2.1 Other companies' marks can be identified with conventional trademark or footnote symbols and referenced in the attribution note. If symbols are used, a registered trademark is identified on first use with the registered trademark symbol (®), and a trademark or service that is not registered is identified with a superscript trademark symbol TM (™) or SM (SM), respectively. The symbols for ®, ™, and SM are not treated as note reference symbols, such as those used in footnotes. Their function is to identify trademark status.

If symbols are not used to identify marks of other companies, the marks can be named in an attribution note.

- 2.2.2 Marks that are held jointly by SLB and other companies as well as other companies' marks may require different treatment from what is stated in Section 2.1.2.2 because of SLB contracts. Refer to ["Writing and Style"](#) for the footnote symbol hierarchy. Note: In joint ventures, SLB has only a portion of ownership, warranting the different usage guidance.

Correct: Alternate Path® is a registered trademark of ExxonMobil Corp.; technology licensed to SLB. (Wording is per SLB-ExxonMobil contract.)

- 2.2.3 In collateral that does not have a title page, attribution notes for other companies' marks are placed at the end of the document.

Correct: Alternate Path® is a registered trademark of ExxonMobil Corp.; technology licensed to SLB. (Wording is per SLB-ExxonMobil contract.)

Other company, product, and service names are the properties of their respective owners.

- 2.2.4 In a document that has a title page, such as a book, attributions for marks that have contractually specified wording can be placed on the back of the title page if the list does not cause a layout problem. They can be either grouped in paragraph style or placed on separate lines. List the marks in order alphabetically by the first mark for each company. Terminal punctuation is used at the end of each line if notes are on separate lines and is used between groups of marks if marks are placed in paragraph style.

Incorrect: AllPAC® and AllFRAC® are registered trademarks of ExxonMobil Corp. for which SLB holds a license.

Correct: AllFRAC[®] and AllPAC[®] are marks of ExxonMobil Corp.; technology licensed to SLB.

- 2.2.5 In a document that has only a few marks that have contractually specified attribution wording, the accompanying notes can be placed at the bottom of the page where the marks appear.

3.0 Marks in tables, titles and headings, figures, digital materials, and slides

Complete information on the correct use of marks and their identifying trademark symbols (or other symbols) and attribution notes is covered in [Section 2.0 SLB marks and companies' marks](#). The following instructions give exceptions to or clarification of the guidelines. Also, refer to SLB [Brand Guidelines](#) located in Media Bank for related graphic standards for marketing materials.

3.1 Tables, table titles, and figure captions

Table copy, table titles, and captions to figures are considered text, and a superscript unregistered trademark symbol is placed after the SLB mark if its first mention occurs in a table or caption. The short version of the identifier following the first mention of a mark in tables may be used to accommodate space limitations.

3.2 Document titles and headings and ad headlines

The superscript nonregistered trademark symbol is not placed after the mark when the mark appears for the first time in a title or heading; rather, it is placed at the first use of the mark in subsequent text, e.g., body copy. An exception is if the mark appears in only a headline, in which case the symbol is placed after its first and only use in the headline. If the first mention of a mark appears in a title, heading, or ad headline, the short version of the identifier on the marks list may be used for readability and conciseness. The paragraph following the head should mention the mark with proper attribution and using its full identifier. If the style of a title, heading, or ad headline is all capital letters, the marks retain their capitalization on the marks list (e.g., initial cap or lowercase).

3.3 Figure callouts

Marks in figure callouts are used as adjectives, and the short version of the identifier may be used if space for type is limited.

If the first use of a mark occurs in a figure callout but the mark is also mentioned elsewhere in the document, place the superscript nonregistered trademark symbol on the first mention in the text or table.

If a mark occurs only in a figure callout and nowhere else in the text, but it is not reasonable to either recast the text to include the mark or rewrite the mark in the callout to a generic description, the superscript nonregistered trademark symbol is used for the mark in the callout.

3.4 Digital materials

- 3.4.1 slb.com webpages: SLB marks are addressed in the link "Terms," which is at the bottom of every public SLB webpage on slb.com. However, still give proper attribution to all

marks discussed in web material. The identifier from the marks list is used per [Section 2.1.2.3](#) in conjunction with the first mention of the mark in the text on a webpage. Use of the short identifier is allowed because of space constraints.

Because web technology facilitates nonlinear access of information, reintroduce the mark and its identifier as possible wherever a reader is most likely to access a digital mention from a link or by using a search engine.

It is recognized that full compliance with principles for first mention of marks (using long identifier) is not possible to achieve in all digital material. It is also recognized that it is difficult to consistently use superscript nonregistered trademark symbols for marks in content that can be used in both digital and printed publications, like software training content. In long technical documentation, such as training guides, the Table of Content can serve as a natural and logical place for the first use of a mark.

- 3.4.2 Digital ads: Digital ads posted on other websites do use the superscript nonregistered trademark symbol. If it is used in a title or text it can use the short identifier to fit space constraints. If the ad has a large graphic area, an attribution statement can optionally be included on it.
- 3.4.3 Posted or linked files: Marks in PDFs or other file formats posted to the web are treated as printed documents (see [Section 2.1.2](#)).
- 3.4.4 Email: Marks in marketing material that is published as email messages are treated identically to marks in printed material (see [Section 2.1.2](#)).

3.5 Slides presentations

Because slides may be rearranged for custom presentations, an attribution note is used to identify SLB marks on each first mention per slide. If marks of other companies are used, they are attributed on a slide that immediately follows the title slide (see [Section 2.2](#)): “Other company, product, and service names are the properties of their respective owners.” If no SLB marks are used in the presentation but other companies’ marks are, use “Company, product, and service names are the properties of their respective owners.” Slide templates can be downloaded from [“Brand Elements and Templates.”](#)

4.0 Copyright Protection and Permissions

This section describes the copyright that applies to materials created by or for SLB and gives guidelines on how to use the copyrighted works of others. It also addresses the permissions required before photographs of people can be used in SLB publications and how to make and respond to requests to reproduce copyrighted materials.

Copyright is a form of intellectual property law that protects original works of authorship—whether unpublished or published in print or electronically or by any other means—as long as they are in a tangible form. Copyright does not protect facts, ideas, systems, or methods of operation, although it may protect the way these things are expressed.

Generally, the owner of a copyright has the exclusive right to protect the copyrighted work from unauthorized use by others. Copyrighted work cannot be reproduced, used as the basis for derivative works, distributed, displayed, or performed without the consent of the owner.

An example of a common copyright violation is cutting and pasting content from one website to another or to another media form without the permission of the copyright owner. Another is scanning a cartoon from the newspaper to include in a newsletter without the permission of the

copyright owner. It is just as much a copyright violation to reproduce the owner's work for internal SLB purposes as it is to reproduce the work for distribution outside SLB.

4.1 SLB copyright identification

All SLB documents identify the copyright owner as SLB; names of Divisions are not included in the copyright line:

Correct: Copyright © [year] SLB. All rights reserved.

Use a copyright date range of © SLB xxxx–yyyy for a derivative work (meaning with many iterations over the years), with xxxx the date of the first publication and yyyy the date of the latest release. If the work is an SLB acquisition, the first publishing date remains the date of first publication and is not modified to the date of first SLB publication.

Because of space constraints and their visual design emphasis, advertisements and tradeshow banners and posters may use a shorter statement:

Correct: © [year] SLB.

For public SLB webpages on slb.com, the link "[Terms](#)" includes the copyright identification.

4.2 Copyright ownership of works for hire

The Legal Department has prepared standard contracts for vendors and contractors to ensure that SLB owns the copyrights to works prepared by people such as writers, artists, designers, and photographers. These contracts are available from Human Resources and must be signed by the person performing a work for hire and by the appropriate SLB authority (designated by Human Resources).

4.3 Permissions to reproduce copyrighted materials

SLB authors, writers, and editors are typically concerned with two types of permissions:

- permission SLB must obtain from a copyright holder before we can legally reproduce copyrighted material
- permission SLB must give to others before they can legally reproduce materials copyrighted by SLB.

A request for permission to reproduce material must be in writing (letter or email) and include sufficient information for an informed decision. In turn, the reply to the request must include the scope of the permission granted, the attribution required, and fee, if applicable. The following information should be included in a request to expedite the permission process:

- title, date of publication, identification of the original work, and specific information (page number, figure number, webpage address, etc.) on what is to be reproduced
- title and information about the publication that will contain the reproduced material (purpose of publication, intended audience and distribution, context that reproduced material will be presented in and, if possible, a copy of the relevant context, such as a book chapter, CD-ROM, or website page)
- scope of permission requested (one use of material, unlimited use, translation, etc.).

SLB personnel who receive requests for permission to reproduce SLB materials should direct the requests to the appropriate Division MarCom Manager. If permission is granted to reproduce, the following acknowledgment is required:

This graphic is copyright SLB, Ltd. Used with permission.

An alternative acknowledgment typically used in media relations work is the following:

Image courtesy of SLB.

SLB has an agreement with the Copyright Clearance Center (CCC) that allows reproduction in printed form of many copyrighted articles and publications without having to seek specific permission to make such copies. The Legal Department should be contacted for specifics of which publications are covered and what is permitted. A major exception to the CCC agreement is SPE papers, which are covered under a separate agreement and for which the reproduction right is subject to several restrictions.

For permission received for SLB to reproduce material, use the attribution requested by the provider as possible.

4.3.1 Reproducing without permission (fair use)

The doctrine of fair use provides for a limited use of copyrighted materials without permission for educational, research, and review and reporting purposes but does not define the limits. The law of fair use is complex and subject to many exceptions. To see if an excerpt from a copyrighted work can be used, contact the Legal Department for advice on whether the use would constitute a fair use.

4.3.2 Retaining SLB copyright to graphics in technical papers or other publications

Many technical societies require authors to assign the copyright to their papers to the society before the paper can be presented at a conference or symposium or published in a proceedings volume. If such papers contain graphics from SLB communications and marketing materials—including material used externally (brochure, ad, presentation, etc.), internally, and for investor relations (annual report, presentation, etc.)—authors must protect SLB's right to reprint the material by excluding such figures from the copyright reassignment.

4.4 Permissions required for photographs and photo shoots

The following permissions apply to all photographs uploaded to the [Media Bank](#).

- 4.4.1 For photo shoots that are not on SLB premises, obtain written permission to conduct the shoot from the appropriate authority, such as the client or property owner. Email permissions are also acceptable.
- 4.4.2 Acquire a signed [release form](#) from each person photographed, both SLB personnel and people not employed by SLB.

Writing and Style

Legal information

Copyright © 2024 SLB, unpublished work. All rights reserved.

This work contains the confidential and proprietary trade secrets of SLB and may not be copied or stored in an information retrieval system, transferred, used, distributed, translated, or retransmitted in any form or by any means, electronic or mechanical, in whole or in part, without the express written permission of the copyright owner.

Trademarks & service marks

"SLB," the SLB logotype, and other words or symbols used to identify the products and services described herein are either trademarks, trade names, or service marks of SLB and its licensors, or are the property of their respective owners. These marks may not be copied, imitated, or used, in whole or in part, without the express prior written permission of their owners. In addition, covers, page headers, custom graphics, icons, and other design elements may be service marks, trademarks, and/or trade dress of SLB and may not be copied, imitated, or used, in whole or in part, without the express prior written permission of SLB.

Overview

Our SLB brand personality inspires everything we do and say. When someone has a great personality, we are naturally drawn to them. It's a trait that makes us want to spend time with another person and what makes them memorable for the right reasons. When it comes to our SLB brand, personality works in the same way. A brand personality is the set of human characteristics and traits that a brand upholds: It makes a company relatable. A strong brand personality can build brand loyalty and equity. It gives our brand an emotional foothold in the mind of our audience.

Your tone and choice of words should give life and meaning to the message to support the SLB brand. As the author of material on slb.com, a technical paper, or other SLB publication, you determine the value, scope, and content of your presentation. You must be able to support your ideas, especially if your assertions contradict accepted scientific knowledge. References and measurements must be accurate; theoretical justifications and examples must be used to substantiate your statements.

Tone is dependent on your audience and the type of marketing collateral being produced: This can range from a more conversational tone to one that is more technical. For technical content, technical accuracy is the backbone of SLB leadership. It bears directly on the impact of the publication and helps uphold the company's reputation for high-quality technology. Writers must be diligent to ensure that all SLB documents are technically accurate. Technical content is directly influenced by the language used to describe it; therefore, weigh your words as carefully as your data.

Appropriate vocabulary and a logical structure help communicate your message and lend credibility to your ideas. That credibility is weakened by inconsistencies in terminology and style as well as by typographical errors, misspelled words, and incorrect grammar. The best technical message may be lost to the audience in a poor presentation. The following guidelines can help you avoid this pitfall.

These guidelines for print materials and slb.com assume that you already know the basics of effective writing. They are intended to steer you through some of the more obscure details of style and to set the standards that present a consistent SLB image. Consult the *SLB Communications Handbook* first if you have a question; if your question is not addressed, then check these additional authorities on which SLB style is based:

- [*Merriam-Webster's Collegiate Dictionary*](#)
- *Oxford Style Manual* (2003), comprising two titles previously published as *The Oxford Dictionary for Writers and Editors* and *The Oxford Guide to Style*
- [*Society of Petroleum Engineers Publications Style Guide*](#)
- [*The Chicago Manual of Style*](#).

Additional resources are listed in the "[Suggested reading](#)" section of this document. Guidance for print templates is in the SLB "[Brand Elements and Templates](#)." Guidance for slb.com elements is in "[Writing for Our Public Websites](#)."

If you have questions, comments, or suggestions about the "Writing and Style" guidelines, email Victoria Morales at vmorales7@slb.com.

Writing for your audience

Most SLB external marketing publications, print and slb.com, are created for a global audience and are distributed internationally. US English is the language used for this audience. The "Writing and Style" guidelines identify style variations for digital and print communications and for local audiences.

For external marketing publications addressing a local audience and distributed only locally or regionally, the appropriate English language variation is used. If you are writing for a local audience, it is best to consult authoritative editorial resources for the area, such as a dictionary and writing and style guides. See resources listed in “[Suggested reading](#)” for important issues in communicating with specific audiences. For the best results and the clearest communications, work with a local resource or vendor trained in SLB “[Brand Elements and Templates](#).”

Grammar

Because sentences are the building blocks of thought, the communication of a message is directly related to the clarity and readability of each sentence. Sentence length is one measure of readability because the reader must read, understand, interpret, and remember information sentence by sentence. Readability studies have shown that sentences of 20 or fewer words are the most easily comprehended.

While shorter sentences are preferred, varying the length of sentences adds interest to your writing. Reserve long sentences for presenting involved concepts and for elaborating on a point that requires some thought. Use short sentences for stating clear, crisp thoughts. Regardless of sentence length, state your point clearly and directly. You can do this by following these rules:

- Position the subject as close as possible to the verb.
- Place modifiers as close as possible to the words they modify.
- Use conjunctions such as **and**, **or**, **but**, and **then** to show connection, contrast, and sequence.

The most emphatic positions within a sentence are at the beginning or end. Key ideas should have important positions, and less important ideas should be subordinated. Transposing words or phrases changes the emphasis in the following sentences:

- Coal mining has been an important industry in this region for three decades.
(Emphasis on coal mining)
- or*
- For three decades coal mining has been an important industry in this region.
(Emphasis on time frame)
- or*
- In this region coal mining has been an important industry for three decades.
(Emphasis on location)

Along with these rules, check your writing for common grammatical errors.

Sentence structure

A complete sentence has at least one independent clause (a group of words containing a subject and verb) that stands by itself to express a complete thought. Ending a sentence too soon results in a sentence	Incorrect: The well that the company drilled. (sentence fragment) Incorrect: This is the well that the company drilled, it is 2,745 m deep. (Incorrectly joining two independent clauses produces a run-on sentence.)
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fragment; ending it too late results in a run-on sentence.	Correct: This is the well that the company drilled; it is 2,745 m deep. Correct: This is the well that the company drilled, and it is 2,745 m deep. Correct: This is the well that the company drilled. It is 2,745 m deep.
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Agreement between sentence elements

The subject of a sentence should agree in number with the sentence verb. Compound subjects and words and phrases between the subject and the verb are common pitfalls in subject-verb agreement. For example, phrases such as with, along with, including, as well as, together with, and plus between the subject and verb do not make a singular verb plural. Collective nouns and pronouns take either a singular or plural verb, depending on whether they mean a unit or separate items, respectively. UK English: The practice of construing collective nouns either with a singular verb to emphasize unity or with a plural verb to emphasize individuality is well established. Names of institutions and political entities are always treated as singular.	Incorrect: The number of workstation users have increased twofold. Correct: The number of workstation users has increased twofold.
	Incorrect: The composition and quality of each sample was impressive. Correct: The composition and quality of each sample were impressive.
	Incorrect: The BHA, plus the tools it requires, were packed in the shipping container. Correct: The BHA, plus the tools it requires, was packed in the shipping container.
	Correct: The driller, as well as the customer's representative, arrives tomorrow.
	Correct: Only 3 m [10 ft] of sandstone was drilled before the bit failed.
Pronouns and the nouns they replace must agree in number, gender, and case.	Incorrect: Smaller, more complex fields are being developed. This requires more log data for reservoir evaluation. (This, which is singular, refers to fields, which is plural.) Correct: Smaller, more complex fields are being developed. These require more log data for reservoir evaluation.
	Incorrect: Everyone is responsible for their safety equipment. (his or her?) Correct: All employees are responsible for their safety equipment.
The antecedent of a pronoun must be unmistakable. (An antecedent is the noun or pronoun to which a following pronoun refers.)	Incorrect: A safety joint is run above the packer. It can be unscrewed to remove the upper part of the string. (The antecedent of the pronoun it is ambiguous.) Incorrect: A safety joint is run above the packer, which can be unscrewed to remove the upper part of the string.

	Correct: A safety joint, which can be unscrewed to remove the upper part of the string, is run above the packer.
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Verb tense

Present, past, or future action should be described consistently within the same sentence.	Incorrect: The geoscientist processes the data and stored the file on tape. Correct: The geoscientist processes the data and stores the file on tape.
Use the past tense to state what was done.	Incorrect: We tested the core samples and confirm the presence of hydrocarbons. Correct: We tested the core samples and confirmed the presence of hydrocarbons.
Use the present tense for scientific facts and truths.	Incorrect: Water will freeze at 0 degC. (future tense) Correct: Water freezes at 0 degC.
Present and past tenses are used as appropriate for results, discussion, and conclusions.	Correct: Pressure measurements demonstrated that six of the zones consisted of reservoir-quality rock.
	Correct: In "Guidelines for Cement Sheath Evaluation," Goodwin pointed out that almost any solid material in the annulus provides adequate support to the casing.
	Correct: The high resolution of vertical seismic profiles enables interpreters to find faults, refine structures, and detect stratigraphic boundaries away from the wellbore.
Use the present tense to distinguish current theory from previous work.	Correct: Jones (1976) argues that the type of material in the annulus must be known to determine whether the zone has good isolation, whereas Smith (1935) believed that the quality of zonal isolation depended solely on the formation characteristics.

Modifiers

The most common word order in sentences is subject, verb, and object. Modifiers added to expand and clarify meaning should be placed next to or near the words they modify. The importance of word position is illustrated by how the meaning of the following sentence changes by moving the adverb only .	During perforating only, qualified personnel are allowed on the rig. (Qualified personnel may not go on the rig at any other time.)
	<i>or</i>
	During perforating, only qualified personnel are allowed on the rig. (No one else may go on the rig at that time.)
	<i>or</i> During perforating, qualified personnel are allowed on the rig only. (They may not go anywhere else.)

	Incorrect: The core samples were evaluated from Texas. Correct: The core samples from Texas were evaluated.
A participial phrase must logically modify its antecedent. (A participle is a verb that functions as an adjective to modify nouns and pronouns. It usually ends in –ed or –ing. A participial phrase is a participle, adjectives that describe it, and complements that complete the meaning of the phrase.)	Incorrect: Comparing laboratory and field results, it is evident that the gas is prevalent in the zone of interest. (It is unable to make comparisons.) Correct: Comparing laboratory and field results, we learned that gas is prevalent in the zone of interest.
	Incorrect: Because they are highly sensitive to changes in pressure, engineers should specify quartz crystals for the pressure sensors. Correct: Because they are highly sensitive to changes in pressure, quartz crystals should be specified for the pressure sensors.

Parallel construction

Sentence elements that are parallel in meaning should also be parallel in style.	Incorrect: The consultant suggested restoring reservoir pressure and resumption of production. Correct: The consultant suggested restoring reservoir pressure and resuming production.
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Complete comparisons

When making a comparison, make sure that the structure of the sentence leads to a comparison of like elements.	Incorrect: These producing wells are similar to the Delaware Basin. Correct: These producing wells are similar to those in the Delaware Basin.
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Verbal identity, tone, and voice

Verbal identity

What you say and how you say it should be crafted carefully. It reflects SLB. (Reference the [“Brand Guidelines”](#))

A clear verbal identity

- helps us to stand out, creating differentiation from competitors
- communicates our personality and shapes audience perception
- builds consistency, trust, and familiarity
- helps us sound our best, always.

Our SLB verbal identity incorporates the language—the words, phrases, names, and terms that we use to describe our brand—as well as the tone, voice, style, and messages we use to

communicate. Both internally and externally. Locally and globally. When writing for the web, our verbal identity is made up of four building blocks:

- **Personality:** how we want to be perceived (e.g., bold, bright blue, balanced). Inspirational, not dry. Think solutions, not problems. Uplifting and positive.
- **Voice:** how we talk to the world. Authentic, not corporate. Write for a person, not a product. Turn up the warmth, connection, and understanding.
- **Language:** how we write. Use simple and clear words. Blatant promotional language is not used. We state facts, not unsubstantiated claims.
- **Tone:** how we modulate our verbal identity. Conversational, not dictatorial. Intelligent, not academic. Connect with the audience and speak to their needs. Draw them into the story, not overwhelm or bore them.

Tone

When writing for slb.com and some ads, in general, use a more conversational tone, where appropriate (Reference the [“Dialing up or down different aspects of our tone”](#) table):

- **We** and **our** are substituted for most mentions of SLB.
- Readers are often addressed as **you**, instead of referring to them in the third person.
- Contractions are freely used.
- Starting sentences with conjunctions is allowed.
- Sentence fragments are allowed, within context, to substitute for complete sentences.

Dialing up or down different aspects of our tone to emphasize a particular aspect of our personality.			
Personality	Tone	Language	Collateral
Bold (leadership and innovation)	Confident (not arrogant) Inspirational (not dry) Intelligent (not academic)	Concise Optimistic Engaging Human Clear	Content in which we clearly have a bold ownership story to tell. Examples: Thought leadership Customer solutions: case studies and tech reports News and press releases Announcements Strategic messaging
Bright blue (personable)	Intelligent (not academic) Conversational (not dictatorial) Authentic (not corporate)	Concise Optimistic Engaging Human Clear	Content and channels where showcasing people and creating dialog is essential. Examples: Social media: LinkedIn, Instagram, etc. People stories

			Careers Brand campaigns Events
Balanced (science, impact, and integrity)	Confident (not arrogant) Intelligent (not academic) Authentic (not corporate)	Concise Optimistic Engaging Human Clear	Content that is essential for demonstrating scientific rigor. Examples: Reports Customer solutions: case studies, tech reports, etc. Technology Sustainability content Impact statistics and infographics

Voice

The choice of verbs can make a great difference in the force of your writing. If the subject of the sentence is the agent of the action, the voice is active; if the subject is the recipient of the action, the voice is passive. Passive voice is generally less emphatic than active voice because the emphasis is shifted to the object acted on rather than the agent of the action. Good writing blends both voices with the active voice predominating.	Passive: The limestone is cut obliquely by fault lines. Active: The fault lines cut the limestone obliquely.
Do not use passive voice to avoid the first person pronouns <i>I</i> , <i>me</i> , <i>we</i> , or <i>us</i> . Although it is sometimes appropriate to avoid mention of the actor, in many cases the passive voice creates ambiguity.	Passive: An experiment was done to show well flowing pressures. Active: We conducted an experiment to show well flowing pressures.
Three techniques for converting passive to active voice are to reverse the word order in the sentence or clause, change the verb, or reword the sentence.	Passive: These techniques are described in great detail in the new manual. Active: The new manual describes these techniques in great detail.
	Passive: Introduction of these techniques is expected to improve productivity. Active: Productivity will probably improve when these techniques are introduced.

Spelling

Correct spelling reinforces both the clarity and credibility of any document. Misspelled words divert the reader's attention away from the message and may cause the writer to appear incompetent, careless, and potentially untrustworthy. Therefore, proofread your documents carefully before publishing.

A computerized spelling-check program may help you find spelling errors in electronic copy; however, spelling checkers should not be substituted for proofreading because they cannot account for meaning and context. For example, a spelling checker will not find an instance where the word **from** is typed instead of **form**. Also, some programs pass over words that are not in the program dictionary.

Challenge the spelling of every word in your document. Oilfield and technical terms, acronyms, and other words are in the list of [“Spellings and Terminology.”](#) If a word is not listed there, consult *Merriam-Webster’s Collegiate Dictionary* (11th ed.), which is one of the most widely accepted authorities for spelling. The first listed spelling of a word is preferred. [“Suggested reading”](#) lists geoscience glossaries, dictionaries, and other resources that can be consulted.

American spelling (US English) is used for slb.com and all SLB documents that have an international distribution or audience (except for proper names of people, companies, and institutions and for direct quotations). UK English or other alternatives can be used for external marketing materials that will be distributed only locally or regionally outside the United States. If you are writing for a local audience, consult an authoritative resource or dictionary for the area to ensure that your spellings are correct.

Compound words

To determine whether a compound word should be closed (**backream**), hyphenated (**back-ream**), or open (**back ream**), first check the list of [“Spellings and Terminology.”](#) The format for various prefixes and suffixes is also on the list. If the compound word is not on the list, then consult *Webster’s*. If the word is not in *Webster’s*, then consult [The Chicago Manual of Style](#), particularly the hyphenation guide in Section 7.85. Open compounds that are well-known terms are usually not hyphenated when they are used as adjectives before the modified noun. If the compound adjective comes after the modified noun, an added hyphen is necessary only to avoid ambiguity.

Special characters, symbols, and word formats

For Microsoft® Office documents, symbols and special characters should be inserted with [American National Standards Institute \(ANSI\) codes](#) because they are more stable than those inserted by using the Insert menu or keyboard shortcuts. To insert an ANSI symbol or character code, NUM LOCK must be on. Hold down the ALT key and use the numeric keypad to type in the four-numeral code. Similar [character codes are used for Macintosh®](#) computers.

Letters that represent shapes (for example, **O-ring**) are capitalized and set in roman type.

Italics

Italics can be used to flag a word, such as a key term on its first mention or definition, but they should be used only sparingly for emphasis in technical writing.

Foreign words and phrases are set in italics only if they are not familiar. However, if an unfamiliar foreign word is used repeatedly throughout a work, it need be italicized only on its first occurrence. Familiar foreign words and phrases (for example, **a priori**, **résumé**) are listed in the English language section of *Webster’s* and are set in roman type.

The names of specific ships or other such vessels are in italics, but the name of a make or class is not (for example, The *DeepSTIM II* offshore stimulation vessel is the flagship of the DeepSTIM™ fleet). The only use of italics on slb.com is for names of specific ships and vessels.

Quotation marks can also be used to identify a special term on its first mention. Do not use both quotation marks and italics for this purpose and maintain consistency of use within the same document by not mixing the two.

Foreign words

Foreign words and phrases familiar to most readers and listed in *Webster's* are not italicized if used in an English context; they should be spelled as in that dictionary. German nouns that are in *Webster's* are lowercased. If following these guidelines could cause confusion, it is best to italicize, spell, and capitalize foreign terms as they are in the original language. If a familiar foreign term occurs in the same context as a less familiar one, maintain consistency by italicizing both or neither.

Word choice and usage

Good writers make all words, sentences, and paragraphs count by removing unnecessary words without sacrificing clarity. They avoid using excessive adjectives, adverbs, and intensifier words such as **very** and **quite**. Effective writers also achieve conciseness by eliminating undesirable repetition in their text. Check your writing for the following examples of wordiness.

Choosing the right words

Avoid using complex words when direct, natural words or phrases are available. Words with many syllables slow the reader and may sound pretentious or stilted.

Instead of	Use
along the lines of	like
assistance	aid, help
at present, at this point in time	now
due to the fact that, in view of the fact that	because
employed	used
facilitate	help, ease
foregoing	this, these
furthermore	also
have to, need to	must
implement	use
in advance of	before
inasmuch as	because
in case of	if
initiate	begin
in order to	to
in spite of the fact that	although
in the event that	if
in the final analysis	finally
in the foreseeable future	soon
in the majority of instances	usually
in the vicinity of, in close proximity to	near
ought to	should
with regard to, with reference to	about, concerning
with the exception of	except

Avoid timeworn clichés and jargon. Be sure the terms you use are readily understood by your reader. If your document will be distributed to different groups (for example, engineers and geologists), avoid the jargon of either group.

Instead of	Use
quick as a flash	quickly
last but not least	finally
as plain as day	obvious
step change	improvement
game changer	revolutionary

Eliminate redundancy.

Instead of	Use
actual measurement	measurement
bisect into two parts	bisect
close proximity	proximity
completely absent	absent
dropped down	dropped
final conclusion	conclusion
first began	began
gray in color	gray
join together	join
low-lying depression	depression
mix together	mix
period of time	period, time
refer back	refer
rise up	rise
surrounded on all sides	surrounded
throughout the entire	throughout

Avoid words that can have a triggering effect for external audiences.

Potentially emotive word or phrase	Alternative wording
environmentally friendly; environmentally acceptable	ecologically sound; environmentally benign
green (unless in the context of an official classification, such as "green chemical" official label in Norway)	ecologically sound
costly; expensive	cost prohibitive (Technology benefits should be priority, e.g., increased production; improved efficiency; reduced capex; reduced opex; reduced cost; reduced risk)
safe, safely (For liability reasons, avoid these absolutes. "Safer" is generally okay when comparing with a conventional approach)	reliable, reliably, secure

Choose the correct word. The following words are frequently misused.

Ability, capacity, able, capable. <i>Ability</i> is the power to do something. <i>Capacity</i> is the power to contain or receive. <i>Able</i> indicates the possession of ability, whereas <i>capable</i> means having adequate capacity to do, make, or receive action.	Incorrect: The capacity to successfully complete the fracturing operation depended on the ability of the tank. Correct: The ability to successfully complete the fracturing operation depended on the capacity of the tank.
	Correct: The tool is able to log rugose boreholes. Correct: The tool is capable of logging rugose boreholes.
About, around. Both <i>about</i> and <i>around</i> are used to describe a physical position, but only <i>about</i> is used to mean approximately.	Correct: The engineer walked about the rig to inspect it. Correct: The engineer walked around the rig to inspect it.
	Incorrect: Around 90 samples were received. Correct: About 90 samples were retrieved.
Accuracy, precision. <i>Accuracy</i> refers to the correctness of a measured or computed value or a statement; <i>precision</i> is used for values reported in very small increments and describes the degree of coincidence (repeatability and reproducibility) of repeated measurements.	Correct: We checked the manufacturer's documentation to confirm the accuracy of our field measurement.
	Correct: The series of identical measurements was precise; however, the value of 5.42 g/cm³ was not necessarily more accurate than the previously determined value of 5.4 g/cm³.
Actually, basically (see “essential, essentially”), literally. <i>Actually</i> and <i>literally</i> mean truly but are usually not necessary as descriptors. <i>Basically</i> means fundamentally; avoid its usage as chiefly or for the most part.	Incorrect: The technology actually increases profits in most new applications. Correct: Despite its touted efficacy, the technology actually increased application time. Incorrect: Market growth basically follows technology growth. Correct: Market growth mainly follows technology growth.
Acute, chronic. <i>Acute</i> means of great importance, intense, or having a rapid onset followed by a short course, which is the opposite of the meaning of <i>chronic</i> as long-lasting or recurrent. The use of <i>chronic</i> for habitual has negative connotations.	Correct: The acute lack of funding stalled the development project. Incorrect: Chronic feedback improved the new hires' performance. Correct: Constant feedback improved the new hires' performance.
Affect, effect. <i>Affect</i> (verb) means to influence. <i>Effect</i> (noun) is the result, and <i>effect</i> (verb) means to achieve or bring about a change.	
Allow, enable, permit. <i>Allow</i> means to let have and implies no restraint. <i>Enable</i> means to give an ability. <i>Permit</i> specifically involves giving permission.	

Alternate, alternative. <i>Alternate</i> means one after the other. <i>Alternative</i> means one or the other.	
Although, whereas. <i>Although</i> means regardless of the fact that. <i>Whereas</i> means while on the contrary and implies a clear contrast between two statements.	
Among, between. Use <i>among</i> when referring to three or more items and <i>between</i> when referring to two. <i>Between</i> can be used for more than two items if each item has an individual relation to the others.	Incorrect: The maps were distributed among the driller and the engineer. Correct: The maps were distributed between the driller and the engineer. Correct: The maps were distributed among the driller, engineer, and geologist. Correct: Obtaining understanding between oil-producing countries is desired.
And, or. Avoid the slashed use of <i>and/or</i> . Either word may be sufficient; the words <i>or both</i> may be added.	Incorrect: The bedding planes may include shale and/or sandstone. Incorrect: The bedding planes may include shale/sandstone. Correct: The bedding planes may include shale or sandstone or both. Correct: The bedding planes may include shale, sandstone, or both.
As much as, up to. <i>As much as</i> is preferred for describing an upper limit. <i>Up to</i> implies a position in space.	
As, since, because. <i>As</i> is often imprecise when used as a conjunction indicating cause. Choose the precise word: <i>while</i> , <i>when</i> , <i>because</i> , or <i>since</i> . When <i>since</i> or <i>because</i> introduce a dependent clause, <i>because</i> always denotes direct cause and <i>since</i> often connotes time and may imply cause, reason, or result.	Incorrect: Since the FMI™ fullbore formation microimager tool provides images of the borehole wall, thin events can be detected. Correct: Because the FMI™ fullbore formation microimager tool provides images of the borehole wall, thin events can be detected.
	Incorrect: As the software was installed, the speed of data interpretation has tripled. Correct: Since the software was installed, the speed of data interpretation has tripled.
Assure, ensure, insure. <i>Assure</i> means to encourage. <i>Ensure</i> means to make certain. <i>Insure</i> is used to refer to underwriting a loss.	
Based on, on the basis of. <i>Based on</i> modifies the noun in the main clause of the sentence; <i>on the basis of</i> modifies the verb.	Incorrect: Based on the measurements, researchers estimate . . . (Researchers are not based on the measurements; the estimates are.) Correct: Researchers' estimates were based on measurements . . .

Can, may. <i>Can</i> suggests ability; <i>may</i> expresses permission or possibility.	Correct: We can evaluate the formation with the new tool. (It is feasible.) Correct: We may evaluate the formation with the tool. (implies that chances are likely that it will be evaluated or that permission will be granted)
Client, customer. Use <i>customer</i> , not <i>client</i> , for business relationships.	Correct: Downhole test data was streamed in real time to the customer's office.
Cloud: Use <i>in the cloud</i> for cloud computing.	Incorrect: Access your data on the Cloud. Correct: Access your data in the cloud.
Compare to, compare with. <i>Compare to</i> implies resemblances between different ideas or things (you and an elephant). <i>Compare with</i> implies contrasts between similar things (you and me). Use <i>similar</i> instead of <i>comparable</i> to indicate likeness.	Correct: The firing of a seismic shot has been compared to a mild earthquake.
	Correct: Winter weather in the Gulf of Mexico is calm compared with that of the North Sea.
	Incorrect: The high rate of data transmission from the LWD tool is comparable to that of conventional wireline tools. Correct: The high rate of data transmission from the LWD tool is similar to that of conventional wireline tools.
Competitor, third party, or conventional We don't acknowledge that we have competitors. Instead of "competitor," mention "third-party company" or "conventional technology."	Incorrect: Competitor company Competitor drill bit Correct: Third-party company Conventional drill bit
Complement, compliment. <i>Complement</i> means to complete or make whole. <i>Compliment</i> means to praise or admire.	
Compose, comprise, consist, constitute, include. It takes A, B, and C to <i>compose</i> or <i>constitute</i> a whole, but the whole <i>comprises</i> A, B, and C. Avoid <i>is comprised of</i> . <i>Include</i> indicates an incomplete listing of constituents.	Incorrect: Four companies comprise the joint venture.
	Correct: The joint venture comprises four companies. Correct: The outcrop includes limestone and sandstone. (implies that other constituents may also be present)
Conspicuous, prominent. These words may both mean immediately noticeable, but <i>prominent</i> suggests a physical jutting out. Rock colors may be <i>conspicuous</i> , but a salt dome may be both conspicuous and prominent.	
Continual, continuous. <i>Continual</i> relates to time and implies unceasingly recurrent activity. <i>Continuous</i> refers to uninterrupted action in time or unbroken extent in space.	

Data, facts, information. <i>Data</i> generally refers to organized information used to reach conclusions or verify facts. <i>Information</i> is knowledge acquired through study, experience, or hearsay and has a more general connotation than <i>data</i>. <i>Data</i> can be used as a singular (for references to the entire data collection as one unit) or plural (for references to individual results) noun with the corresponding verb form.	Correct: Is the field data for last month available? Correct: The data from the latest perforating tests are compared with the previous results. Correct: Monitoring data from all wells in the field were streamed over one connection.
Digitization, digitalization. <i>Digitization</i> is the process of changing from analog to digital form. <i>Digitalization</i> is the use of digital technologies to change a business model and provide new revenue and value-producing opportunities.	
Due to. <i>Due to</i> is correctly used to modify a noun or pronoun that directly precedes it or following a form of the verb <i>to be</i>. Avoid its use as an adverb and use <i>since</i>, <i>through</i>, <i>because of</i>, <i>caused by</i>, <i>owing to</i>, or <i>resulting from</i> instead.	Incorrect: We were unable to obtain samples due to the collapsed wellbore. Correct: We were unable to obtain samples because of the collapsed wellbore. Correct: Delays due to bad weather are common in North Sea operations. Correct: Cancellation of the well test was due to the weather.
e.g., i.e. <i>E.g.</i> means for example; <i>i.e.</i> means that is. (See guidance on using commas with i.e. and e.g.)	
Employ, use, utilize. These words are often interchangeable. <i>Utilize</i> implies a practical or profitable use, and <i>employ</i> also means to hire. Both are more stilted than <i>use</i>.	
Entitled, titled. <i>Entitled</i> means the right to do or have something. <i>Titled</i> means to have a title.	Incorrect: He was titled to the promotion because he wrote the report entitled “Downhole Testing Safety.” Correct: He was entitled to the promotion because he wrote the report titled “Downhole Testing Safety.”
Essential, essentially. Both words mean part of the nature or essence, or a basic property. Do not substitute these words for such words as <i>almost</i>, <i>chiefly</i>, <i>mainly</i>, <i>virtually</i>, or <i>principally</i>.	

etc. Etc. is an abbreviation of the Latin et cetera and means and so forth. Etc. at the end of a list implies that the list is not complete, so it cannot be used with a specific list introduced by for example or such as.	Incorrect: Downhole measurements provide petrophysical information (e.g., rock properties, detailed lithology, etc.). Correct: Downhole measurements provide petrophysical information (e.g., rock properties and detailed lithology). Correct: Rock properties, detailed lithology, etc., are petrophysical information provided by downhole measurements.
Farther, further. Use farther when distance is implied and further when referring to time or quantity.	
Forego, forgo. Forego means to precede. Forgo means to do without.	
If, when, where, whether. If implies uncertainty or a conditional relation. When implies certainty and is used for situations requiring preparation or the passage of time. Where is related to place and is not a substitute for when, that, for which, in which, or on which. Whether implies alternatives. Whether or not means regardless of whether and is usually used to emphasize doubt; the phrase should be avoided in technical writing.	Incorrect: Whether he leaves this job, he will not be eligible for severance pay. Correct: If he leaves this job, he will not be eligible for severance pay.
	Incorrect: The log will not be representative when the wrong inputs are selected. Correct: The log will not be representative if the wrong inputs are selected.
	Incorrect: I read in the report where a multizone completion was recommended. Correct: I read in the report that a multizone completion was recommended.
	Incorrect: To learn if the equipment is available, check the inventory list. Correct: To learn whether the equipment is available, check the inventory list.
	Incorrect: The engineer was not sure whether or not to rerun the log. Correct: The engineer was not sure whether to rerun the log.
Intermittent, occasional, sporadic. Intermittent means starting and stopping at intervals. Occasional and sporadic imply more random occurrences.	
Its, it's. Its is the possessive form of it. It's is the contraction of it is.	
Less, fewer. Less refers to quantity. Fewer refers to number and is generally used with items that can be counted.	Correct: less material Correct: fewer drill collars
Lie, lay. The intransitive verb lie means to recline or be situated and takes no direct	Incorrect: The limestone lays on the shale. Correct: The limestone lies on the shale.

object. Its forms are lie , lying , lay , and lain . The transitive verb lay means to put or place and requires an object. Its forms are lay , laying , and laid .	Incorrect: Lie the tool on the catwalk. Correct: Lay the tool on the catwalk.
Like, as. Use like for a conjunction with noun phrases (a noun phrase is a noun or pronoun and any modifiers); use as , as if , or as though for comparing clauses.	Correct: The formation absorbed the treatment like a sponge. Incorrect: Flow from the formation decreased like a choke had been applied. Correct: Flow from the formation decreased as if a choke had been applied.
Like, such as. A like comparison refers to a person or thing as similar to a group of people or things but excludes that person or thing from the group being discussed. A such as comparison typically includes persons or things as part of the group being discussed.	Incorrect: Building a new deepwater platform like Troll A would not be possible today because of changes in legislation. Correct: Oil from aging deepwater developments such as Troll and Ekofisk continues to form a substantial portion of world production.
Master, slave. Avoid this configuration terminology that could be interpreted as discriminatory.	Incorrect: Signals from the master controller synchronize the slave controllers. Correct: Signals from the main controller synchronize the subsidiary controllers.
Majority, most. Majority means the greater part of something. Most is preferred for indicating greater degree, number, quantity, or size.	
Number of. A number of is ambiguous and connotes several. Use the exact number if it is known. About or approximately may be used to show uncertainty.	Incorrect: A number of productive zones were perforated. Correct: Three productive zones were perforated.
Oral, verbal. Verbal refers to what is written, but it also can mean spoken. Oral means spoken only. Ambiguity can be avoided by using written or a similar term instead of verbal .	Incorrect: The publication explained the problem orally and mathematically. Correct: The publication explained the problem verbally and mathematically. Correct: The publication explained the problem in words and mathematically.
Over, under. Avoid using these words as the equivalent of phrases such as more than and less than or greater than and fewer than , respectively, if they cause confusion about physical position.	Incorrect: The producing zone dips under over 30 m [100 ft] of massive shale. Correct: The producing zone dips beneath more than 30 m [100 ft] of massive shale.
Percent, percentage. Percent is used with a spelled-out number (the percent symbol % is used as the unit symbol with measured numerical values). Percentage is used to express a proportion.	Incorrect: 20 percentage of the class Correct: 20% of the class Correct (beginning a sentence): Twenty percent of the class
	Incorrect: a large percent of the samples Correct: a large percentage of the samples

Presence, present. These words are often superfluous. Presently does not mean at present; it means soon.	Incorrect: The metallic minerals present in the sample . . . Correct: The metallic minerals in the sample . . .
Principal, principle. Principal (noun) means chief person or thing; principal (adjective) means first in rank or importance. Principle is a law, fundamental truth, or doctrine.	Correct: The principal investor in the venture is a person of principle.
Proved, proven. Use proved for the past tense and past participle of the verb prove. Use proven as an adjective.	Incorrect: A leak in the seal was proven responsible for failure of the pump. Correct: A leak in the seal was proved responsible for failure of the pump.
	Incorrect: The pump has a proved record of reliability. Correct: The pump has a proven record of reliability.
Quite, rather, somewhat. Avoid these vague descriptors.	
Significant. The word means important or to have meaning. It is often misused to mean large.	Incorrect: The oil contained significant asphaltene content. Correct: The oil had a high asphaltene content.
So, so that. So may be used alone to join clauses of equivalent rank. When used to introduce a clause that gives a reason for an action, so that should be used.	Correct: The results were inconclusive, so more tests were required. Correct: We described the results in detail so that everyone would understand them.
That, which. That distinguishes its subject from all others, whereas which adds detail about the subject. That introduces restrictive (defining) clauses; without these clauses, the essential meaning of the sentence is lost. Which introduces nonrestrictive (parenthetical) clauses; the sentence still makes sense if the clause is deleted. A clause beginning with which is set off with commas. (See guidance on using commas with nonrestrictive clauses.)	Correct: The Senate reached a new agreement that will provide price incentives to find and develop more gas reserves. Correct: This region, which is the largest, contains the principal reservoir.
This, that, these, those. These demonstrative pronouns must point directly to their antecedents and should not be used alone if what they refer to is in doubt. This and its plural, these, are used for references near in time, space, or thought. That and those are for references that are not so close.	Incorrect: The treatment required to improve production involved reservoir analysis. This was an expensive proposition. (Is the antecedent of the pronoun this the treatment as a whole or the analysis component?)

Toward, towards. In US English, the preferred form is without the s. In UK English the s is used. The same is true for other directional words, such as upward, downward, forward, backward, and afterward.	Correct for US: The bit was guided toward the target bed. Correct for UK: The bit was guided towards the target bed.
Unique. Unless you are certain it is one of a kind, use unusual or rare. Do not use unique with qualifiers such as very, rather, or quite.	
Use, utilize. Utilize suggests a new or practical use for something. Do not use it routinely for use.	Incorrect: The wrench was utilized to tighten the bolt. Correct: The wrench was used to tighten the bolt. Correct: Because the nearest medical facility was a half day's drive, a wrench was utilized as a splint to set the broken leg.
Varied, various. Varied means altered or changed. Various means different or diverse.	Incorrect: Deposition of varied types of scale deposits resulted from the various rate of production. Correct: Deposition of various types of scale deposits resulted from the varied rate of production.
Via is not restricted to meaning by way of in a geographical sense. It can be used as by means of, such as for conveyance or to describe electronic communications.	Correct: The tool was returned from the base via truck. Correct: The tool was returned from the base via Kuala Lumpur. Correct: The well report was sent via e-mail.
While, when. Both while and when may be used to introduce a contrast, but when has a sense of although. While implies contemporaneity (at the time that).	
Who, whom. Use who when the pronoun is the subject of the verb that follows and whom when it is the object of the verb.	Incorrect: The supervisor determined whom would write the report. Correct: The supervisor determined who would write the report. Correct: The committee would not reveal whom they nominated to be officers.

Gender-neutral language

Words are preferably not gender specific.

Instead of	Use
chairman	chair
fireman, policeman	firefighter, police officer
layman	laypeople, layperson, lay public, nonprofessionals
man (verb)	staff, cover, operate, run, work

Instead of	Use
manhole	access or access hole, entry or entry hole, utility hole
man-hour	staff-hour, labor-hour, labor time, work-hour
mankind	humanity, humankind, people
manmade	artificial, fabricated, manufactured, synthetic, constructed
manpower	staff, workforce, personnel, employees, workers

Routine use of the pronouns **he** and **his** is usually inaccurate. The numerous ways of achieving gender-neutral language are demonstrated for the following sentence: The lead engineer should familiarize himself with the well data before beginning his job design. (The noun **engineer** has no implied gender.)

Change singular forms to plural (they and theirs).	Incorrect: The lead engineer should familiarize themselves with the well data before beginning their job design. (singular antecedent and plural pronouns) Correct: The lead engineers should familiarize themselves with the well data before beginning their job designs.
Change to the first person (for example, I , we , us , our) or to the second person (for example, you , yourself).	Correct: As the lead engineer, I should familiarize myself with the well data before beginning my job design. Correct: If you are the lead engineer, you should familiarize yourself with the well data before beginning your job design.
Use the imperative mood.	Correct: Be familiar with the well data before beginning the job design.
Eliminate the pronouns.	Correct: The lead engineer should be familiar with the well data before beginning the job design. Correct: The well data should be familiar to the lead engineer before the job design is begun.
Use he or she and his or her sparingly.	Correct: The lead engineer should familiarize him- or herself with the well data before beginning his or her job design.

Similarly, gender-specific pronouns are not used to personify inanimate objects or ideas because English does not assign grammatical gender. For example, do not use **she** for ships or natural phenomena.

Names of people, places, and things

People

The use and position of surnames, family names, and given names are not the same in all languages and cultures. Check an authority, such as *The Chicago Manual of Style* or *Oxford Style*

Manual, when using non-English personal names (French, Russian, Arabic, Chinese, etc.) in an English-language context.

Initials: Close up initials in a person's name.	Incorrect: P. D. Smith Correct: P.D. Smith Correct: W.N.T. Doran
Titles or rank or position: Capitalize titles that precede a name. Do not capitalize titles that follow a name. Titles used without the name or generically are usually lowercased. Exceptions include displayed lists of names and titles and certain promotional materials.	Correct: George Wilson, field engineer, began his career with SLB in 1954. Correct: Field Engineer George Wilson began his career with SLB in 1954. Correct: The field engineer began his career in 1954.
Signatures: Individuals' signatures are not used in publications or on the Web because of security concerns. Exceptions are as required by law or regulation.	

Places

US states and territories and Canadian provinces and territories: Names of the states and territories of the United States and provinces and territories of Canada are normally spelled out in text.	Incorrect: The field office moved to Laramie, WY. Correct: The field office moved to Laramie, Wyoming. Correct: Kingston, Ontario
Countries and country codes: The ISO maintains the official and most up-to-date list of countries. Refer to ISO 3166 for official spellings. For space-constrained uses, use the options available on this platform.	Correct: France Correct: FR (Alpha-2 code) Correct: FRA (Alpha-3 code)
the Netherlands: A lowercase definite article is used except in display text, such as stacked address lines, lists of contributors, and references.	Correct: Production from tight gas plays in the Netherlands is growing. Correct: Operators International 29 Production Row 17 AJ Amsterdam The Netherlands Correct: LN Homer, Consultant, The Netherlands

Cities: Except for the listed cities, which are well known in the oil industry, follow the name of a city with the state (in the United States) or province (in Canada) and the country on first mention. Exceptions are references and listings that require parallel formatting, such author affiliations that each include the city, state or province, and country. Use commas to separate the parts of addresses and names of geographical places or political divisions. English names for foreign cities should be used (for example, Rome instead of Roma and Belgrade instead of Beograd). If a city is renamed, place names that incorporate the old name are not automatically renamed; however, Bombay was changed to Mumbai and Bombay High field is now called Mumbai High field.	Cities that can be used alone: <table> <tr><td>Aberdeen</td><td>London</td></tr> <tr><td>Amsterdam</td><td>Los Angeles</td></tr> <tr><td>Anchorage</td><td>Mexico City</td></tr> <tr><td>Athens</td><td>Moscow</td></tr> <tr><td>Beijing</td><td>Mumbai</td></tr> <tr><td>Berlin</td><td>New Orleans</td></tr> <tr><td>Buenos Aires</td><td>New York City</td></tr> <tr><td>Cairo</td><td>Paris</td></tr> <tr><td>Calcutta</td><td>Riyadh</td></tr> <tr><td>Calgary</td><td>Rio de Janeiro</td></tr> <tr><td>Caracas</td><td>Rome</td></tr> <tr><td>Chicago</td><td>San Francisco</td></tr> <tr><td>Dallas</td><td>Shanghai</td></tr> <tr><td>Denver</td><td>Singapore</td></tr> <tr><td>Doha</td><td>Stavanger</td></tr> <tr><td>Dubai</td><td>Sydney</td></tr> <tr><td>Hong Kong</td><td>The Hague</td></tr> <tr><td>Houston</td><td>Tokyo</td></tr> <tr><td>Jakarta</td><td>Tulsa</td></tr> <tr><td>Kuala Lumpur</td><td>Washington, DC</td></tr> </table>	Aberdeen	London	Amsterdam	Los Angeles	Anchorage	Mexico City	Athens	Moscow	Beijing	Mumbai	Berlin	New Orleans	Buenos Aires	New York City	Cairo	Paris	Calcutta	Riyadh	Calgary	Rio de Janeiro	Caracas	Rome	Chicago	San Francisco	Dallas	Shanghai	Denver	Singapore	Doha	Stavanger	Dubai	Sydney	Hong Kong	The Hague	Houston	Tokyo	Jakarta	Tulsa	Kuala Lumpur	Washington, DC
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Hong Kong	The Hague																																								
Houston	Tokyo																																								
Jakarta	Tulsa																																								
Kuala Lumpur	Washington, DC																																								
Compass points: Compass points and terms that indicate direction or location are lowercased. Regional terms that are proper names are capitalized. Abbreviations are allowed to describe fault orientation. UK English: Compass points used as spelled-out terms are hyphenated, except for wind directions.	Incorrect: Data from the Jene field in South Sumatra Indonesia were submitted to the US Geological Survey's National Center in Reston Virginia and the customer's office in Houston, Texas. Correct: Data from the Jene field in South Sumatra, Indonesia, were submitted to the US Geological Survey's National Center in Reston, Virginia, and the customer's office in Houston.																																								
	Correct: The beds strike east–west and dip 45° to the south. Correct: the Middle East Correct: a southwest wind Correct: NNW–SSE																																								
Geographic or generic terms in place names: Capitalize geographic or generic terms if they are part of place names. Capitalize descriptive terms that denote a defined region. Do not capitalize generic terms when used alone or descriptively.	Correct for US: Geologists identified a fault in the northeast part of the field. Correct for UK: Geologists identified a fault in the north-east part of the field.																																								
	Correct: Trinity River Correct: Trinity and Sabine Rivers Correct: US Continental Shelf (subsequent mentions can be just “Shelf”) Correct: Mounts McKinley and Olympus Correct: West Coast Correct: western Harris County Correct: the valley of the Mississippi River																																								
	Incorrect: the Rio Grande River Correct: the Rio Grande																																								

Non-English terms in place names: When a foreign generic term forms part of a geographic name, the English term should not be included. Foreign names of thoroughfares and buildings are not italicized and may be preceded by the English word the if the definite article is used in the original language. Do not use both the foreign and English terms.	Incorrect: the Le Champ-Elysées Correct: the Champs-Elysées Incorrect: The SLB office is on rue St. Dominique Street. Correct: The SLB office is on rue St. Dominique.
Geologic place names: Do not capitalize descriptive uses of words such as basin, belt, delta, field, pay, play, pool, reservoir, sand, shale, and trend. They are capitalized as part of a proper name.	Correct: San Juan Basin Correct: A famous East Texas field is Luling Field Correct: These wells in the Eagle Ford Play target the Eagle Ford Shale and Woodbine Formation.

Things

Companies, institutions, buildings, thoroughfares, and organizations: Capitalize the names of companies, institutions, buildings, thoroughfares, and organizations. Capitalize Geounit and Technology (external) or Business Line (internal) for SLB entities. Exceptions are names that begin with a lowercase letter. Generic mentions are lowercased. Foreign names of thoroughfares and buildings are not italicized and may be preceded by the English word the if the definite article is used in the original language. Do not use both the foreign and English terms.	Correct: Gulf Coast Division Incorrect: the Division Correct: the division Correct: The EUR Geounit is headquartered in Aberdeen. Correct: eBay Correct: the Champs-Elysées				
Geologic unit names: Capitalize both parts of the name of a lithostratigraphic unit that is formally named in accordance with a recognized classification code. The first part of the name is usually a place name; the second part indicates the unit or stratigraphic rank (for example, supergroup, group, formation, member, bed, supersuite, suite, complex). Unit names for the United States, Canada, and Mexico can be researched at the National Geologic Map Database. For informally named units, which are not named in accordance with a recognized classification code, lowercase the rank or unit part of the name, which should not be a formal stratigraphic rank term.	Correct: The source rocks of Jonah field are probably the coal layers in the lower Lance Formation and upper part of the Mesaverde Group. Correct: The Barnett Formation in the field consists primarily of 100-ft-thick Barnett Shale with intercalated lenses of Barnett sand.				
Casing and tubing grades: Grades are set solid.	Common grades from API Spec 5CT: <table> <tr> <td>C90, C110</td> <td>N80</td> </tr> <tr> <td>H40</td> <td>P110</td> </tr> </table>	C90, C110	N80	H40	P110
C90, C110	N80				
H40	P110				

	J55 R95 K55 T95 L80 Q125 M65
<i>Ships or other vessels:</i> The names of specific ships or other vessels are in italics, but the name of a make or class is not.	Correct: The <i>DeepStim II</i> offshore stimulation vessel is the flagship of the DeepSTIM™ fleet.
Trademarks: Trademarks must be properly formatted, identified with the appropriate trademark symbol, and used as adjectives only, never as nouns or verbs. A generic term typically can be used instead of a trademarked name. For more information on trademark usage, see the “Legal and Marketing” chapter of the <i>Communications Handbook</i>. Trademark symbols are not used on slb.com pages.	Incorrect: The surface was Teflon. Correct: The surface had a Teflon® coating. Correct for slb.com pages: The surface had a Teflon coating. Correct: The surface was coated with polytetrafluoroethylene (PTFE).
Nonmarked or generic SLB offerings: Do not capitalize nonmarked or generic SLB service offerings in body copy. If you want to add a level of importance in web copy, you can bold the name.	Incorrect: Intelligent Power Management is a predictive power management system that reduces emissions, fuel consumption, and engine run time. Correct: Intelligent power management is a predictive power management system that reduces emissions, fuel consumption, and engine run time. Correct: Intelligent power management is a predictive power management system that reduces emissions, fuel consumption, and engine run time.
Cloud computing: Application software and data storage are described as <i>in the cloud</i>.	Incorrect: Users access their data on the Cloud. Correct: Users access their data in the cloud.
Uniform resource locators (URLs) and Web addresses: Promotional materials can use a short redirect form that is more easily remembered. Include the full server name and the protocol as necessary in a URL or Web address. References include full URLs (see guidance on formatting electronic references). Lowercase all portions of the short URL with the exception of marks and some acronyms. For URLs that include marks, capitalize the short URL ending as the mark appears in the marks list.	Correct: slb.com Correct: onesubsea.slb.com/subsea-boosting Correct: slb.com/ADD (for “autonomous directional drilling”) Correct: slb.com/PowerDriveArcher Correct: slb.com/PowerDrive-Archer Correct: slb.com/QuantaGeo Correct: slb.com/Quanta-Geo Incorrect: slb.com/quantageo Incorrect: slb.com/Quanta_Geo

Do not use underscores (_) in short URLs. You can choose to use a hyphen or remove the space, whichever is more legible. If a mark is very long, you may choose to condense the mark only for the short URL.	Correct: slb.com/QGeo (for Quanta Geo) Correct: slb.com/NATCODF (for NATCO DUAL FREQUENCY)
Parts of a publication: Capitalize Appendix , Chapter , Equation , Fig. , Figure , Section , and Table for specific numbered, lettered, or named components of a work. Lowercase these terms referencing other works to differentiate them.	Correct: The data plotted in Fig. 4 of the following chapter contrast with Waxman and Smit's (1968) fig. 2.

Punctuation

Period

A period (also called a full stop or full point in UK English) is used as terminal punctuation for a complete declarative or imperative sentence. Place a period inside parentheses enclosing a complete and independent sentence. When a complete sentence enclosed by parentheses is within another sentence, it does not receive a period. If a sentence ends with an abbreviation that ends with a period, that period also serves as the terminal punctuation for the entire sentence.	Correct: Cemented casing with selective perforation (a 7-in casing is recommended) is the prescribed completion. Correct: Cemented casing with selective perforation is the prescribed completion. (A 7-in casing is recommended.)
	Incorrect: The total production rate of 460 m³/d during drawdown came from a fraction of the openhole interval. (50 m) Correct: The total production rate of 460 m³/d during drawdown came from a fraction of the openhole interval (50 m).
	Incorrect: Sunrise was at 5 a.m.. Correct: Sunrise was at 5 a.m.
	Incorrect: Specifications are at slb.com/oilfield. Correct: Specifications are at slb.com/oilfield. Correct: Request more information on stimulation from stimulation@slb.com.
UK English: A full stop is used to separate the day of the week, month, and year in dates presented numerically in day-month-year format. UK English: A full stop separates the hour and minutes when 24-h time is not used.	Correct for UK: 16.7.42 (16 July 1942) Correct for UK: 9.15 p.m.
Use of a period as terminal punctuation for titles and headings varies with the context and layout template. Such usage is common in	Typical for ad: New depths. New horizons. Typical for report: Applications in Wells with Low Water Cut

advertisement headlines but infrequent in other headings. A period is used as terminal punctuation for all figure captions, regardless of whether they are complete sentences. A period is not used as terminal punctuation for a callout, key, or figure title (defined in figure terminology) on a figure unless the element is a complete sentence.	Incorrect: Figure 3. Job design flowchart Correct: Figure 3. Job design flowchart.
Omit the period in abbreviations of foreign-language terms that are contractions.	Correct: Cie (French “Compagnie”) Correct: LP (limited partnership) Correct: PLC (public limited company)
Always place periods inside footnote reference numbers. Always place periods outside superscript trademark symbols or subscripts that are part of a term. The period follows the superscript nonregistered trademark symbol identifying the first use of a SLB mark that is a company slogan (which is the only kind of mark not used exclusively as an adjective but stands alone) at the end of a sentence. The same arrangement applies to other companies’ slogan marks.	Incorrect: The nonlinear rise in the inertial resistance has long been recognized¹³. Correct: The nonlinear rise in the inertial resistance has long been recognized.¹³ Incorrect: The volume of fluid withdrawn during a single test may be adjusted to a maximum volume of 100 cm.³ Correct: The volume of fluid withdrawn during a single test may be adjusted to a maximum volume of 100 cm³. Correct: Production success depends on Connecting you to your reservoir™. Correct: Although the laptop fell through the balcony railing to the sidewalk below, the only damage was a scratch on the logo decal proclaiming Intel Inside®.
Always place periods inside single or double quotation marks. UK English: If the full stop is not part of the term set off by quotation marks, place it outside the closing single or double quotation marks.	Correct for US: The years between 1865 and 1985 may be called “the century of exploration.” Correct for UK: The years between 1865 and 1985 may be called ‘the century of exploration’.

Question mark and exclamation point

A question mark marks a query or expresses uncertainty about an element; it is not used extensively in technical writing.	
An exclamation point provides emphasis at the end of a sentence. It is only rarely used in technical writing.	
Question marks and exclamation points are used sparingly as necessary in titles and headings.	Correct: What is Value-Added Development? Incorrect: New Acid Diverter Technology! Correct: New Acid Diverter Technology
Question marks and exclamation points are placed within single or double quotation marks	Incorrect: She asked, “Why?” Correct: She asked, “Why?”

or within parentheses only if they are part of the quoted or parenthetical material.	Incorrect: Who initialed this form with an “X?” Correct: Who initialed this form with an “X”?
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Comma

Place a comma between independent clauses that are joined by coordinating or correlative conjunctions. The comma may be omitted if the clauses are short and closely related. Two or more verbs that have the same subject are not separated by a comma. If a comma may be necessary to prevent misreading in an extremely complex sentence in which two or more verbs have the same subject, it is preferable to rewrite the sentence for clarity. (Coordinating conjunctions join clauses of equal status—for example, and, but, for, nor, or, so, yet. Correlative conjunctions are used in pairs to correlate two parts of a sentence—for example, either . . . or, neither . . . nor, both . . . and, not only . . . but also.)	Correct: The position of the bottom boundary was fixed at 439 m, and the top boundary location was adjusted to 431 m.
	Correct: The dual packers isolated the interval and the sample was collected. <i>or</i> The dual packers isolated the interval, and the sample was collected.
	Incorrect: The transient surge of fluid cleans the perforation damage, and yields a perforation damage skin of zero. Correct: The transient surge of fluid cleans the perforation damage and yields a perforation damage skin of zero.
	Correct: Movement of the tool was adversely affected by heave and cable stretch, and was recorded as flagged values for later correction. <i>or</i> Movement of the tool was adversely affected by heave and cable stretch. The recorded values were flagged for later correction.
	Correct: The driller’s assistant was injured when the still-inflated packer element was removed from the wellbore, and collapsed. (The person collapsed, not the packer.)
A comma is needed after an introductory phrase or clause if its absence would confuse the reader.	Incorrect: If data are inaccurate or incomplete results will not be representative. Correct: If data are inaccurate or incomplete, results will not be representative.
A comma is usually used to set off an introductory participial phrase. (See guidance on participial phrases .)	Correct: Working quickly as the floodwaters rose, the crew protected the wellsite with sandbags and a pump.

Unless they are restrictive, words that identify a noun or pronoun (appositives) are set off by commas if they follow the noun or pronoun. Restrictive words or phrases are necessary for the sentence to make sense; nonrestrictive components can be removed from a sentence without changing the meaning. (See guidance on <i>that</i> and <i>which</i> clauses.) Restrictive: The computer in my office needs a new modem. (There is more than one computer; only the computer in my office needs a modem.) Nonrestrictive: The computer, which I got last week, needs a new modem. (There is only one computer.) Words that identify a noun or pronoun are not set off by commas if they precede the noun or pronoun.	Correct: George Wilson, field engineer, began his career with SLB in 1954. Correct: Field Engineer George Wilson began his career with SLB in 1954. Correct: Single-phase, or monophasic, flow plots as a straight line.
Place a comma between two adjectives that modify the same noun if the word and could replace the comma without altering the meaning.	Incorrect: The array is used to detect hydrocarbons in thin highly porous formations. Correct: The array is used to detect hydrocarbons in thin, highly porous formations.
Use commas to set off phrases that give descriptive information that is not essential to the meaning of the sentence. Do not use commas to set off phrases that are essential to the meaning of the sentence.	Correct: The governing equation for the flow in the skin cells, which includes the inertial resistance effects, is developed next.
	Incorrect: All samples, which came from unfractured sections, were obtained with 22-g charges. Correct: All samples that came from unfractured sections were obtained with 22-g charges.
Use a comma before the conjunction in the last element in a series of three or more elements. UK English: Although use of this comma is often debated for US English, it is used so consistently in UK English that it is referred to as ‘the Oxford comma’. Press releases and articles for trade journals follow AP style and use the serial comma only as necessary to avoid ambiguity. Use a semicolon to punctuate elements in a series of three or more that has commas within the elements.	Incorrect: The fluid modulus for water depends on pressure, temperature and salinity. Correct: The fluid modulus for water depends on pressure, temperature, and salinity.
	Incorrect: The parameters that dictate horizontal well success include permeability anisotropy, formation porosity, saturation, and pressure along the horizontal length, and location of the wellbore in relation to the formation, gas/oil contact, and oil/water contact. Correct: The parameters that dictate horizontal well success include permeability anisotropy; formation porosity, saturation, and pressure along the horizontal length; and location of the wellbore in relation to the formation, gas/oil contact, and oil/water contact.

A serial comma is not used where an ampersand (&) is used instead of the word and. (See guidance on use of the ampersand.)	
Continental European English: In some parts of continental Europe, a comma is used in place of a period to indicate the decimal point.	Correct for continental Europe: EUR 155,25
The Latin abbreviations i.e. and e.g. must always be followed by a comma. If i.e. or e.g. connects two independent clauses, precede it with a semicolon. If i.e. or e.g. introduces an incomplete clause, enclose the clause within parentheses. Clauses beginning with i.e. or e.g. can also be set off with an em dash. The Latin terms i.e. and e.g. may be replaced by their English equivalents, that is and for example, respectively. (See i.e. vs. e.g.) UK English: A comma may be used but is not required after that is, for example, and namely. To avoid double punctuation, no comma follows the abbreviations i.e. and e.g.	Incorrect: This validation takes place in inverse mode, i.e., a synthetic test data set is generated with the rigorous model. Correct: This validation takes place in inverse mode; i.e., a synthetic test data set is generated with the rigorous model. Incorrect: The interpretation can be attempted only if reliable starting conditions, e.g., a good stabilization, can be found for each transient. Correct for US: The interpretation can be attempted only if reliable starting conditions (e.g., a good stabilization) can be found for each transient. Correct for US: The interpretation can be attempted only if reliable starting conditions—e.g., a good stabilization—can be found for each transient. Correct for UK: The interpretation can be attempted only if reliable starting conditions—e.g. a good stabilization—can be found for each transient. Correct for UK: Several trucks, namely the 15 ton and crew cab pickup, are due for maintenance.
Use commas to separate the parts of addresses and names of geographical places or political divisions. In an address formatted as a postal address, do not use commas at the ends of lines. UK English: In text, in is employed to separate city and county instead of a comma.	Incorrect: Data from the Jene field in South Sumatra, Indonesia are used to illustrate the concepts explained in this paper. Correct: Data from the Jene field in South Sumatra, Indonesia, are used to illustrate the concepts explained in this paper. Correct for UK: The drilling data from Croydon in Surrey were used for testing the database.

Place commas between the day of the week, the month and day, and the year. Place a comma after the year preceded by the month and day if the year is not followed by the terminal punctuation of the sentence. Do not use a comma between the month and the year or after the year if no date is given. Do not use a comma if the date precedes the month and year, such as in the day-month-year system commonly used in English-speaking countries other than the United States.	Incorrect: This well was drilled on Friday March 14 1986. Correct: This well was drilled on Friday, March 14, 1986. Correct: This well was drilled on March 14, 1986, in the Austin chalk. Correct: This well was drilled in March 1986 in the Austin chalk. Correct: Friday, 14 March 1986
Always place commas inside footnote reference numbers. Always place commas outside superscript trademark symbols or subscripts that are part of a term.	Incorrect: The nonlinear rise in the inertial resistance has long been recognized¹³, but its significance was greatly underestimated. Correct: The nonlinear rise in the inertial resistance has long been recognized,¹³ but its significance was greatly underestimated.
	Incorrect: The maximum withdrawal volume is 1,000 cm,³ but the capacity of the transport container is only 600 cm³. Correct: The maximum withdrawal volume is 1,000 cm³, but the capacity of the transport container is only 600 cm³.
Always place commas inside single or double quotation marks. UK English: If the comma is not part of the term set off by quotation marks, place it outside the closing quotation marks.	Incorrect: The years between 1865 and 1985 have been called “the century of exploration”, but what should we call the years since then? Correct for US: The years between 1865 and 1985 have been called “the century of exploration,” but what should we call the years since then? Correct for UK: The years between 1865 and 1985 have been called ‘the century of exploration’, but what should we call the years since then?

Semicolon

Use a semicolon in compound sentences that are not joined by a conjunction.	Incorrect: Sample chambers are available in three sizes, the number of samples recovered per trip is limited by tool weight. Correct: Sample chambers are available in three sizes; the number of samples recovered per trip is limited by tool weight.
UK English: Use a semicolon to impart greater emphasis to the second clause in a sentence in which the two clauses are linked by a conjunction.	Correct for UK: Integrity is essential; and woe be unto the company without it in today’s business climate.

Use a semicolon to punctuate elements in a series that has commas within the elements.	Incorrect: The parameters that dictate horizontal well success include permeability anisotropy, formation porosity, saturation, and pressure along the horizontal length, and location of the wellbore in relation to the formation, gas/oil contact, and oil/water contact. Correct: The parameters that dictate horizontal well success include permeability anisotropy; formation porosity, saturation, and pressure along the horizontal length; and location of the wellbore in relation to the formation, gas/oil contact, and oil/water contact.
If a conjunctive adverb is used between clauses of a compound sentence, it is preceded with a semicolon and usually followed by a comma. Conjunctive adverbs are punctuated with commas if they are in the middle or at the end of a clause. (Conjunctive adverbs are adverbs that are used like conjunctions to introduce clauses—for example, <i>besides, consequently, finally, first, furthermore, hence, however, indeed, likewise, moreover, next, nevertheless, then, therefore, thus</i>. Some adverb phrases also have this function—for example, <i>for example, for instance, on the contrary, on the other hand</i>.)	Incorrect: The epithermal ratio was sensitive to formation porosity, however attempts to characterize the response from field data were subject to the conditions under which the data were acquired. Correct: The epithermal ratio was sensitive to formation porosity; however, attempts to characterize the response from field data were subject to the conditions under which the data were acquired. Correct: The blowout was spectacular; it was, however, short-lived.
Always place semicolons outside single or double quotation marks.	Incorrect: The years between 1865 and 1985 may be called “the century of exploration;” since then, improved recovery has been the primary means of increasing reserves. Correct: The years between 1865 and 1985 may be called “the century of exploration”; since then, improved recovery has been the primary means of increasing reserves.

Colon

Use a colon to introduce a series or list unless the introductory clause ends in a verb or preposition.	Incorrect: The trends that occurred were: wellbore storage, partial penetration, and radial flow. Correct: The trends that occurred were wellbore storage, partial penetration, and radial flow. Correct: Three trends were observed in the data: wellbore storage, partial penetration, and radial flow.
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Capitalize the first word after a colon if it begins a complete sentence or independent clause. UK English: Use a colon to direct the reader forward: from cause to effect, from general statement to example, etc. It has the same function as <i>namely</i>, <i>that is</i>, <i>because</i>, and <i>therefore</i>. The first word following the colon is not capitalized.	Correct: Examination of the map made us question our plans: Would standard logging trucks be able to negotiate the steep path to the wellsite? Correct for UK: The pressure increased: the driller increased the mud weight.
Always place colons outside single or double quotation marks.	Incorrect: Each person is ultimately responsible for protecting the “human machinery:” eyes, ears, fingers, and brain. Correct: Each person is ultimately responsible for protecting the “human machinery”: eyes, ears, fingers, and brain.
UK English: Use a colon to introduce quoted or paraphrased material more formally or emphatically than a comma would.	Correct for UK: Sir Galahad said: ‘Without the PowerDrive™ system we would have drilled all year.’

Dashes

Use an em dash, or “long” dash (about as wide as a capital <i>M</i> in a typeface), without spaces before or after to set off parenthetical material. Never precede or follow a dash with a comma. Never begin a line of text with a dash.	Correct: The response is characterized for the three basic lithologies—limestone, sandstone, and dolomite—over a wide range of borehole sizes, mud weights, and fluid salinities.
Use an em dash to introduce a clause that gives emphasis or explanation.	Correct: The author describes coal as a solid colloid—a solid, highly porous dispersion of discrete particles.
Use an en dash, or “short” dash (about as wide as a capital <i>N</i> in a typeface), without spaces before or after to indicate a range such as continuing or inclusive dates, times, or reference numbers. Do not use an en dash with the word <i>from</i> or <i>between</i>.	Correct: 1990–1991 <i>or</i> 1990–91 Incorrect: 1998–02 Correct: 1998–2002 Correct: September–December 1991 Correct: 10 a.m.–3 p.m. Correct: pages 72–78 Incorrect: We met from 10 a.m.–3 p.m. Correct: We met from 10 a.m. to 3 p.m.
Use the word to when expressing ranges other than dates and times. An en dash may be used for ranges if there are space limitations (for example, tabular material). (See guidance on the use of to and from.)	Incorrect: The skin factor varied from 1– –2. Correct: The skin factor varied from 1 to –2. Incorrect: The producing interval was at 1,180–2,000 ft. Correct: The producing interval was at 1,180 to 2,000 ft.
Use multiple hyphens in a compound adjective when the terms have internal hyphens or are open compounds. UK English: An en dash (also called an en rule) is used to express to or and between words of equal importance and to join names	Correct: by-product-hydrocarbon mixture Incorrect: 4–20-mA loop-powered display Correct: 4- to 20-mA-loop-powered display Incorrect: New York-Paris flight Correct: New York–Paris flight Correct: Dover–Calais crossing

of joint authors or creators to distinguish them from a single hyphenated name.	Correct for UK: Lloyd-Jones theory (one person) Correct for UK: Lloyd–Jones theory (two people) Correct for UK: A cost–benefit analysis was conducted for installation of the new on–off switch.
An en dash is used for a minus or negative sign; for that reason, avoid ambiguous constructions.	Incorrect: The temperature range was –15—5 degF. Correct: The temperature range was –15 to –5 degF. Incorrect: 6 - 3 = 3 Correct: 6 – 3 = 3
Hyphens are used to avoid ambiguity in unit modifiers comprising multiple words that precede the modified noun or pronoun. A hyphen is not necessary if each word in the unit modifier individually modifies the noun, if the first word is an adverb ending in <i>–ly</i> , or if the term is well known. A hyphen is not used in currency terms.	Correct: oil-bearing formation Correct: well log analysis Correct: higher-producing zone (a zone that has higher production than others, whereas a “higher producing zone” is physically higher than the others) Correct: An additional USD 5 million was added to the USD 10 million commitment.
Use a hyphen in each of a series of compound adjectives or numeric values and units that modify the same noun and precede it. Multiple hyphens are sometimes necessary to avoid ambiguity.	Correct: Oil- and water-base muds were used in both small- and large-scale tests in single- and multiwell fields.
	Incorrect: 2,000–3,000–4,000-psi pressure Correct: 2,000-, 3,000-, and 4,000-psi pressure Correct: 2,000-psi, 3,000-psi, and 4,000-psi pressure Incorrect: 30-m [100-ft]/h rate Correct: 30-m/h [100-ft/h] rate
	Correct: 3-year-old equipment
A joined prefix uses a hyphen if the prefix is placed before a capitalized word, numeral, or hyphenated compound term. A hyphen is also used to separate letter or syllable combinations that may confuse readers or to separate repeated terms in a double prefix.	Correct: The sub-Saharan drilling campaign in the pre-1950s was non-unique except for the use of mini-DSTs.

Apostrophe and quotation marks

An apostrophe is used to form a possessive of most singular nouns by adding it and the letter s. Exceptions are certain indefinite pronouns that do not have a possessive form (for example, any, few, none). Only an apostrophe is used to form the possessive of plural nouns if the added s would produce an awkward double sibilant sound, or the term can be rewritten to a phrase using of.	Correct: Texas's production was greater than Kansas's. Correct: France and Italy's border Correct: France's and Italy's transportation systems Correct: the surfactants' properties or the properties of the surfactants
Only an apostrophe is used to form the possessive of singular proper nouns that look like plural nouns because they end in the letter s.	Correct: the United States' participation Correct: Beverly Hills' history
An apostrophe is not used when a plural term is formed by adding an s unless it is necessary to avoid confusion. For lowercase letters used as letters, form the plural with an apostrophe and a lowercase letter s.	Incorrect: Three BHA's were shipped. Correct: Three BHAs were shipped. Correct: x's and y's Correct: All the PhDs in the class earned straight As. Correct: Production steadily declined through the 1950s. Correct: How many "not applicable's" were entered on the form?
An apostrophe is placed where letters are omitted in contractions. Contractions (typically one word is a verb and the other is a subject or the word not) should be avoided in formal writing.	Incorrect: Periodic huff 'n' puff operations weren't successful. Correct: Periodic huff 'n' puff operations were not successful.
Quotation marks can be used to set off a key term on its first mention. Do not use both quotation marks and italics for this purpose.	Correct: The pulleys on the crown block of the rig are called sheaves (pronounced "shivs"). Incorrect: The shale shaker does not remove the "fines" from the mud. Desilters and desanders are used to remove these very small particles because "fines" can erode drilling equipment or make the mud too heavy. Correct: The shale shaker does not remove the "fines" from the mud. Desilters and desanders are used to remove these very small particles because fines can erode drilling equipment or make the mud too heavy.
Use single quotation marks only inside double quotation marks. UK English: The use of single and double quotation marks (also called inverted commas) is the opposite of that in US English. Single quotation marks are used to enclose quoted material. Double quotation marks are used to emphasize words or to enclose a quotation within a quotation.	Incorrect: The axial receivers are grouped in two pairs: the 'far' and the 'near.' Correct for US: The axial receivers are grouped in two pairs: the "far" and the "near." Correct for US: The instructor said, "The axial receivers are grouped in two pairs: the 'far' and the 'near'." Correct for UK: The instructor said, 'The axial receivers are grouped in two pairs: the "far" and the "near".'

Always place periods and commas inside single or double quotation marks. Always place semicolons and colons outside single or double quotation marks. UK English: Only those punctuation points that appeared in the original material should be included within the quotation marks; all others follow the closing quotation mark.	Correct: The years between 1865 and 1985 may be called “the century of exploration.” Correct for US: The years between 1865 and 1985 may be called “the century of exploration”; since then, improved recovery has been the primary means of increasing reserves. Correct for UK: The years between 1865 and 1985 may be called ‘the century of exploration’. Correct for UK: The name of the SLB communities of practice came from Archimedes’s cry of discovery: ‘Eureka!’
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Other punctuation marks

Use a solidus (slash) to express a ratio of word terms or a boundary relation.	Incorrect: length:width ratio Correct: length/width ratio
	Correct: oil/water contact
Ellipsis points are used to indicate omissions within and between sentences, typically in quotations to remove unnecessary material. Ellipsis points can also be used to suggest faltering or fragmented speech. If the omission is in the middle of a sentence, three dots are used and are separated from the text and each other by spaces. If the omission is between sentences and the remaining text is grammatically complete, four dots are used—the first is the period for terminal punctuation.	Incorrect: Pressure profiles...are crucial for optimizing recovery. Correct: Pressure profiles . . . are crucial for optimizing recovery.
	Incorrect: In virgin reservoirs, vertical pressure profiles are obtained to . . . determine completion strategy . . . Correct: In virgin reservoirs, vertical pressure profiles are obtained to . . . determine completion strategy. . . .

Capitalization

Capital letters are used to call attention to certain words (proper nouns and the first word of a sentence) and can help eliminate ambiguity in the meaning of words (for example, **china** and **China**). Down style, or the conservative use of capital letters, is preferred for all SLB documents and publications. The following rules deal with specific examples, but when in doubt about whether to capitalize a particular word, consult the list of “[Spellings and Terminology](#).” If a word is not listed there, consult *Merriam-Webster’s Collegiate Dictionary* (11th ed.) or another resource (see “[Suggested reading](#)”).

As addressed in the following two sections, capitalization conventions differ for titles and text headings, body text, and figure titles and labels. Capitalization guidance for tabular material is in “[Table examples](#).” This capitalization guidance is used in conjunction with template styles, such as in the “[Brand Elements and Templates](#)” and “[Writing for Our Public Websites](#).”

Words and text

Capitalize only the first word in a complete sentence and proper nouns. Avoid using initial capital letters to imply uniqueness, importance, or authority for nouns that are not proper nouns or for generic mentions of proper nouns. Do not capitalize a parenthetical sentence within another sentence. A term or proper name that begins with a lowercase letter is capitalized at the beginning of a sentence unless it is a company name or a mark of SLB that begins with a lowercase letter. In references, follow the author's preference for capitalization. A sentence can also be rewritten to avoid having an initial lowercase letter. The term pH is always lowercased.	Incorrect: The tool was assembled with a Centralizer and additional Sub. Correct: The tool was assembled with a centralizer and additional sub. Incorrect: Personnel were assigned to groups on the basis of their training (The length of their employment with the company was not a factor). Correct: Personnel were assigned to groups on the basis of their training (the length of their employment with the company was not a factor). Incorrect: de Luna's seminal research led to the development of the innovative eFire™ electronic firing head system. Correct: De Luna's seminal research led to the development of the innovative eFire™ electronic firing head system. Correct: eFire™ electronic firing head systems use field-proven technology. Incorrect: PH is not a reliable indicator. Correct: pH is not a reliable indicator. Correct: The pH is not a reliable indicator. Correct: Do not rely on pH as an indicator.
Capitalize the first word after a colon if it begins a complete sentence or independent clause. UK English: The first word following the colon is not capitalized.	Correct: Examination of the map made us question our plans: Would standard logging trucks be able to negotiate the steep path to the wellsite? Correct for UK: The pressure increased: the driller increased the mud weight.
Capitalize titles of rank or position that precede a name. Do not capitalize titles of rank or position that follow a name. Titles used without the name or generically are usually lowercased. Title capitalization exceptions include displayed lists of names and titles and certain promotional materials.	Incorrect: Division manager George Wilson was awarded . . . Correct: Division Manager George Wilson was awarded . . . Incorrect: George Wilson, Division Manager, was awarded . . . Correct: George Wilson, division manager, was awarded . . . Incorrect: George Wilson was named Division Manager. Correct: George Wilson was named division manager.
Capitalize names of companies, institutions, and organizations. Capitalize Geounit and Technology (external) or Business Line (internal) for SLB entities. Capitalize geographic or generic terms if they are part of	Correct: Gulf Coast Division Incorrect: the Division Correct: the division Correct: The EUR Geounit is headquartered in Aberdeen.

place names. Capitalize descriptive terms that denote a defined region. But do not capitalize generic terms when used alone or descriptively.	Correct: Trinity River Correct: Trinity and Sabine Rivers Correct: US Continental Shelf (subsequent mentions can be just “Shelf”) Correct: Mounts McKinley and Olympus Correct: West Coast Correct: western Harris County Correct: the valley of the Mississippi River
Capitalize the names of academic degrees used in complete form. Do not capitalize academic degrees when referred to in general terms.	Correct: Bachelor of Science degree Correct: bachelor’s degree Correct: doctor of philosophy
Capitalize nouns and adjectives designating parts of the world or regions. Do not capitalize adjectives that refer to a direction or general location.	Correct: Middle East Correct: South Pacific
	Incorrect: The fault is downthrown to the West. Correct: The fault is downthrown to the west.
Geologic place names: Do not capitalize descriptive uses of words such as basin, belt, delta, field, pay, play, pool, reservoir, sand, shale, and trend . They are capitalized as part of a proper name.	Correct: San Juan Basin Correct: A famous East Texas field is Luling Field Correct: These wells in the Eagle Ford Play target the Eagle Ford Shale and Woodbine Formation.
Capitalize only eponyms, not the accompanying names of devices, methods, theories, techniques, systems, or laws.	Correct: Darcy’s law Correct: Reynolds number
Capitalize Well, Track, and Booth used with specific alphanumeric identification.	Correct: Well 238 was fractured but Bessie Smith 1, the offset well, was shut in. Correct: The numerous fractures imaged in Track 3 of the log are in the same interval as the gas crossover in the neutron density track. Correct: Visit SLB at Booth 435.
Do not capitalize the names of chemical elements or chemical compounds.	Incorrect: The pressure drop increased the precipitation of Calcium Carbonate scale. Correct: The pressure drop increased the precipitation of calcium carbonate scale.
Capitalize the names of formally named geochronologic (time) and chronostratigraphic	Correct: Jurassic Period Correct: Early Pliocene Epoch

(time-rock) units. Time scales are available from the International Commission on Stratigraphy and the US Geological Survey. The preceding modifiers, such as early, describe the position of the unit and must be used carefully. Early, middle, and late are restricted to time contexts and are used with the rank units eon, era, period, epoch, and age (listed in order of decreasing magnitude). The corresponding modifiers lower, middle, and upper are restricted to the rocks themselves and their position and are used with the rank units eonothem, erathem, system, series, and stage.	Incorrect: The sediments were deposited in the Lower Jurassic. Incorrect: The sediments were deposited in the early Jurassic. Correct: The sediments were deposited in the Early Jurassic. Correct: Early Jurassic highstands resulted in the deposition of sediments.
Capitalize earth when it is used as the proper name of the planet.	Correct: Gravitational effects between the moon and the Earth influenced measurement of the wellbore fluids. Correct: earth-moving equipment Correct: His attitude was down to earth.
Never use all capital or initial capital letters for emphasis in text.	Incorrect: The Electric Log was a Revolutionary development. Correct: The electric log was a revolutionary development.

Titles, headings, and figures

Templates for print materials are described in the SLB “[Brand Elements and Templates](#).” Guidance for the slb.com framework and page components is in “[Writing for Our Public Websites](#).” The capitalization guidance in this section is used in conjunction with the styles specified for titles, headings, and figures.

When using initial-capitalization style (also called headline-style capitalization), capitalize the first word, nouns, pronouns, verbs, adverbs, adjectives, and all other words of five or more letters. Exceptions are company names or marks of SLB or other companies and unit symbols that begin with a lowercase letter.	Incorrect: An Innovative Approach To Correct Density Measurements In Clays Correct: An Innovative Approach to Correct Density Measurements in Clays Incorrect: EBay is Selling EFire Systems Correct: eBay Is Selling eFire Systems Incorrect: EFire Systems Available on EBay Correct: eFire Systems Available on eBay Incorrect: Increased Capacity 500 Bbl/d Correct: Increased Capacity 500 bbl/d
Capitalize the first word after a colon in a title or heading.	Correct: Calculating Density from Clays: A Maximum Porosity Correct: Calculating density from clays: A maximum porosity
Capitalize both parts of a compound adjective in a title using initial-capitalization style.	Correct: Ratio-Based Epithermal Neutron Porosity for Air-Filled Boreholes

For figure titles, capitalize the first word, nouns, pronouns, adverbs, and all other words of five or more letters.

For figure callouts, capitalize only the first word and proper nouns. If the first element of a callout is a numeral, the first following word is not capitalized (for example, **15-ft interval**).

Exceptions are company names or marks of SLB that begin with a lowercase letter.

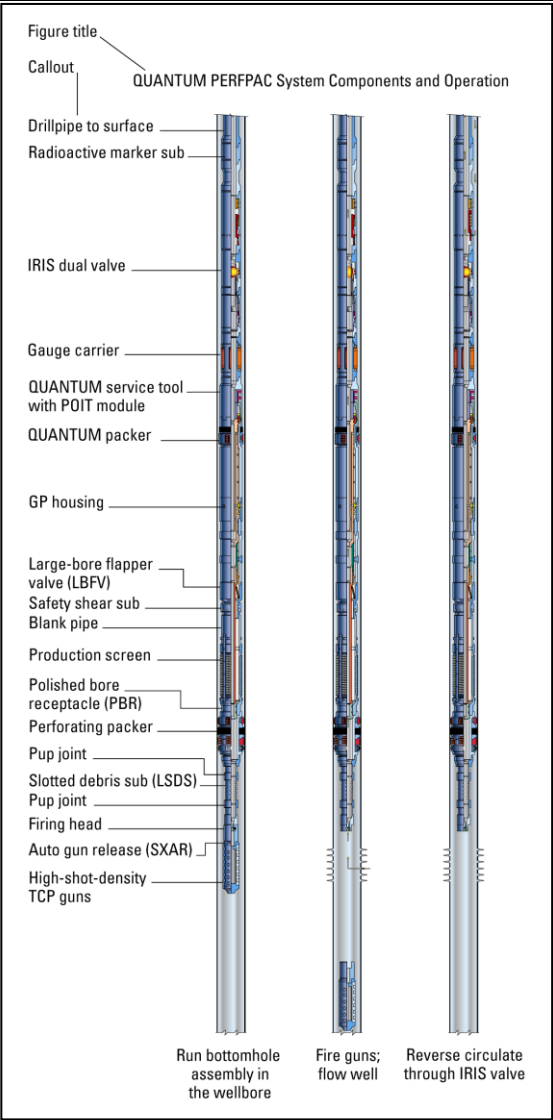
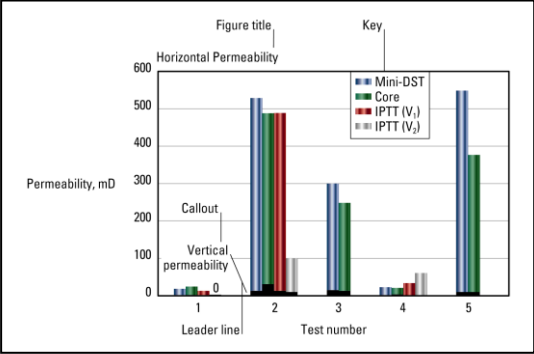


Figure 3. The QUANTUM™ PERFPAC™ sand control system provides one-trip perforating and gravel packing.

For axis labels on graphs, capitalize only the first word and proper nouns. Unit values follow, separated from the label by a comma.



Comparison of horizontal permeability measured by mini-DSTs, cores, and IPTTs shows generally good agreement of the different measurements.

Abbreviations, acronyms, and symbols

Abbreviations and acronyms are important space savers in technical writing and help provide the maximum amount of information in limited space. Consider the audience's level of knowledge of the subject when deciding whether to use abbreviations. Then, use abbreviations only if you are sure the reader will understand them.

Oilfield, technical, and nontechnical abbreviations are in the list of "[Spellings and Terminology](#)." If an abbreviation is not listed there, consult *Merriam-Webster's Collegiate Dictionary* (11th ed.).

The term **symbol** is used for the abbreviations of units of measure that are used with numeric values. SLB uses the American Petroleum Institute–Le Système International d'Unités (API-SI) unit symbols specified in Recommended Practice (RP) 66, which is under the stewardship of the Petrotechnical Open Software Corporation (POSC). Only units of the SI and the API-SI supplementary unit symbols and derived unit symbols are used to express the values of quantities. Exceptions and their usage are explicitly identified in this guidance and in the set of SLB "[Unit Symbols and Prefixes](#)." Additional exceptions for the use of nonpreferred unit symbols are as required by contract, law, or regulation or as appropriate for the material. "[Guidelines for Unit Symbols](#)" further details the use of unit symbols and unit names.

General format and usage

Consult the list of "Spellings and Terminology" for a list of abbreviations that can be used as stand-alone terms. Otherwise, spell out an abbreviated term the first time it is used with the abbreviation enclosed in parentheses following the term. Thereafter, the abbreviation may be used alone. In tables, titles, and mark identifiers, an abbreviation that is not stand-alone may be used without spelled-out introduction in consideration of space constraints. These abbreviations must be defined in accompanying text or a table footnote.	Incorrect: The United States (US) rig count rose 2% over the previous quarter. Correct: The US rig count rose 2% over the previous quarter. Incorrect: Today, efficient drilling includes PDC bits, oil-base mud, and measurements while drilling. Correct: Today, efficient drilling includes polycrystalline diamond compact (PDC) bits, oil-base mud, and measurements while drilling. In recent years, the use of PDC bits has boosted the average penetration rate to more than 100 m/d.
Use a spelled-out term (for example, and instead of & [ampersand] and at instead of @ ["at" symbol]) instead of a symbol for the word unless the symbol is a component of a company name, abbreviation, or e-mail address. See Section 5.1.1 in the "Writing for Our Public Websites" guidelines of the <i>Communications Handbook</i> on use of ampersand in writing for the Web. Do not use the number symbol (#), also called the hash, hashtag, or pound sign, as a garnishment but keep it in official names. An exemption is using # in "#1" for superlative, informal statements.	Correct: Audits of D&A wells are conducted by Gain & Loss Inc. Correct: E-mail drilling@slb.com for information. Incorrect: Valves #1 and #3 failed. Correct: Valves 1 and 3 failed. Correct: The MC 941 #3 well is in approximately 4,000 ft of water. Incorrect: The no. 1 name in rock-ready bits Incorrect: The number-one name in rock-ready bits Correct: The #1 name in rock-ready bits
	Incorrect: Casing Collar Locator (CCL) Correct: casing collar locator (CCL)

Do not use initial capital letters for the spelled-out form of an abbreviation unless the words are proper nouns.	Correct: Petrotechnical Open Software Corporation (POSC)
An abbreviation retains its capitalization in figure callouts and table headings and entries.	
Most acronyms are pronounced as words; most initialisms are pronounced as a series of letters. Use a or an as the indefinite article depending on the first sound of the abbreviation.	Correct: an FAQ Correct: a NACE standard
Minor variations to what an abbreviation stands for can be read in context and do not require reintroduction. However, plurals are formed from the abbreviation itself (for example, the plural of BHA for bottomhole assembly is BHAs).	Correct: The cost to plug and abandon (P&A) the well was included in the contract. However, test results were so promising that the well was completed instead of P&A [read as “plugged and abandoned”].
The plural forms of abbreviations for singular terms are formed by adding an s to the abbreviation. An apostrophe is not used for a plural form of an abbreviation unless it is necessary to avoid confusion.	Incorrect: A new mobile offshore drilling unit (MODU) was designed for the unique drilling conditions, and 15 MODU's were quickly shipped to the field. Correct: A new mobile offshore drilling unit (MODU) was designed for the unique drilling conditions, and 15 MODUs were quickly shipped to the field.
Close up initials in a person's name.	Incorrect: P. D. Smith Correct: P.D. Smith
	Correct: W.N.T. Doran
Abbreviate academic degrees without periods or spaces.	Correct: BS, BSc, BFA Correct: MBA, PhD, DSc
The names of months can be spelled out or abbreviated; use one style consistently in text. Abbreviations are useful in graphics and tables because of space constraints. See guidance on the use of months in references .	Correct: Aug. 1988 <i>or</i> Aug 1988 <i>or</i> August 1988 Correct: Aug. 24, 1988 <i>or</i> 24 Aug. 1988 Correct: Aug 24, 1988 <i>or</i> 24 Aug 1988
Symbols for chemical elements may be used without introducing them following the spelled-out element name. Chemical symbols have an initial capital letter. Do not mix symbols with words in the name of chemical compounds.	Correct: The fluid was saturated with Fe and Mn. Correct: The fluid was saturated with iron (Fe) and manganese (Mn). Incorrect: The fluid was rich in Na chloride. Correct: The fluid was rich in sodium chloride. Correct: The fluid was rich in NaCl.

Time zones are stand-alone abbreviations and given in parentheses immediately following the time when the time is used. For promotion of trade shows or events (e.g., OTC, ADIPEC, etc.), it is recommended to use the time zone promoted by the trade show or event for consistency.	GMT—Greenwich mean time EST—eastern standard time EDT—eastern daylight time CST—central standard time CDT—central daylight time CT—central time MST—mountain standard time MDT—mountain daylight time PST—Pacific standard time PDT—Pacific daylight time UTC—coordinated universal time Incorrect: The event starts at 4:45 p.m. CST. Correct: The event starts at 4:45 p.m. (CST). Correct: The event starts at 13:00 (UTC–05:00). Correct: All times are UTC+04:00. Correct: All times are Houston local.												
Common oilfield chemical compounds may be referred to by their formulas (element symbols) without first introducing them following the spelled-out name: CaCO₃ (calcium carbonate) CH₄ (methane) CO₂ (carbon dioxide) HCl (hydrochloric acid) HF (hydrofluoric acid) H₂O (water) H₂S (hydrogen sulfide) KCl (potassium chloride) N₂ (nitrogen molecule or molecular nitrogen) NaCl (sodium chloride) H₂ (hydrogen molecule) O₂ (oxygen molecule or molecular oxygen) SiO₂ (silica) CaBr₂ (calcium bromide) NaBr₂ (sodium bromide) ZnBr₂ (zinc bromide) Ca(OH)₂ (calcium hydroxide, lime) Na(OH)₂ (sodium hydroxide, caustic soda) BaSO₄ (barite). Do not mix symbols with words in the name of chemical compounds.	Correct: The well was treated with first HCl and then HF. Correct: The well was treated with first hydrochloric acid (HCl) and then hydrofluoric acid (HF).												
Hydrocarbon compounds are expressed by the number of carbon atoms.	<table> <tr><td>methane</td><td>C₁</td></tr> <tr><td>ethane</td><td>C₂</td></tr> <tr><td>propane</td><td>C₃</td></tr> <tr><td>butane</td><td>C₄</td></tr> <tr><td>pentane</td><td>C₅</td></tr> <tr><td>hexane and higher</td><td>C₆₊</td></tr> </table>	methane	C ₁	ethane	C ₂	propane	C ₃	butane	C ₄	pentane	C ₅	hexane and higher	C ₆₊
methane	C ₁												
ethane	C ₂												
propane	C ₃												
butane	C ₄												
pentane	C ₅												
hexane and higher	C ₆₊												

Words such as figure , equation , and volume may be abbreviated when followed by a number except when used at the beginning of a sentence. Do not abbreviate table , chapter , or appendix . These words and their allowed abbreviations are capitalized if they precede a numeral or letter used for specific identification.	Incorrect: This is obvious in Figure 8. Correct: This is obvious in Fig. 8. Correct: Figure 8 shows the obvious relation.
	Correct: Eq. 4 Correct: Vol. 39 Correct: Table 1-7 Correct: Chapter 4 Correct: Appendix C

Unit symbols of measure

When accompanied by a numeric value, units of measure are formatted as symbols, not the unit names. The SLB "Unit Symbols and Prefixes" list is based on API-SI RP 66 units. Do not mix unit names with unit symbols. The exceptions to the use of symbols include day, hour, minute, second, and year, which are spelled out for nontechnical uses. Week and month are not abbreviated. Do not use the pound sign (#). Powers in unit symbols can be superscript trademark symbol or baseline; one format should be used consistently in a document. The unit symbol is separated by a space from the number or joined with hyphen if the number-symbol pair is used as a compound adjective preceding the modified noun. (See guidance on compound words.) Exceptions are the degree symbol used for angles and the percent symbol or as required by contract, law, or regulation.	Incorrect: 10 meters Correct: 10 m
	Incorrect: We met from 8:30 a.m. to 10 p.m. every Thursday for 5 yr. Correct: We met from 8:30 a.m. to 10 p.m. every Thursday for 5 years.
	Incorrect: 3 kilogram/m, 3 kg/meter, or 3 kg per m Correct: 3 kg/m
	Incorrect: 250 # Incorrect: 250 pounds Incorrect: 250 lb Correct: 250 lbm
	Incorrect: The 8, 11, and 14-in wellbores developed only minimal scale. Correct: The 8-, 11-, and 14-in wellbores developed only minimal scale. Correct: The 8-in, 11-in, and 14-in wellbores developed only minimal scale.
	Incorrect: The average density was 5 g/cm3, and the maximum was 10 g/cm ³ . Correct: The average density was 5 g/cm ³ , and the maximum was 10 g/cm ³ . Correct: The average density was 5 g/cm3, and the maximum was 10 g/cm3.
For a range or two or more serial values with the same unit, the unit symbol can be listed with each value or with the final value; whichever format is used must be consistently applied in a document. It must be clear which symbol belongs to a number, and a number and its accompanying symbol may not be typographically separated. Repeat nonalphabetic symbols with each numeric value in a range or series.	Correct: alternating increments of 5 cm and 10 cm Correct: alternating increments of 5 and 10 cm Correct: For a saturation range of 7% to 19%, angle variance was from 10° to 25°.
	Incorrect: 35 × 48 cm Correct: 35 cm × 48 cm
	Incorrect: 70 g ± 2 Correct: 70 g ± 2 g
	Incorrect: The value is expressed in km. Correct: The value is expressed in kilometers.

Do not use a symbol for a unit of measure that is not accompanied by a number unless the unit is complex.	Incorrect: The thermal conductivity was recorded in watts per meter kelvin. Correct: The thermal conductivity was recorded in W/(m.K).
Unit symbols do not have ending punctuation except at the end of a sentence. Use numerals, not letters, for square and cube (an exception is psi for pounds per square inch). Do not add an s to a unit symbol to indicate plural.	Incorrect: 5 inches Incorrect: 5 in. Correct: 5 in
	Incorrect: The interval drilled was 5 m Correct: The interval drilled was 5 m. Incorrect: A 5-m. interval was drilled. Correct: A 5-m interval was drilled.
	Incorrect: square feet or sq ft Correct: ft ² or ft ²
	Incorrect: 10 cms Correct: 10 cm
SI units are preferred to US customary oilfield units that are accepted for use with RP 66 (for example, m ³ or m3 for cubic meter is preferred to bbl for barrel). Exceptions are nominal diameters of pipe (tubing and casing), bit sizes, and nominal tool diameters, for which the preferred unit is the inch (See guidance on specific usage conventions .) Equivalent values are given in secondary units in square brackets following the primary units only if necessary for the intended audience (for example, as historical secondary units). The structure of multiple sets of units should be parallel.	Correct: [20 cm] 8 in
	Incorrect: 61 m [20-ft] trajectory Correct: 61-m [20-ft] trajectory Incorrect: The 1920 calculation of reserves was 5 Bcf. Correct: The 1920 calculation of reserves was 140 million m ³ [5 Bcf]. Incorrect: The wellbore was reamed with an 11.43-cm [4 1/2-in] bit to 4,500 m [14,760 ft] before the 7.62-cm [3-in] casing was run. Correct: The wellbore was reamed with a 4 1/2-in bit to 4,500 m [14,760 ft] before the 3-in casing was run.
Do not mix other information with unit symbols.	Incorrect: 20-mL H ₂ O/kg or 20 mL of water/kg Correct: water content is 20 mL/kg
For scientific or statistical writing, express percentages with the % symbol. In other text spell out the word percent .	Incorrect: The outcrop cores were vacuum saturated with 2 percent NaCl solution. Correct: The outcrop cores were vacuum saturated with 2% NaCl solution.
	Correct: The crew was urged to give 100 percent on the job.

Numbers

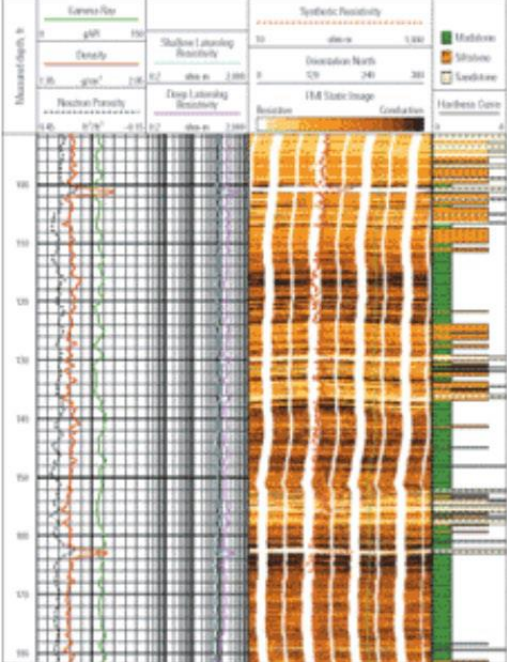
Spelled-out and numeral forms

Spell out the numbers one through nine except when used with units of measure or in expressions with a mathematical sense. (Do not repeat a spelled-out number in numerals, which is a legal convention.) Spell out ordinal numbers from first through ninth; use integers for 10th and higher. Use only numerals for statistical and tabular data, dates, addresses, and time. Very large or very small numbers in a technical context can be expressed in scientific notation. Large, rounded numbers can be spelled out or expressed as a combination of numerals and words, but this format is not used for precise values with units of measure.	Incorrect: five (5) wells Correct: five wells Correct: 10 wells Correct: 5 ft Correct: Rinsing the sample two times reduced the likelihood of contamination by a factor of 4. Correct: second Correct: 22nd Correct: 3 p.m.
If numerals are used for one of the numbers in a specific category or series in a sentence, use numerals for all category mentions in the sentence.	Incorrect: August 12th, 2000 Correct: August 12, 2000 Correct: The value of Avogadro's number for the number of atoms or other particles in a gram mole of a substance is 6.023×10^{23}. Correct: almost a thousand years Correct: The country's population is about 2 million.
Spell out a number that begins a sentence or recast the sentence.	Correct: Together the three fields have six producing wells and nine injection wells. Correct: Together the three fields have 6 producing wells and 12 injection wells. Correct: The reservoir sand was 19 m thick in one well.
Spell out the first number to distinguish between two juxtaposed numbers.	Incorrect: 45 days later . . . Correct: Forty-five days later . . . Correct: After 45 days . . .
Spell out the first number to distinguish between two juxtaposed numbers.	Incorrect: 10 2-min tests Correct: ten 2-min tests

There are several options for presenting time of day in numerical format, but one style should be used consistently throughout a document. Digital clocks have increased the use of four-digit numbers in all countries. In the US, a colon separates the hour and minutes and a.m. or p.m. is used. Use a colon to separate the hour and minutes in 24-h time. If writing for a local audience, select the option appropriate to the region. In the UK, a full stop separates the hour and minutes. When exact times are emphasized, the 24-h system is customary, and a colon separates the hour and minutes. In Norway, the 24-h system is used. There is no punctuation between hours and minutes, and a.m. and p.m. are not used. In some parts of Southeast Asia and the Middle East, four digits are used, and a colon or period separates the hour and minutes. For promotion of trade shows or events (e.g., OTC, ADIPEC, etc.), it is recommended to use the time (i.e., 24-h time system vs. 12-h time system) promoted by the trade show or event for consistency.	Customary for US: Cookies are served in the cafeteria at 9:15 a.m. and 4:30 p.m. Customary for US: Operations resumed at 08:00 when the weather cleared. Customary for UK: The cafeteria begins tea service at 4.30 p.m. Customary for UK: Trains depart for London at 08:12 and 16:33. Customary for Norway: 0800 (8 a.m.), 2400 (midnight) Other options: 08:30 or 08.30
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Numeral formats

In the US, which is transitioning to the metric system, numbers with four or more digits are typically separated into groups of three digits by a comma. US monetary values and US patent numbers are always written with commas. In countries that use the SI system, numbers with four digits typically are set closed. Numbers with five or more digits typically are separated into groups of three digits by either set spaces or commas. When set spaces are used before the decimal point (which can also be represented by a comma), spaces are also used, if necessary, after the decimal. Commas are never used after the decimal point. Dates, pages, and addresses never use commas or spaces.	Incorrect for US: 4000 Incorrect for US: 4 000 Correct for US: 4,000
	Incorrect: 10000 Correct for US: 10,000 Correct for UK: 10 000
	Incorrect for US: 10,000 strokes in 1500 min Correct for US: 10,000 strokes in 1,500 min
Form plurals of numerals by adding an s.	Incorrect: During the 1990's Correct: During the 1990s

Use a colon to express a ratio between numerals.	Incorrect: The ratio of engineers to drillers at the wellsite was 3/1. Correct: The ratio of engineers to drillers at the wellsite was 3:1.
If depths require anonymity, use letters to alias the digits. For example, on the depth track of a log if all depths have the same number of digits (for example, all four-digit or all five-digit numbers), replace the thousands digits with an X for 1,000 to 9,999 (X,000 to X,999) or with XX for 10,000 and deeper (XX,000). Do not have X,000 and XX,000 in the same depth track because that reveals that the corresponding depths are obviously 9,000 and 10,000. If a specific depth or depth interval must be differentiated in an aliased depth track, use different letters in alphabetic sequence. For example: XX,000 XX,500 XY,000 XY,500 XZ,000 XZ,500	
Spell out common fractions when they are used alone in the text. A spelled-out fraction uses an internal hyphen unless it has a hyphenated component. Use numbers when the fraction is combined with a whole number or when used with a unit of measure. For web or digital collateral, nonstacked (e.g., 3/8) vs stacked (e.g., ⅜) fractions (are preferred due to limited formatting.	Correct: one-third the cost Correct: The performance rating increased by thirty-two hundredths of a point. Incorrect: Two and three-fourths in Correct: 5 1/2 days
A decimal point must be preceded and followed by a numeral. The decimal point typically is represented by a period or baseline dot. Continental Europe: The decimal point typically is represented by a comma.	Incorrect: .35 cm Correct: 0.35 cm Incorrect: 5. Correct: 5.0 Correct: 36,333.33 Correct for continental Europe: 36 333,33
An en dash is used for a minus or negative sign; for that reason, avoid ambiguous constructions.	Incorrect: The temperature range was –15—5 degF. Correct: The temperature range was –15 to –5 degF. Incorrect: 6 - 3 = 3 Correct: 6 – 3 = 3

Numerals and unit symbols

Formats

When a unit of measure is accompanied by a numeric value and used as an adjective, a hyphen is placed between the number and the unit.	Incorrect: 5 in casing Correct: 5-in casing Correct: 3-ft zone
	Incorrect: 30-m [100-ft]/h rate Correct: 30-m/h [100-ft/h] rate
The SLB “Unit Symbols and Prefixes” list of RP 66–based unit symbols gives preference to the SI unit symbols over US customary oilfield units. Exceptions are nominal diameters of pipe (tubing and casing), bit sizes, and nominal tool diameters, for which the preferred unit is the inch. (See guidance on specific usage conventions.) If desired, SI and US customary units may both be given; usually the SI unit symbol is listed first with the secondary unit system given within brackets. The conversion factors for the SLB unit symbols are exact values or to seven-digit accuracy and are primarily from <i>The SI Metric System of Units and SPE Metric Standard</i> (June 1984, 2nd printing). For guidance on rounding numbers in making unit conversions, see page 57 of Guide for the Use of the International System of Units (SI). This US National Institute of Standards and Technology (NIST) publication is based on International Organization for Standardization (ISO) conventions.	Correct: Measurements were taken at 5-m intervals. Correct: The second dome was drilled in 60 m [200 ft] of water. Correct: For use in high-temperature conditions (i.e., >204 degC [>400 degF]), the tool must be modified.
	Incorrect: 30-m [100-ft]/h rate Correct: 30-m/h [100-ft/h] rate
	Correct: The 5.5-in [139.7-mm] tool became stuck in 7-in casing. Correct: The 6 5/8-in casing used in the completion had a wall thickness of 0.288 in [7.32 mm].

Specific usage conventions

Tubing diameters for sizes 1.050 through 2.063 in are presented in decimal inches to three decimal places. Sizes 2 3/8 in and larger are presented in fractions to designate API nominal size, but measured diameters and wall thickness, etc., are presented in decimal inches to three decimal places. Casing diameters for API nominal sizes are presented in fractions, but measured diameters and wall thickness, etc., are presented in decimal inches to three decimal places.	Correct: The 1.315-in tubing has a wall thickness of 0.133 in [33.78 mm]. Correct: This 6 5/8-in casing has a wall thickness of 0.288 in. Correct: 4.5-in 12.75-lbm/ft casing (no comma in industry usage)
Bit sizes and drill collars are expressed as nominal sizes in inches and fractions.	Correct: Although the final section was drilled with an 8 1/2-in bit, the wellbore diameter measured by the 4.5-in caliper tool was 12 in.

Tool outside and inside diameters (ODs and IDs, respectively) are expressed in decimal inches to the number of decimal places required to correctly express the fraction. For example, 1 11/16 in is expressed as 1.6875 in, 4 1/2 in is expressed as 4.5 in, 6 3/4 in is expressed as 6.75 in, and 9 5/8 in is expressed as 9.625 in.	Correct: Even the slim 2.125-in tool could not pass the restriction where the borehole had collapsed.
Conversion of tool diameters and measured dimensions of casing and tubing in US customary units is to millimeters only and displayed to two decimal places.	Correct: Corrosion and wear had reduced the thickness of the 4 1/2-in casing to 0.199 in [5.05 mm] from the original 0.205 in [5.21 mm].
API connections are expressed as a fraction and API type, but the unit (inch) is not included and no SI conversion is made.	Incorrect: 6 5/8-in Reg Incorrect: 6 5/8-in [168.27-mm] Reg Correct: 6 5/8 Reg
Length dimensions are usually presented in centimeters to one decimal place or in meters to one or two decimal places. Critical lengths such as measure points may be expressed to three decimal places. In table columns or rows listing like dimensions, use only one convention. For example, if the tool dimensions are all in meters, even short sections should be given in meters (0.43 m, not 43 cm) to avoid the unintentional calculation error of adding centimeters to meters. Similarly, in a table or figure use only one convention for related dimensions that may be part of a calculation (for example, all lengths may be centimeters, but all diameters may be decimal inches).	Incorrect: Removing the 30-cm sub reduced the tool length to 3.1 m. Correct: Removing the 30-cm sub reduced the tool length to 310 cm.
Tool resolution is expressed in centimeters or meters without decimals.	Correct: The first tool has a vertical depth resolution of 15 cm and a depth of investigation of 25 to 125 cm.
Units of temperature do not use unit symbols with a degree symbol (°). Kelvin temperatures are conventionally written without the degree symbol. Unsigned temperature values are assumed to be positive, so a plus sign closed up to the temperature is used only as necessary to avoid ambiguity. The equivalence of the temperature units is 1 degC = 1.8 degF or 1 degF = 0.56 degC. Temperature conversions are in the SLB “Unit Symbols and Prefixes” list of RP 66–based unit symbols.	Incorrect: 109°C [228°F] Correct: 109 degC [228 degF] Correct: 114 K
	Incorrect: The temperature range was –15 to +5 degF. Correct: The temperature range was –15 to 5 degF. Correct: –10 degC to 150 degC [14 degF to 302 degF]

Nuclide descriptors are arranged around an element symbol as follows: - number of atoms as a right subscript - mass number of an isotope as a left superscript trademark symbol (or hyphenate it to the element name) - atomic number of an atom (used only for nuclear chemistry) as a left subscript - ionic charge as a right superscript trademark symbol (if the charge number is 1, only the sign is shown) - oxidation number (roman numeral) as a right superscript trademark symbol or in line and closed up in parentheses following the element name or symbol.	Correct: H₂S Correct: The large C₁ peak corresponds to the methane content.
	Correct: ¹³⁷Cs Incorrect: ¹³⁷cesium Correct: cesium-137
	Incorrect: Na¹⁺ Correct: Na⁺ Incorrect: Ca⁺⁺, CO₃⁻⁻ Correct: Ca²⁺, CO₃²⁻
	Correct: Mn^{III} Correct: Mn(III) Correct: manganese(III)
Monetary amounts are formatted as numerals preceded by the three-letter currency code from ISO Standard ISO 4217. The currency code typically consists of a two-letter country code followed by a one-letter currency code. Very large or rounded amounts can be written as numerals preceding a numeric unit, but do not use the ambiguous terms billion or trillion because they have multiple meanings. A hyphen is not used in currency terms.	Correct: USD 5,000 (US dollar) Correct: USD 10 million Correct: CAD 800 (Canadian dollar) Correct: GBP 14 (UK pound sterling) Correct: EUR 620 (European Union euro) Correct: JPY 80 (Japanese yen) Correct: An additional USD 5 million was added to the USD 10 million commitment.

Mathematics

Equations follow the same rules of grammar and punctuation that apply to sentences of text. Mathematics is written in sentences; terms and operational symbols function as subjects, verbs, conjunctions, or adjectives and are formatted accordingly. Correct grammatical structure must also be maintained in text sentences that contain, precede, or follow equations. Complete sentences preceding equations may end in a period or colon.

Symbols and special characters should be inserted in Microsoft Office documents with [ANSI codes](#) because they are more stable than those inserted in a document by using the Insert menu or keyboard shortcuts. To insert a symbol or character code, NUM LOCK must be on. Hold down the ALT key and use the numeric keypad to type in the four-numeral code. Similar [character codes are used for Macintosh](#) computers.

In long documents or those that are mathematically intensive, all letter symbols should be defined in a formal, alphabetical nomenclature in the backmatter.

Set all letters used as mathematical terms (either as quantities or numerical values) in italic type except for whole words, contiguous letters that form abbreviations, or letters that correspond to proper nouns. Numerals, Greek alphabet characters, and abbreviations such as log (logarithm), sin (sine), and exp (exponential function) are also set in roman type. Letter symbols must be single letters; their sub- and superscript trademark symbols can be multiple letters, numerals, or both.	Correct: $(59)\log_a x$ Correct: vertical permeability = k_v or k_{vertical} Correct: maximum temperature = T_{max} Correct: Reynolds number = N_{Re} Correct: longitudinal relaxation time = T_1 Incorrect: The tool response (TR) is defined as $TR = f(V_1, V_2, \dots, V_n, V_f)$. Correct: The tool response (TR) is defined as $X = f(V_1, V_2, \dots, V_n, V_f)$.
If commas or operational signs are required in elisions, they should come after each term and after the three ellipsis dots when followed by a final term.	Incorrect: $a_1, a_2, \dots a_n$ Correct: $a_1, a_2, \dots, a_n$
	Incorrect: $x = 0, 1, 2 \dots$ Correct: $x = 0, 1, 2, \dots$
An operational symbol used as an adjective is not spaced apart from the term it modifies; a symbol used as a verb or conjunction has a space before and after it.	Correct: an error of ± 3 cm Correct: $-x^2(a + b) = -2$ Correct: Eq. 5 is true if $g < 4$.
Do not use an asterisk (*) as an operational symbol for multiplication because the asterisk is reserved as a footnote. Multiplication may be indicated by a times sign ($\times$) or multiplier dot ($\cdot$) (a baseline dot [$\cdot$] is used within unit symbols), but only one of these operational symbols should be consistently applied in a document.	Incorrect: $A = 2B * F^2$ Correct: $A = 2B \times F^2$ Correct: $A = 2B \cdot F^2$ Correct: $A = 2B(F^2)$ Correct: 5 ft $\times$ 5 ft Correct: 5(20)
An en dash is used for a minus or negative sign; for that reason, avoid ambiguous constructions.	Incorrect: The temperature range was $-15\text{—}5$ degF. Correct: The temperature range was -15 to -5 degF. Incorrect: $6 - 3 = 3$ Correct: $6 - 3 = 3$
Use a colon to separate parts of a ratio or proportion expressed in numerals.	Correct: a 3:1 mixture

Expressions displayed on separate lines should be centered within the margins. The equation number should be placed flush with the right-hand margin on the last line of the equation. Enclose the number within parentheses. Equations are generally numbered in sequence throughout the body of the text: (1), (2), (3), etc. Label two-part equations as (1a) and (1b) rather than (1) and (1a). If equations appear in appendices, number them in independent sequence corresponding to each appendix. Equations in Appendix A are labeled (A-1), (A-2), etc., and those in Appendix B are (B-1), (B-2), etc. If there are chemical reactions displayed in the same document, they may be numbered with a separate sequence of letters or roman numerals.	Correct: Therefore, Eq. 6 may be reduced to $abc = def, \quad (7)$ where a equals 2.
When referring to numbered equations in text, omit the parentheses from the number.	Correct: We substitute Eq. 6 into Eq. 7.
When an equation will not fit on one line, try to break it at the equals sign. If this is not possible, break the equation after a complete term and begin the next line with a sign. If the equation requires more than two lines, align each line on operational or descriptive signs. If possible, do not break fractions or expressions in parentheses, brackets, and braces or after radical signs.	Correct: $c_t = (0.7)(10^{-5}) + 0.3(3 \times 10^{-6})$ $+ (3 \times 10^{-6})$ $= 1.09 \times 10^{-5} \text{ psi}^{-1}. \quad (2)$
Use the solidus (/) or format the denominator with a negative exponent for equations that occur in text rather than as display. Terms in display equations in a document should be consistently formatted stacked or in a one-line form (solidus or negative exponent).	Correct: It was assumed that $a/b = c/d = 1$. Correct: It was assumed that $ab^{-1} = 1$.
	Incorrect: $x = a/b + \left(\frac{y}{z}\right)^2$ Correct: $x = \frac{a}{b} + \left(\frac{y}{z}\right)^2$
The preferred order for signs of aggregation is parentheses, brackets, and braces, working from the inside of the equation outward: $\{[(\{[(\{())\})]\}] \}$	Correct: $3\{[A(B - C/D) + E]^2\} = F$

Lists in text

List types

Lists within text can be used to present a series of analogous elements. All items of a list should have the same syntactic structure (i.e., parallel construction).

Long lists can overwhelm readers, who need to be able to comprehend a list as a whole and see how the parts are related. Research on short-term “working” memory indicates that most people can remember between three and nine items for short periods. To improve the readability and usability of lists, the recommended limits for the number of items are

- nine items maximum for information presented in print form
- seven items maximum for information presented online.

If a list contains more than 10 list items, reorganize the list into multiple separate lists of no more than 10 items each or use sublists to group related items. Long lists of reference information (usually presented in alphabetical order) can be acceptable because readers look up selected information from the list rather than read the list from end to end.

List types are as follows:

- ordered (consisting of items that are sequential in hierarchy or action, such as steps to a procedure)
- unordered
- defining (used for reference or appendix material, such as a glossary or dictionary).

For ordered lists, introduce each item with a numeral. Typically, a list with sequential items is individually numbered (i.e., the first item on each list is numbered “1”), but the lists can be sequentially numbered within a whole document or part of a document.

A single-step list that implies sequence can have the item numbered “1” or begin with a bullet; another option is to rewrite to avoid the single step. A consistent approach should be followed within a single document. For example, single-step lists emphasize the action item for procedures that require only a single step. An additional consideration is that numbering or bulleting a single list item helps differentiate the list item from the surrounding text.

For unordered lists, introduce each item with a bullet.

Lists can be run in a paragraph or arranged vertically. Vertical arrangement is preferred for long lists or if the list items are lengthy or complex. The enumeration and demarcation of a list depends on its format and type.

List Format and Type	Enumeration and Demarcation
Run in paragraph and not sequential or hierarchical	No enumeration: Separate items with commas or semicolons as appropriate for the complexity of the items
Run in paragraph and sequential or hierarchical	Enumerate with numerals within parentheses and separate items with commas or semicolons as appropriate for the complexity of the items
Vertical and not sequential or hierarchical	Begin each item with a bullet (•)
Vertical and sequential or hierarchical	Enumerate each item with a numeral followed by a period. A bullet can be used for a list with a single item.

To emphasize subsidiary items in vertically arranged lists (e.g., substeps in procedures), use sublists. No more than two levels of sublists are recommended. Ordered sublists should have numbered subitems; unordered sublists introduce subitems with a bullet.

List formats

Use a vertical format if items in a list are complete sentences. Begin each item with a capital letter and end each item with a period. The introductory sentence ends in either a colon or a period.	Correct: The following precautions are recommended during the run-in phase: • The gun should be lowered as fast as local practice dictates. • All restrictions or ID changes should be noted and speed reduced as the gun passes them. Correct: Two precautions are recommended during the run-in phase. • The gun should be lowered as fast as local practice dictates. • All restrictions or ID changes should be noted and speed reduced as the gun passes them.
When items in a list are syntactic continuations of the introductory clause, begin each item with a lowercase letter unless the item begins with a proper noun. Place a period at the end of the last item only.	Correct: Each center provides a range of activities to support oil companies: data management, interpretation, workstations, data transfer, and communication. Correct: Each center provides a range of activities to support oil companies: • data processing, interpretation, and archival • Interpretation Bureau Services • workstation software and equipment • worldwide transmission and communication.
	Correct: Functionally test the motor as follows: 1. Open the rams and lower the motor below the rotary table. 2. Secure the motor and the kelly. 3. Turn on the pumps and slowly increase the stroke speed.
If the list is embedded within a sentence, do not place a period at the end of the last item in the list.	Correct: The procedure for data exchange, reset, clear, and restart has a local meaning. Correct: The procedure for • data exchange • reset • clear • restart has a local meaning.

Do not place a colon at the end of an introductory clause that ends in a verb or preposition.	Correct: Surface testing equipment must • safely contain well pressures • accurately measure volume, temperature, and flow rate • separate produced gas, oil, and water phases • allow collection of representative PVT-quality samples.
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Visual aids

Use visual aids (tables, figures, and graphs) to emphasize important ideas and to compare data. For the latter purpose, figures or tables to be compared should be of similar scale; however, the same data should not be presented in both tables and figures. In print materials, visuals that are not merely decorative are introduced in the text. The introduction should precede the visual and help the reader interpret the visual.

For examples and guidelines on how to use visuals for print and online communications, see the [“Brand Elements and Templates.”](#) Additional resources on visuals are listed in [“Suggested reading.”](#)

Figure captions and table titles

Except for visuals that are purely decorative elements, each figure should have a caption and each table should have a title. For figures, both complete sentences and fragments may be used as captions in the same document, although consistency is recommended. Figure captions—even sentence fragments—have sentence-style capitalization and terminal punctuation. Table titles should be brief phrases; explanatory material can be in one or more footnotes. Table titles are usually formatted with headline-style capitalization and no terminal punctuation.

If the figures and tables are sequentially numbered, separate sequences of Arabic numerals are used and are consecutive throughout the text and within each appendix. Within each appendix, use Arabic numerals and the same letter designation as that of the appendix (A-1, A-2, etc.).

All citations of numbered illustrations and nontabular material use the abbreviation **Fig.** unless the word begins a sentence, in which case it is spelled out **Figure**. Do not use the words **chart**, **exhibit**, **graph**, or **photograph** when naming a specific figure. The word **Figure** is always spelled out with the number in the caption. Designate numbered tabular text visuals by **Table**. Designate one portion of a multipart figure with uppercase letters when the figures are separate (Fig. 2A, Fig. 2B). Use lowercase letters when referring to distinct parts of a single figure (Fig. 2a, Fig. 2b).

Figures

Figure terminology

The following terms are used in these figure guidelines.

Figure Element	Definition
Callout	Descriptive term or symbol label within the figure
Caption	Explanatory material that is separate from but immediate to the figure

Figure Element	Definition
Figure title	Informative description (equivalent to a text heading) within the figure that applies to all or part of figure content and does not merely duplicate the figure caption
Key	Part of the figure that identifies the symbols used in the figure (also called a legend)
Leader line	Line indicating where a callout applies

Figure labeling

Capitalize the first word, nouns, pronouns, adverbs, and all other words of five or more letters in a figure title.

Capitalize only the first word and proper nouns in a callout. An exception is abbreviations, which retain their capitalization. If the first element of a callout is a numeral, the first following word is not capitalized (for example, **15-ft interval**).

Capitalization exceptions are company names or marks of SLB or other companies that begin with a lowercase letter.

A period is not used as terminal punctuation for figure titles and callouts except for those that are complete sentences.

Callout leader lines do not have arrowheads, which are reserved to depict motion or indicate continuing data, a directional relationship, or north.

Delete or recalculate scale ratios (for example, 1 in = 1 ft, magnification 5×) because they are not correct if the size of the graphic is different from that of the original log printout.

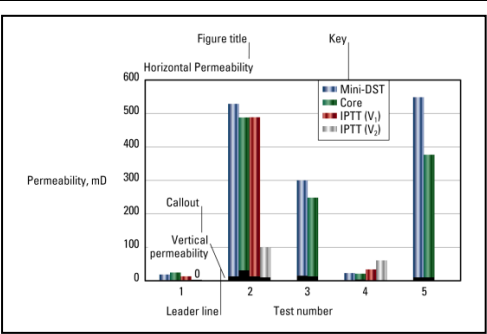
Figure 3. QUANTUM™ PERFPAC™ single-trip perforating and gravel- and frac-pack system provides one-trip perforating and gravel packing.

Capitalize only the first word and proper nouns in an axis label on a graph. An exception is abbreviations, which retain their capitalization. Unit values follow, separated by a comma. In rare instances an axis label consists solely of a unit value.

The axis labels are usually horizontally aligned; on three-dimensional “cube” graphs a label may be parallel to the axis if horizontal alignment distracts from the graph itself.

All symbols (including other relevant components, such as lines of calculated values) used on a graph must be identified in the text or caption or labeled on the graph, either directly with callouts or in an inset key. Data symbols are identified in a key in hierarchical order, if one applies.

Capitalize only the first word and proper nouns in a key. An exception is abbreviations, which retain their capitalization.



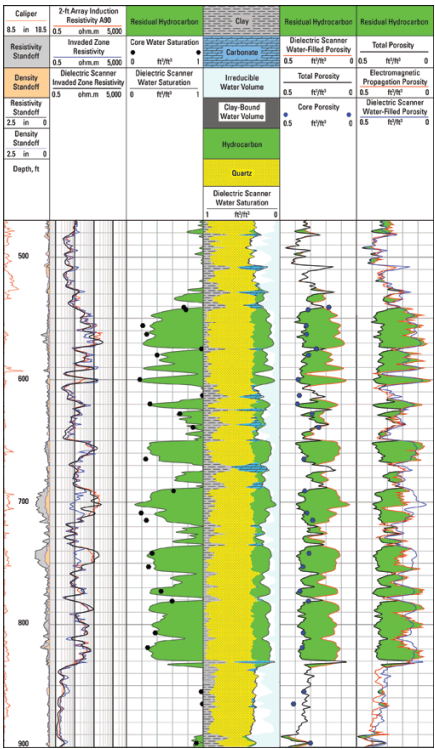
Comparison of horizontal permeability measured by mini-DSTs, cores, and IPTTs shows generally good agreement of the different measurements.

Logs

The track headers in the insert section of a log are edited following text guidelines. Tracks are not numbered or lettered; they are referred to by number, from left to right, although the depth track is typically not numbered but referred to as the **depth track**.

The log name is spelled out (do not use the mnemonic or an abbreviation) with headline-style initial capitalization and centered above an example of the log measurement curve with the scale endpoints labeled. The unit symbol is centered beneath the example curve.

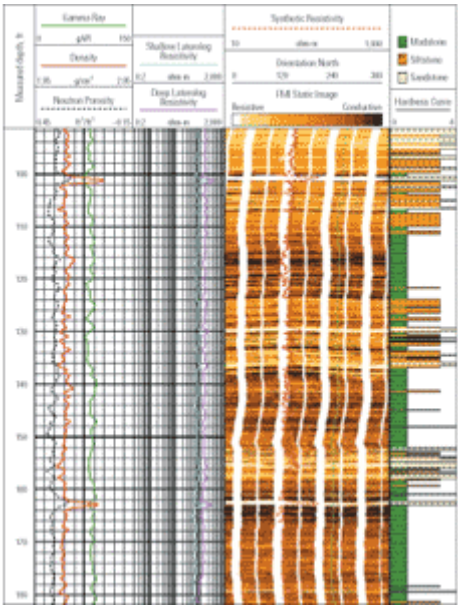
Delete scale ratios (for example, 1:40 m) because they are not correct if the size of the graphic is different from that of the original log printout.



If log depths require anonymity, use letters to alias the thousands-place digits. If all depths on the depth track are the same number of digits (for example, all four-digit or all five-digit numbers), replace the thousands digits with an **X** for 1,000 to 9,999 (X,000 to X,999) or with **XX** for 10,000 and deeper (XX,000). Do not have X,000 and XX,000 in the same depth track because that reveals that the corresponding depths are obviously 9,000 and 10,000.

If a specific depth or depth interval must be differentiated in an aliased depth track, use different letters in alphabetic sequence. For example:

- XX,000
- XX,500
- XY,000
- XY,500
- XZ,000
- XZ,500

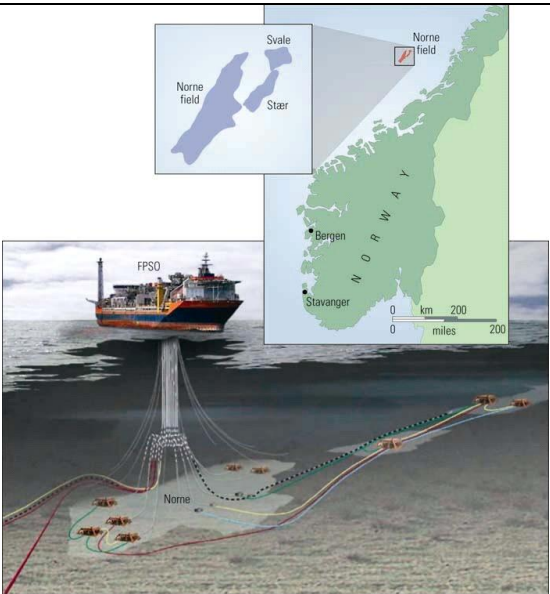


Maps

This guidance also applies to satellite imagery, orthophotos, and similar metric photography. Maps used for specific technical purposes, such as positioning wells and seismic lines, typically require more detailed coordinate information.

A map should present multiple sets of information, not just location details. For example, oil and gas fields should be depicted by their approximate boundaries rather than by dot symbols. Graphic elements can also be added into a map, but it's important to avoid information overload.

In this example, two maps are used to locate the field offshore Norway. Light shading links the detailed view to the boxed area on the larger scale map. The shape of the field is consistently represented across all three elements. However, two satellite fields to the right in the FPSO graphic should also be labeled. Blue color used for fields in the detailed view implies water rather than hydrocarbons.



Location names on maps follow these conventions:

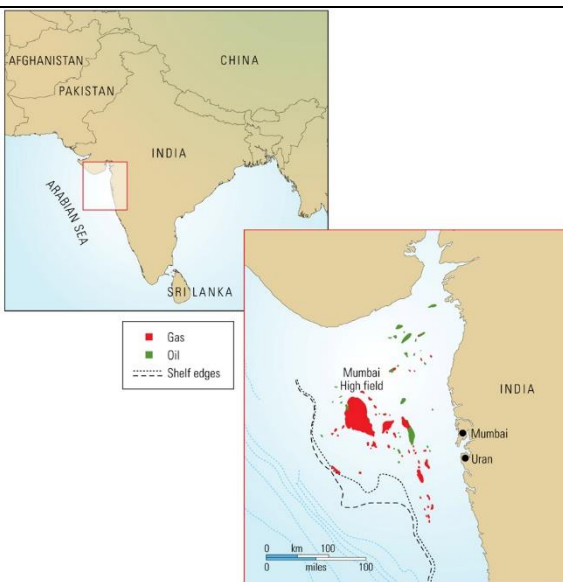
All names are spelled out unless space constraints require abbreviations (see [Spellings and Terminology](#) list for abbreviations such as USA, UK, and UAE.)

Country names are written in all capital letters.

State, province, basin, and field names use headline-style capitalization.

Major bodies of water are written in all capital letters; headline-style capitalization for all others. The text should curve to parallel the shoreline or river.

See [guidance on names of places](#) and [guidance on capitalization](#).

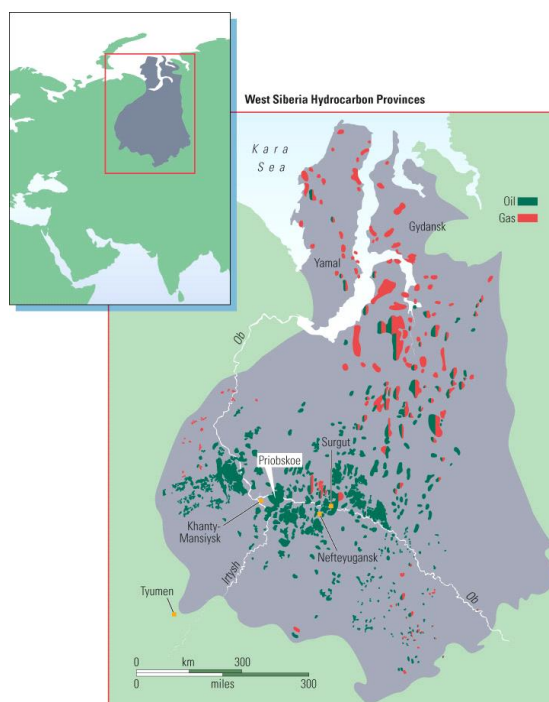


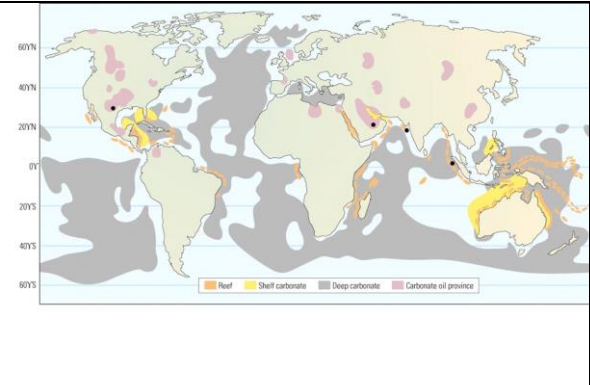
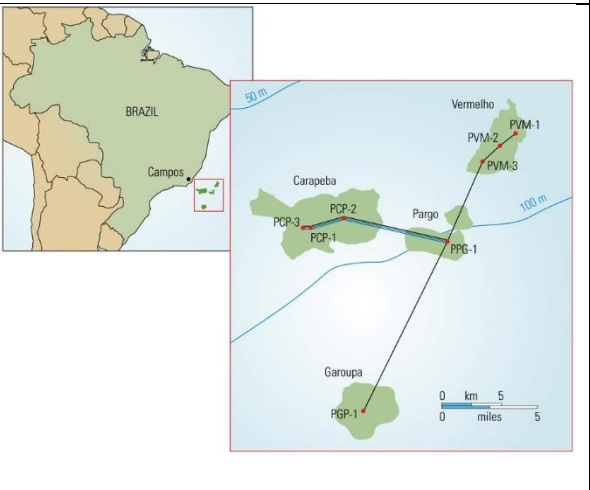
Map orientation assumes north is toward the top of the map. If it is not, a north arrow labeled "N" should be included.

Map scales are not required for global and continental maps or other large geopolitical entities familiar to the readers. If multiple maps are in the same graphic, at least one must have a scale. Scales should be provided in both SI and oilfield customary units, with the SI units on top. Alternating colored and white bars indicate scale subunits. The unit name is centered for its scale.

Map keys must use colors, codings, and line types that exactly match the corresponding features on the map. All symbols must be explained in the key or caption. Contour units can be included on the map or in the map key.

In the key, the first word and proper nouns should be capitalized, except for abbreviations, which retain their capitalization.



Color can be used to code important map entities, such as areas in a country, fields, fault blocks, or types of production. However, states and provinces should not be color-coded; they should be defined by boundary lines.	
Contours on a map must be clearly defined. At least two contours should be labeled, with every fifth interval typically shown in a heavier line weight and labeled. Contour units should be included either on the map itself or in the map key.	

Tables

The guidelines for table elements generally follow those for text, with concessions to the space limitations endemic to most tables. [Click here for example tables.](#)

Table terminology

The following terms are used in these table guidelines.

Table Element	Definition
Body	All vertical columns below the column headings and to the right of the stub
Cut-in heading	Heading that cuts across table columns and applies to all columns below it
Decked heading	Multilevel heading consisting of a spanner heading and at least two column subheadings with a horizontal spanner rule between the spanner and the subheadings
Headline-style capitalization	Capitalization of the first word, nouns, pronouns, verbs, adverbs, adjectives, and all other words of five or more letters
Rule	Printed line, horizontal or vertical
Sentence-style capitalization	Capitalization of only the first word, proper nouns, and proper adjectives

Table Element	Definition
Spanner heading	Heading that applies to at least two columns and may have subheadings
Spanner rule	Horizontal line between a spanner heading and the column subheadings to which it applies
Stub	First (left-hand) column that lists “row headings” of categories or individual entries for which information is given in the other columns
Title	Caption or heading for the entire table

Rules in tables

Horizontal rules are typically used between table elements. If entries in the stub are indented, the corresponding rules have the same indent. A rule is used beneath a spanner or a cut-in heading to show which subheadings and columns it applies to. Vertical rules are used for complicated tables.

Column and stub headings

Column headings have headline-style capitalization and no routine terminal punctuation. The only capitalization exception is abbreviations and unit symbols, which retain their capitalization. Headings can be singular or plural. Headings are bold and left justified except for centered spanner and cut-in headings. Headings are stacked “down” from the first line.

A column heading is not required for the stub if its contents are clear from the table caption. The stub heading uses sentence-style capitalization and can be singular or plural in number. The first level of stub entries is flush left; successive sublevels are indented to the right.

If the unit symbol is the same for all entries in a column or row, it is listed at the end of the column heading or row stub, respectively, set off by a comma. If entries are presented in two sets, or dual units, the equivalent secondary unit symbol is listed in brackets following the primary unit symbol. If the table consists of more than two columns (including the stub) and the entries have dual values in primary and equivalent secondary units, the unit can be listed with each value in brackets (for example, 20 cm [8 in]) to aid the reader instead of only in the heading or stub. Tables in a document with similar subject matter or with values that are meant to be compared should use a consistent format, listing the units in the headings or stub or with the values. In rare instances a heading consists solely of a unit.

Tool Specifications		
	Unit A Example	Unit B Example
Length—max., m [ft]	1.5 [5]	1.8 [6]
Weight—max., kg [lbm]	24 [55]	36 [80]

Tool Specifications		
	Length—Max.	Weight—Max.
Unit A example	1.5 m [5 ft]	24 kg [55 lbm]
Unit B example	1.8 m [6 ft]	36 kg [80 lbm]

Tool Specifications		
	Unit A Example	Unit B Example
Length—max.	1.5 m [5 ft]	1.8 m [6 ft]
Weight—max.	24 kg [55 lbm]	36 kg [80 lbm]

Tool Specifications		
	Length—Max., m [ft]	Weight—Max., kg [lbm]
Unit A example	1.5 [5]	24 [55]
Unit B example	1.8 [6]	36 [80]

Table body

Entries in the table body columns have sentence-style capitalization, with only the first word and proper nouns capitalized. If the first element of an entry is a numeral, the first following word is not capitalized (e.g., **15-ft interval**). They are left justified, including wrapped lines, to the column heading. The exception is columns of comparable (i.e., the same measured variable), single-entry numerical values in a column, which are decimal aligned to each other with the widest entry left justified to the column heading. Columns of comparable double-entry numerical values, ranges, or entries with operational signs are left justified, not decimal aligned. Units are preferentially in the column headings or stub to save space and avoid repetition, but they may be listed instead with each value. Placement of the units in the headings or stub or with the values should be consistent within a document for similar or comparable tables.

If the table has a stub, row entries are vertically aligned to the first line of the corresponding stub entry.

Entries should be parallel for both grammar and punctuation within a column:

- Words or sentence fragments on individual lines do not have terminal punctuation (including commas or semicolons).
- Words or sentence fragments grouped in a paragraph are separated by commas or semicolons; there is no terminal punctuation for the paragraph.
- Complete sentences on individual lines each have terminal punctuation.
- Complete sentences grouped in a paragraph either each have terminal punctuation or are separated by commas or semicolons. The last sentence should have terminal punctuation.

Do not use ditto marks for repeated entries; repeat the entry or list it once, spanning the applicable columns or rows. Either a blank entry or **na** (for **not applicable**) is used if a heading does not apply or an entry would not be logical; blank entries or **na** must be used consistently within a document. An en dash (–) for the entry indicates that no data were available or collected. A footnote should be used to explain the meaning of blank, **na**, or en dash entries.

Note: When tables are created using the PIMS, some of these guidelines may not be implemented due to the system’s limitations. For example, the inability to use three-place commas, proper decimal usage, nonstacked fractions, and inline use only for superscripts and subscripts.

Numbers in tables

- Use only numerals for statistical and tabular data, dates, addresses, and time.
- Numbers with four or more digits are separated into groups of three digits with a comma. Commas are not used to mark off groups of three digits in dates, pages, and addresses. Commas are used in all monetary values with four or more digits.

Abbreviations in tables

Check the list of “[Spellings and Terminology](#)” for the format for abbreviations and which abbreviations can be used as stand-alone terms. Abbreviations that cannot stand alone and that are

first mentioned in a table must be defined in a footnote except for common abbreviations. Most technical abbreviations and unit symbols retain their capitalization.

Footnotes in tables

Footnotes should be used in the table instead of letters or numerals, which could be mistaken for exponents or chemical symbols. [Footnote symbols](#) used in tables should be used in continuous sequence when footnotes are also used in the text, and do not begin anew with each table. The symbols are assigned beginning at the upper left-hand corner of the table and reading across, row by row. A footnote symbol in a table stub entry is placed following the category, not the unit symbol at the end (for example, Yield point,[†] psi).

The footnote symbol is placed to the left of the margin-aligned footnote and is separated from the following text by a space. Multiple-symbol footnotes build to the left. Footnotes for a multiple-page table are placed together at the end of the table.

If the table has a **Source:** footnote acknowledging the source of information in a table, it is the first footnote. General table footnotes are listed next, preceded with **Note:** or **Notes:** if more than one concept is covered. Caption-, column- or entry-specific footnotes follow in the order assigned in the table.

The capitalization, punctuation, and parallelism conventions for table column entries apply to footnotes.

Table examples

Specifications	
	ReSOLVE iX Anchor-Linear Actuator Tool[†]
Output	Anchor: radial force, radial position, displacement Linear actuator: axial force, extension and retract measurement Head tension Casing collar locator (CCL) and optional gamma ray Wellbore pressure Wellbore temperature Accelerometer for anchor slippage indication
Anchor module	Max. opening diameter: [‡] 5.2 in [132 mm] Opening diameter measurement resolution: [‡] 0.01 in [0.254 mm] Max. anchoring force: ^{‡,§} 50,000 lbf [222,411 N] Max. closing force: ^{‡,§} 25,000 lbf [111,205 N]
Linear actuator module	Stroking length: 12 in [305 mm] Min. linear actuator displacement measurement resolution: ^{††} Indirect measurement: 0.05 in [1.27 mm] Direct measurement: 0.35 in [8.89 mm] Max. axial force: ^{††} 16,000 lbf [71,171 N]
Mud type and weight	All
Pressure rating	20,000 psi [138 MPa]
Temperature range	32 to 350 degF [0 to 177 degC]
Hole size—min.	2.2 in [55.9 mm]
Outside diameter [‡]	2½ in [53.97 mm]
Length—min. [‡]	35.2 ft [10.73 m]
Weight [‡]	307 lbm [139 kg]
Tension	25,000 lbf [111,205 N]
Compression	18,000 lbf [80,068 N]
Special applications	NACE MR0175 compliant for H ₂ S and CO ₂ resistance Fishing capability: Optional 3¼- or 2½-in [79.38- or 53.97-mm] WIRED wireline inline release devices above and below tractors

[†] All values are for standard specifications and are subject to change without notice. A dedicated engineering team is available for customizing tools to address your challenges.

[‡] Configuration dependent

[§] Measured and controlled in real time in the acquisition software

^{††} Linked to stroking speed

SIMplus Bridge Plug Technical Information							
Nominal Tubing Size, in	Weight, lbm/ft [kg/m]	Plug OD, in [mm]	Plug ID, in [mm]	Length, in [mm]	ISO 14310:2008 V3 Qualification		
					Differential Pressure, psi [MPa]	Temperature Range, [†] degF [degC]	Max. Setting ID, in [mm]
2½	6.4 [9.5]	2.22 [56.4]	0.781 [19.8]	54 [1,372]	7,500 [51.7]	104–300 [40–149]	2.493 [63.3]
3½	9.2–10.2 [13.7–15.2]	2.72 [69.1]	1.259 [32]	61 [1,549]	5,000 [34.5]	104–300 [40–149]	3.046 [77.4]
4½	11.6–15.1 [17.3–22.5]	3.65 [92.7]	2.000 [50.8]	59 [1,499]	5,000 [34.5]	104–300 [40–149]	4.030 [102.4]
5½	15.5 [23]	4.53 [115.1]	2.362 [60]	61 [1,549]	5,000 [34.5]	104–300 [40–149]	5.030 [127.8]
	17–20 [25.3–29.8]	4.53 [115.1]	2.362 [60]	61 [1,549]	5,000 [34.5]	104–300 [40–149]	4.975 [126.4]
7	23 [34.2]	4.45 [113.0]	2.362 [60]	61 [1,549]	5,000 [34.5]	104–300 [40–149]	4.765 [121.0]
	29–32 [43.2–47.6]	5.72 [145.3]	2.362 [60] [‡]	63 [1,600]	5,000 [34.5]	104–300 [40–149]	6.293 [159.8]
	23–26 [34.2–38.7]	5.84 [148.3]	2.362 [60] [‡]	63 [1,600]	5,000 [34.5]	104–300 [40–149]	6.465 [164.2]

[†] Contact a Schlumberger representative for operations outside this temperature range.
[‡] 3.15-in [80.01-mm] ID available on request.

BREAKOUT SESSIONS				
Field Development Breakout Sessions	Well Construction Breakout Sessions	Production Breakout Sessions	Business Sessions	Live Moderated Chat Hours
Collaborative Subsea Field Development Service	Allegro Casing- and Liner-While-Drilling Services	Subsea Integration Alliance Amplify Subsea Performance	DELFI Cognitive E&P Environment	Houston (GMT -5) Monday, June 29— Wednesday, July 1 07:00–11:00 18:00–22:00 Paris (GMT +2) 14:00–18:00 01:00–05:00* Questions submitted outside of these hours will be answered as soon as the domain experts are available. <i>*Times fall into the following day.</i>
Enriching Geoscience Workflows in the DELFI Cognitive E&P Environment	DrillOps On-Target Well Delivery Solution	Powered Production Avoid Lost Production	GAIA Digital Subsurface Platform	
FDPlan Agile Field Development Planning Solution	PowerDrive Orbit G2 Rotary Steerable System	NATCO DUAL FREQUENCY Electrostatic Treater (Fourth Generation)	Sensia The First Fully Integrated Digital Oilfield Automation Solutions Provider	
Multiclient Nodal Solution Unlocking Remaining Exploration Potential	RheoProfiler Automated Rheometer	Process Live Data-Enriched Performance Service	Digital Hardware How We Are Adding Digital Intelligence to Our Entire Portfolio	
Ora Intelligent Wireline Formation Testing Platform	DrillPlan Coherent Well Construction Planning Solution	REDA Continuum Extended-Life ESP Pump		
Modernize Your Flow Measurement		Manara Production and Reservoir Management System		

Active Power, Allegro, DELFI, DrillOps, DrillPlan, FDPlan, PowerDrive, GAIA, Sensia, Manara, NATCO DUAL FREQUENCY, Ora, Powered Production, Process Live, REDA Continuum, and RheoProfiler are marks of Schlumberger. Other company, product, and service names are the properties of their respective owners. Copyright © 2020 Schlumberger. All rights reserved. 20-36765550

BluePack ESP Packer Specifications						
Casing Size, in [mm]	Casing Weight Range, lbm/ft [kg/m]	Tubing Size, in [mm]	Penetrator Types	Feedthrough Diameter Options, in [mm]	Max. Working Temperature, degF [degC]	Differential Pressure Rating, psi [MPa]
5½ [139.7]	15.5–20 [23.1–29.8]	2½ [60.3]	3-legged	¼ [6.35] ⅝ [9.525]	325 [163]	5,000 [34.5]
7 [177.8]	23–32 [34.2–47.6]	2½ [73.0] 3½ [88.9]	Single, 3-legged	¼ [6.35] ⅝ [9.525]	325 [163]	5,000 [34.5]
7½ [193.7]	26.4–33.7 [39.3–50.2]	2½ [73.0] 3½ [88.9]	Single, dual, 3-legged	¼ [6.35] ⅝ [9.525]	325 [163]	5,000 [34.5]
9½ [244.5]	36–53.5 [53.6–79.6]	3½ [88.9] 4½ [114.3]	Single, dual, 3-legged	¼ [6.35] ⅝ [9.525]	325 [163]	5,000 [34.5]
10½ [273.1]	55.5–65.7 [82.6–97.8]	4½ [114.3] 5½ [139.7]	Single, dual, 3-legged	¼ [6.35] ⅝ [9.525]	325 [163]	5,000 [34.5]
13½ [339.7]	68–72 [101.2–107.1]	5½ [139.7] 7 [177.8]	Single, dual, 3-legged	¼ [6.35] ⅝ [9.525]	257 [125]	3,000 [20.7]

References

In technical publications, always identify your sources of information. References to published works (in print or electronic) are fully formatted in print or electronic documents. The “Related resources” and similar links on slb.com pages are a succinct description of the link and typically its format and file size, as described in [“Writing for Our Public Websites.”](#)

References should include the following ordered information.

- Author’s last name and initial(s)—List the names of all authors if there are fewer than four. If the source has four or more authors, use the first author’s name and **et al.** (meaning “and others”). Follow an author’s preference for capitalization of his or her name. Initials are not spaced apart.
- Title of the paper or chapter within a book or electronic document—Paper and chapter titles are in quotation marks and capitalized headline style.

- Name of the publication—Titles of journals, books, and electronic documents are italicized and capitalized headline style. If a journal title has changed, use the journal title as it was when the article was published.
- Edition and name of editor (if applicable) and followed by **(ed.)** or **(eds.)**.
- Place of publication, name of the publisher, and date or year of publication or presentation.
- Page numbers.

Do not include references for “personal communication” and unpublished works because the reader cannot readily locate these materials. Footnotes may be used to provide information for these unpublished sources.

If author and date citations are used in the text, list the references alphabetically, with all work attributed to the same author listed chronologically.

The basic elements of author and date citations are the last name of the author or authors and year of publication.

Referenced Document	Author and Date Citation
One author	(Jones, 1995) Simulation of the field by Jones (1995) revealed that the two zones were in communication.
Two or three authors	(Ehlers and Blatt, 1981)
Four or more authors: List the first author's name and <i>et al.</i>	(Johnson et al., 2002)
Multiple documents published by the same author in the same year: Differentiate chronologically by adding letters beginning with a to the year	Pettijohn and Simmons' (1996b) second evaluation contradicted their initial findings (Pettijohn and Simmons, 1996a).
Corporate, group, or government agency : Can be cited with an acronym or abbreviation that is defined only in the reference	(US EPA, 2000)
Personal communication (not included in the references)	According to Owens (pers. comm., 1965), junk remained in the borehole after fishing.

Technical reference format

Book	<i>Formation MicroScanner Applications in Italy</i> , Milano, Italy, Schlumberger Italiana S.p.A. (March 1988).
	Krumbein, W.C., and Sloss, L.L.: <i>Stratigraphy and Sedimentation</i> (2nd ed.), San Francisco, California, USA, W.H. Freeman and Company (1951).
Book chapter	Elbel, J., and Britt, L.: “Fracture Treatment Design,” <i>Reservoir Stimulation</i> (3rd ed.), M.J. Economides and K.G. Nolte (eds.), Chichester, England, John Wiley & Sons (2000), 10-1–10-50.

	Levorsen, A.I.: "Reservoir Conditions—Pressure and Temperature," <i>Geology of Petroleum</i> (2nd ed.), San Francisco, California, USA, W.H. Freeman and Company (1951), 389–432.
Book in a series	Dott, R.H., Sr., and Reynolds, M.J. (eds.): <i>Sourcebook for Petroleum Geology</i> , Tulsa, Oklahoma, USA, The American Association of Petroleum Geologists, AAPG Memoir 5 (September 1969).
Book with no author	<i>Drilling</i> (rev. ed.), Dallas, Texas, USA, Society of Petroleum Engineers, SPE Reprint Series 6a (1973).
Article in journal	Gardner, G.H.F., Gardner, L.W., and Gregory, A.R.: "Formation Velocity and Density: The Diagnostic Basics of Stratigraphic Traps," <i>Geophysics</i> (1974) 39, 770–780.
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Article in press in a journal	Zeng, H., Song, Y.-Q., Johnson, D.L., and Mullins, O.C.: "Critical Nanoaggregate Concentration in Asphaltenes by DC-Electrical," <i>Energy & Fuels</i> (in press).
Place double references in reverse chronological order unless circumstances dictate otherwise.	Fertl, W.: "Gamma Ray Spectral Data Assists in Complex Formation Evaluation," <i>The Log Analyst</i> (September–October 1979) 20, No. 5, 3–37; also in <i>Transactions of the 6th SPWLA European Formation Evaluation Symposium</i> , London, England (March 26–27, 1979), paper X.
For foreign references with French, Italian, or Spanish titles, capitalize first words, proper names, and abbreviations that are normally capitalized. For German titles, capitalize only nouns. If it is unclear whether a reference is in a foreign language (for example, the title has been translated into English), put the language in parentheses after the page numbers.	. . . : 34–46 (in Russian).
	Hassan, M., and Hossin, A.: "Contribution à l'étude des comportements du thorium et du potassium dans les roches sédimentaires," <i>Comptes rendus hebdomadaires des séances, Académie des Sciences</i> (1975) 280, 533–535.

Electronic documents may not have all the typical elements of a traditional publication. The reference should include the protocol and its address, access path, or directory and the date the document was accessed.	Ahmed, U., Johnston, D., and Colson, L.: "An Advanced and Integrated Approach to Coal Formation Evaluation," <i>Petrophysics SIG S&BP Best Practice Articles</i> (December 1990), http://i2s.houston.oilfield.slb.com/ROCKS/web/coal_bed_methane.pdf (accessed September 5, 2002).
	"Amerada Hess Names Schlumberger as Core Software Provider," <i>Oil and Gas Online</i> (February 17, 2000), http://www.oilandgasonline.com/content/news/article.asp?docid={b7ea2eb6-e548-11d3-8c1e-009027de0829} (accessed March 15, 2000).
	Association of Earth Science Editors, http://www.aese.org/ (accessed November 24, 2002).
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Patent	Fessenden, R.A.: "Method and Apparatus for Locating Ore Bodies," US Patent No. 1,240,328 (September 18, 1917).
Corporate, group, or government agency authors can be cited with an acronym or abbreviation that is defined only in the reference.	RRC (Railroad Commission of Texas): <i>Waste Minimization in the Oil Field</i> , Austin, TX, USA, (1994).
Proceedings with editors	Parham, W.E.: "Lateral Variations of Clay Mineral Assemblages in Modern and Ancient Sediments," <i>Proceedings of the International Clay Conference</i> , L. Heller and A. Weiss (eds.), Jerusalem, Israel, Israel Program for Scientific Translations (1966), 1, 135–145.
	Simaan, M. (ed.): <i>Advances in Geophysical Data Processing: Vertical Seismic Profiles</i> , Greenwich, Connecticut, USA, JAI Press Inc. (1984).
Paper presented at SPE meeting Format for meetings up to and including 1974: SPE Annual Meeting. Format for 1975 onward: SPE Annual Technical Conference and Exhibition.	Hertzog, R.C.: "Laboratory and Field Evaluation of an Inelastic Neutron Scattering and Capture Gamma Ray Spectroscopy Tool," paper SPE 7430 presented at the SPE Annual Technical Conference and Exhibition, Houston, Texas, USA (October 1–3, 1978).
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	<i>Unsolicited paper</i> Golapudi, U.K., Ramakrishnan, V., and Islam, M.R.: "Improvement of Properties and Microstructure of Oil-Well Cements," paper SPE 26312 available from SPE, Richardson, Texas, USA (1993).
	<i>Article from a paper</i> Arnold, D.R.: "Planning and Progress on Drilling a Record-Depth Well in the Rocky Mountains," <i>Journal of Petroleum Technology</i> (April 1980), 694–702; also presented as paper SPE 7853 at the SPE Deep Drilling and Production Symposium, Amarillo, Texas, USA (April 13, 1979).
	<i>Article assigned a new number from a paper</i> Shepard, S.F., Reiley, R.H., and Warren, T.M.: "Casing Drilling: An Emerging Technology," <i>SPE Drilling & Completions</i> (March 2002), 17, No. 1, 414 (paper SPE 76640, revised from paper SPE 67731).
Paper presented at SPWLA meeting	Marett, G., Chevalier, P., and Souhaite, P.: "Shaly Sand Evaluation Using Gamma Ray Spectrometry, Applied to the North Sea Jurassic," <i>Transactions of the SPWLA 17th Annual Logging Symposium</i> , Denver, Colorado, USA (June 9–12, 1976), paper DD.
Thesis or dissertation	Mazzullo, E.K.: "Depositional Environment of Lower Cretaceous Mitchell Sandstone, St. Mary and Duty Fields, Lafayette County, Arkansas," MS thesis, Texas A&M University, College Station, Texas, USA (1983).

Footnotes

Footnotes are used to direct readers to supplementary or explanatory material at the bottom of a page or table. Text footnotes are continuous for a document; [table footnotes](#) are independent of the text footnotes, and the footnote sequence starts anew for each table.

Use the following symbols, in order, to designate footnotes. Superscript trademark symbol numerals and letters are not preferred because they could be mistaken for exponents or chemical symbols. Footnote symbols are separated by a comma when more than one used.	Asterisk	*
	Double asterisk	**
	Dagger	†
	Double dagger (diesis)	‡
	Section mark	§
	Two daggers	††
	Two double daggers	‡‡
	Two section marks	§§

Footnote symbols follow all punctuation except an em dash. They go within the closing parenthesis or bracket if exclusive to the parenthetical material.	Correct: The trend was not identifiable in the legacy dataset*—another survey was required to define the seismic salt boundaries.
	Correct: The wells were initially treated with an acid flush,* which subsequent production logging found to have reduced permeability instead of improving it. (The footnoted material applies to the initial acid flush only.) Correct: The wells were initially treated with an acid flush, which subsequent production logging found to have reduced permeability instead of improving it.* (The footnoted material applies to the entire statement.)
	Incorrect: J. Smith [†] , G. Gillis [‡] , and K. Goy [†] Correct: J. Smith,* G. Gillis,** and K. Goy [†]
In table stub entries , the footnote symbol is placed following the category, not the unit symbol at the end.	Incorrect: Yield point, psi [†] Correct: Yield point,* psi

Clarity

The hallmark of a good scientist—as well as a good writer—is attention to detail. Examine every detail from the reader’s point of view, then rewrite to sharpen, simplify, and clarify. As you rewrite, ask yourself these questions:

- Are my ideas logically ordered from the simple to the complex?
- Have I written simple, succinct sentences?
- Have I used verbs in the active rather than passive voice?
- Have I avoided shifts in subject, number, verb tense, and voice?
- Have I placed subjects close to their verbs?
- Have I placed verbs close to their objects?
- Have I chosen words that express my meaning precisely?
- Have I avoided jargon and clichés?
- Have I explained all abbreviations and acronyms that cannot stand alone?
- Have I eliminated needless words, especially modifiers?
- Have I used correct grammar and punctuation? Have I verified numerical values, quantities, and unit symbols? Have I proofread for spelling and typographical errors?

Suggested reading

Mention of these resources does not imply endorsement by SLB.

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The book references are examples of the abridged reference format that was used by *Oilfield Review*.

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- Style Manual*. Washington, DC, USA: US GPO (United States Government Printing Office), 2000. (<http://www.access.gpo.gov/styleman/2000/browse-sm-00.html>)

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- Dictionary.com*, Lexico Publishing Group, <http://dictionary.reference.com/>.
Grammar Slammer, English Plus+, <http://englishplus.com/grammar/contents.htm>.
HyperGrammar, Writing Centre at the University of Ottawa,
<http://www.uottawa.ca/academic/arts/writcent/hypergrammar/grammar.html>.
McGovern G: <http://www.gerrymcgovern.com/>.
Purdue University Online Writing Lab (OWL): *Grammar, Punctuation, and Spelling*,
<http://owl.english.purdue.edu/handouts/grammar/index.html>.
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The International System of Units (SI), 8th ed., Bureau International des Poids et Mesures (2006), https://www.bipm.org/documents/20126/41483022/si_brochure_8.pdf/632d85cd-cd9f-cfae-21d2-9ffcfc2bb198?version=1.6&download=true.
The SI Metric System of Units and SPE Metric Standard, 2nd printing, Richardson, Texas, USA, Society of Petroleum Engineers (June 1984),
https://www.spe.org/authors/docs/metric_standard.pdf.

Spellings and Terminology

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Overview

Many technical words commonly used in SLB documents are listed here. For definitions and illustrations of many of the oilfield terms, see the [SLB Energy Glossary](#). For spelling and usage of nontechnical words, including place names, consult [Merriam-Webster's Collegiate Dictionary](#). SLB documents use American spelling. For additional guidance on spelling and word choice and usage, see the "Writing and Style" chapter of the *Communication Handbook*.

In words designated as "joined prefixes" or "joined suffixes," the hyphen is dropped when the prefix or suffix is added to the root word (for example, "multioperation" and "fieldwide"). In words designated as "hyphenated prefixes" or "hyphenated suffixes," the hyphen is retained when the prefix or suffix is added to the root word (for example, "self-closing" and "unit-based"). The hyphen should always be retained when joining a prefix with a word that begins with a capital letter (for example, "pro-European") or when identical vowels or consonants come together (for example, "multi-investment" and "non-negotiable"), except with one-syllable prefixes ending in e or o (for example, "reentry") and compound words listed as closed in *Webster's* (for example, "underrun").

For additional guidance on spelling compound words and words with prefixes and suffixes, consult Section 7.85 in the *Chicago Manual of Style* (16th edition). For example, most compound unit modifiers that are hyphenated when they precede a noun are not joined by a hyphen when they come after the modified noun ("decision-making guidance" and "guidance for decision making").

Except for commonly used nontechnical abbreviations (for example, "US" and "a.m.") and abbreviations identified as "stand alone" by the dagger footnote, always spell out the term or phrase at first use with the abbreviation enclosed in parentheses following the term. Thereafter, the abbreviation is used alone. Stand-alone abbreviations do not require introduction with the spelled-out term or phrase unless readers will not readily recognize them. Minor variations to what an abbreviation stands for can be read in context, not reintroduced (for example, "POOH" is spelled out on first mention as "pull out of hole" but used later in the document for "pulled out of hole"). However, plurals must be formed from the abbreviation itself (for example, the plural of "BHA" for "bottomhole assembly" is "BHAs").

The abbreviations on this list are repeated in a sublist for ease of reference.

A

A-annulus	aeromagnetic
absolute open flow—AOF	after hot rolling—AHR
acetic acid (noun)	afterflow
acid in oil (noun)	afterproduction (adj.)
acid-in-oil (adj.)	aftershock
acknowledgment	air drill
Acme (thread)	air lance
acquisition and divestiture—A&D	airgun
add-on (noun, adj.)	airwave
adviser	all- (hyphenated prefix)
American Association of Petroleum Geologists—AAPG [†]	alternating current—AC [†]
	a.m. [†]
	American Bureau of Shipping—ABS [†]
	American National Standards Institute—ANSI [†]

[†] A stand-alone abbreviation that can be used without being spelled out to introduce it on first mention.

[‡] A stand-alone abbreviation only in tables and sidebar bullet listings; spell out on first mention in body text.

American Petroleum Institute—API[†]
 American Society of Mechanical Engineers—
 ASME[†]
 American Society of Testing and Materials (see
 “ASTM International”)
 American Standard Code for Information
 Interchange—ASCII[†]
 American Wire Gauge—AWG[†] (used without
 hyphen or No. or #; e.g., AWG 4)
 amplitude variation with azimuth—AVA[†]
 amplitude variation with offset—AVO[†]
 analog
 analog-to-digital—A/D (adj.)
 anno Domini—AD (do not use for Common Era—
 CE[†])
 annular pressure while drilling—APWD
 annulus
 Ant Tracking
 anti- (joined prefix)
 API Class
 API free-water content
 API gravity
 API Monogram
 API Recommended Practice—(API RP[†])
 API Specification—Spec (API Spec[†])
 API Standard
 appendixes (for document sections)
 application programming interface—API[†]
 application service provider—ASP
 application—app[†]
 arcuate
 Area (capitalized for a specific SLB area;
 superseded by Hemispheres)
 artificial intelligence—AI[†]
 artificial lift (noun, adj.)
 asset performance management—APM
 ASTM International (formerly the American Society
 for Testing and Materials; do not use just
 “ASTM”)
 at-bit steerable system—ABSS (stand-alone
 abbreviation only when space constrained)
 ATEX
 authorization for expenditure—AFE[†]
 auto- (joined prefix)
 auto-curve
 auto-inclination
 auto-tangent
 auto-vertical
 average—avg
 azimuthal

B

-based (hyphenated suffix)
 B-annulus

back allocation (noun)
 back-allocation (adj.)
 back error (noun)
 back-error (adj.)
 back off (verb)
 back up (verb)
 backflow
 backflush
 backoff (adj.)
 backpressure (noun, adj.)
 backream
 backscatter
 backup (noun, adj.)
 backwash
 ball sealer
 ballout (noun)
 band-pass
 bandwidth
 bandwidth on demand—BOD[†] (noun)
 bandwidth-on-demand—BOD[†] (adj.)
 barchan
 barite
 base map
 base station
 baseline
 basepipe
 basic sediment and water (also “bottom sediment
 and water”)—BS&W
 batch (noun, verb)
 batch mix (noun, verb)
 batch-mix (adj.)
 baud—Bd[†]
 bedrock
 before Common Era—BCE[†]
 before hot rolling—BHR
 below rotary table—BRT
 benzene, toluene, ethylbenzene, and xylene—
 BTEX[†]
 Berea Sandstone
 berm
 bi- (joined prefix)
 bi-mill
 bicenter
 bid round, bidding round
 (interchangeable as keywords)
 bit—b[†] (use K = 1,024 for binary multiplier)
 bitbus
 bits per second—bps[†]
 bleed off (verb)
 bleedoff (noun, adj.)
 blog
 blowback (noun, adj.)
 blowdown (noun, adj.)
 blowout (noun, adj.)
 blowout preventer—BOP[†]
 bobtail

[†] A stand-alone abbreviation that can be used without being spelled out to introduce it on first mention.

[‡] A stand-alone abbreviation only in tables and sidebar bullet listings; spell out on first mention in body text.

borehole
 borehole acoustic reflection survey—BARS
 borehole compensated—BHC
 borehole televiewer—BHTV
 bottom sediment and water (variant of “basic sediment and water”)—BS&W
 bottomhole (adj.)
 bottomhole assembly—BHA[†]
 bottomhole circulating pressure—BHCP
 bottomhole circulating temperature—BHCT
 bottomhole flowing pressure—BHFP
 bottomhole flowing temperature—BHFT
 bottomhole pressure—BHP
 bottomhole static pressure—BHSP
 bottomhole static temperature—BHST
 bottomhole temperature—BHT
 bottomwater (noun, adj.)
 bound-fluid volume—BFV
 bradenhead
 break down (verb)
 break out (verb)
 breakdown (noun, adj.)
 breaker
 breakout (noun, adj.)
 breakthrough
 bright spot
 brownfield (noun, adj.)
 brute stack
 bubblepoint (noun, adj.)
 build out (verb)
 build up (verb)
 buildout (noun, adj.)
 buildup (noun, adj.)
 bull plug
 bullhead
 buoyant
 burn-in (adj.)
 burn over (verb)
 burnover (noun, adj.)
 by volume of solids—BVOS[‡]
 by weight of cement—BWOC[‡]
 by weight of proppant—BWOP[‡]
 by weight of water—BWOW[‡]
 bypass
 by-product
 byte—B[†] (use K = 1,024 for binary multiplier)
 bytes per second—Bps[†]

C

cable head
 calculated absolute open flow—CAOF
 Cambrian (geochronologic [time] unit: period; chronostratigraphic [time-rock] unit: system)
 Canadian Electrical Code—CEC[†]

Canadian Registration Number—CRN[†]
 Canadian Standards Association—CSA[‡]
 capital expenditure—capex[†]
 caprock
 capscrew
 carbon capture and sequestration—CCS[†] (also known as “carbon capture and storage,” but SLB collateral should use “sequestration” for the initialism)
 carbon capture, utilization, and sequestration—CCUS[†] (also known as “carbon capture, utilization, and storage,” but SLB collateral should use “sequestration” for the initialism)
 Cartesian
 cascading style sheet—CSS
 cased hole
 casing collar locator—CCL[†]
 casinghead (adj.)
 catalog
 cathode ray tube—CRT[†]
 cation exchange capacity—CEC
 cement head
 cement plug
 Cenozoic (geochronologic [time] unit: era; chronostratigraphic [time-rock] unit: erathem)
 center (verb) (not “centralize”)
 centerline
 centi- (joined prefix)
 central processing unit—CPU[†]
 centralizer (terms such as “centering standoffs” and “centering devices” preferred)
 chainwheel
 changeout (noun, adj.)
 changeover (noun, adj.)
 channeling
 checklist—CL
 checkshot (noun, adj.)
 chert
 chi-square
 chokeline (noun)
 Christmas tree (avoid; if known, specify “production tree” or “frac tree”; otherwise, use “tree”)
 circle shooting
 Class—CL[†] (restricted to ASME Class; CL 150)
 clayey
 clean out (verb)
 clean up (verb)
 cleanout (noun, adj.)
 cleanup (noun, adj.)
 clear-brine fluid—CBF
 client (do not use for “customer” in a business relationship except for investor highlights for SIS and WesternGeco)
 closed-circuit television—CCTV
 close-up (noun, adj.)
 cloud (use “in the cloud,” not “on the cloud”)

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CO₂ equivalent—CO₂e[†]
 co- (joined prefix)
 coalbed
 coalbed methane—CBM
 coastline
 coauthor
 coiled tubing (noun, adj.)—CT[†]
 coiled tubing unit—CTU
 cold heavy-oil production with sand—CHOPS
 cold working pressure—CWP
 collocate (use for precise placement, not just proximity; see “colocate”)
 colocate (use for proximity, not precise placement; see “collocate”)
 combinability
 commingle
 common depth point—CDP
 Common Era—CE[†]
 common midpoint—CMP
 common offset
 common reflection point—CRP
 compact disc—CD[†]
 compact disc read-only memory—CD-ROM[†]
 compressed natural gas—CNG
 compressional wave—also P-wave
 compressional-wave (adj.)—also P-wave
 compute (adj., verb)
 Computer Graphics Metafile—CGM
 computer-aided design—CAD[†]
 computerized axial tomography—CAT
 computerized maintenance management system—CMMS
 computerized tomography—CT[†]
 configuration—Fig[†] (for end connection names, typically for hammer unions)
 Conformité Européenne—CE[†]
 continuous mix (noun, verb)
 continuous-mix (adj.)
 controlled-source electromagnetic—CSEM
 coreflood (noun, adj.)
 Coriolis
 cosecant—cosec
 cosine—cos
 cotangent—cot
 count rate
 counter- (joined prefix, except “counter-ion”)
 Cretaceous (geochronologic [time] unit: period; chronostratigraphic [time-rock] unit: system)
 crisscross
 cross coupling (shortened from “across the coupling”)
 cross section (noun)
 cross-sectional (adj.)
 crossbed
 crosscorrelation (noun, adj.)
 crossfault

crossflow
 crosshole (adj.)
 crossline
 crosslink (noun, verb)
 crosslinked
 crosslinker
 crossover (noun, adj.)
 crossplot
 crosswell
 cusplate
 customer (not “client” for a business relationship except for investor highlights for SIS and WesternGeco)
 customer-centric
 cut off (verb)
 cutoff (noun, adj.)
 cuttings (preferred to “drill cuttings” or “drilling cuttings”)
 cyan, magenta, yellow, and key (black)—CMYK[†]
 cyclic steam stimulation—CSS
 cyclotetramethylene tetranitramine—HMX
 cyclotrimethylene trinitramine—RDX

D

-drive (joined suffix)
 darcies
 darcy
 Darcy’s law
 Darwin Information Typing Architecture—DITA
 data (plural for individual results, singular for the entire data collection as one unit)
 data room
 data source
 data store
 database
 datalogger
 dataset
 datasheet
 de- (joined prefix except “de-aeration”)
 deaerate
 decision making (noun)
 decision-making (adj.)
 decline curve analysis—DCA
 décollment (accent acute on the first e)
 deconvolution
 deconvolve
 deep water (noun)
 deepsea (adj.)
 deepset
 deepwater (adj.)
 deghost
 deliquification (use for the artificial lift removal of liquid)
 depth migration

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depth point
 derisk (verb)
 Det Norske Veritas—DNV[†]
 Devonian (geochronologic [time] unit: period;
 chronostratigraphic [time-rock] unit: system)
 dewpoint (noun, adj.)
 dewpointing
 dial up (verb)
 dial-up (noun, adj.)
 dialog
diamètre nominal—DN[†] (nominal pipe sizing; DN 8,
 DN 40, etc.)
 diamond-enhanced insert—DEI[†]
 Digital Log Information Standard—DLIS
 digital oilfield (use only generically because there
 is a similar trademark)
 digital subscriber line—DSL[†]
 digital twin
 digital video disc—DVD[†]
 DIN (for “Deutsches Institut für Normung”)
 dip log
 dip map
 dip moveout—DMO
 dipmeter
 dipole
 direct current—DC[†]
 direction and inclination—D&I
 directional driller—DD
 disc (for optical media storage or uses in zoology
 and botany; see “rupture disc”)
 disk (for “diskette” and magnetic media storage)
 distributed acoustic sensing—DAS
 distributed temperature sensing or sensor—DTS[†]
 distributed vibration sensing—DVS
 District of Columbia—DC[†]
 dogleg
 dogleg severity—DLS
 dot-com
 down- (joined prefix)
 downdip
 downhole
 downhole fluid analysis—DFA
 downstroke
 drainhole
 drawdown
 drawworks
 drill bit (noun)
 drill collar
 drill cuttings (see “cuttings”)
 drill down (verb)
 drill floor
 drill-in (noun, adj.)
 drillability
 drillbit (adj.)
 drilldown (noun, adj.)
 drilled but uncompleted—DUC

drilled solids
 driller’s log
 drilling cuttings (see “cuttings”)
 drillout (noun, adj.)
 drillpipe
 drillship
 drillsite
 drillstem
 drillstem test—DST
 drillstring
 drivehead
 drivetrain
 dropoff (noun, adj.)
 dry and abandoned—D&A
Durchmesser nach Norm—DN[†] (nominal pipe
 sizing; DN 8, DN 40, etc.)
 Dynamic HTML—DHTML

E

e- (hyphenated prefix for “electronic”) (Use a
 capital “E-” at the beginning of a sentence, or
 rewrite the sentence to avoid having “e-” as
 the first word.)
 e-business
 e-coil
 e-commerce
 email
 early production facility—EPF[†]
 easterly (from the east)
 eccentric (verb) (not “eccentralize”)
 eccentralizer (terms such as “eccentering
 standoffs” and “eccentering devices”
 preferred)
 edge water (noun)
 edgewater (adj.)
 e.g. (means “for example”)
 electric line
 electric submersible progressing cavity pump (also
 progressive cavity pump)—ESPCP[†]
 electric submersible pump—ESP[†]
 electrical stability—ES
 electrically erasable, programmable, read-only
 memory—EEPROM
 electro- (joined prefix)
 electromagnetic—EM[†]
 electromagnetic radiation—EMR
 electromotive force—emf
 emergency checklist—ECL
 emergency shutdown—ESD
 en echelon
 encapsulated PostScript®—EPS (lowercase “eps”
 used in a filename extension)
 end member
 end user (noun)

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end-user (adj.)
 endpoint
 Energy Institute—EI[†]
 energy-dispersive spectroscopy—EDS
 engineering, procurement, construction, and
 installation—EPCI[†]
 engineering, procurement, construction,
 installation, and commissioning—EPCIC[†]
 engineering prototype—ENP
 enhanced oil recovery—EOR
 ensure (means “to make certain”)
 enterprise resource planning—ERP
 Environmental Impact Factor—EIF
 environmental, social, and governance (ESG)
 eolian (not “aeolian”)
 equals (spell out except in equations)
 equation of state—EOS (noun)
 equation-of-state—EOS (adj.)
 equivalent circulating density—ECD[†]
 equivalent static density—ESD[†]
 estimated ultimate recovery—EUR
 et al. (means “and others”)
 Ethernet
 Eurasian Conformity—EAC[†]
 European Union—EU[†] (superseded European
 Community)
 experimental prototype—EXP
 exploration and production—E&P[†]
 explorationist
 extended operating system—EOS
 extended-reach (adj.)
 extended-reach drilling—ERD
 extender
 Extensible HTML—XHTML[†]
 Extensible Markup Language—XML[†]
 external upset ends—EUE[†]
 extreme overbalance—EOB (noun)
 extreme overbalanced perforating—EOP

F

-fold (joined suffix except following a numeral: “20-
 fold.” Avoid using because of confusion
 whether it refers to incremental change or
 total amount.)
 -free (hyphenated suffix)
 fail-safe (noun, adj.)
 fall back (verb)
 fallback (noun)
 falloff
 farm in (verb)
 farm out (verb)
 farmout (noun, adj.)
 fast-track (adj., verb)
 fast track (noun)

feedstream
 feedthrough (noun, adj.)
 feedwater (noun)
 fiber-optic (adj.)
 fiber optics (noun)
 fiberglass (generic term)
 field (capitalize for a specifically named one)
 fieldwide (adj.)
 file server
 File Transfer Protocol—FTP[†] (lowercase “ftp” used
 for an Internet location)
 filename
 fill up (verb)
 fill-up (noun, adj.)
 filtercake (noun, adj.)
 finite-difference (adj.)
 fire brick
 fire tube
 fire wall
 fireflood
 firewall (for computer terms)
 fiscal model library—FML
 fishtail (adj.)
 five-spot (noun, adj.)
 fixed cutter bit
 flame ionization detector—FID
 flash point
 flex-lay
 floating production, storage, and offloading—
 FPSO[†]
 floating storage and offloading—FSO[†]
 flocculant (noun, substance that promotes the
 clumping of particles)
 flocculent (adj., for containing, consisting of, or
 occurring in the form of loosely aggregated
 particles or flakes)
 floodwater
 flow back (verb)
 flow check (verb, noun)
 flow control (noun, adj.)
 flow-in (noun, adj.)
 flow loop
 flow rate
 flow ring
 flowback (noun, adj.)
 flowchart
 flowhead
 flowline (noun, adj.)
 flowmeter
 fluid loss
 Fluid Controls Institute—FCI
 fluoroelastomer—FKM[†]
 fluvial
 foamed cement
 foldback (adj.)
 follow up (verb)

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follow-up (noun, adj.)
 footwall
 Forchheimer
 forklift
 formation (capitalize as part of a geologic unit name)
 formation evaluation while drilling—FEWD
 formation integrity test—FIT
 formation volume factor—FVF
 formulas (plural)
 four-component—4C
 four-dimensional—4D[†]
 Fourier transform
 Fourier transform infrared—FTIR
 frac
 frac ball
 frac head
 frac iron
 frac manifold
 frac pack (noun, verb)
 frac-pack (adj.)
 frac plug
 frac port
 frac tree
 frac valve
 fracking (“fracturing” is the standard usage; avoid this casual usage)
 fracturability (do not use “fracability”)
 fracture
 fracturing flowback
 free-fluid volume—FFV
 free water (noun)
 free-water level
 freestanding
 frequency domain—FD
 frequently asked questions—FAQ[†]
 freshwater (noun, adj.)
 front end (noun)
 front-end (adj.)
 front-end engineering and design—FEED[†]
 Full Hole—FH[†] (API screw threads; e.g., 3½ FH)
 full-azimuth—FAZ[†]
 full bore (noun)
 fullbore (adj.)

G

gamma ray—GR[†] (noun, adj.)
 gas chromatography—GC
 gas chromatography—mass spectrometry—GCMS
 gas-bearing
 gas field (noun, adj.)
 gas lift (noun, adj.)
 gas/liquid ratio—GLR
 gas/oil contact—GOC

gas/oil ratio—GOR[†]
 Gas Research Institute—GRI (see “Gas Technology Institute—GTI”)
 gas shale (rock type; see “shale gas”)
 Gas Technology Institute—GTI (successor to “Gas Research Institute—GRI”)
 gas volume fraction—GVF
 gas/water contact—GWC
 gases
 gasflood
 gas-to-liquid—GTL (adj.)
 gas-to-value—(adj.)
 gas-to-X—(noun, adj.)
 gauge
 Geiger counter
 gel
 gel permeation chromatography—GPC
 gel strength (noun, adj.)
 geo- (joined prefix)
 geographic information system—GIS[†]
 geological and geophysical *or* geology and geophysics—G&G
 geophone
 geothermal
 geounit
 glass-reinforced epoxy—GRE
 global positioning system—GPS[†]
 gooseneck
 GOST R
 grain-size distribution
 graphical user interface—GUI[†]
 graphics interchange format—GIF[†] (lowercase “gif” used in a filename extension)
 gravel pack (noun)
 gravel-pack (adj.)
 gray (not “grey”)
 greenfield
 gridblock
 gridpoint
 Gröningen
 groundwater (noun, adj.)
 guar
 Gulf of Mexico (preferred spelled out; use of “Gulf” or “GOM” depends on context)
 gun shock (noun)
 gunshock (adj.)
 gunstring (noun, adj.)

H

hairline
 half-length
 half-life (noun, adj.)
 half-lives
 half-width

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handwheel
 hangoff (noun, adj.)
 hard copy (noun)
 hard-copy (adj.)
 hardfacing
 Harmonised Offshore Chemical Notification
 Format—HOCNF[†]
 Highway Addressable Remote Transducer—
 HART[†]
 health, safety, and environment—HSE[†]
 heating, ventilation, and air conditioning—HVAC[†]
 heavy oil (noun, adj.)
 heavy weight (noun)
 heavyweight (adj.)
 Hemisphere (capitalized for a specific region of the
 world)
 heptacable
 heterogeneous
 hexanitrostilbene—HNS
 high- (hyphenated as first element in a compound
 adjective)
 high definition (noun)—HD[†]
 high-definition (adj.)—HD[†]
 high-density (adj.)
 high-pressure—HP[†] (adj.)
 high performance computing—HPC
 high pressure, high temperature—HPHT[†] (noun)
 high-pressure, high-temperature—HPHT[†] (adj.)
 high-temperature—HT[†] (adj.)
 highstand
 Hilbert transform
 hindcast
 history match (noun, adj.)
 history-matching (noun, adj.)
 hold-down
 hold inclination and azimuth—HIA
 hold up (verb)
 holdup (noun, adj.)
 Holocene (geochronologic [time] unit: epoch;
 chronostratigraphic [time-rock] unit: series)
 home page
 homogeneous
 hook up (verb)
 hookload
 hookup (noun, adj.)
 horizontal pumping system (also “horizontal pump
 system”)—HPS[†]
 hot spot (noun)
 hot-spot (adj.)
 hot-water (adj.)
 huff ‘n’ puff
 human machine interface—HMI
 hydraulic fracturing
 hydrocarbon in place—HIP
 hydrocarbon pore volume—HCPV

hydrochloric acid—HCl
 hydrofluoric acid—HF
 hydrogen index—HI
 hydrogenated nitrile butadiene rubber—HNBR[†]
 hydrophone
 hydrotransport
 hydroxyethyl cellulose—HEC
 hydroxypropyl guar—HPG
 hyperbolic sine, cosine, etc.—sinh, cosh, etc.
 Hypertext Markup Language—HTML[†]
 Hypertext Transfer Protocol—HTTP[†] (lowercase
 “http” used for an Internet location)

I
i- (hyphenated italic prefix for “iso-”)
 i.e. (means “that is”)
 improved oil recovery—IOR
 in-field (adj.)
 in-house
 in situ (adj., adv.)
 indexes (for document sections)
 Industrial Internet of Things—IIoT[†]
 infill (adj.)
 inflow (adj.)
 inflow control device—ICD[†]
 inflow performance relationship—IPR
 information technology—IT[†]
 infrastructure as a service—IaaS[†]
 inhibitor
 inhole
 initial shut-in pressure—ISIP
 inline (adj.)
 input (collective noun, adj.) (Avoid use as verb.)
 input/output—I/O[†]
 inside diameter—ID[†]
 Institute of Electrical and Electronics
 Engineers, Inc.—IEEE[†]
 insulated gate bipolar transistor—IGBT[†]
 Integrated Services Digital Network—ISDN[†]
 Intelligent Well Interface Standardisation—IWIS
 inter- (joined prefix)
 intermediate-strength proppant—ISP
 Internal Flush—IF[†] (API screw thread; e.g., 3½ IF)
 International Association of Drilling Contractors—
 IADC
 International Electrotechnical Commission—IEC[†]
 International Organization for Standardization—
 ISO[†] (do not use generic terms “ISO
 certification” and “ISO certified” but specify
 which unique standard [e.g., “ISO 9001:2000
 certification” and “ISO 14001:2004 certified,”
 respectively])
 Internet
 Internet of Things—IoT[†]

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Internet Protocol—IP[†]
 interval pressure transient test *or* interval pressure
 transient testing—IPTT
 intra- (joined prefix)
 intranet
 intromitter
 invert-emulsion (adj.)
 iso- (joined prefix)
 isotope ratio mass spectrometry—IRMS[‡]

J

jack up (verb)
 jackup (noun, adj.)
 Joint Photographic Experts Group—JPEG[†] *or*
 JPG[†] (lowercase “jpg” used in a filename
 extension)
 Judgment
 Junk slot area
 Jurassic (geochronologic [time] unit: period;
 chronostratigraphic [time-rock] unit: system)

K

kelly bushing—KB
 kerosene
 key performance indicator—KPI[†]
 keypad
 keypunch
 keyseat
 kick off (verb)
 kickoff (noun, adj.)
 kick-start (verb, adj.)
 kilobit—Kb[†]
 kilobyte—KB[†]
 Kirchhoff
 knock out (verb)
 knockout (noun, adj.)
 knowledge base
 knowledge management
 kriging

L

laboratory (preferred in formal contexts to “lab”)
 labor-hour (this *or* “work-hour” preferred to “man-
 hour”)
 large-scale (adj.)
 Larmor frequency
 laterolog deep—LLD
 laterolog shallow—LLS
 leak off (verb)
 leakoff (noun, adj.)
 leakoff test—LOT
 least squares (noun)

least-squares (adj.)
 least-time (adj.)
 lidar
 life cycle (noun, adj.)
 lifespan
 light-emitting diode—LED[†]
 light weight (noun)
 lightweight (adj.)
 limy (not “limey”)
 line drive (noun)
 line-drive (adj.)
 line pipe
 liquefied natural gas—LNG[†]
 liquefied petroleum gas—LPG[†]
 liquid crystal display—LCD[†]
 liter (not “litre”)
 lithology
 load out (noun)
 load-out (adj.)
 local area network—LAN[†]
 lock up (verb)
 lock washer (noun, adj.)
 lockout/tagout
 lockup (noun, adj.)
 log ASCII standard—LAS[†]
 log in (verb)
 log information standard—LIS
 log into
 log-log
 log-normal
 log off (verb)
 log on (verb)
 log onto
 logarithm—log[†]
 logarithm natural—ln[†]
 logging while drilling—LWD[†] (noun)
 logging-while-drilling—LWD[†] (adj.)
 login (noun, adj.) (closed form used only in
 reference to the access or exit procedure for a
 computer, network, or program)
 logoff (noun, adj.) (closed form used only in
 reference to the access or exit procedure for a
 computer, network, or program)
 logon (noun, adj.) (closed form used only in
 reference to the access or exit procedure for a
 computer, network, or program)
 long-reach (adj.)
 long-standing (adj.)
 long-term (adj.)
 longitudinal relaxation time— T_1
 look-ahead (adj.)
 lost circulation material—LCM
 lost in hole—LIH (noun)
 lost-in-hole—LIH (adj.)
 lost-time incident ratio—LTIR
 lost-time incident—LTI

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low- (hyphenated as first element in a compound adjective)
 low-temperature (adj.)
 lump sum turnkey—LSTK

M

machine learning—ML
 macro- (joined prefix)
 magnetic resonance imaging—MRI
 magnetotelluric—MT
 main bore (use instead of “motherbore”)
 mainframe
 make up (verb)
 makeup (noun, adj.)
 man-hour (“labor-hour” or “work-hour” preferred)
 Manufacturers Standardization Society of the Valve and Fittings Industry—MSS[†]
 man-year (“work-year” preferred)
 managed pressure drilling—MPD[†]
 map convolution
 map migration
 marine magnetotelluric—MMT
 massive hydraulic fracturing—MHF
 Material Safety Data Sheet—MSDS[†]
 maximum—max.[‡]
 mean sea level—MSL
 mean time between failure(s)—MTBF
 measured depth—MD[†]
 measured depth from rotary table—MDRT
 measurement(s) and logging while drilling—MLWD[†] (noun)
 measurement(s)- and logging-while-drilling—MLWD[†] (adj.)
 measurement(s) while drilling—MWD[†] (noun)
 measurement(s)-while-drilling—MWD[†] (adj.)
 mechanical earth model—MEM
 mediumweight
 megabyte—MB[†]
 Mesozoic (geochronologic [time] unit: era; chronostratigraphic [time-rock] unit: erathem)
 meter (not “metre”)
 micro- (joined prefix, except “micro-hydraulic fracture”)
 micro-hydraulic fracture
 microCT (for “microncomputed tomography”)
 microelectromechanical systems—MEMS[†]
 mid- (joined prefix)
 milled-tooth bit
 Miller Dyes and Hutchinson—MDH
 millidarcies
 millout
 milltooth
 mineback
 minimum—min.[‡]

mis- (joined prefix)
 misrun
 Mississippian (geochronologic [time] unit: period; chronostratigraphic [time-rock] unit: system)
 mis-tie
 mix water
 mobile offshore drilling unit—MODU
 modeled
 modeling
 monocable
 monoethylene glycol—MEG[†]
 moonpool
 motor head (noun)
 motorhead (adj.)
 motor lead extension—MLE
 mousehole
 moveout
 Moving Picture Experts Group—MPG[†] or MPEG[†] (lowercase “mp3” used in a filename extension)
 MPEG-4[†]—MP4[†] (lowercase “mp4” used in a filename extension)
 mud-based
 mud-gas separator—MGS
 mud motor
 mud pulse (noun, adj.)
 mud weight (noun)
 mud-weight (adj.)
 mudcake
 mudcap
 mudline
 mud log (also “mud logging”)
 mudmat
 mudrock
 muleshoe
 multi- (joined prefix)
 multiclient
 multicomponent
 multiphase flowmeter—MPFM
 multiplex
 multipurpose float—MPF (“multipurpose trailer-mounted” preferred)
 multipurpose skid—MPS (“multipurpose skid-mounted” preferred)

N

n- (hyphenated italic prefix for “normal”)
 NACE International (name of association, formerly “National Association of Corrosion Engineers—NACE”)
 NACE[†] (use for standards)
 NACE MR0175
 nano- (joined prefix)
 naphtha

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National Association of Corrosion Engineers (see “NACE International”)
 National Electrical Code—NEC[†]
 National Electrical Manufacturers Association—NEMA[‡]
 National Institute of Standards and Technology—NIST
 natural gas (noun, adj.)
 natural gas liquids—NGL[†]
 naturally occurring radioactive material—NORM
 nearshore (adj.)
 net pay (noun)
 net-pay (adj.)
 net present value—NPV[†]
 net-to-gross—NTG
 net zero (noun)
 net-zero (adjective)
 neutron gamma density—NGD
 Newtonian
 nitrile butadiene rubber—NBR[†]
 nominal pipe size—NPS
 non- (joined prefix)
 non-governmental organization—NGO[†]
 non-guar
 non-Newtonian
 non-pressure-sealed thread—NPST
 nonproductive time—NPT[†]
 nonupset end—NUE[†]
 normal moveout—NMO
 NORSOK
 northerly (from the north)
 nuclear magnetic resonance—NMR[†]
 Numbered Connector—NC[†] (API screw threads; e.g., NC50)

O

object linking and embedding—OLE[†]
 Occupational Safety and Health Administration—OSHA[†]
 ocean bottom cable—OBC
 ocean bottom node—OBN
 ocean bottom seismic—OBS
 off-bottom
 off line (adv.) (closed “offline” used for disconnection from a computer system)
 off-load
 off site (adv.)
 off-take (noun)
 offline (adj.)
 offroad (adj.)
 offset
 offshore (adj., adv.)
 offshoring
 offsite (adj.)

oil and gas—O&G[†]
 oil-based (adj.)
 oil-based mud—OBM[†]
 oil-based mud filtrate—OBMF
 oil-bearing
 oil country tubular goods—OCTG
 oil field (noun)
 oil in place—OIP
 oil in water—OIW
 oil/water contact—OWC
 oil well (noun)
 oil-wet
 oilfield (adj.)
 oilflood
 oilwell (adj.)
 on site (adv.)
 one-dimensional—1D[†]
 online (adj.)
 on-stream (adj.)
 onshore (adj., adv.)
 onsite (adj.)
 open-ended (adj.)
 open-flow potential—OFP
 open hole (noun)
 Open Software Foundation—OSF
 openhole (adj.)
 operating system—OS[†]
 operational expenditure—opex[†]
 Ordovician (geochronologic [time] unit: period; chronostratigraphic [time-rock] unit: system)
 orient (not “orientate”)
 original equipment manufacturer—OEM[†]
 original oil in place—OOIP
 O-ring
 OSDU® Technical Standard—See [official guidelines](#) for The Open Group Open Subsurface Data Universe™ (OSDU) Forum. In technical papers, use as part of first mention, “Aligned with implementing the Open Group’s OSDU® Technical Standard,” followed by an attribution in the Acknowledgments section of the paper. Further use should not mention OSDU directly; use “the technical standard.”
 OSPAR
 out- (joined prefix)
 outcrop
 output (collective noun, adj.) (Avoid use as verb.)
 outside diameter—OD[†]
 outside screw and yoke—OS&Y
 over- (joined prefix)
 over/under—O/U
 overlay

[†] A stand-alone abbreviation that can be used without being spelled out to introduce it on first mention.

[‡] A stand-alone abbreviation only in tables and sidebar bullet listings; spell out on first mention in body text.

P

- pack off (verb)
- packoff (noun)
- Paleozoic (geochronologic [time] unit: era;
 chronostratigraphic [time-rock] unit: erathem)
- partial pressure of CO₂—pCO₂[†]
- partial pressure of H₂S—pH₂S[†]
- particle-size distribution
- pay out (verb)
- pay zone
- payout (noun)
- Pennsylvanian (geochronologic [time] unit: period;
 chronostratigraphic [time-rock] unit: system)
- Pensky-Martens closed cup—PMCC[‡]
- percent
- perforate, jet, and cement—PJC
- perforate, wash, and cement—PWC
- period
- Permian (geochronologic [time] unit: period;
 chronostratigraphic [time-rock] unit: system)
- Permian Basin
- permittivity
- personal computer—PC[†]
- Petróleos de Venezuela S.A.—PDVSA[†]
- Petroleum Industry Data Dictionary—PIDD
- petrotechnical
- pH
- phase out (verb)
- phaseout (noun)
- phase-shift (adj.)
- pick up (verb)
- pickup (noun, adj.)
- picture description system—PDS
- pileup (noun)
- pinch out (verb)
- pinch point (noun)
- pinchout (noun)
- pinch-point (adj.)
- pipe handling
- pipe-lay
- pipeline
- pipeline end manifold—PLEM
- plane wave
- plaster of paris
- plastic viscosity—PV
- platform as a service—PaaS[†]
- Pleistocene (geochronologic [time] unit: epoch;
 chronostratigraphic [time-rock] unit: series)
- Pliocene—Pleistocene (not “Plio-Pleistocene”
 (geochronologic [time] unit: epochs,
 chronostratigraphic [time-rock] unit: series)
- plug and abandon—P&A[†] (also “plugged and
 abandoned”)
- plug and perf
- plug and play (noun, verb)
- plug-and-play (adj.)
- plug-in (noun, adj.)
- plug back (verb)
- plugback (noun, adj.)
- plugged and abandoned—P&A[†] (also “plug and
 abandon”)
- p.m.
- podcast
- Poisson’s ratio
- polycrystalline diamond compact—PDC[†] (noun,
 adj.)
- polyetheretherketone—PEEK[†]
- polytetrafluoroethylene—PTFE[†]
- polyvinyl chloride—PVC[†]
- pop-up (noun, adj.)
- pore to pipeline
- pore volume—PV
- pore-water fluid
- poroelastic
- portable document format—PDF[†] (lowercase “pdf”
 used in a filename extension)
- Portable Network Graphics—PNG[†] (lowercase
 “png” used in a filename extension)
- portland cement
- Pose Little or No Risk—PLONOR (shortened from
 OSPAR’s Preparations Used and Discharged
 Offshore Which Are Considered to Pose Little
 or No Risk to the Environment)
- positive displacement—PD
- positive displacement motor—PDM
- positron emission tomography—PET
- post- (joined prefix)
- post-stimulation (adj. exception due to Google
 SEO)
- postsalt
- PostScript® (lowercase “ps” used in a filename
 extension)
- poststack
- postack time migration—PSTM
- power law
- pozzolan
- pre- (joined prefix)
- preferred hydraulic fracture plane—PFP
- presalt
- pressure and temperature—PT[†]
- Pressure Equipment Directive—PED[†]
- pressure test (noun, verb, adj.)
- pressure-volume-temperature—PVT[†]
- prestack
- prestack time migration—PSTM
- preventive (not “preventative”)
- print out (verb)
- printout (noun)
- pro- (joined prefix)
- producibility

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production data management—PDM
 production logging tool—PLT
 production tree (also called “Christmas tree”)
 productivity index—PI
 program (not “programme”)
 programmable logic control or programmable logic controller—PLC
 progressing cavity pump (also progressive cavity pump)—PCP[†]
 prove-up (noun, adj.)
 pseudo- (joined prefix)
 pseudopressure
 pseudosection
 pseudosteady state (noun)
 pseudosteady-state (adj.)
 Public Petroleum Data Model—PPDM
 pull-off (noun)
 pull out of hole—POOH
 pulsed neutron generator—PNG
 pulse-loading
 pump down (verb)
 pump-in
 pumpdown (noun, adj.)
 pumphead
 pumpoff (adj.)
 pumpout
 P-wave—also “compressional wave” (noun) and “compressional-wave” (adj.)

Q

quad-combo
 quadrupole
 quality assurance and quality control—QA/QC[†]
 quality assurance—QA[†]
 quality control—QC[†]
 quality, health, safety, and environment—QHSE[†]
 quasi (spaced apart prefix for compound nouns)
 quasi- (hyphenated prefix for compound adjectives)
 Quaternary (geochronologic [time] unit: period; chronostratigraphic [time-rock] unit: system)
 quicklook
 quick-setting (adj.)

R

radii (plural)
 radio frequency—RF
 radio frequency identification—RFID
 ramp-up (noun, adj.)
 random-access memory—RAM[†]
 rate of penetration—ROP[†]
 rate-normalized pressure—RNP
 rathole

ray tracing
 Rayleigh
 raypath
 re- (joined prefix)
 read back (verb)
 read out (verb)
 read-only memory—ROM[†]
 readback (adj.)
 readout (noun, adj.)
 readout port—ROP
 real time (noun)
 real-time (adj.)
 really simple syndication—RSS[†]
 red-green-blue—RGB[†]
 reduced instruction set computer or computing—RISC
 Redundant Array of Independent Disks—RAID
 reflection point
 reflection survey
 refraction statics
 refraction survey
 Regular—Reg[†] (API screw threads; e.g., 3½ Reg)
 remote terminal unit—RTU[†]
 remotely operated vehicle—ROV[†]
 repeatability
 research and development—R&D[†]
 reservoir-centric
 reservoir drill-in fluid—RDF
 resin-coated proppant—RCP
 resistive temperature detector—RTD
 restrictor
 retarder
 retentate
 return on investment—ROI[†]
 reverse time migration—RTM[†]
 Reynolds number
 rheology
 rig down (verb)
 rig-down (noun, adj.)
 rig site (noun)
 rig up (verb)
 rig-up (noun, adj.)
 rigsite (adj.)
 Risk Identification Report—RIR
 rodstring
 roentgen equivalent man—REM
 roller cone (noun, adj.)
 rollout (noun, adj.)
 root-mean-square—rms[†]
 Rostekhnadzor—RTN[‡]
 rotary kelly bushing—RKB
 rotary steerable system—RSS[†]
 rotary table—RT
 routine core analysis—RCAL

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RSVP[†] (abbreviation for *répondez s'il vous plaît* [please reply])
 run in hole—RIH
 run life (noun)
 run-life (adj.)
 run time (noun)
 run-time (adj.)
 runnel
 runtime (solid for software uses)
 rupture disk (“disc” is preferred for some industry-specific terms)

S

sabkha
 saddle point
 Safety Data Sheet—SDS[†]
 safety integrity level—SIL[†]
 salt bed
 saltwater (noun, adj.)
 sand control (noun, adj.)
 sand count
 sand line
 sandface
 sandout
 sandpack
 saturates, aromatics, resins, and asphaltenes—SARA
 sawtooth
 Scalable Vector Graphics—SVG (lowercase “svg” used in a filename extension)
 scanning electron microscope (also “scanning electron microscopy”)—SEM[†]
 screen out (verb)
 screenout (noun, adj.)
 sea level (noun)
 seabed
 seafloor
 sealbore
 sea-level (adj.)
 seastate (noun, adj.)
 seawater
 secant (spell out)
 Secure File Transfer Protocol—SFTP
 Segment (capitalized for internal mention of a specific SLB segment, but has been replaced by Business Line; use “Technology” for external communications)
 seismic (adj. only, never as noun)
 seismicity (noun)
 selective inflow performance—SIP
 self- (hyphenated prefix)
 self-potential—SP
 semi- (joined prefix)
 service level agreement—SLA

set cement (noun)
 set-cement (adj.)
 set up (verb)
 setdown (noun, adj.)
 setup (noun)
 shale gas (hydrocarbon content; see “gas shale”)
 shallow water (noun)
 shallow-water (adj.)
 shaly
 shapefile
 shear wave—S-wave
 shear-wave—S-wave (adj.)
 shock wave
 shoreline
 short message service—SMS[†]
 short-term (adj.)
 shot depth
 shot hole
 shot noise
 shotpoint—SP
 shots per foot—spf
 shoulder-bed (adj.)
 shut down (verb)
 shut in (verb)
 shut-in (noun, adj.)
 shut-in casing pressure—SICP
 shut-in pressure—SIP
 shut-in tubing pressure—SITP
 shut off (verb)
 shutdown (noun)
 shutoff (noun, adj.)
 side pocket (noun, adj.)
 side pocket mandrel—SPM
 side-scan (adj.)
 sidetrack
 sidewall
 signal-to-noise ratio—S/N
 Silicon Graphics Inc.—SGI[†] (lowercase “sgi” used in a filename extension)
 simul-frac
 sine—sin[†]
 skin (noun, adj.)
 slack off (verb)
 slack-off (noun, adj.)
 slickline (noun, adj.)
 slickwater
 slimhole (adj.)
 slowness-time-coherence—STC
 Small Computer System Interface—SCSI[†]
 Society of Exploration Geophysicists—SEG[†]
 Society of Petroleum Engineers—SPE[†]
 Society of Petrophysicists and Well Log Analysts—SPWLA[†]
 software as a service—SaaS[†]
 Software Assurance Reliability Automation—SARA
 software development kit—SDK[†]

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solid volume fraction—SVF
 source point—SP
 southerly (from the south)
 space out (verb)
 space-out (noun, adj.)
 special core analysis—SCAL
 spill point
 specific productivity index—SPI
 spontaneous potential—SP
 spud-in
 spurt loss (noun)
 spurt-loss (adj.)
 staged field experiment—SFE
 stainless steel (noun, adj.)
 stand-alone (adj.)
 stand by (verb)
 stand off (verb)
 standard—std
 standard deviation—SD
 standard work instructions—SWI
 Standardized Generalized Markup Language—
 SGML
 standby (adj.)
 standoff (noun, adj.)
 standpipe
 start up (verb)
 startup (noun, adj.)
 static corrections
 static self-potential—SSP
 steady state (noun)
 steady-state (adj.)
 steam-assisted gravity drainage—SAGD[†]
 steam/oil ratio—SOR
 steamdrive (noun, adj.)
 steamflood
 steerable
 step change (noun, avoid; use “improvement”
 instead)
 step-change (adj., void)
 step-down (noun, adj.)
 step out (verb)
 step-out (noun, adj.)
 stick-slip
 stock tank (noun)
 stock-tank (adj.)
 stock-tank oil initially in place—STOIIP
 Stokes law
 Stoneley
 straightedge
 straight-hole (adj.)
 stratigraphic
 streamtube
 strike-slip (fault)
 strokes per minute—spm
 Structured Query Language—SQL[†]

Stub Acme (thread)
 sub- (joined prefix)
 subarray
 submersible pumping system—SPS
 subsalt
 Subsea7
 subsea monitoring and control—SMC
 subsea production system—SPS
 subsea umbilicals, risers, and flowlines—SURF
 subsurface safety valve—SSV
 sulfate
 sulfide
 sulfur
 super- (joined prefix)
 superlateral
 supervisory control and data acquisition—SCADA[†]
 suprasalt
 surface casing pressure—SCP
 surface-controlled subsurface safety valve—
 SCSSV
 surface residual multiple elimination—SRME
 sustained casing pressure—SCP
 S-wave
 sweepout (noun, adj.)
 synfuel
 synthetic-based mud—SBM

T

-type (hyphenated suffix)
 Tagged Image File Format—TIFF[†] (lowercase “tif”
 used for a filename extension)
 tailbuoy
 tangent—tan[†]
 Technical Readiness Level—TRL[†]
 Technology (use instead of “Segment” and
 capitalized for a specific SLB segment in
 external communications)
 technology advancement for multilaterals—TAML[†]
 (capitalize a specific level; e.g., Level 6)
 television—TV[†]
 telltale
 Tertiary (geochronologic [time] unit: period;
 chronostratigraphic [time-rock] unit: system)
 tetrafluoroethylene—TFE
 thalweg
 thickening time
 thin-bed (adj.)
 thin-bedded (adj.)
 thin layer (noun)
 thin-layer (adj.)
 thin-section (noun, as in laboratory tests)
 thixotropic
 three-dimensional—3D[†]

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through (interchangeable with “thru” for through-tubing mentions)
 through-tubing (adj.) (interchangeable with “thru-tubing”)
 through-tubing rotary drilling—TTRD
 throughbore (adj., noun)
 throughput
 thru-tubing (adj.) (interchangeable with “through-tubing”)
 tie in (verb)
 tie-in (noun)
 tie line (noun)
 tie-line (in mathematics)
 tieback (noun, adj.)
 tight gas sands—TGS
 time domain
 time-depth-velocity
 time frame
 time-lapse (adj.)
 time slice
 timestep (noun, adj.)
 tip-screenout (adj.)—TSO
 tool joint
 toolbox
 toolface
 toolpusher
 toolset
 toolstring
 top drive (noun, adj.)
 top of cement—TOC
 tophole (noun, adj.)
 topside
 total cost of ownership—TCO
 total depth—TD[†]
 total dissolved solids—TDS
 total organic carbon—TOC
 touchpad
 touchscreen
 toward
 towout (noun, adj.)
 trade show (noun)
 tradeshow (adj.)
 transistor-transistor logic—TTL[‡]
 Transmission Control Protocol/Internet Protocol—TCP/IP[†]
 Transmission Control Protocol—TCP[†]
 transverse relaxation time— T_2
 traveltime
 Triassic (geochronologic [time] unit: period; chronostratigraphic [time-rock] unit: system)
 tricone
 triethylene glycol—TEG[†]
 tri-mill
 trip in hole—TIH
 triple-combo
 true measured depth—TMD

true stratigraphic thickness—TST
 true vertical depth—TVD[†]
 trunkline
 tubing-conveyed perforating—TCP
 tubinghead (adj.)
 tubinghead pressure—THP
 tungsten carbide insert—TCI[†]
 turbodrilling
 turn around (verb)
 turnaround (noun)
 turnkey
 twenty-four/seven/three hundred sixty-five—24/7/365[†]
 twenty-four/seven—24/7[†]
 twistoff (adj.)
 two-dimensional—2D[†]
 type curve (noun)
 type-curve (adj.)

U

UK Conformity Assessment—UKCA
 ultra- (joined prefix)
 ultradeep water (noun)
 ultradeepwater (adj.)
 ultra-HD (adj.)
 ultrahigh
 ultrahigh definition (n.)
 ultrahigh-definition (adj.)
 ultrahigh pressure and temperature (noun)
 ultrahigh-pressure and -temperature (adj.)
 ultraviolet—UV[†]
 un- (joined prefix)
 unconfined compressive strength—UCS
 under- (joined prefix)
 under way (adv.)
 underbalanced drilling—UBD
 underbalanced perforating
 underreaming
 underway (adj.)
 Underwriters Laboratory—UL[‡]
 uni- (joined prefix)
 Uniform Resource Locator—URL[†]
 uninterruptible power supply—UPS
 unique well identifier—UWI
 United Arab Emirates—UAE[†]
 United Kingdom—UK[†] (noun, adj., addresses)
 United Nations—UN[†] (noun, adj.)
 United States—US[†] (noun, adj.; use “USA”[†] in addresses)
 universal serial bus—USB[†]
 unmigrated
 up- (joined prefix)
 upcone
 updip

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uphole
upstroke
user interface—UI
U-tubing

V

vapor extraction—vapex
variable frequency drive—VFD[†]
variable speed drive—VSD[†]
venturi
versus *or* vs.
vertical exaggeration—VE
vertical seismic profile—VSP
very small aperture terminal—VSAT[†]
videocassette recorder—VCR[†]
virtual LAN—VLAN[†]
virtual private network—VPN[†]
viscoelastic
viscoelastic surfactant—VES
viscometer
viscosity
voice over Internet protocol—VoIP[†]
volatile organic compound—VOC[†]
vs. *or* versus

W

water, oil, gas—WOG
-wide (closed if listed as such in Merriam-Webster's; if not in Merriam-Webster's, hyphenated)
-wise (joined suffix)
wadi
waiting on cement—WOC[†]
waiting-on-cement—WOC[†] (adj.)
walkaway
warm back (v)
warmback (noun, adj.)
wash out (verb)
Washington, DC
washout (noun)
washover
washpipe
washwater
wastewater (noun, adj.)
water-alternating-gas—WAG (adj.)
water-based (adj.)
water-based mud—WBM[†]
water-bearing
water cut (noun)
water pack (noun)
water-pack (adj.)
water soluble (adv.)
water-soluble (adj.)

water well (exception for M-I SWACO HDD, Mining & Waterwell)
water/oil contact—WOC
water/oil ratio—WOR
water-wet
waterdrive
waterflood
waterfrac
wave amplitude
wave band
wave equation
wave-equation migration
wave train
wavefield
waveform
wavefront
wavelength
wavelet
wavenumber
weakpoint (noun, adj.)
Web (capitalized only as shortened form of “World Wide Web”)
web page
Web Services for Remote Portlets—WSRP
webcam
webcast
webinar
webmaster
website
weight on bit—WOB[†]
well (initial cap if naming a specific one: “Well 2”)
well field (noun)
well-field (adj.)
well log (noun, adj.)
well logging (noun, adj.)
well pad (noun)
well path
well stream (noun)
well-pad (adj.)
wellbore
wellhead
wellhead pressure—WHP
wellpoint
wellsite
wellsite information transfer specifications—WITS
wellstream (adj.)
westerly (from the west)
whipstock
wide area network—WAN[†]
wide-azimuth—WAZ[†]
wiki
wind field (noun)
wind speed (noun)
windfield (adj.)
wire mesh (screens)

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[‡] A stand-alone abbreviation only in tables and sidebar bullet listings; spell out on first mention in body text.

wireless fidelity—WiFi[†]
wireline
wiremesh (model)
work group
work over (verb)
work site (noun)
workboat
workflow
workforce
work-hour (this or “labor-hour” preferred to “man-hour”)
workover (noun, adj.)
work-site (adj.)
workstation
workstring
work-year (preferred to “man-year”)
World Wide Web—WWW[†] (lowercase “www” used for an Internet location)
worldwide
worm gear
wormhole
write-back (adj.)
Wyllie

X

X (alias for depth numeral)
x-axis
X-ray
X-ray diffraction—XRD
X-ray fluorescence—XRF

Y

y-axis
Y-block
Y-tool
yield point—YP
Young’s modulus

Z

z-axis

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Spellings and Terminology

Abbreviations

Legal information

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Numerals

1D[†]—one-dimensional
2D[†]—two-dimensional
3D[†]—three-dimensional
4C—four-component
4D[†]—four-dimensional
24/7/365[†]—twenty-four/seven/three hundred
sixty-five
24/7[†]—twenty-four/seven

A

A/D—analogue-to-digital
A&D—acquisition and divestiture
AAPG[†]—American Association of Petroleum
Geologists
ABS[†]—American Bureau of Shipping
ABSS[†]—at-bit steerable system (stand-alone
abbreviation only when space constrained)
AC[†]—alternating current
AD (do not use; see CE for Common Era)
AFE[†]—authorization for expenditure
AHR—after hot rolling
AI[†]—artificial intelligence
a.m.[†]
ANSI[†]—American National Standards Institute
AOF—absolute open flow
API RP[†] (subsequently RP)—API
Recommended Practice
API Spec[†] (subsequently Spec)—API Specification
API[†]—American Petroleum Institute
API[†]—application programming interface
app[†]—application
ANSI/FCI[†]—American National Standards
Institute and the Fluid Controls Institute
APWD—annular pressure while drilling
ASCII[†]—American Standard Code for
Information Interchange
ASME[†]—American Society of Mechanical
Engineers
ASP—application service provider
ASTM International (formerly the American
Society of Testing and Materials)
AVA[†]—amplitude variation with azimuth
avg—average
AVO[†]—amplitude variation with offset

[†] Abbreviation that can stand alone

[‡] Abbreviation that can stand alone only in tables; spell out on first mention in body text.

AWG[†]—American Wire Gauge

B

b[†]—bit
B[†]—byte
BARS—borehole acoustic reflection survey
BC (do not use; see BCE for before Common
Era)
BCE[†]—before Common Era
Bd[†]—baud
BFV—bound-fluid volume
BHA[†]—bottomhole assembly
BHC—borehole compensated
BHCP—bottomhole circulating pressure
BHCT—bottomhole circulating temperature
BHFP—bottomhole flowing pressure
BHFT—bottomhole flowing temperature
BHP—bottomhole pressure
BHR—before hot rolling
BHSP—bottomhole static pressure
BHST—bottomhole static temperature
BHT—bottomhole temperature
BHTV—borehole televiewer
BOD[†]—bandwidth on demand (noun),
bandwidth-on-demand (adj.)
BOP[†]—blowout preventer
bps[†]—bits per second
Bps[†]—bytes per second
BRT—below rotary table
BS&W—basic sediment and water (*also*
bottom sediment and water)
BTEX[†]—benzene, toluene, ethylbenzene, and
xylene
BVOS[‡]—by volume of solids
BWOC[‡]—by weight of cement
BWOP[‡]—by weight of proppant
BWOW[‡]—by weight of water

C

CAD[†]—computer-aided design
CAOF—calculated absolute open flow
capex[†]—capital expenditure
CAT—computerized axial tomography
CBF—clear-brine fluid

CBM—coalbed methane
 CCL[†]—casing collar locator
 CCS[†]—carbon capture and sequestration
 (also known as “carbon capture and storage,” but SLB collateral should use “sequestration” for the initialism)
 CCTV—closed-circuit television
 CCUS[†]—carbon capture, utilization, and sequestration (also known as “carbon capture, utilization, and storage,” but SLB collateral should use “sequestration” for the initialism)
 CD[†]—compact disc
 CDP—common depth point
 CD-ROM[†]—compact disk read-only memory
 CE[†]—Common Era
 CE[‡]—Conformité Européenne
 CEC[†]—Canadian Electrical Code
 CEC—cation exchange capacity
 CGM—Computer Graphics Metafile
 CHOPS—cold heavy-oil production with sand
 CL[†]—Class (restricted to ASME Class)
 CL—checklist
 CMMS—computerized maintenance management system
 CMP—common midpoint
 CMYK[†]—cyan, magenta, yellow, and key (black)
 CNG—compressed natural gas
 CO₂e[†]—CO₂ equivalent
 CPU[†]—central processing unit
 CRN[†]—Canadian Registration Number
 CRP—common reflection point
 CRT[†]—cathode ray tube
 CSA[‡]—Canadian Standards Association
 CSEM—controlled-source electromagnetic
 CSS—cascading style sheet
 CSS—cyclic steam stimulation
 CT[†]—coiled tubing
 CT[†]—computerized tomography
 CTU—coiled tubing unit
 CWP—cold working pressure

D

D&A—dry and abandoned

D&I—direction and inclination
 DAS—distributed acoustic sensing
 DC[†]—direct current
 DC[†]—District of Columbia (Washington)
 DCA—decline curve analysis
 DD—directional driller
 DEI[†]—diamond-enhanced insert
 DFA—downhole fluid analysis
 DHTML—Dynamic HTML
 DITA—Darwin Information Typing Architecture
 DLIS—Digital Log Information Standard
 DLS—dogleg severity
 DMO—dip moveout
 DN[†]—*diamètre nominal* or *Durchmesser nach Norm* (nominal pipe sizing; DN 8, DN 40, etc.)
 DNV[†]—Det Norske Veritas
 DRIFTS—diffuse reflectance infrared Fourier transform spectroscopy
 DSL[†]—digital subscriber line
 DST—drillstem test
 DTS[†]—distributed temperature sensing or sensor
 DUC—drilled but uncompleted
 DVD[†]—digital video disc
 DVS—distributed vibration sensing

E

E&P[†]—exploration and production
 EAC[‡]—Eurasian Conformity
 EC (do not use, superseded by the “EU”)
 ECD—equivalent circulating density
 ECL—emergency checklist
 EDS—energy-dispersive spectroscopy
 e.g. [†] (means “for example”)
 EEPROM—electrically erasable, programmable, read-only memory
 EI[†]—Energy Institute
 EIF—Environmental Impact Factor
 EM[†]—electromagnetic
 emf—electromotive force
 EMR—electromagnetic radiation
 ENP—engineering prototype
 EOB—extreme overbalance
 EOP—extreme overbalanced perforating
 EOR—enhanced oil recovery

[†] Abbreviation that can stand alone

[‡] Abbreviation that can stand alone only in tables; spell out on first mention in body text.

EOS—equation of state (noun), equation-of-state (adj.)
 EOS—extended operating system
 EPCI[†]—engineering, procurement, construction, and installation
 EPCIC[†]—engineering, procurement, construction, installation, and commissioning
 EPF[†]—early production facility
 EPS—encapsulated PostScript® (lowercase “eps” used in a filename extension)
 ERD—extended-reach drilling
 ERP—enterprise resource planning
 ES—electrical stability
 ESD[†]—equivalent static density
 ESD—emergency shutdown
 ESP[†]—electric submersible pump
 ESPCP[†]—electric submersible progressing cavity pump
 et al. (means “and others”)
 EU[†]—European Union
 EUE[†]—external upset ends
 EUR—estimated ultimate recovery
 EXP—experimental prototype

F

FAQ[†]—frequently asked questions
 FAZ[†]—full-azimuth
 FD—frequency domain
 FEED[†]—front-end engineering and design
 FEWD—formation evaluation while drilling
 FFV—free-fluid volume
 FH[†]—Full Hole (API screw threads; e.g., 3½ FH)
 FID—final investment decision
 FID—flame ionization detector
 Fig[†]—configuration (for end connection names, typically for hammer unions)
 FIT—formation integrity test
 FKM[†]—fluoroelastomer
 FML—fiscal model library
 FPSO[†]—floating production, storage, and offloading
 FTIR—Fourier transform infrared
 FTP[†]—File Transfer Protocol (lowercase “ftp” used for an Internet location)

FVF—formation volume factor
 FWI—full-waveform inversion
 FSO[†]—floating storage and offloading

G

G&G—geological and geophysical *or* geology and geophysics
 GC—gas chromatography
 GCMS—gas chromatography–mass spectrometry
 GHG—greenhouse gas
 GIF[†]—graphics interchange format (lowercase “gif” used in a filename extension)
 GIS[†]—geographic information system
 GLR—gas/liquid ratio
 GOC—gas/oil contact
 GOM—Gulf of Mexico (spelled-out “Gulf of Mexico” preferred; use of “GOM” depends of context)
 GOR[†]—gas/oil ratio
 GPC—gel permeation chromatography
 GPS[†]—global positioning system
 GR[†]—gamma ray
 GRI—Gas Research Institute (see “GTI—Gas Technology Institute”)
 GTI—Gas Technology Institute (successor to “GRI—Gas Research Institute”)
 GTL—gas-to-liquid
 GUI[†]—graphical user interface
 GVF—gas volume fraction
 GWC—gas/water contact

H

HART[†]—Highway Addressable Remote Transducer
 HCl—hydrochloric acid
 HCPV—hydrocarbon pore volume
 HD[†]—high definition (noun), high-definition (adj.)
 HEC—hydroxyethyl cellulose
 HF—hydrofluoric acid
 HI—hydrogen index
 HIA—hold inclination and azimuth
 HIP—hydrocarbon in place

[†] Abbreviation that can stand alone

[‡] Abbreviation that can stand alone only in tables; spell out on first mention in body text.

HMI—human machine interface
 HMX—cyclotetramethylene tetranitramine
 HNBR[†]—hydrogenated nitrile butadiene rubber
 HNS—hexanitrostilbene
 HOCNF[‡]—Harmonised Offshore Chemical Notification Format
 HP[†]—high-pressure
 HPC—high performance computing
 HPG—hydroxypropyl guar
 HPHT[†]—high pressure, high temperature (noun) and high-pressure, high-temperature (adj.)
 HPS[†]—horizontal pumping system (also “horizontal pump system”)
 HSE[†]—health, safety, and environment
 HT[†]—high temperature (noun), high-temperature (adj.)
 HTML[†]—Hypertext Markup Language
 HTTP[†]—Hypertext Transfer Protocol (lowercase “http” used for an Internet location)
 HVAC[†]—heating, ventilation, and air conditioning

I

I/O[†]—input/output
 IaaS[†]—infrastructure as a service
 IADC—International Association of Drilling Contractors
 ICD[†]—inflow control device
 ID[†]—inside diameter
 i.e.[†] (means “that is”)
 IEC[†]—International Electrotechnical Commission
 IEEE[‡]—Institute of Electrical and Electronics Engineers, Inc.
 IF[†]—Internal Flush (API screw thread; e.g., 3½ IF)
 IGBT[‡]—insulated gate bipolar transistor
 IIoT[†]—Industrial Internet of Things
 IOC[†]—international oil company
 IOR—improved oil recovery
 IoT[†]—Internet of Things
 IP[†]—Internet Protocol
 IPR—inflow performance relationship
 IPTT—interval pressure transient test or interval pressure transient testing
 IRMS[‡]—isotope ratio mass spectrometry

ISDN[†]—Integrated Services Digital Network
 ISIP—initial shut-in pressure
 ISO[†]—International Organization for Standardization (do not use generic terms “ISO certification” and “ISO certified” but specify which unique standard [e.g., “ISO 9001:2000 certification” and “ISO 14001:2004 certified,” respectively])
 ISP—intermediate-strength proppant
 IT[†]—information technology
 IWIS—Intelligent Well Interface Standardisation

J

JPEG[†] or JPG[†]—joint photographic experts group (lowercase “jpg” used in a filename extension)
 JV[†]—joint venture

K

Kb[†]—kilobit
 KB[†]—kilobyte
 KB—kelly bushing
 KPI[†]—key performance indicator

L

LAN[†]—local area network
 LAS[†]—log ASCII standard
 LCD[†]—liquid crystal display
 LCM—lost circulation material
 LED[†]—light-emitting diode
 LIH—lost in hole (noun), lost-in-hole (adj.)
 LIS—log information standard
 LLD—laterolog deep
 LLS—laterolog shallow
 ln[†]—logarithm natural
 LNG[†]—liquefied natural gas
 log[†]—logarithm
 LOT—leakoff test
 LPG[†]—liquefied petroleum gas
 LTI—lost-time incident
 LTIR—lost-time incident ratio

[†] Abbreviation that can stand alone

[‡] Abbreviation that can stand alone only in tables; spell out on first mention in body text.

LWD[†]—logging while drilling (noun), logging-while-drilling (adj.)

M

max.[‡]—maximum

MB[†]—megabyte

MD[†]—measured depth

MDH—Miller Dyes and Hutchinson

MDRT—measured depth from rotary table

MEG[†]—monoethylene glycol

MEM—mechanical earth model

MEMS[†]—microelectromechanical systems

MGS—mud-gas separator

MHF—massive hydraulic fracturing

min.[‡]—minimum

ML—machine learning

MLE—motor lead extension

MLWD[†]—measurement(s) and logging while drilling (noun), measurement(s)- and logging-while-drilling (adj.)

MMT—marine magnetotelluric

MODU—mobile offshore drilling unit

MPD[†]—managed pressure drilling

MPEG[†] or MPG[†]—Moving Picture Experts Group

MPEG-3[†] or MP3[†]—MPEG-1 Audio Layer 3 (lowercase “mp3” used in a filename extension)

MPEG-4[†] or MP4[†] (lowercase “mp4” used in a filename extension)

MPF—multipurpose float (“multipurpose trailer-mounted” preferred)

MPS—multipurpose skid (“multipurpose skid-mounted” preferred)

MRI—magnetic resonance imaging

MSDS[†]—Material Safety Data Sheet

MSL—mean sea level

MSS[†]—Manufacturers Standardization Society of the Valve and Fittings Industry

MTBF—mean time between failure(s)

MT—magnetotelluric

MWD[†]—measurement(s) while drilling (noun), measurement(s)-while-drilling (adj.)

N

NACE[†] (use for standards)

NACE International (formerly National Association of Corrosion Engineers—NACE)

NAZ[†]—narrow azimuth

NBR[†]—nitrile butadiene rubber

NC[†]—Numbered Connector (API screw threads; e.g., NC50)

NEC[†]—National Electrical Code

NEMA[†]—National Electrical Manufacturers Association

NGD—neutron gamma density

NGL[†]—natural gas liquids

NIST—National Institute of Standards and Technology

NMO—normal moveout

NMR[†]—nuclear magnetic resonance

NOC[†]—national oil company

NGO[†]—non-governmental organization

NORM—naturally occurring radioactive material

NPST—non-pressure-sealed thread

NPT[†]—nonproductive time

NPV[†]—net present value

NTG—net-to-gross

NUE[†]—nonupset end

O

O/U—over/under

O&G[†]—oil and gas

OBC—ocean bottom cable

OBMF—oil-based mud filtrate

OBM[†]—oil-based mud

OBN—ocean bottom node

OBS—ocean bottom seismic

OCTG—oil country tubular goods

OD[†]—outside diameter

OEM[†]—original equipment manufacturer

OFP—open-flow potential

OIP—oil in place

OIW—oil in water

OLE[†]—object linking and embedding

OOIP—original oil in place

opex[†]—operational expenditure

OS[†]—operating system

[†] Abbreviation that can stand alone

[‡] Abbreviation that can stand alone only in tables; spell out on first mention in body text.

OSF—Open Software Foundation
OSHA[†]—Occupational Safety and Health
Administration
OWC—oil/water contact

P

P&A[†]—plug and abandon (also “plugged and
abandoned”)
PaaS[†]—platform as a service
PC[†]—personal computer
pCO₂[†]—partial pressure of CO₂
PCP[†]—progressing cavity pump (also
“progressive cavity pump”)
PDC[†]—polycrystalline diamond compact
(noun, adj.)
PDF[†]—portable document format (lowercase
“pdf” used in a filename extension)
PDM—positive displacement motor
PDM—production data management
PD—positive displacement
PDS—picture description system
PDVSA[†]—Petróleos de Venezuela S.A.
PED[†]—Pressure Equipment Directive
PEEK[†]—polyetheretherketone
PET—positron emission tomography
PFP—preferred hydraulic fracture plane
pH₂S[†]—partial pressure of H₂S
PIDD—Petroleum Industry Data Dictionary
PI—productivity index
PLC—programmable logic control or
programmable logic controller
PLEM—pipeline end manifold
PLONOR—Pose Little or No Risk (shortened from
OSPAR’s Preparations Used and Discharged
Offshore Which Are Considered to Pose Little
or No Risk to the Environment)
PLT—production logging tool
p.m.[†]
PMCC[‡]—Pensky-Martens closed cup
PNG[†]—Portable Network Graphics (lowercase
“png” used in a filename extension)
PNG—pulsed neutron generator
POOH—pull out of hole
PPDM—Public Petroleum Data Model
PSTM—prestack time migration
PSTM—poststack time migration

PT[†]—pressure and temperature
PTFE[†]—polytetrafluoroethylene
PV—plastic viscosity
PV—pore volume
PVC[†]—polyvinyl chloride
PVT[†]—pressure-volume-temperature
P-wave—compressional wave or
compressional-wave

Q

Q&A[†]—question and answer
QA[†]—quality assurance
QA/QC[†]—quality assurance and quality
control
QC[†]—quality control
QHSE[†]—quality, health, safety, and
environment

R

R&D[†]—research & development
RAID—Redundant Arrays of Independent
Disks
RAM[†]—random-access memory
RCAL—routine core analysis
RCP—resin-coated proppant
RDF—reservoir drill-in fluid
Reg[†]—Regular (API screw threads; e.g., 3½ Reg)
REM—roentgen equivalent man
RF—radio frequency
RFID—radio frequency identification
RGB[†]—red-green-blue
RIH—run in hole
RIR—Risk Identification Report
RISC—reduced instruction set computer or
computing
RKB—rotary kelly bushing
rms[†]—root mean square
RNP—rate-normalized pressure
ROI[†]—return on investment
ROM[†]—read-only memory
ROP[†]—rate of penetration
ROP—readout port
ROV[†]—remotely operated vehicle
RP—API Recommended Practice

[†] Abbreviation that can stand alone

[‡] Abbreviation that can stand alone only in tables; spell out on first mention in body text.

RSS[†]—really simple syndication
RSS[†]—rotary steerable system
RSVP[†] (abbreviation for *répondez s'il vous
plaît* [please reply])
RT—rotary table
RTD—resistive temperature detector
RTM[†]—reverse time migration
RTN[†]—Rostekhnadzor
RTU[†]—remote terminal unit

S

S/N—signal-to-noise ratio
SaaS[†]—software as a service
SAGD[†]—steam-assisted gravity drainage
SARA—saturates, aromatics, resins, and
asphaltenes
SARA—Software Assurance Reliability
Automation
SBM—synthetic-based mud
SCADA[†]—supervisory control and data
acquisition
SCAL[†]—special core analysis
SCP—surface casing pressure
SCP—sustained casing pressure
SCSI[†]—Small Computer System Interface
SCSSV—surface-controlled subsurface safety
valve
SD—standard deviation
SDK[†]—software development kit
SDS[†]—Safety Data Sheet
SEG[†]—Society of Exploration Geophysicists
SEM[†]—scanning electron microscope (also
“scanning electron microscopy”)
SFE—staged field experiment
SFTP—Secure File Transfer Protocol
SGI[†]—Silicon Graphics Inc. (lowercase “sgi”
used in a filename extension)
SGML—Standardized Generalized Markup
Language
SICP—shut-in casing pressure
SIL[†]—safety integrity level
sin[†]—sine
SIP—selective inflow performance
SIP—shut-in pressure
SITP—shut-in tubing pressure
SLA—service level agreement

SMC—subsea monitoring and control
SMS[†]—short message service
SOR—steam/oil ratio
SPE[†]—Society of Petroleum Engineers
Spec—API Specification
spf—shots per foot
SPI—specific productivity index
SPM—side pocket mandrel
spm—strokes per minute
SP—self-potential
SP—shotpoint
SP—source point
SP—spontaneous potential
SPS—submersible pumping system
SPWLA[†]—Society of Petrophysicists and Well Log
Analysts
SQL[†]—Structured Query Language
SRME—surface residual multiple elimination
SSP—static self-potential
SSV—subsurface safety valve
STC—slowness-time-coherence
STOIIIP—stock-tank oil initially in place
SVF—solid volume fraction
SVG—Scalable Vector Graphics (lowercase
“svg” used in a filename extension)
S-wave—shear wave or shear-wave
SWI—standard work instructions

T

T₁—longitudinal relaxation time
T₂—transverse relaxation time
TAML[†]—technology advancement for
multilaterals
tan[†]—tangent
TCI[†]—tungsten carbide insert
TCO—total cost of ownership
TCP[†]—Transmission Control Protocol
TCP—tubing-conveyed perforating
TCP/IP[†]—Transmission Control
Protocol/Internet Protocol
TD[†]—total depth
TDS—total dissolved solids
TEG[†]—triethylene glycol
TFE—tetrafluoroethylene
TGS—tight gas sands
THP—tubinghead pressure

[†] Abbreviation that can stand alone

[‡] Abbreviation that can stand alone only in tables; spell out on first mention in body text.

TIFF[†]—Tagged Image File Format (lowercase “tif” used for a filename extension)
TIH—trip in hole
TMD—true measured depth
TOC—top of cement
TOC—total organic carbon
TRL[†]—Technical Readiness Level
TSO—tip-screenout
TST—true stratigraphic thickness
TTL[‡]—transistor-transistor logic
TTRD—through-tubing rotary drilling
TV[†]—television
TVD[†]—true vertical depth
TVDSS—true vertical depth subsea

U

UAE[†]—United Arab Emirates
UBD—underbalanced drilling
UCS—unconfined compressive strength
UI—user interface
UK[†]—United Kingdom (noun, adj., addresses)
UKCA—UK Conformity Assessment
UL[‡]—Underwriters Laboratory
UN[†]—United Nations (noun, adj.)
UPS—uninterruptible power supply
URL[†]—Uniform Resource Locator
US[†]—United States (noun, adj.; use “USA” in addresses)
USA[†]—United States of America (use in addresses; use “US” as noun and adj.)
USB[†]—universal serial bus
UV[†]—ultraviolet
UWI—unique well identifier

V

vapex—vapor extraction
VCR[†]—videocassette recorder
VE—vertical exaggeration
VES—viscoelastic surfactant
VFD[†]—variable frequency drive
VLAN[†]—virtual LAN
VOC[†]—volatile organic compound
VoIP[†]—voice over Internet protocol
VPN[†]—virtual private network
vs. *or* versus

[†] Abbreviation that can stand alone

[‡] Abbreviation that can stand alone only in tables; spell out on first mention in body text.

VSAT[†]—very small aperture terminal
VSD[†]—variable speed drive
VSP—vertical seismic profile

W

WAG—water-alternating-gas (adj.)
WAN[†]—wide area network
WAZ[†]—wide-azimuth
WBM[†]—water-based mud
WHP—wellhead pressure
WiFi[†]—wireless fidelity
WITS—wellsite information transfer specifications
WOB[†]—weight on bit
WOC[†]—waiting on cement, waiting-on-cement (adj.)
WOC—water/oil contact
WOG—water, oil, gas
WOR—water/oil ratio
WSRP—Web Services for Remote Portlets
WWW[†]—World Wide Web (lowercase “www” used for an Internet location)

X

XHTML[†]—Extensible HTML
XML[†]—Extensible Markup Language
XRD—X-ray diffraction
XRF—X-ray fluorescence

Y

YP—yield point

Spellings and Terminology

Stand-Alone Abbreviations

Legal information

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Hypertext Transfer Protocol—HTTP
(lowercase “http” used for an Internet location)

I

i.e. (means “that is”)
Industrial Internet of Things—IIoT
inflow control device—ICD
information technology—IT
infrastructure as a service—IaaS
input/output—I/O
inside diameter—ID
Institute of Electrical and Electronics Engineers, Inc.—IEEE[†]
insulated gate bipolar transistor—IGBT[†]
Integrated Services Digital Network—ISDN
Internal Flush—IF (API screw thread; e.g., 3½ IF)
International Electrotechnical Commission—IEC
international oil company—IOC
International Organization for Standardization—ISO (do not use generic terms “ISO certification” and “ISO certified” but specify which unique standard [e.g., “ISO 9001:2000 certification” and “ISO 14001:2004 certified,” respectively])
Internet of Things—IoT
Internet Protocol—IP
isotope ratio mass spectrometry—IRMS[†]

J

Joint Photographic Experts Group—JPEG or JPG (lowercase “jpg” used in a filename extension)
joint venture—JV

K

key performance indicator—KPI
kilobit—Kb
kilobyte—KB

L

light-emitting diode—LED
liquefied natural gas—LNG
liquefied petroleum gas—LPG
liquid crystal display—LCD

local area network—LAN
log ASCII standard—LAS
logarithm—log
logarithm natural—ln
logging while drilling—LWD (noun)
logging-while-drilling—LWD (adj.)

M

managed pressure drilling—MPD
Manufacturers Standardization Society of the Valve and Fittings Industry—MSS
Material Safety Data Sheet—MSDS
maximum—max.[†]
measured depth—MD
measurement(s) and logging while drilling—MLWD (noun)
measurement(s)- and logging-while-drilling—MLWD (adj.)
measurement(s) while drilling—MWD (noun)
measurement(s)-while-drilling—MWD (adj.)
megabyte—MB
microelectromechanical systems—MEMS
minimum—min.[†]
monoethylene glycol—MEG
Moving Picture Experts Group—MPG or MPEG
MPEG-1 Audio Layer 3—MPEG-3 or MP3 (lowercase “mp3” used in a filename extension)
MPEG-4—MP4 (lowercase “mp4” used in a filename extension)

N

NACE (use for standards)
narrow azimuth—NAZ
National Electrical Code—NEC
National Electrical Manufacturers Association—NEMA[†]
national oil company—NOC
natural gas liquids—NGL
net present value—NPV
nitrile butadiene rubber—NBR
non-governmental organization—NGO
non-pressure-sealed thread—NPST[†]
nonproductive time—NPT
nonupset end—NUE
nuclear magnetic resonance—NMR

[†] Stand-alone abbreviation only in tables and sidebar bulleted listings; spell out on first mention in body text.

Overview

Along with commonly used nontechnical abbreviations (for example, “US” and “a.m.”), the following “stand-alone” abbreviations do not require spelling out the technical term or phrase at first use with the abbreviation enclosed in parentheses following the term. However, if readers will not readily recognize an abbreviation, introduce it with the spelled-out term or phrase as a courtesy. The abbreviations identified with a dagger footnote can stand alone in tables or sidebar bullet lists but should be fully introduced on the first text mention.

The list is in two parts. The first is organized by the spelled-out terms; the second, by the abbreviations.

Listed by spelled-out terms

A

alternating current—AC
a.m.
American Association of Petroleum Geologists—AAPG
American Bureau of Shipping—ABS
American National Standards Institute—ANSI
American Petroleum Institute—API
American Society of Mechanical Engineers—ASME
American Standard Code for Information Interchange—ASCII
American Wire Gauge—AWG
amplitude variation with azimuth—AVA
amplitude variation with offset—AVO
API Recommended Practice—API RP (subsequently RP)
API Specification—API Spec (subsequently Spec)
application programming interface—API application—app
artificial intelligence—AI
at-bit steerable system[†]—ABSS (stand-alone abbreviation only when space constrained)
authorization for expenditure—AFE

B

bandwidth on demand—BOD (noun)
bandwidth-on-demand—BOD (adj.)
baud—Bd
before Common Era—BCE

benzene, toluene, ethylbenzene, and xylene—BTEX
bit—b
bits per second—bps
blowout preventer—BOP
bottomhole assembly—BHA
by volume of solids—BVOS[†]
by weight of cement—BWOC[†]
by weight of proppant—BWOP[†]
by weight of water—BWOW[†]
byte—B
bytes per second—Bps

C

Canadian Electrical Code—CEC
Canadian Registration Number—CRN
Canadian Standards Association—CSA[†]
capital expenditure—capex
carbon capture and sequestration—CCS (also known as “carbon capture and storage,” but SLB collateral should use “sequestration” for the initialism)
carbon capture, utilization, and sequestration—CCUS (also known as “carbon capture, utilization, and storage,” but SLB collateral should use “sequestration” for the initialism)
casing collar locator—CCL
cathode ray tube—CRT
central processing unit—CPU
Class—CL (restricted to ASME Class)
CO₂ equivalent—CO₂e[†]
coiled tubing—CT
Common Era—CE
compact disc—CD

[†] Stand-alone abbreviation only in tables and sidebar bulleted listings; spell out on first mention in body text.

compact disc read-only memory—CD-ROM
computer-aided design—CAD
computerized tomography—CT
configuration—Fig (for end connection names, typically for hammer unions)
Conformité Européenne—CE[†]
cyan, magenta, yellow, and key (black)—CMYK

D

Det Norske Veritas—DNV
diamètre nominal—DN[†] (nominal pipe sizing; DN 8, DN 40, etc.)

diamond-enhanced insert—DEI
digital subscriber line—DSL
digital video disc—DVD
direct current—DC
distributed temperature sensing or sensor—DTS
District of Columbia—DC
Durchmesser nach Norm—DN[†] (nominal pipe sizing; DN 8, DN 40, etc.)

E

early production facility—EPF
e.g. (means “for example”)
electric submersible progressing cavity pump—ESPCP
electric submersible pump—ESP
electromagnetic—EM
Energy Institute—EI[†]
engineering, procurement, construction, and installation—EPCI
engineering, procurement, construction, installation, and commissioning—EPCIC
equivalent circulating density—ECD
equivalent static density—ESD
et al. (means “and others”)
Eurasian Conformity—EAC[†]
European Union—EU (superseded European Community)
exploration and production—E&P
Extensible HTML—XHTML
Extensible Markup Language—XML
external upset ends—EUE

F

File Transfer Protocol—FTP (lowercase “ftp” used for an Internet location)
floating production, storage, and offloading—FPSO
floating storage and offloading—FSO
fluoroelastomer—FKM
four-dimensional—4D
frequently asked questions—FAQ
front-end engineering and design—FEED
full-azimuth—FAZ
Full Hole—FH (API screw threads; e.g., 3½ FH)

G

gamma ray—GR
gas/oil ratio—GOR
geographic information system—GIS
global positioning system—GPS
graphical user interface—GUI
graphics interchange format—GIF (lowercase “gif” used in a filename extension)

H

Harmonised Offshore Chemical Notification Format—HOCNF[†]
Highway Addressable Remote Transducer—HART
health, safety, and environment—HSE
heating, ventilation, and air conditioning—HVAC
high definition (noun)—HD
high-definition (adj.)—HD
high-pressure (adj.)—HP
high pressure, high temperature (noun)—HPHT
high-pressure, high-temperature (adj.)—HPHT
high temperature (noun)
high-temperature (adj.)—HT
hydrogenated nitrile butadiene rubber—HNBR
horizontal pumping system (also “horizontal pump system”)—HPS
Hypertext Markup Language—HTML

[†] Stand-alone abbreviation only in tables and sidebar bulleted listings; spell out on first mention in body text.

Numbered Connector—NC (API screw threads; e.g., NC50)

O

object linking and embedding—OLE
Occupational Safety and Health Administration—OSHA

oil and gas—O&G
oil-based mud—OBM
one-dimensional—1D
operating system—OS
operational expenditure—opex
original equipment manufacturer—OEM
outside diameter—OD

P

partial pressure of CO₂—pCO₂
partial pressure of H₂S—pH₂S
Pensky-Martens closed cup—PMCC[†]
personal computer—PC
Petróleos de Venezuela S.A.—PDVSA
platform as a service—PaaS
plug and abandon—P&A (also “plugged and abandoned”)
plugged and abandoned—P&A (also “plug and abandon”)
p.m.
polycrystalline diamond compact—PDC (noun, adj.)
polyetheretherketone—PEEK
polytetrafluoroethylene—PTFE
polyvinyl chloride—PVC
portable document format—PDF (lowercase “pdf” used in a filename extension)
Portable Network Graphics—PNG (lowercase “png” used in a filename extension)
pressure and temperature—PT
Pressure Equipment Directive—PED
pressure-volume-temperature—PVT
progressing cavity pump (also “progressive cavity pump”)—PCP

Q

quality assurance—QA

quality assurance and quality control—QA/QC
quality control—QC
quality, health, safety, and environment—QHSE
question and answer—Q&A

R

random-access memory—RAM
rate of penetration—ROP
read-only memory—ROM
really simple syndication—RSS
red-green-blue—RGB
Regular (API screw threads; e.g., 3½ Reg)—Reg
remote terminal unit—RTU
remotely operated vehicle—ROV
research and development—R&D
return on investment—ROI
reverse time migration—RTM
root-mean-square—rms
Rostekhnadzor—RTN[†]
rotary steerable system—RSS
RSVP (abbreviation for *répondez s’il vous plait* [please reply])

S

Safety Data Sheet—SDS
safety integrity level—SIL
scanning electron microscope (also “scanning electron microscopy”)—SEM
short message service—SMS
Silicon Graphics Inc.—SGI (lowercase “sgi” used in a filename extension)
sine—sin
Small Computer System Interface—SCSI
Society of Exploration Geophysicists—SEG
Society of Petroleum Engineers—SPE
Society of Petrophysicists and Well Log Analysts—SPWLA
software as a service—SaaS
software development kit—SDK
special core analysis—SCAL
steam-assisted gravity drainage—SAGD
Structured Query Language—SQL
supervisory control and data acquisition—SCADA

[†] Stand-alone abbreviation only in tables and sidebar bulleted listings; spell out on first mention in body text.

T

Tagged Image File Format—TIFF
(lowercase “tif” used for a filename extension)
tangent—tan
Technical Readiness Level—TRL
technology advancement for multilaterals—
TAML (capitalize a specific level; e.g.,
Level 6)
television—TV
three-dimensional—3D
total depth—TD
transistor-transistor logic—TTL[†]
Transmission Control Protocol—TCP
Transmission Control Protocol/Internet
Protocol—TCP/IP
true vertical depth—TVD
triethylene glycol—TEG
tungsten carbide insert—TCI
twenty-four/seven—24/7
twenty-four/seven/three hundred sixty-
five—24/7/365
two-dimensional—2D

U

ultraviolet—UV
Underwriters Laboratory—UL[†]
Uniform Resource Locator—URL
United Arab Emirates—UAE
United Kingdom—UK (noun, adj.,
addresses)
United Nations—UN (noun, adj.)
United States—US (noun, adj.; use “USA” in
addresses)
universal serial bus—USB

V

variable frequency drive—VFD
variable speed drive—VSD
versus *or* vs.
very small aperture terminal—VSAT
videocassette recorder—VCR
virtual LAN—VLAN
virtual private network—VPN
voice over Internet protocol—VoIP
volatile organic compound—VOC

W

waiting on cement—WOC
waiting-on-cement—WOC (adj.)
water-based mud—WBM
weight on bit—WOB
wide area network—WAN
wide-azimuth—WAZ
wireless fidelity—WiFi
World Wide Web—WWW (lowercase “www”
used for an Internet location)

[†] Stand-alone abbreviation only in tables and sidebar bulleted listings; spell out on first mention in body text.

Listed by abbreviations

Numerals

1D—one-dimensional
2D—two-dimensional
3D—three-dimensional
4D—four-dimensional
24/7—twenty-four/seven
24/7/365—twenty-four/seven/three hundred
sixty-five

A

AAPG—American Association of Petroleum Geologists
ABS—American Bureau of Shipping
ABSS[†]—at-bit steerable system (stand-alone abbreviation only when space constrained)
AC—alternating current
AFE—authorization for expenditure
AI—artificial intelligence
a.m.
ANSI—American National Standards Institute
API RP (subsequently RP)—API Recommended Practice
API Spec (subsequently Spec)—API Specification
API—American Petroleum Institute
API—application programming interface
app—application
ASCII—American Standard Code for Information Interchange
ASME—American Society of Mechanical Engineers
AVA—amplitude variation with azimuth
AVO—amplitude variation with offset
AWG—American Wire Gauge

B

b—bit
B—byte
BCE—before Common Era
Bd—baud
BHA—bottomhole assembly
BOD—bandwidth on demand (noun),
bandwidth-on-demand (adj.)

BOP—blowout preventer
bps—bits per second
Bps—bytes per second
BTEX—benzene, toluene, ethylbenzene, and xylene
BVOS[†]—by volume of solids
BWOC[†]—by weight of cement
BWOP[†]—by weight of proppant
BWOW[†]—by weight of water

C

CAD—computer-aided design
capex—capital expenditure
CCL—casing collar locator
CCS—carbon capture and sequestration (also known as “carbon capture and storage,” but SLB collateral should use “sequestration” for the initialism)
CCUS—carbon capture, utilization, and sequestration (also known as “carbon capture, utilization, and storage,” but SLB collateral should use “sequestration” for the initialism)
CD—compact disc
CD-ROM—compact disc read-only memory
CE—Common Era
CE[†]—Conformité Européenne
CEC—Canadian Electrical Code
CL—Class (restricted to ASME Class)
CMYK—cyan, magenta, yellow, and key (black)
CO₂e[†]—CO₂ equivalent
CPU—central processing unit
CRN—Canadian Registration Number
CRT—cathode ray tube
CSA[†]—Canadian Standards Association
CT—coiled tubing
CT—computerized tomography

D

DC—direct current
DC—District of Columbia
DEI—diamond-enhanced insert
DN—*diamètre nominal* or *Durchmesser nach Norm* (nominal pipe sizing; DN 8, DN 40, etc.)

[†] Stand-alone abbreviation only in tables and sidebar bulleted listings; spell out on first mention in body text.

DNV—Det Norske Veritas
DSL—digital subscriber line
DTS—distributed temperature sensing or sensor
DVD—digital video disc

E

E&P—exploration and production
EAC[†]—Eurasian Conformity
ECD—equivalent circulating density
e.g. (means “for example”)
EI[†]—Energy Institute
EM—electromagnetic
EPCI—engineering, procurement, construction, and installation
EPCIC—engineering, procurement, construction, installation, and commissioning
EPF—early production facility
ESD—equivalent static density
ESP—electric submersible pump
et al. (means “and others”)
EU—European Union
EUE—external upset ends

F

FAQ—frequently asked questions
FAZ—full-azimuth
FEED—front-end engineering and design
FH—Full Hole (API screw threads; e.g., 3½ FH)
Fig—configuration (for end connection names, typically for hammer unions)
FKM—fluoroelastomer
FPSO—floating production, storage, and offloading
FSO—floating storage and offloading
FTP—File Transfer Protocol (lowercase “ftp” used for an Internet location)

G

GIF—graphics interchange format (lowercase “gif” used in a filename extension)
GIS—geographic information system
GOR—gas/oil ratio
GPS—global positioning system
GR—gamma ray

GUI—graphical user interface

H

HART—Highway Addressable Remote Transducer
HD—high definition (noun), high-definition (adj.)
HNBR—hydrogenated nitrile butadiene rubber
HOCNF[†]—Harmonised Offshore Chemical Notification Format
HP—high-pressure (adj.)
HPHT—high pressure, high temperature (noun); high-pressure, high-temperature (adj.)
HPS—horizontal pumping system (also “horizontal pump system”)
HSE—health, safety, and environment
HT—high temperature (noun) and high-temperature (adj.)
HTML—Hypertext Markup Language
HTTP—Hypertext Transfer Protocol (lowercase “http” used for an Internet location)
HVAC—heating, ventilation, and air conditioning

I

I/O—input/output
IaaS—infrastructure as a service
ICD—inflow control device
ID—inside diameter
i.e. (means “that is”)
IEC—International Electrotechnical Commission
IEEE[†]—Institute of Electrical and Electronics Engineers, Inc.
IF—Internal Flush (API screw thread; e.g., 3½ IF)
IGBT[†]—insulated gate bipolar transistor
IIoT—Industrial Internet of Things
IOC—international oil company
IoT—Internet of Things
IP—Internet Protocol
ISDN—Integrated Services Digital Network
ISO—International Organization for Standardization (do not use generic terms “ISO certification” and “ISO certified” but specify which standard [e.g., “ISO 9001 certification” and “ISO 14001 certified,” respectively])

[†] Stand-alone abbreviation only in tables and sidebar bulleted listings; spell out on first mention in body text.

IT—information technology

J

JPEG or JPG—Joint Photographic Experts Group (lowercase “jpg” used in a filename extension)

JV—joint venture

K

Kb—kilobit

KB—kilobyte

KPI—key performance indicator

L

LAN—local area network

LAS—log ASCII standard

LCD—liquid crystal display

LED—light-emitting diode

LNG—liquefied natural gas

ln—logarithm natural

log—logarithm

LPG—liquefied petroleum gas

LWD—logging while drilling (noun), logging-while-drilling (adj.)

M

max.[†]—maximum

MB—megabyte

MD—measured depth

MEG—monoethylene glycol

MEMS—microelectromechanical systems

min.[†]—minimum

MLWD—measurement(s) and logging while drilling (noun), measurement(s)- and logging-while-drilling (adj.)

MPD—managed pressure drilling

MPEG or MPG—Moving Picture Experts Group

MPEG-3 or MP3—MPEG-1 Audio Layer 3 (lowercase “mp3” used in a filename extension)

MPEG-4 or MP4 (lowercase “mp4” used in a filename extension)

MSDS—Material Safety Data Sheet

MSS—Manufacturers Standardization Society of the Valve and Fittings Industry

MWD—measurement(s) while drilling (noun), measurement(s)-while-drilling (adj.)

N

NACE (use for standards)

NAZ—narrow azimuth

NBR—nitrile butadiene rubber

NC—Numbered Connector (API screw threads; e.g., NC50)

NEC—National Electrical Code

NEMA[†]—National Electrical Manufacturers Association

NGL—natural gas liquids

NGO—non-governmental organization

NMR—nuclear magnetic resonance

NOC—national oil company

NPST[†]—non-pressure-sealed thread

NPT—nonproductive time

NPV—net present value

NUE—nonupset end

O

O&G—oil and gas

OBM—oil-based mud

OD—outside diameter

OEM—original equipment manufacturer

OLE—object linking and embedding

opex—operational expenditure

OS—operating system

OSHA—Occupational Safety and Health Administration

P

P&A—plug and abandon (also “plugged and abandoned”)

PaaS—platform as a service

PC—personal computer

pCO₂—partial pressure of CO₂

PCP—progressing cavity pump (also “progressive cavity pump”)

PDC—polycrystalline diamond compact (noun, adj.)

PDF—portable document format (lowercase “pdf” used in a filename extension)

PDVSA—Petróleos de Venezuela S.A.

[†] Stand-alone abbreviation only in tables and sidebar bulleted listings; spell out on first mention in body text.

PEEK—polyetheretherketone
PED—Pressure Equipment Directive
p.m.
PMCC[†]—Pensky-Martens closed cup
PNG—Portable Network Graphics (lowercase
“png” used in a filename extension)
PT—pressure and temperature
PTFE—polytetrafluoroethylene
PVC—polyvinyl chloride
PVT—pressure-volume-temperature

Q

Q&A—question and answer
QA—quality assurance
QA/QC—quality assurance and quality control
QC—quality control
QHSE—quality, health, safety, and
environment

R

R&D—research & development
RAM—random-access memory
Reg—Regular (API screw threads; e.g., 3½ Reg)
RGB—red-green-blue
rms—root mean square
ROI—return on investment
ROM—read-only memory
ROP—rate of penetration
ROV—remotely operated vehicle
RP (subsequent mention after API RP)—API
Recommended Practice
RSS—really simple syndication
RSS—rotary steerable system
RSVP (abbreviation for *répondez s’il vous plait*
[please reply])
RTM—reverse time migration
RTN[†]—Rostekhnadzor
RTU—remote terminal unit

S

SaaS—software as a service
SAGD—steam-assisted gravity drainage
SCADA—supervisory control and data
acquisition
SCAL—special core analysis

SCSI—Small Computer System Interface
SDK—software development kit
SDS—Safety Data Sheet
SEG—Society of Exploration Geophysicists
SEM—scanning electron microscope (also
“scanning electron microscopy”)
SGI—Silicon Graphics Inc. (lowercase “sgi”
used in a filename extension)
SIL—safety integrity level
sin—sine
SMS—short message service
SPE—Society of Petroleum Engineers
Spec—API Specification (subsequent mention
after API Spec)
SPWLA—Society of Petrophysicists and Well Log
Analysts
SQL—Structured Query Language

T

TAML—technology advancement for
multilaterals
tan—tangent
TCI—tungsten carbide insert
TCP—Transmission Control Protocol
TCP/IP—Transmission Control
Protocol/Internet Protocol
TEG—triethylene glycol
TD—total depth
TIFF—Tagged Image File Format (lowercase
“tif” used for a filename extension)
TRL—Technical Readiness Level
TTL[†]—transistor-transistor logic
TV—television
TVD—true vertical depth

U

UAE—United Arab Emirates
UK—United Kingdom (noun, adj., addresses)
UL[†]—Underwriters Laboratory
UN—United Nations
URL—Uniform Resource Locator

[†] Stand-alone abbreviation only in tables and sidebar bulleted listings; spell out on first mention in body text.

US—United States (noun, adj.; use “USA” in geographic location)
USA—United States of America (use for physical ; use “US” as noun and adj.)
USB—universal serial bus
UV—ultraviolet

V

VCR—videocassette recorder
VFD—variable frequency drive
VLAN—virtual LAN
VOC—volatile organic compound
VoIP—voice over Internet protocol
VPN—virtual private network
vs. *or* versus
VSAT—very small aperture terminal
VSD—variable speed drive

W

WAN—wide area network
WAZ—wide-azimuth
WBM—water-based mud
WiFi—wireless fidelity
WOB—weight on bit
WOC—waiting on cement, waiting-on-cement (adj.)
WWW—World Wide Web (lowercase “www” used for an Internet location)

X

XHTML—Extensible HTML
XML—Extensible Markup Language

[†] Stand-alone abbreviation only in tables and sidebar bulleted listings; spell out on first mention in body text.

Unit Names

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Unit Name	Unit Symbol	Comment
acceleration due to gravity (g-force)	g_n	Use this SPE- and IEEE-accepted unit symbol; do not use G for acceleration due to gravity $1 g_n = 9.806650E+00 \text{ m/s}^2$
acre	acre (see “m ² or m ² ” or “ha”)	Use of m ² or m ² for square meter or ha for hectare preferred; $1 \text{ acre} = 4.046856E+03 \text{ m}^2 = 4.046856E-01 \text{ ha}$ $1 \text{ acre (US survey)} = 4.046873E+03 \text{ m}^2 = 4.046873E-01 \text{ ha}$
acre-foot	acre.ft (see “m ³ or m ³ ” or “ha.m”)	Use of m ³ or m ³ for cubic meter or ha.m for hectare-meter preferred; $1 \text{ acre.ft} = 1.233489E+03 \text{ m}^3 = 1.233489E-01 \text{ ha.m}$
ampere	A	
ampere per meter	A/m	Do not use Oe for oersted; $1 \text{ Oe} = 7.957747E+01 \text{ A/m}$
ampere-hour	A.h (see “kC”)	Use of kC for kilocoulomb preferred; $1 \text{ A.h} = 3.6E+00 \text{ kC}$
ångstrom (10 ⁻¹⁰ m)	See “nm”	Use nm for nanometer; $1 \text{ Å} = 1.0E-01 \text{ nm}$ Do not use Å for ångstrom
annum (sidereal year)	a	Use only in combined units for measured time values (e.g., m/a); use allowable customary unit yr for year only in combined units (e.g., m ³ /yr) Spell out “year” for values used in a descriptive or literary manner
API (American Petroleum Institute) gamma ray	gAPI	
API gravity	API	Do not use °API, dAPI, or degAPI for API gravity
API neutron	nAPI	
atmosphere (standard)	atm (see “MPa” or “kPa”)	Use of MPa for megapascal or kPa for kilopascal preferred; $1 \text{ atm} = 1.01325E-01 \text{ MPa} = 1.01325E+02 \text{ kPa}$ Standard conditions are 760-mm Hg at 0 degC or 14.696 lbf/in ²
atomic mass unit	See “u”	Use u for unified atomic mass unit, which is equal to 1/12 of the mass of an atom of the nuclide ¹² C $1 u = 1.6605402 \times 10^{-27} \text{ kg}$ with a combined standard uncertainty of $0.0000010 \times 10^{-27} \text{ kg}^\dagger$ Do not use amu for atomic mass unit
bar	bar (see “MPa” or “kPa”)	Use of MPa for megapascal or kPa for kilopascal preferred; $1 \text{ bar} = 1.0E-01 \text{ MPa} = 1.0E+02 \text{ kPa}$
barn	b (see “fm ² or fm ² ”)	The unit of area used to measure the reaction cross section of atomic nuclei and subatomic particles. Not to be confused with “B” for bel, the unit of sound intensity, or for byte, the unit of information in computer storage and processing. Use of fm ² or fm ² for square femtometer preferred; $1 \text{ b} = 1.0E+02 \text{ fm}^2^\dagger$

Unit Name	Unit Symbol	Comment
barrel	bbl (see “m ³ or m3”)	Use of m³ or m3 for cubic meter preferred; 1 bbl = 42 galUS = 1.589873E–01 m³ Do not modify the unit symbol bbl (e.g., reservoir bbl, stock-tank bbl)—specify the conditions at which the value was measured Do not use B for barrel (e.g., BFPD) except for BOE—specify the fluid type and composition and the conversion factor as necessary
barrel at reservoir conditions	See “bbl” or “m ³ or m3”	Do not modify the unit symbol bbl—specify the reservoir conditions at which the value was measured Use of m³ or m3 for cubic meter preferred to bbl for barrel; 1 bbl = 42 galUS = 1.589873E–01 m³ Do not use res bbl or bbl (reservoir) for reservoir barrel
barrel at stock-tank conditions	See “bbl” or “m ³ or m3”	Do not modify the unit symbol bbl—specify the stock-tank conditions at which the value was measured Use of m³ or m3 for cubic meter preferred to bbl for barrel; 1 bbl = 42 galUS = 1.589873E–01 m³ Do not use STB for stock-tank barrel
barrel of fluid per day	See “bbl/d” or “m ³ /d or m3/d”	Use of m³/d or m3/d for cubic meter per day preferred to bbl/d for barrel per day; 1 bbl/d = 42 galUS/d = 1.589873E–01 m³/d Specify the conditions at which the value was measured Specify the fluid type as necessary Do not use BFPD for barrel of fluid per day
barrel of liquid per day	See “bbl/d” or “m ³ /d or m3/d”	Use of m³/d or m3/d for cubic meter per day preferred to bbl/d for barrel per day; 1 bbl/d = 42 galUS/d = 1.589873E–01 m³/d Specify the conditions at which the value was measured Specify the fluid type as necessary Do not use BLPD for barrel of liquid per day
barrel of oil equivalent	BOE (see “bbl” or “m ³ or m3”)	Allowable modification of the unit symbol bbl—can also be expressed as “x bbl of oil equivalent” with specification as necessary the conditions at which the value was measured, the fluid type and composition, and the equivalency conversion factor Use of m³ or m3 for cubic meter preferred to bbl for barrel; 1 bbl = 42 galUS = 1.589873E–01 m³
barrel of oil equivalent per day	See “bbl/d” or “m ³ /d or m3/d”	Do not modify the unit symbol bbl—express as “x bbl/d of oil equivalent” and specify as necessary the conditions at which the value was measured, the fluid type and composition, and the equivalency conversion factor Use of m³/d or m3/d for cubic meter per day preferred to bbl/d for barrel per day; 1 bbl/d = 42 galUS/d = 1.589873E–01 m³/d Do not use BOEPD for barrel of oil equivalent per day

Unit Name	Unit Symbol	Comment
barrel of oil per day	See “bbl/d” or “m ³ /d or m3/d”	Use of m ³ /d or m3/d for cubic meter per day preferred to bbl/d for barrel per day; 1 bbl/d = 42 galUS/d = 1.589873E–01 m ³ /d Specify the conditions at which the value was measured Specify the fluid type as necessary Do not use BOPD for barrel of oil per day
barrel of water per day	See “bbl/d” or “m ³ /d or m3/d”	Use of m ³ /d or m3/d for cubic meter per day preferred to bbl/d for barrel per day; 1 bbl/d = 42 galUS/d = 1.589873E–01 m ³ /d Specify the conditions at which the value was measured Specify the fluid type as necessary Do not use BWPD for barrel of water per day
barrel per day	bbl/d (see “m ³ /d or m3/d”)	Use of m ³ /d or m3/d for cubic meter per day preferred; 1 bbl/d = 42 galUS/d = 1.589873E–01 m ³ /d Specify the conditions at which the value was measured Do not use B/D, BFPD, BLPD, BOEPD, or BOPD for barrel per day—specify the fluid type and composition and the conversion factor as necessary
barrel per day at reservoir conditions	See “bbl/d” or “m ³ /d or m3/d”	Do not modify the unit symbol bbl/d—specify the reservoir conditions at which the value was measured Use of m ³ /d or m3/d for cubic meter per day preferred to bbl/d for barrel per day; 1 bbl/d = 42 galUS/d = 1.589873E–01 m ³ /d Do not use RB/d or bbl/d (reservoir) for reservoir barrel per day
barrel per day at stock-tank conditions	See “bbl/d” or “m ³ /d or m3/d”	Do not modify the unit symbol bbl/d—specify the stock-tank conditions at which the value was measured Use of m ³ /d or m3/d for cubic meter per day preferred to bbl/d for barrel per day; 1 bbl/d = 42 galUS/d = 1.589873E–01 m ³ /d Do not use STB/d for stock-tank barrel per day
barrel per minute	bbl/min (see “m ³ /min or m3/min”)	Use of m ³ /min or m3/min for cubic meter per minute preferred to bbl/min for barrel per minute; 1 bbl/min = 42 galUS/min = 1.589873E–01 m ³ /min Specify the conditions at which the value was measured
becquerel	Bq	Do not use Ci for curie; 1 Ci = 3.7E+10 Bq
bel	B	The unit of sound intensity. Note: the unit symbol “B” is also used for “byte,” the basic unit of information in computer storage and processing. Consider converting bel to decibel (dB): 1 dB = 0.1 B; 1 B = 10 dB) to avoid confusion.

Unit Name	Unit Symbol	Comment
billion (10 ⁹) cubic feet	Bcf	Use for gas preferred to m ³ or m3 for cubic meter or dm ³ or dm3 for cubic decimeter; values are expressed without scientific notation 1 Bcf = 1,000,000,000 ft ³ = 2.831685E–07 m ³ = 2.831685E+10 dm ³
billion (10 ⁹) cubic feet per day	Bcf/d	Use for gas preferred to m ³ /d or m3/d for cubic meter per day or dm ³ /d or dm3/d for cubic decimeter per day; values are expressed without scientific notation 1 Bcf/d = 1,000,000,000 ft ³ /d = 2.831685E–07 m ³ /d = 2.831685E+10 dm ³ /d Bscf/d for billion standard cubic feet per day is used only if the conditions at which the value was measured are specified
billion (10 ⁹) standard cubic feet per day	Bscf/d (use restricted: see “Bcf/d”)	Use Bcf/d for billion cubic feet per day Use Bscf/d only if the conditions at which the value was measured are specified
brake horsepower	bhp (see “kW”)	Brake horsepower is the power available for work. It is determined by multiplying the nominal horsepower by an efficiency factor specific to the equipment Use of kW for kilowatt preferred; 1 bhp = 7.46E–01 kW
British thermal unit (international)	Btu (see “J”)	Use of J for joule preferred; 1 Btu (international) = 1.055056E+03 J
byte	B	The basic unit of information in computer storage and processing. The unit symbol “B” is also used for “bel,” the unit of sound intensity.
candela	cd	
capture unit	cu	Use of 1/m for per meter preferred; 1 cu = 1.0E–01 1/m
centimeter	cm	Use of this unit symbol or m for meter preferred to ft for foot; 1 ft = 3.048E–01 m = 3.048E+01 cm Use of this unit symbol or mm for millimeter preferred to in for inch; 1 in = 2.54E+01 mm = 2.54E+00 cm
centimeter per second	cm/s	Use of this unit symbol or m/s for meter per second preferred to ft/min for foot per minute; 1 ft/min = 5.08E–03 m/s = 5.08E–01 cm/s Use of this unit symbol or mm/s for millimeter per second preferred to in/s for inch per second; 1 in/s = 2.54E+01 mm/s = 2.54E+00 cm/s Use of this unit symbol, mm/s for millimeter per second, m/h for meter per hour, or min/m for minute per meter preferred to ft/h for foot per hour; 1 ft/h = 8.466667E–03 cm/s = 8.466667E–02 mm/s = 3.048E–01 m/h = 1.968504E+02 min/m

Unit Name	Unit Symbol	Comment
centimeter per second squared	cm/s ² or cm/s2	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use preferred to Gal for galileo; 1 Gal = 1.0E+00 cm/s ²
centipoise	cP (see "Pa.s")	Use of Pa.s for pascal-second preferred; 1 cP = 1.0E−03 Pa.s
centistokes	See "mm ² /s or mm2/s"	Use mm ² /s or mm2/s for square millimeter per second; 1 cSt = 1.0E+00 mm ² /s Do not use cSt for centistokes
coulomb	C	
coulomb per kilogram	C/kg	Do not use R for roentgen; 1 R = 2.58E−04 C/kg
counts per second	counts/s	Use counts/s for counts per second Do not use cps for counts per second
cubic centimeter	cm ³ or cm3	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Do not use cc for cubic centimeter Do not use fl ozUK for fluid ounce (UK); 1 fl ozUK = 2.841308E+01 cm ³ Do not use fl ozUS for fluid ounce (US); 1 fl ozUS = 2.957353E+01 cm ³ Use of this unit symbol or mm ³ or mm3 for cubic millimeter preferred to in ³ or in3 for cubic inch; 1 in ³ = 1.638706E+04 mm ³ = 1.638706E+01 cm ³
cubic decimeter	dm ³ or dm3	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Except for gas, use of this unit symbol or m ³ or m3 for cubic meter preferred to ft ³ or ft3 for cubic foot; 1 ft ³ = 2.831685E−02 m ³ = 2.831685E+01 dm ³ For gas, use Bcf for billion cubic feet, Mcf for thousand cubic feet, MMcf for million cubic feet, and Tcf for trillion cubic feet and express the value without scientific notation
cubic decimeter per kilogram	dm ³ /kg or dm3/kg	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use preferred to ft ³ /lbm or ft3/lbm for cubic foot per pound (mass); 1 ft ³ /lbm = 6.242796E+01 dm ³ /kg

Unit Name	Unit Symbol	Comment
cubic decimeter per second	dm ³ /s or dm3/s	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use of this unit symbol or L/s for liter per second preferred to - ft³/min or ft3/min for cubic foot per minute; 1 ft³/min = 4.719474E–01 dm³/s = 4.719474E–01 L/s - ft³/s or ft3/s for cubic foot per second; 1 ft³/s = 2.831685E+01 dm³/s = 2.831685E+01 L/s
cubic foot	ft ³ or ft3 (see “m ³ or m3” or “dm ³ or dm3”)	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Do not modify the unit symbol ft³ or ft3 (e.g., standard ft³)—specify the conditions at which the value was measured Except for gas, use of m³ or m3 for cubic meter or dm³ or dm3 for cubic decimeter preferred; 1 ft³ = 2.831685E–02 m³ = 2.831685E+01 dm³ For gas use “cf” for cubic foot in the unit symbols Bcf, Mcf, MMcf, and Tcf and express the value without scientific notation
cubic foot at standard conditions	See “ft ³ or ft3”	Do not modify the unit symbol ft³ or ft3—instead, specify the conditions at which the value was measured Except for gas, use of m³ or m3 for cubic meter or dm³ or dm3 for cubic decimeter preferred; 1 ft³ = 2.831685E–02 m³ = 2.831685E+01 dm³ Do not use scf for standard cubic foot except in Bscf for billion standard cubic feet, Mscf for thousand standard cubic feet, MMscf for million cubic feet, and Tscf for trillion cubic feet with the conditions at which the value was measured specified and the value expressed without scientific notation
cubic foot per barrel	ft ³ /bbl or ft3/bbl (see “L/m ³ or L/m3” or “m ³ /m ³ or m3/m3”)	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use of L/m³ or L/m3 for liter per cubic meter or m³/m³ or m3/m3 for cubic meter per cubic meter preferred; 1 ft³/bbl = 1.78073E+02 L/m³ = 1.78073E–01 m³/m³

Unit Name	Unit Symbol	Comment
cubic foot per day	ft ³ /d or ft3/d (see “m ³ /h or m3/h” or “m ³ /d or m3/d”)	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Except for gas, use of m³/h or m3/h for cubic meter per hour or m³/d or m3/d for cubic meter per day preferred; 1 ft³/d = 1.179869E–03 m³/h = 2.831685E–02 m³/d For gas, use Bcf/d for billion cubic feet per day, Mcf/d for thousand cubic feet per day, MMcf/d for million cubic feet per day, and Tcf/d for trillion cubic feet per day (do not use cf/d for cubic feet per day without a prefix) and express the value without scientific notation
cubic foot per day at standard conditions	See “ft ³ /d or ft3/d”	Do not modify the unit symbol—specify the standard conditions at which the value was measured Except for gas, use of m³/h or m3/h for cubic meter per hour or m³/d or m3/d for cubic meter per day preferred; 1 ft³/d = 1.179869E–03 m³/h = 2.831685E–02 m³/d For gas, use Bscf/d for billion standard cubic feet per day, Mscf/d for thousand standard cubic feet per day, MMscf for million cubic feet per day, and Tscf for trillion cubic feet per day (do not use scf/d for standard cubic foot per day without a prefix) with the conditions at which the value was measured specified and the value expressed without scientific notation
cubic foot per minute	ft ³ /min or ft3/min (see “dm ³ /s or dm3/s” or “L/s”)	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use of dm³/s or dm3/s for cubic decimeter per second or L/s for liter per second preferred; 1 ft³/min = 4.719474E–01 dm³/s = 4.719474E–01 L/s
cubic foot per pound (mass)	ft ³ /lbm or ft3/lbm (see “dm ³ /kg or dm3/kg”)	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use of dm³/kg or dm3/kg for cubic decimeter per kilogram preferred; 1 ft³/lbm = 6.242796E+01 dm³/kg
cubic foot per second	ft ³ /s or ft3/s (see “dm ³ /s or dm3/s” or “L/s”)	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use of dm³/s or dm3/s for cubic decimeter per second or L/s for liter per second preferred; 1 ft³/s = 2.831685E+01 dm³/s = 2.831685E+01 L/s
cubic inch	in ³ or in3 (see “mm ³ or mm3” or “cm ³ or cm3”)	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use of mm³ or mm3 for cubic millimeter or cm³ or cm3 for cubic centimeter preferred; 1 in³ = 1.638706E+04 mm³ = 1.638706E+01 cm³

Unit Name	Unit Symbol	Comment
cubic kilometer	km ³ or km3	Power can be superscript trademark symbol or in line; one format must be used consistently in a document
cubic meter	m ³ or m3	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use preferred to - bbl for barrel; 1 bbl = 42 galUS = 1.589873E-01 m³ - galUK or galUS; 1 galUK = 4.546092E-03 m³ and 1 galUS = 3.785412E-03 m³ Except for gas, use of this unit symbol or dm³ or dm3 for cubic decimeter preferred to ft³ or ft3 for cubic foot; 1 ft³ = 2.831685E-02 m³ = 2.831685E+01 dm³ For gas, use Bcf for billion cubic feet, Mcf for thousand cubic feet, MMcf for million cubic feet, and Tcf for trillion cubic feet per day, and express the value without scientific notation Use of this unit symbol or ha.m for hectare-meter preferred to acre.ft for acre-foot; 1 acre.ft = 1.233489E+03 m³ = 1.233489E-01 ha.m Use preferred to allowable customary unit yd³ for cubic yard; 1 yd³ = 7.645549E-01 m³ 1 L = 1.0E-03 m³
cubic meter per cubic meter	m ³ /m ³ or m3/m3	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use of this unit symbol or L/m³ or L/m3 for liter per cubic meter preferred to ft³/bbl or ft3/bbl for cubic foot per barrel; 1 ft³/bbl = 1.78073E+02 L/m³ = 1.78073E-01 m³/m³ Use preferred to allowable customary unit vol % for volume percent; 1 vol % = 1.0E-02 m³/m³ or express as a volume fraction of x % Do not use vol/vol for volume per volume
cubic meter per day	m ³ /d or m3/d	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use preferred to galUS/d for gallon per day; 1 galUS/d = 3.785412E-03 m³/d Except for gas, use of this unit or m³/h or m3/h for cubic meter per hour preferred to ft³/d or ft3/d for cubic foot per day; 1 ft³/d = 1.179869E-03 m³/h = 2.831685E-02 m³/d For gas, use Bcf/d for billion cubic feet per day, Mcf/d for thousand cubic feet per day, MMcf/d for million cubic feet per day, and Tcf/d for trillion cubic feet per day and express the value without scientific notation

Unit Name	Unit Symbol	Comment
cubic meter per hour	m ³ /h or m3/h	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use preferred to galUS/min for gallon per minute; 1 galUS/min = 2.271247E-01 m³/h Do not use gpm for gallon per minute Except for gas, use of this unit or m³/d or m3/d for cubic meter per day preferred to ft³/d or ft3/d for cubic foot per day; 1 ft³/d = 1.179869E-03 m³/h = 2.831685E-02 m³/d For gas, use Bcf/d for billion cubic feet of gas per day, Mcf/d for thousand cubic feet per day, MMcf/d for million cubic feet per day, and Tcf/d for trillion cubic feet per day and express the value without scientific notation
cubic millimeter	mm ³ or mm3	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use of this unit symbol or cm³ or cm3 for cubic centimeter preferred to in³ or in3 for cubic inch; 1 in³ = 1.638706E+04 mm³ = 1.638706E+01 cm³
cubic yard	yd ³ (see “m ³ or m3”)	Allowable customary unit or preferably use m ³ or m3 for cubic meter; 1 yd ³ = 7.645549E-01 m ³
curie	See “Bq”	Use Bq for becquerel; 1 Ci = 3.7E+10 Bq Do not use Ci for curie
cycle	c	
cycle per min	c/min	Use preferred to allowable customary unit rpm for revolution per minute; 1.0 rpm = 1.0 c/min
darcy	D	
day	d	Use spelled-out unit name when not used with a measured value or depending on context
decibel	dB	The unit of sound intensity. decibel (dB) preferred to bel (B), which can be confused with “byte,” the basic unit of information in computer storage and processing.
degree (angle)	° or dega	Close up the unit symbol to the value Use of ° preferred for degree (angle); one format must be used consistently in a document Use of °/100 ft, °/30 m, and °/10 m allowed for expressing well curvature with °/m at 0.01 or 1/30 scale preferred; use of rad/m for radian per meter preferred; 1 °/m = 1.745329E-02 rad/m Use API for API gravity Use degC, degF, or degR for temperature values; use degB only as necessary as a historical secondary unit

Unit Name	Unit Symbol	Comment
degree baumé	degB	Use degB only as necessary as a historical secondary unit for fluid density and specify the conversion from conventional relative density (S) as Light liquids: $d = (144.3/S) - 144.3$ Heavy liquids: $d = 144.3 - (144.3/S)$, where d = relative density in degree baumé
degree Celsius	degC	Do not use °C for degree Celsius $(\text{degF} - 32)/1.8 = \text{degC}$ For one-degree increments: $1 \text{ degC} = 1.8 \text{ degF}$
degree Fahrenheit	degF	Do not use °F for degree Fahrenheit $1.8 \text{ degC} + 32 = \text{degF}$ For one-degree increments: $1 \text{ degF} = 0.56 \text{ degC}$
degree Rankine	degR	Do not use °R for degree Rankine
electromotive force	See "V"	Use V for volt (the unit symbol for electromotive force) Do not use emf for electromotive force
electronvolt	eV	
equivalent	eq (use restricted: see "mmol")	Use mmol for millimole; can be used as a secondary unit with the "equivalent" defined $1 \text{ meq} = [1 \text{ mmol}/(\text{valence of the substance})]^\dagger$
equivalent per liter	See "mol/m ³ or mol/m ³ "	Use mol/m ³ or mol/m ³ to express the normality as a whole-number multiple of the molarity dependent on the valence of the solution; $(\text{mol}/\text{m}^3)/\text{valence}^\dagger$ Do not use eqL for equivalent per liter or N for normality
erg	erg	The symbol for erg is the name of the unit Use of J for joule preferred; $1 \text{ erg} = 1\text{E}-07 \text{ J}$
farad	F	
fluid ounce (UK)	See "cm ³ or cm ³ "	Use cm ³ or cm ³ for cubic centimeter $1 \text{ fl ozUK} = 2.841308\text{E}+01 \text{ cm}^3$ Do not use fl ozUK for fluid ounce (UK)
fluid ounce (US)	See "cm ³ or cm ³ "	Use cm ³ or cm ³ for cubic centimeter $1 \text{ fl ozUS} = 2.957353\text{E}+01 \text{ cm}^3$ Do not use fl ozUS for fluid ounce (US)
foot	ft (see "m" or "cm")	Use of m for meter or cm for centimeter preferred; $1 \text{ ft} = 3.048\text{E}-01 \text{ m} = 3.048\text{E}+01 \text{ cm}$ Do not use ' (prime) for foot
foot per hour	ft/h (see "min/m," "m/h," "mm/s," or "cm/s")	Use of min/m for minute per meter, m/h for meter per hour, mm/s for millimeter per second, or cm/s for centimeter per second preferred to ft/h for foot per hour; $1 \text{ ft/h} = 1.968504\text{E}+02 \text{ min/m} = 3.048\text{E}-01 \text{ m/h} = 8.466667\text{E}-02 \text{ mm/s} = 8.466667\text{E}-03 \text{ cm/s}$ Do not use fph or ft/hr for foot per hour
foot per minute	ft/min (see "m/s" or "cm/s")	Use of m/s for meter per second or cm/s for centimeter per second preferred; $1 \text{ ft/min} = 5.08\text{E}-03 \text{ m/s} = 5.08\text{E}-01 \text{ cm/s}$

Unit Name	Unit Symbol	Comment
foot per second	ft/s (see “m/s”)	Use of m/s for meter per second preferred; 1 ft/s = 3.048E–01 m/s
foot-pound (force) (use for work or energy values)	ft.lbf (see “kJ”)	Use of kJ for kilojoule preferred for work or energy unit values; 1 ft.lbf = 1.355818E–03 kJ Do not use for torque unit values, for which use of N.m for newton-meter preferred to lbf.ft for pound (force)-foot; 1 lbf.ft = 1.355818E+00 N.m
galileo	Gal (see “cm/s ² or cm/s2”)	Use of cm/s ² or cm/s2 for centimeter per second squared preferred; 1 Gal = 1.0E+00 cm/s ²
gallon (UK)	galUK (see “m ³ or m3”)	Use of m ³ or m3 for cubic meter preferred 1 galUK = 4.546092E–03 m ³
gallon (US)	galUS (see “m ³ or m3”)	Use of m ³ or m3 for cubic meter preferred 1 galUS = 3.785412E–03 m ³
gallon (US) per sack	galUS/sack (use restricted, see “L/m ³ or L/m3” or “L/t”)	Use of L/m ³ or L/m3 for liter per cubic meter preferred to galUS/sack for gallon per sack to express the volume of fluid per volume of solid (concentration). Use of L/metric ton for liter per metric ton preferred to galUS/sack for gallon (US) per sack to express the volume of fluid per mass of solid (specific volume); for portland cement: 1 galUS/sack = 88.78 L/t To use sack as a unit, specify the material and its volume in m ³ or m3 for cubic meter (preferable to characterizing it by its mass) Do not use sk for sack
gallon per day	galUS/d (see “m ³ /d or m3/d”)	Use of m ³ /d or m3/d for cubic meter per day preferred 1 galUS/d = 3.785412E–03 m ³ /d
gallon per minute	galUS/min (see “m ³ /h or m3/h”)	Use of m ³ /h or m3/h for cubic meter per hour preferred; 1 galUS/min = 2.271247E–01 m ³ /h Do not use gpm for gallon per minute
gauss	See “T”	Use T for tesla; 1 G = 1 Gs = 1.0E–04 T Do not use G or Gs for gauss
g-force	g _n	Use this SPE- and IEEE-accepted unit symbol; do not use G for g-force. 1 g _n = 9.806650E+00 m/s ²
gigabyte	GB	One gigabyte equals one billion bytes.
gigahertz	GHz	
gigapascal	GPa	
gigawatt	GW	Use for geothermal power measurement. Preferred for large scale output.

Unit Name	Unit Symbol	Comment
gigawatt (thermal)	GWth (see "MWth")	A unit of power measurement, representing one billion watts of heat energy delivered. Use preferred to measure the thermal energy produced by a power plant. Often used in the context of nuclear reactors where thermal energy is converted to electrical energy. Example: The reactor produces 3.0 GWth of thermal energy.
gigawatt-hour	GWh (see "MWh")	A unit of energy measurement, representing one billion watt-hours. Use preferred to quantify the total energy delivered by large power generation projects over extended periods.
gram	G	Use preferred to ozm for ounce (mass); 1 ozm = 2.834952E+01 g
gram per cubic centimeter	g/cm ³ or g/cm3	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Do not use g/cc for gram per cubic centimeter
gram per cubic meter	g/m ³ or g/m3	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use of this unit symbol or kg/m³ or kg/m3 for kilogram per cubic meter preferred to lbm/ft³ or lbm/ft3 for pound (mass) per cubic foot; 1 lbm/ft³ = 1.601846E+01 kg/m³ = 1.601846E+04 g/m³ Use of this unit symbol preferred to lbm/1,000 galUS or lbm/1,000 galUK for pound (mass) per thousand gallons; 1 lbm/1,000 galUS = 1.198264E+02 g/m³ or 1 lbm/1,000 gal UK = 9.977633E+01 g/m³ Use of this unit symbol preferred to PTB for pound per thousand barrels (salt content); 1 PTB = 2.853010E+00 g/m³ = 2.853010E+00 mg/dm³ Do not use ppt for pound per thousand (gallons)
gram per kilogram	g/kg	Use preferred to allowable customary unit wt % for weight percent or use kg/kg for kilogram/kilogram or express as a mass fraction of x % 1 wt % = 1.0E–02E kg/kg = 1.0E+01 g/kg
gray	Gy	Do not use rad for radiation absorbed dose to avoid confusion with SI unit symbol rad for radian 1 rad = 1.0E–02 Gy
hectare	ha (see "m ² or m2")	Use of m ² or m2 preferred; 1 ha = 1.0E+04 m ²
hectare-meter	ha.m (See "m ³ or m3")	Use of this unit symbol or m ³ or m3 for cubic meter preferred to acre.ft for acre-foot; 1 acre.ft = 1.233489E+03 m ³ = 1.233489E–01 ha.m
henry	H	
hertz	Hz	

Unit Name	Unit Symbol	Comment
horsepower (550 ft.lbf/s) (also called “mechanical” or “standard” horsepower)	hp (see “W” or “kW”)	Use of W for watt or kW for kilowatt preferred; 1 hp = 7.456999E+02 W = 7.456999E–01 kW
horsepower (electric)	ehp (see “kW”)	Use of kW for kilowatt preferred; 1 ehp = 7.46E–01 kW
horsepower-hour	hp.h (see “MJ”)	Use of MJ for megajoule preferred; 1 hp.h = 2.684520E+00 MJ
hour	h	Use spelled-out unit name when not used with a measured value or depending on context Do not use hr for hour
hydraulic horsepower	hhp (see “kW”)	Use of kW for kilowatt preferred; 1 hhp = 7.46043E–01 kW
inch	in (see “mm” or “cm”)	The unit symbol in is not followed by a period unless it is at the end of a sentence Use of mm for millimeter or cm for centimeter preferred; 1 in = 2.54E+01 mm = 2.54E+00 cm Do not use “ (double prime) for inch
inch per second	in/s	Use of mm/s for millimeter per second or cm/s for centimeter per second preferred; 1 in/s = 2.54E+01 mm/s = 2.54E+00 cm/s
joule	J	Do not use kgf.m for kilogram (force)–meter; 1 kgf.m = 9.806650E+00 J Use preferred to Btu for British thermal unit (international); 1 Btu (international) = 1.055056E+03 J Use preferred to erg; 1 erg = 1E–07 J
kelvin	K	
kilocoulomb	kC	Use preferred to A.h for ampere-hour; 1 A.h = 3.6E+00 kC
kilogram	Kg	Use preferred to lbm for pound (mass); 1 lbm = 4.535924E–01 kg Exemption: Only M-I SWACO collateral uses lb for pound (mass) and in combined unit symbols Use preferred to sack, for which the material must be specified and its volume in m ³ or m3 for cubic meter (preferable to characterizing it by its mass); for portland cement, 1 sack = ~42.64 kg Do not use sk for sack
kilogram (force)	See “N”	Use N for Newton; 1 kgf = 9.80665E+00 N Do not modify the unit symbol kg Do not use kgf for kilogram (force) Do not use kp for kilopond
kilogram (force)–meter	See “J”	Use J for joule; 1 kgf.m = 9.806650E+00 J Do not use kgf.m for kilogram (force)–meter

Unit Name	Unit Symbol	Comment
kilogram per cubic meter	kg/m ³ or kg/m3	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use of this unit symbol or g/m³ or g/m3 for gram per cubic meter preferred to lbm/ft³ or lbm/ft3 for pound (mass) per cubic foot; 1 lbm/ft³ = 1.601846E+01 kg/m³ = 1.601846E+04 g/m³ Use preferred to lbm/galUS for pound (mass) per gallon (US); 1 lbm/galUS = 1.198264E+02 kg/m³
kilogram per kilogram	kg/kg	Use preferred to allowable customary unit wt % for weight percent or use g/kg for gram/kilogram or express as a mass fraction of x %; 1 wt % = 1.0E–02E kg/kg = 1.0E+01 g/kg
kilohertz	kHz	
kilojoule	kJ	Use preferred to kW.h for kilowatt-hour; 1 kW.h = 3.6E+03 kJ Use preferred to ft.lbf for foot-pound (force) for energy or work values; 1 ft.lbf = 1.355818E–03 kJ
kilometer	Km	
kilometer per hour	km/h	Use preferred to allowable customary unit knot for knot to express nautical speed values; 1 knot = 1.852E+00 km/h Use preferred to mi/h for mile per hour to express non-nautical speed values; 1 mi/h = 1.609344E+00 km/h
kilopascal	kPa	Use of this unit symbol or MPa for megapascal preferred to - atm for atmosphere (standard); 1 atm = 1.01325E–01 MPa = 1.01325E+02 kPa at standard conditions of 760-mm Hg at 0 degC or 14.696 lbf/in² - bar for bar; 1 bar = 1.0E–01 MPa = 1.0E+02 kPa - psi for pound (force) per square inch; 1 psi = 6.894757E+00 kPa = 6.894757E–03 MPa Do not use kpsi for 1,000 pounds (force) per square inch or ksi for kip (1,000 lbf) per square inch; 1 ksi = 1 kpsi = 6.894757E+03 kPa
kilopond	See “N”	Use N for Newton; 1 kp = 9.80665E+00 N Do not use kp for kilopond Do not use kgf for kilogram (force)
kilopound (force) (= 1,000 lbf)	See “lbf”	Use of N for Newton preferred; 1 klbf = 4.448222E+03 N Use lbf instead of klbf (SI prefixes cannot be used with non-SI unit symbols); 1E+03 lbf = 1 klbf Do not use klbf for kilopound (force) Do not use kip for 1,000 lbf
kilovolt	kV	

Unit Name	Unit Symbol	Comment
kilowatt	kW	A unit of power that quantifies the rate at which electrical energy is delivered on a smaller scale, such as residential use. It represents one thousand watts of power. Use preferred to ehp for horsepower (electric); 1 ehp = 7.46E-01 kW Use preferred to hhp for hydraulic horsepower; 1 hhp = 7.46043E-01 kW Use preferred to hp for horsepower (550 ft.lbf/s) (also called “mechanical” or “standard” horsepower); 1 hp = 7.456999E-01 kW
kilowatt-hour	kW.h (see “kJ”)	Use of kJ for kilojoule preferred; 1 kW.h = 3.6E+03 kJ
kip (= 1,000 lbf)	See “N”	Use N for newton; 1 kip = 4.448222E+03 N Do not use kip for 1,000 lbf Do not use klbf for 1,000 lbf
kip per square inch (1 kip = 1,000 lbf)	See “psi” or “kPa”	Use of kPa for kilopascal preferred to psi for pound (force) per square inch; 1 ksi = 6.895E+03 kPa Do not use ksi for kip per square inch
knot	knot (see “km/h”)	Allowable customary unit for nautical speed values or preferably use km/h for kilometer per hour; 1 knot = 1.852E+00 km/h
liter	L	Use restricted to the measurement of liquids and gases Do not use lowercase or script letter l for the unit symbol Do not spell the unit name “litre” (SLB style uses American spelling: <i>Merriam Webster’s Collegiate Dictionary</i>, 11th ed.) 1 L = 1.0E-03 m³ or m³ for cubic meter = 1.0E+00 dm³ or dm³ for cubic decimeter; 1 kL = 1.0E+00 m³ Do not use pt for pint; 1 pt = 4.731765E-01 L Do not use qt for quart; 1 qt = 9.463529E-01 L Use preferred to sack, for which the material must be specified and its volume in m³ or m³ for cubic meter (preferable to characterizing it by its mass); for portland cement, 1 sk = ~1 ft³ = ~28.3 L Do not use sk for sack
liter per cubic meter	L/m ³ or L/m ³	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use of this unit symbol or m³/m³ or m³/m³ for cubic meter per cubic meter preferred to ft³/bbl or ft³/bbl for cubic foot per barrel; 1 ft³/bbl = 1.78073E+02 L/m³ = 1.78073E-01 m³/m³
liter per metric ton	L/metric ton	Use of L/metric ton for liter per metric ton preferred to galUS/sack for gallon (US) per sack to express the volume of fluid per mass of solid (specific volume); for portland cement: 1 galUS/sack = ~88.78 L/metric ton Do not use sk for sack

Unit Name	Unit Symbol	Comment
liter per minute	L/min	Do not use Lpm
liter per second	L/s	Use of this unit symbol or dm ³ /s or dm ³ /s for cubic decimeter per second preferred to - ft³/min or ft³/min for cubic foot per minute; 1 ft³/min = 4.719474E–01 dm³/s = 4.719474E–01 L/s - ft³/s or ft³/s for cubic foot per second; 1 ft³/s = 2.831685E+01 dm³/s = 2.831685E+01 L/s
lumen	Lm	
lux	Lx	
megaelectronvolt	MeV	
megagram	Mg (see “metric ton”)	Can be confused with the chemical symbol for magnesium. Convert to lbm, kg, metric ton, or US ton. 1 Mg = 1,000 kg = 1 metric ton 1 US ton = 0.902 Mg = 902 kg
megahertz	MHz	
megajoule	MJ	Use preferred to hp.h for horsepower-hour; 1 hp.h = 2.684520E+00 MJ
megampere	MA	
megapascal	MPa	Use of this unit symbol or kPa for kilopascal preferred to - atm for atmosphere (standard); 1 atm = 1.01325E–01 MPa = 1.01325E+02 kPa at standard conditions of 760-mm Hg at 0 degC or 14.696 lbf/in² - bar for bar; 1 bar = 1.0E–01 MPa = 1.0E+02 kPa - psi for pound (force) per square inch; 1 psi = 6.894757E–03 MPa = 6.89477E+00 kPa Do not modify the unit symbol (e.g., MPa (absolute))—specify the conditions at which the value was measured or use an expression such as “gauge pressure = 15 MPa”
megapascal (absolute)	See “MPa”	Do not modify the unit symbol MPa by using MPa (absolute) for megapascal (absolute)—specify the conditions at which the value was measured
megapascal (gauge)	See “MPa”	Do not modify the unit symbol MPa by using MPa (gauge) for megapascal (gauge)—specify the conditions at which the value was measured
megarayl	Mrayl	Use for acoustic impedance unit
megawatt	MW	A unit of power equivalent to one million watts. It is commonly used to express the output power of engines, the power consumption of large-scale electrical devices, and the capacity of power plants and other electrical infrastructure.

Unit Name	Unit Symbol	Comment
megawatt (thermal)	MWth (see “GWth”)	A unit of power that quantifies the rate at which heat energy is delivered. It is used to measure the heat output of geothermal and other thermal power plants.
megawatt-hours	MWh (see “GWh”)	A unit of energy representing the total amount of electrical energy delivered over one hour. It is used to quantify energy delivery over specified periods such as daily, monthly, or yearly. For geothermal power, plant output is measured in megawatts (MW) or gigawatts (GW), depending on the scale. MW and GW are preferred for larger geothermal projects, while kW is typically used for smaller-scale applications.
meter	m	Do not spell the unit name “metre” (SLB style uses American spelling: <i>Merriam Webster’s Collegiate Dictionary</i> , 11th ed.) Use of this unit symbol or cm for centimeter preferred to ft for foot; 1 ft = 3.048E–01 m = 3.048E+01 cm Use preferred to allowable customary unit yd for yard; 1 yd = 3 ft = 9.144E–01 m
meter per hour	m/h	Use of this unit or min/m for minute per meter, mm/s for millimeter per second, or cm/s for centimeter per second preferred to ft/h for foot per hour; 1 ft/h = 3.048E–01 m/h = 1.968504E+02 min/m = 8.466667E–02 mm/s = 8.466667E–03 cm/s
meter per second	m/s	Use of this unit symbol or cm/s for centimeter per second preferred to ft/min for foot per minute; 1 ft/min = 5.08E–03 m/s = 5.08E–01 cm/s Use preferred to ft/s for foot per second; 1 ft/s = 3.048E–01 m/s
metric ton; long-ton; UK ton; tonUK (1,000 kg; 2,204 lbm)	metric ton (see “ton” for use in space-constrained copy)	Do not use tonne, tonUK, dead-weight ton, or long-ton for unit name. Transition Technologies and EPRs prefer “metric ton,” which follows the U.N.’s Sustainable Development Goal terminology. 1 metric ton = 1,000 kg = 2,204 lbm
metric ton per hour	t/h	For geothermal applications only. Customary for Europe and Asia. Use galUS/min for US audience.
mho	See “S”	Use S for siemens for mho; 1 mho = 1.0E+00 S Do not use mho or Ω (inverted Greek letter Omega) for mho
mho per meter	See “S/m”	Use S/m for siemens per meter; 1.0 mho/m = 1.0E+00 S/m Do not use Ω (inverted Greek letter Omega) for mho
microfarad	uF	Do not use μF for microfarad
microgram	ug	Do not use μg for microgram
microliter	uL	Do not use μL for microliter
micrometer	um	Do not use μ or μm for micrometer Do not use micron for the unit name

Unit Name	Unit Symbol	Comment
micron	See “um”	Do not use μ or u prefix alone; use um for micrometer
microsecond	us	Do not use μ s for microsecond
microstrain	ue	Do not use μ e or μ strain for microstrain
microvolt	uV	Do not use μ V for microvolt
mil (0.001 in) per year	See “mm/a”	Use mm/a for millimeter per year; 1 mil per year = $2.54\text{E}-02$ mm/a Do not use mil/yr or mpy for mil per year
mile	mi	
mile per hour	mi/h (use restricted: see “km/h”)	Use of km/h for kilometer per hour preferred for non-nautical speed values; 1 mi/h = $1.609344\text{E}+00$ km/h Do not use mph for mile per hour
milliampere	mA	
millicurie	mCi	
millidarcy	mD	
milliequivalent	meq (use restricted: see “mmol”)	Use mmol for millimole; eq for equivalent or meq for milliequivalent can be used as a secondary unit with the “equivalent” defined $1 \text{ meq} = [1 \text{ mmol}/(\text{valence of the substance})]^{\dagger}$
milligal	mGal (see “cm/s ² or cm/s ² ”)	Use of cm/s ² or cm/s ² for centimeter per second squared preferred; 1 mGal = $1.0\text{E}-03$ cm/s ²
milligram	mg	
milligram per cubic decimeter	mg/dm ³ or mg/dm3	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use of this unit symbol preferred to PTB for pound per thousand barrels (salt content); 1 PTB = $2.853010\text{E}+00$ g/m ³ = $2.853010\text{E}+00$ mg/dm ³
milliliter	mL	
millimeter	mm	Use of this unit symbol or cm for centimeter preferred to in for inch; 1 in = $2.54\text{E}+01$ mm = $2.54\text{E}+00$ cm
millimeter per second	mm/s	Use of this unit symbol or cm/s for centimeter per second preferred to in/s for inch per second; 1 in/s = $2.54\text{E}+01$ mm/s = $2.54\text{E}+00$ cm/s Use of this unit symbol, cm/s for centimeter per second, m/h for meter per hour, or min/m for minute per meter preferred to ft/h for foot per hour; 1 ft/h = $8.466667\text{E}-02$ mm/s = $8.466667\text{E}-03$ cm/s = $3.048\text{E}-01$ m/h = $1.968504\text{E}+02$ min/m
millimeter per year	mm/a	Do not use mil/yr or mpy for mil (0.001 in) per year; 1 mil per year = 1 mpy = $2.54\text{E}-02$ mm/a
millimho	See “mS”	Use mS for millisiemens; 1 mmho = $1.0\text{E}+00$ mS Do not use mmho or m Ω (inverted Greek letter Omega) for millimho
millimole	mmol	Use preferred to eq for equivalent or meq for milliequivalent, which can be used as necessary as a secondary unit with the “equivalent” defined; 1 meq = $[1 \text{ mmol}/(\text{valence of the substance})]^{\dagger}$

Unit Name	Unit Symbol	Comment
million (10 ⁶) cubic feet	MMcf	Use for gas preferred to m ³ or m3 for cubic meter or dm ³ or dm3 for cubic decimeter; values are expressed without scientific notation 1 MMcf = 1,000,000 ft ³ = 2.831685E+04 m ³ = 2.831685E+07 dm ³
million (10 ⁶) cubic feet per day	MMcf/d	Use for gas preferred to m ³ /h or m3/h for cubic meter per hour or m ³ /d or m3/d for cubic meter per day; values are expressed without scientific notation 1 MMcf/d = 1,000,000 ft ³ /d = 1.179869E+03 m ³ /h = 2.831685E+04 m ³ /d MMscf/d for million standard cubic feet per day used only if the conditions at which the value was measured are specified
million (10 ⁶) standard cubic feet per day	MMscf/d (use restricted: see “MMcf/d”)	Use MMcf/d for million cubic feet per day; express the value without scientific notation Use MMscf/d only if the conditions at which the value was measured are specified
million years	Ma	Do not use MY for million years
millipascal	mPa	
millisecond	ms	Do not use msec for millisecond
millisiemens	mS	Do not use mmho for millimho; 1 mmho = 1.0E+00 mS Do not use mΩ for millimho
millitesla	mT	
millivolt	mV	
minute	min	Use spelled-out unit name when not used with a measured value or depending on context Do not use m or ms for minute
minute (angle)	mina	Do not use ‘ (prime) for minute (angle)
minute (prime)	See “mina” or “ft”	Use mina for minute (angle) or use ft for foot Do not use ‘ (prime) for minute (angle) or for foot
minute per meter	min/m	Use of this unit or m/h for meter per hour, mm/s for millimeter per second, or cm/s for centimeter per second preferred to ft/h for foot per hour; 1 ft/h = 1.968504E+02 min/m = 3.048E–01 m/h = 8.466667E–02 mm/s = 8.466667E–03 cm/s
molarity	See “mol/m ³ or mol/m3”	Use mol/m ³ or mol/m3 for mole per cubic meter to express as the amount-of-substance concentration (volume based); 1 mol/L = 1.0E+03 mol/m ³ [†] Do not use M for molarity or mol/L for mole per liter
mole	mol	

Unit Name	Unit Symbol	Comment
mole per cubic meter	mol/m ³ or mol/m3	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Do not use M for molarity or mol/L for mole per liter—express as the amount-of-substance concentration (volume based); 1 mol/L = 1.0E+03 mol/m ³ [†] Do not use N for normality or eq/L for equivalent per liter—express the normality as a whole-number multiple of the molarity dependent on the solution valence; (mol/m ³)/valence [†]
mole per liter	See “mol/m ³ or mol/m3”	Use mol/m ³ or mol/m3 for mole per cubic meter to express as the amount-of-substance concentration (volume based); 1 mol/L = 1.0E+03 mol/m ³ Do not use mol/L for mole per liter or M for molarity
month	month	Always spelled out Do not use m or mon for month
nanohm	nohm	Do not use nanoΩ
nanometer	nm	Do not use Å for angstrom; 1 Å = 1.0E–01 nm
nanosecond	ns	
nanotesla	nT	
newton	N	Use preferred to lbf for pound (force); 1 lbf = 4.448222E+00 N Do not use kgf for kilogram (force) or kp for kilopond; 1 kgf = 1 kp = 9.80665E+00 N Do not use kip or klbf (= 1,000 lbf); 1 kip = 1 klbf = 4.448222E+03 N Use preferred to ozf for ounce (force); 1 ozf = 2.780139E–01 N
newton-meter	N.m	Use preferred to lbf.ft for pound (force)-foot for torque values; 1 lbf.ft = 1.355818E+00 N.m Do not use for energy or work unit values, for which use of kJ for kilojoule preferred to ft.lbf for foot-pound (force); 1 ft.lbf = 1.355818E–03 kJ
normality	See “mol/m ³ or mol/m3”	Use mol/m ³ or mol/m3 to express the normality as a whole-number multiple of the molarity dependent on the solution valence; (mol/m ³)/valence [†] Do not use N for normality or eq/L for equivalent per liter
oersted	See “A/m”	Use A/m for ampere per meter; 1 Oe = 7.957747E+01 A/m Do not use Oe for oersted
ohm	ohm	Do not use Ω (Greek letter Omega) for ohm
ohm-centimeter	ohm.cm	Do not use Ω.cm
ohm-meter	ohm.m	Do not use Ω.m
ounce (force)	ozf (see “N”)	Use of N for newton preferred; 1 ozf = 2.780139E–01 N

Unit Name	Unit Symbol	Comment
ounce (mass)	ozm (see “g”)	Use of g for gram preferred; 1 ozm = 2.834952E+01 g
part per billion	ppb	Allowable customary unit or use scientific notation ($\times 10^{-9}$) or a ratio of unit symbols (e.g., ug/kg)
part per million	ppm	Allowable customary unit or use scientific notation ($\times 10^{-6}$) or a ratio of unit symbols (e.g., ug/g) 1 ppm = 1 ug/g
part per million by volume	ppmv	Allowable customary unit
part per thousand	ppk or ppt	Allowable customary unit or use scientific notation ($\times 10^{-3}$) or a ratio of unit symbols (e.g., g/kg) If ppk or ppt is used for part per thousand, consistently use one
pascal	Pa	Use preferred to lbf/100 ft ² for pound (force) per 100 square feet; 1 lbf/100 ft ² = 4.788026E-01 Pa
pascal-second	Pa.s	Use preferred to P for poise and cP for centipoise; 1 cP = 1.0E-03 Pa.s and 1 P = 1.0E-01 Pa.s
percent	%	Spacing apart the value and unit symbol is preferred for consistency and integrity of data transfer. Marketing material can use the symbol closed up; one format must be used consistently in a document Use preferred to allowable customary unit pu for porosity unit and should be specified this is not the “true” porosity (e.g., use terms such as uncorrected porosity, limestone porosity, or sandstone matrix porosity) Use instead of su for saturation unit to express the percentage of fluid in the pore space; express as “a fluid fraction of x %” Preferred to allowable customary unit vol % for volume percent and expressed as “a volume fraction of x %” or use the ratio m ³ /m ³ or m3/m3; 1 vol % = 1.0E-02 m ³ /m ³ Do not use vol/vol for volume per volume Preferred to allowable customary unit wt % for weight percent and expressed as “a mass fraction of x %” or use a ratio of mass unit symbols: kg/kg for kilogram per kilogram or g/kg for gram/kilogram; 1 wt % = 1.0E-02E kg/kg = 1.0E+01 g/kg
percent per annum	%/a	
picofarad	pF	
pint	See “L”	Use L for liter; 1 pt = 4.731765E-01 L Do not use pt for pint
poise	P (see “Pa.s”)	Use of Pa.s for pascal-second preferred; 1 P = 1.0E-01 Pa.s
porosity unit	pu (see “%”)	Allowable customary unit or use % for percent and should be specified that this is not the “true” porosity (e.g., use terms such as uncorrected porosity, limestone porosity, or sandstone matrix porosity)

Unit Name	Unit Symbol	Comment
pound (force)	lbf (see “N”)	Use of N for Newton preferred; 1 lbf = 4.448222E+00 N Do not use klbf for kilopound (force); 1 klbf = 1.0E+03 lbf
pound (force) per 100 square feet (unit for yield point)	See “Pa”	Use Pa for pascal; 1 lbf/100 ft ² = 4.788026E–01 Pa Do not use lbf/100 ft ² or lbf/100 ft2 for pound (force) per 100 square feet
pound (force) per square inch	psi (see “kPa” or “MPa”)	Use of kPa for kilopascal or MPa for megapascal preferred; 1 psi = 6.894757E–03 MPa = 6.894757E+00 kPa Do not modify the psi unit symbol (e.g., psi (absolute), psig)—specify the conditions at which the value was measured or use an expression such as “gauge pressure = 230 kPa” Do not use kpsi for 1,000 pounds (force) per square inch or ksi for kip (1,000 lbf) per square inch; 1 ksi = 1 kpsi = 6.894757E+03 kPa
pound (force) per square inch (absolute)	See “psi”	Do not modify the unit symbol psi by using psia or psi (absolute) for pound (force) per square inch (absolute)—specify the conditions at which the value was measured Use of kPa for kilopascal or MPa for megapascal preferred to psi; 1 psi = 6.894757E–03 MPa = 6.894757E+00 kPa
pound (force) per square inch (gauge)	See “psi”	Do not modify the unit symbol psi by using psig or psi (gauge) for pound (force) per square inch (gauge)—specify the conditions at which the value was measured—specify the conditions at which the value was measured or use an expression such as “gauge pressure = 230 kPa” Use of kPa for kilopascal or MPa for megapascal preferred to psi; 1 psi = 6.894757E–03 MPa = 6.894757E+00 kPa
pound (force)-foot (unit for torque)	lbf.ft (see “N.m”)	Use of N.m for newton-meter preferred; 1 lbf.ft = 1.355818E+00 N.m Do not use for energy or work unit values, for which use of kJ for kilojoule preferred to ft.lbf for foot-pound (force); 1 ft.lbf = 1.355818E–03 kJ
pound (mass)	lbm (see “kg”)	Use of kg for kilogram preferred; 1 lbm = 4.535924E–01 kg Use of lbm preferred to allowable customary unit symbol ppa for pound of proppant added per barrel of clean fluid with expression as “x lbm of proppant added per barrel of clean fluid” Do not use # for pound (mass) Exemption: Only M-I SWACO collateral uses lb for pound (mass) and in combined unit symbols

Unit Name	Unit Symbol	Comment
pound (mass) per cubic foot	lbm/ft ³ or lbm/ft3 (see “kg/m ³ or kg/m3” or “g/m ³ or g/m3”)	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use of kg/m ³ or kg/m3 for kilogram per cubic meter or g/m ³ or g/m3 for gram per cubic meter preferred; 1 lbm/ft ³ = 1.601846E+01 kg/m ³ = 1.601846E+04 g/m ³
pound (mass) per gallon	lbm/galUS (see “kg/m ³ or kg/m3”)	Use of kg/m ³ or kg/m3 for kilogram per cubic meter preferred; 1 lbm/galUS = 1.198264E+02 kg/m ³
pound (mass) per thousand gallons	lbm/1,000 galUK or lbm/1,000 galUS (see “g/m ³ or g/m3”)	Use of g/m ³ or g/m3 for gram per cubic meter preferred; 1 lbm/1,000 galUK = 0.977633E+01 g/m ³ and 1 lbm/1,000 galUS = 1.198264E+02 g/m ³ Do not use ppt for pound per thousand (gallons)
pound of proppant added per barrel of clean fluid	ppa (see “lbm”)	Allowable customary unit or use simple unit symbol lbm for pound (mass)—express as “x lbm of proppant added per barrel of clean fluid” Use of kg for kilogram preferred; 1 lbm = 4.535924E–01 kg
pound per thousand (gallons)	See “lbm/1,000 galUS” or “lbm/1,000 galUK”	Use lbm/1,000 galUS pound (mass) per 1,000 galUS or lbm/1,000 gal UK for pound (mass) per 1,000 galUK Use of g/m ³ or g/m3 for gram per cubic meter preferred; 1 lbm/1,000 galUS = 1.198264E+02 g/m ³ or 1 lbm/1,000 gal UK = 9.977633E+01 g/m ³ Do not use ppt for pound per thousand (gallons)
pound per thousand barrels (salt content)	See “g/m ³ or g/m3” or “mg/dm ³ or mg/dm3”)	Use of g/m ³ or g/m3 for gram per cubic meter or mg/dm ³ or mg/dm3 for milligram per cubic decimeter preferred; 1 PTB = 2.853010E+00 g/m ³ = 2.853010E+00 mg/dm ³
quart	See “L”	Use L for liter; 1 qt = 9.463529E–01 L Do not use qt for quart
radian	rad	
radian per meter	rad/m	Use preferred to °/m or dega/m at 0.01 or 1/30 scale for degree (angle) per meter; 1 °/m = 1.745329E–02 rad/m
radiation absorbed dose	See “Gy”	Use Gy for gray; 1 rad = 1.0E–02 Gy Do not use rad for radiation absorbed dose
rem	rem	Use Sv for sievert; 1 rem = 1.0E–02 Sv†
reservoir barrel	See “bbl” or “m ³ or m3”	Do not modify the unit symbol bbl—specify the reservoir conditions at which the value was measured Use of m ³ or m3 for cubic meter preferred to bbl for barrel; 1 bbl = 42 galUS = 1.589873E–01 m ³ Do not use bbl (reservoir) or res bbl for reservoir barrel

Unit Name	Unit Symbol	Comment
reservoir barrel per day	See “bbl/d” or “m ³ /d or m3/d”	Do not modify the unit symbol bbl/d—specify the reservoir conditions at which the value was measured Use of m ³ /d or m3/d for cubic meter per day preferred to bbl/d for barrel per day; 1 bbl/d = 42 galUS/d = 1.589873E–01 m ³ /d Do not use bbl/d (reservoir) or RB/d for reservoir barrel per day
revolution per minute	rpm (see “c/min”)	Allowable customary unit or preferably use c/min for cycle per minute; 1.0 rpm = 1.0 c/min
roentgen	See “C/kg”	Use C/kg for coulomb per kilogram; 1 R = 2.58E–04 C/kg Do not use R for roentgen
sack	sack (use restricted: see “kg” or “L”)	Use of kg for kilogram and L for liter preferred to sack; for portland cement, 1 sack = ~42.64 kg and 1 sack = ~1 ft ³ = ~28.3 L The absolute volume of a 94-lbm sack of portland cement is 3.59 galUS. To use sack as a unit, specify the material and its volume in m ³ or m3 for cubic meter (preferable to characterizing it by its mass). Use of L/metric ton for liter per metric ton is preferred to galUS/sack for gallon per sack to express the volume of fluid per mass of solid (specific volume). Use of L/m ³ or L/m3 for liter per cubic meter is preferred to galUS/sack for gallon per sack to express the volume of fluid per volume of solid (concentration). Do not use sk for sack
saturation unit	See “%”	Use % for percent to express the percentage of fluid in the pore space (e.g., “a fluid fraction of x %”) Do not use su for saturation unit
second	s	Use spelled-out unit name when not used with a measured value or depending on context Do not use sec or ” (double prime) for second
second (angle)	seca	Do not use ” (double prime) for second (angle)
siemens	S	Do not use mho for mho; 1 mho = 1.0E+00 S Do not use Ū (inverted Greek letter Omega) for mho
siemens per meter	S/m	Do not use mho/m for mho per meter; 1 mho/m = 1.0E+00 S/m Do not use Ū/m for mho/m
sievert	Sv	Do not use rem for rem; 1 rem = 1.0E–02 S [†]
specific gravity	sg	Specific gravity uses water as the reference density (1 g/cm ³ = 1.0E+03 kg/m ³); other materials used for the reference density must be specified “Relative density” is preferred to “specific gravity” for the unitless ratio of the density of a material divided by a reference density
square	sq (not permitted: use the power ² or 2)	Use the power ² or 2 instead of sq for square; one format (superscript trademark symbol or in line) must be used consistently in a document

Unit Name	Unit Symbol	Comment
square centimeter	cm ² or cm2	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use of this unit symbol or m² or m2 for square meter preferred to ft² or ft2 for square foot; 1 ft² = 9.290304E-02 m² = 9.290304E+02 cm² Use of this symbol or mm² or mm2 for square millimeter preferred to in² or in2 for square inch; 1 in² = 6.5416E+02 mm² = 6.5416E+00 cm²
square femtometer	fm ² or fm2	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use preferred to b for barn; 1 b = 1.0E+02 fm²†
square foot	ft ² or ft2 (see “m ² or m2” or “cm ² or cm2”)	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use of m² or m2 for square meter or cm² or cm2 for square centimeter preferred; 1 ft² = 9.290304E-02 m² = 9.290304E+02 cm²
square foot per second	ft ² /s or ft2/s (see “m ² /s or m2/s”)	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use of m²/s or m2/s for square meter per second preferred; 1 ft²/s = 9.290304E-02 m²/s
square inch	in ² or in2 (see “mm ² or mm2” or “cm ² or cm2”)	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use of mm² or mm2 for square millimeter or cm² or cm2 for square centimeter preferred; 1 in² = 6.5416E+02 mm² = 6.5416E+00 cm²
square kilometer	km ² or km2	Power can be superscript trademark symbol or in line; one format must be used consistently in a document
square meter	m ² or m2	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use preferred to • acre for acre; 1 acre = 4.046856E+03 m² and 1 acre (US survey) = 4.046873E+03 m² • ha for hectare; 1 ha = 1.0E+04 m² Use of this unit symbol or cm² or cm2 for square centimeter preferred to ft² or ft2 for square foot; 1 ft² = 9.290304E-02 m² = 9.290304E+02 cm²
square meter per second	m ² /s or m2/s	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use of m²/s or m2/s for square meter per second preferred to ft²/s or ft2/s for square foot per second; 1 ft²/s = 9.290304E-02 m²/s Do not use St for Stokes; 1 St = 1.0E-04 m²/s

Unit Name	Unit Symbol	Comment
square millimeter	mm ² or mm2	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use of this symbol or cm ² or cm2 for square centimeter preferred to in ² or in2 for square inch; 1 in ² = 6.5416E+02 mm ² = 6.5416E+00 cm ²
square millimeter per second	mm ² /s or mm2/s	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Do not use cSt for centistokes; 1 cSt = 1.0E+00 mm ² /s
standard cubic feet per day	scf/d (use restricted: see “ft ³ /d or ft3/d,” “m ³ /d or m3/d,” or “m ³ /h or m3/h”)	Use ft ³ /d or ft3/d and specify the standard conditions at which the value was measured Except for gas, use of m ³ /h or m3/h for cubic meter per hour or m ³ /d or m3/d for cubic meter per day preferred; 1 ft ³ /d = 1.179869E–03 m ³ /h = 2.831685E–02 m ³ /d For gas, use Bscf/d for billion standard cubic feet per day, Mscf/d for thousand standard cubic feet per day, MMscf/d for million standard cubic feet per day, and Tscf/d for trillion standard cubic feet per day only if the standard conditions at which the value was measured are specified; the value is expressed without scientific notation If standard conditions are not specified for gas, use Bcf/d for billion cubic feet per day, Mcf/d for thousand cubic feet per day, MMcf/d for million cubic feet per day, and Tcf/d for trillion cubic feet per day
standard cubic foot	scf (use restricted: see “ft ³ or ft3” or “m ³ or m3”)	Use ft ³ or ft3 and specify the standard conditions at which the value was measured Except for gas, use of m ³ or m3 for cubic meter or dm ³ or dm3 for cubic decimeter preferred; 1 ft ³ = 2.831685E–02 m ³ = 2.831685E+01 dm ³ For gas, use Bscf for billion standard cubic feet, Mscf for thousand standard cubic feet, MMscf for million standard cubic feet, and Tscf for trillion standard cubic feet only if the standard conditions at which the value was measured are specified; express the value without scientific notation If standard conditions are not specified for gas, use Bcf for billion cubic feet, Mcf for thousand cubic feet, MMcf for million cubic feet, and Tcf for trillion cubic feet
steradian	sr	
stock-tank barrel	See “bbl” or “m ³ or m3”	Do not modify the unit symbol bbl—specify the stock-tank conditions at which the value was measured Use of m ³ or m3 for cubic meter preferred to bbl for barrel; 1 bbl = 42 galUS = 1.589873E–01 m ³ Do not use bbl (stock tank) for stock-tank barrel

Unit Name	Unit Symbol	Comment
stock-tank barrel per day	See “bbl/d” or “m ³ /d or m3/d”	Do not modify the unit symbol bbl/d—specify the stock-tank conditions at which the value was measured Use of m ³ /d or m3/d for cubic meter per day preferred to bbl/d for barrel per day; 1 bbl/d = 42 galUS/d = 1.589873E–01 m ³ /d Do not use STB/d for stock-tank barrel per day
stock-tank original oil in place	See “bbl” or “m ³ or m3”	Do not modify the implied unit symbol bbl—specify the stock-tank conditions at which the value was measured Use of m ³ or m3 for cubic meter preferred to bbl for barrel; 1 bbl = 42 galUS = 1.589873E–01 m ³ Do not use STOOIP for stock-tank original oil in place
Stokes	See “m ² /s or m2/s”	Use m ² /s or m2/s for square meter per second 1.0E–E04 St = 1 m ² /s Do not use St for Stokes
terabyte	TB	One terabyte equals 1,000,000,000,000 bytes.
tesla	T	Use instead of G or Gs for gauss; 1 G = 1 Gs = 1.0E–04 T
thousand cubic feet	Mcf	Use for gas preferred to m ³ or m3 for cubic meter or dm ³ or dm3 for cubic decimeter; values are not expressed in scientific notation 1 Mcf = 1,000 ft ³ = 2.831685E+01 m ³ = 2.831685E+04 dm ³
thousand cubic feet per day	Mcf/d	Use for gas preferred to m ³ /h or m3/h for cubic meter per hour or m ³ /d or m3/d for cubic meter per day; values are not expressed in scientific notation 1 Mcf/d = 1,000 ft ³ /d = 1.179869E m ³ /h = 2.831685E+01 m ³ /d Mscf/d for thousand standard cubic feet per day is used only if the conditions at which the value was measured are specified
thousand pounds (force) per square inch	See “psi” or “kPa”	Use psi for pound (force) per square inch (SI prefixes cannot be used with non-SI unit symbols so do not use kpsi for thousand pounds [force] per square inch); 1.0E+03 psi = 1 kpsi Use of kPa for kilopascal or MPa for megapascal preferred to psi; 1 psi = 6.894757E–03 MPa = 6.894757E+00 kPa
thousand standard cubic feet per day	Mscf/d (use restricted: see “Mcf/d”)	Use Mcf/d for thousand cubic feet per day; express the value without scientific notation Use Mscf/d only if the conditions at which the value was measured are specified
ton (in space-constrained copy)	t	For space constraints, you can use t for metric ton or US ton as long as, further into the document, you specify “metric ton” or “US ton”. Transition Technologies and EPRs prefer “metric ton,” which follows the U.N.’s Sustainable Development Goal terminology.

Unit Name	Unit Symbol	Comment
trillion cubic feet (trillion = 10^{12})	Tcf	Use for gas preferred to m^3 or m^3 for cubic meter or dm^3 or dm^3 for cubic decimeter; values are expressed without scientific notation $1 \text{ Tcf} = 1,000,000,000,000 \text{ ft}^3 = 2.831685E+10 \text{ m}^3 = 2.831685E+13 \text{ dm}^3$
trillion cubic feet per day	Tcf/d	Use for gas preferred to m^3/h or m^3/h for cubic meter per hour or m^3/d or m^3/d for cubic meter per day; values are expressed without scientific notation $1 \text{ Tcf/d} = 1,000,000,000,000 \text{ ft}^3/d = 1.179869E+09 \text{ m}^3/h = 2.831685E+10 \text{ m}^3/d$
unified atomic mass unit	u	Do not use amu for atomic mass unit, which is equal to 1/12 of the mass of an atom of the nuclide ^{12}C $1 \text{ u} = 1.6605402 \times 10^{-27} \text{ kg}$ with a combined standard uncertainty of $0.0000010 \times 10^{-27} \text{ kg}^\dagger$
US ton	US ton	Do not use standard ton, short-ton, or tonUS; 1 US ton = 2,000 lbm = 907 kg Transition Technologies and EPRs prefer “metric ton,” which follows the United Nation’s Sustainable Development Goal terminology.
volt	V	Do not use the abbreviation emf as a unit symbol for electromotive force
volume per volume	vol/vol (not permitted: see “vol %” or “%” or “ m^3/m^3 or m^3/m^3 ”)	Use allowable customary unit vol % or preferably % for percent (e.g., “a volume fraction of x %”) or Use the ratio m^3/m^3 or m^3/m^3 ; $1 \text{ % vol/vol} = 1.0E-02 \text{ m}^3/m^3$ Do not use vol/vol for volume per unit volume
volume percent	vol % (see “%” or “ m^3/m^3 or m^3/m^3 ”)	Allowable customary unit or preferably use % for percent (e.g., “a volume fraction of x %”) or Use the ratio m^3/m^3 or m^3/m^3 ; $1 \text{ vol %} = 1.0E-02 \text{ m}^3/m^3$
watt	W	Use of this unit symbol or kW for kilowatt preferred to hp for horsepower (550 ft.lbf/s) (also called “mechanical” or “standard” horsepower); $1 \text{ hp} = 7.456999E+02 \text{ W} = 7.456999E-01 \text{ kW}$
weber	Wb	
week	week	Always spelled out Do not use w or wk for week
weight percent	wt % (see “%,” “kg/kg,” or “g/kg”)	Allowable customary unit or preferably use % for percent (e.g., “a mass fraction of x %”) or Use ratio of mass unit symbols: kg/kg for kilogram per kilogram or g/kg for gram/kilogram $1 \text{ wt %} = 1.0E-02E \text{ kg/kg} = 1.0E+01 \text{ g/kg}$
yard	yd (see “m”)	Allowable customary unit or preferably use m for meter; $1 \text{ yd} = 3 \text{ ft} = 9.144E-01 \text{ m}$

Unit Name	Unit Symbol	Comment
year	yr (see “a”)	Restrict use of yr to combined units (e.g., m ³ /yr) Spell out “year” for values used in a descriptive or literary manner Use a for annum only in combined units for measured time values (e.g., m/a)

Source:

This list of unit symbols is based on the [Petrotechnical Open Software Corporation \(POSC\) Recommended Practice \(RP\) 66 V2 specification](#), part 5.

Notes:

Except as noted, conversion factors are from *The SI Metric System of Units and SPE Metric Standard* (Society of Petroleum Engineers (June 1984, 2nd printing) and are to exact values or seven-digit accuracy.

[†]Taylor, B.N.: *Guide for the Use of the International System of Units (SI)*: NIST Special Publication 811, (1995).

[‡]Recommended Practices for Exploration & Production Data Digital Interchange: Part 5: API-SI Unit Symbols (1st ed): API Recommended Practice 66, V2 (May 1, 1996).

Unit Symbols and Prefixes

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Unit Symbols

Unit Symbol	Unit Name	Comment
%	percent	Spacing apart the value and unit symbol is preferred for consistency and integrity of data transfer. Marketing material can use the symbol closed up; one format must be used consistently in a document Preferred to allowable customary unit symbol pu for porosity unit and should be specified this is not the “true” porosity (e.g., use terms such as uncorrected porosity, limestone porosity, or sandstone matrix porosity) Use instead of su for saturation unit to express the percentage of fluid in the pore space; express as “a fluid fraction of x %”) Preferred to allowable customary unit symbol vol % for volume percent and expressed as “a volume fraction of x %” or use the ratio m^3/m^3 or m^3/m^3; $1 \text{ vol } \% = 1.0\text{E}-02 \text{ m}^3/\text{m}^3$ Do not use vol/vol for volume per volume Preferred to allowable customary unit wt % for weight percent and expressed as “a mass fraction of x %” or use a ratio of mass unit symbols: kg/kg for kilogram per kilogram or g/kg for gram/kilogram; $1 \text{ wt } \% = 1.0\text{E}-02 \text{ kg/kg} = 1.0\text{E}+01 \text{ g/kg}$
%/a	percent per annum	
° (used only for angle; for uses not permitted see “API,” “degB,” “degC,” “degF,” or “degR”)	degree	Close up unit symbol to the value Use of ° preferred to dega for degree (angle); one format must be used consistently in a document Use of °/100 ft, °/30 m, and °/10 m allowed for expressing well curvature with °/m at 0.01 or 1/30 scale preferred; use of rad/m for radian per meter preferred; $1 \text{ °/m} = 1.745329\text{E}-02 \text{ rad/m}$ Use API for API gravity Use degC, degF, or degR for temperature values; use degB only as necessary as a historical secondary unit
‘ (not permitted: see “mina” or “ft”)	minute (prime)	Use mina for minute (angle) Use ft for foot
” (not permitted: see “seca” or “in”)	second (double prime)	Use seca for second (angle) Use in for inch
# (not permitted: see “lbm”)	pound (mass)	Use lbm for pound (mass)
A	ampere	
Å (not permitted: see “nm”)	ångstrom (10^{-10} m)	Use nm for nanometer; $1 \text{ Å} = 1.0\text{E}-01 \text{ nm}$

Unit Symbol	Unit Name	Comment
a (see “yr”)	annum (sidereal year)	Use only in combined units for measured time values (e.g., m/a); use allowable customary unit yr for year only in combined units (e.g., m ³ /yr) Spell out “year” for values used in a descriptive or literary manner
acre (see “m ² or m2” or “ha”)	acre	Use of m ² or m2 for square meter or ha for hectare preferred; 1 acre = 4.046856E+03 m ² = 4.046856E–01 ha 1 acre (US survey) = 4.046873E+03 m ² = 4.046873E–01 ha
acre.ft (see “m ³ or m3” or “ha.m”)	acre-foot	Use of m ³ or m3 for cubic meter or ha.m for hectare-meter preferred; 1 acre.ft = 1.233489E+03 m ³ = 1.233489E–01 ha.m
A.h	ampere-hour	Use of kC for kilocoulomb preferred; 1 A.h = 3.6E+00 kC
A/m	ampere per meter	Do not use Oe for oersted; 1 Oe = 7.957747E+01 A/m
amu (not permitted: see “u”)	atomic mass unit	Use u for unified atomic mass unit, which is equal to 1/12 of the mass of an atom of the nuclide ¹² C 1 u = 1.6605402 × 10 ^{–27} kg with a combined standard uncertainty of 0.0000010 × 10 ^{–27} kg [†]
API	API (American Petroleum Institute) gravity	Do not use °API, dAPI, or degAPI for degree API gravity
atm (see “MPa” or “kPa”)	atmosphere (standard)	Use of MPa for megapascal or kPa for kilopascal preferred; 1 atm = 1.01325E–01 MPa = 1.01325E+02 kPa Standard conditions are 760-mm Hg at 0 degC or 14.696 lbf/in ²
B	bel	The unit of sound intensity. Note: the unit symbol “B” is also used for “byte,” the basic unit of information in computer storage and processing. Consider converting bel to decibel (dB): 1 dB = 0.1 B; 1 B = 10 dB) to avoid confusion.
b (see “fm ² or fm2”)	barn	The unit of area used to measure the reaction cross section of atomic nuclei and subatomic particles. Not to be confused with “B” for bel, the unit of sound intensity, or for byte, the unit of information in computer storage and processing. Use of fm2 or fm2 for square femtometer preferred; 1 b = 1.0E+02 fm2 [†]
B	byte	The basic unit of information in computer storage and processing. The unit symbol “B” is also used for “bel,” the unit of sound intensity.

Unit Symbol	Unit Name	Comment
bar (see “MPa” or “kPa”)	bar	Use of MPa for megapascal or kPa for kilopascal preferred; 1 bar = 1.0E–01 MPa = 1.0E+02 kPa
bbl (see “m ³ or m3”)	barrel	Use of m ³ or m3 for cubic meter preferred; 1 bbl = 42 galUS = 1.589873E–01 m ³ Do not modify the unit symbol bbl (e.g., reservoir bbl, stock-tank bbl)—specify the conditions at which the value was measured Do not use B for barrel (e.g., BFPD) except for BOE—specify the fluid type and composition and the conversion factor as necessary
bbl (reservoir) (not permitted: see “bbl”)	barrel at reservoir conditions	Do not modify the unit symbol bbl—specify the reservoir conditions at which the value was measured Use of m ³ or m3 for cubic meter preferred to bbl for barrel; 1 bbl = 42 galUS = 1.589873E–01 m ³ Do not use res bbl for reservoir barrel
bbl (stock tank) (not permitted: see “bbl”)	barrel at stock-tank conditions	Do not modify the unit symbol bbl—specify the stock-tank conditions at which the value was measured Use of m ³ or m3 for cubic meter preferred to bbl for barrel; 1 bbl = 42 galUS = 1.589873E–01 m ³ Do not use STB for stock-tank barrel
bbl/d (see “m ³ /d or m3/d”)	barrel per day	Use of m ³ /d or m3/d for cubic meter per day preferred; 1 bbl/d = 42 galUS/d = 1.589873E–01 m ³ /d Specify the conditions at which the value was measured Do not use B/D, BFPD, BLPD, BOEPD, or BOPD for barrel per day—specify the fluid type and composition and the conversion factor as necessary
bbl/d (reservoir) (not permitted: see “bbl/d”)	barrel per day at reservoir conditions	Do not modify the unit symbol bbl/d—specify the reservoir conditions at which the value was measured Use of m ³ /d or m3/d for cubic meter per day preferred to bbl/d for barrel per day; 1 bbl/d = 42 galUS/d = 1.589873E–01 m ³ /d Do not use RB/d for reservoir barrel per day
bbl/d (stock tank) (not permitted: see “bbl/d”)	barrel per day at stock-tank conditions	Do not modify the unit symbol bbl/d—specify the stock-tank conditions at which the value was measured Use of m ³ /d or m3/d for cubic meter per day preferred to bbl/d for barrel per day; 1 bbl/d = 42 galUS/d = 1.589873E–01 m ³ /d Do not use STB/d for stock-tank barrel per day

Unit Symbol	Unit Name	Comment
bbl/min (see “m ³ /min or m ³ /min”)	barrel per minute	Use of m ³ /min or m ³ /min for cubic meter per minute preferred to bbl/min for barrel per minute; 1 bbl/min = 42 galUS/min = 1.589873E–01 m ³ /min Specify the conditions at which the value was measured
Bcf	billion (10 ⁹) cubic feet	Use for gas preferred to m ³ or m ³ for cubic meter or dm ³ or dm ³ for cubic decimeter; values are expressed without scientific notation 1 Bcf = 1,000,000,000 ft ³ = 2.831685E–07 m ³ = 2.831685E+10 dm ³
Bc	Bearden units of consistency	Use for pumpability or consistency of a slurry, measured in Bc, a dimensionless quantity with no direct conversion factor to more common units of viscosity.
Bcf/d	billion (10 ⁹) cubic feet per day	Use for gas preferred to m ³ /d or m ³ /d for cubic meter per day or dm ³ /d or dm ³ /d for cubic decimeter per day; values are expressed without scientific notation 1 Bcf/d = 1,000,000,000 ft ³ /d = 2.831685E–07 m ³ /d = 2.831685E+10 dm ³ /d Bscf/d for billion standard cubic feet per day is used only if the conditions at which the value was measured are specified
B/D (not permitted: see “bbl/d”)	barrel per day	Use of m ³ /d or m ³ /d for cubic meter per day preferred to bbl/d for barrel per day; 1 bbl/d = 42 galUS/d = 1.589873E–01 m ³ /d
BFPD (not permitted: see “bbl/d”)	barrel of fluid per day	Use of m ³ /d or m ³ /d for cubic meter per day preferred to bbl/d for barrel per day; 1 bbl/d = 42 galUS/d = 1.589873E–01 m ³ /d Specify the conditions at which the value was measured Specify the fluid type as necessary
bhp (see “kW”)	brake horsepower	Brake horsepower is the power available for work. It is determined by multiplying the nominal horsepower by an efficiency factor specific to the equipment Use of kW for kilowatt preferred; 1 bhp = 7.46E–01 kW
BLPD (not permitted: see “bbl/d”)	barrel of liquid per day	Use of m ³ /d or m ³ /d for cubic meter per day preferred to bbl/d for barrel per day; 1 bbl/d = 42 galUS/d = 1.589873E–01 m ³ /d Specify the conditions at which the value was measured Specify the fluid type as necessary

Unit Symbol	Unit Name	Comment
BOE (see “bbl” or “m ³ or m3”)	barrel of oil equivalent	Allowable modification of the unit symbol bbl—can also be expressed as “x bbl of oil equivalent” with specification as necessary the conditions at which the value was measured, the fluid type and composition, and the equivalency conversion factor Use of m ³ or m3 for cubic meter preferred to bbl for barrel; 1 bbl = 42 galUS = 1.589873E–01 m ³
BOEPD (not permitted: see “bbl/d”)	barrel of oil equivalent per day	Do not modify the unit symbol bbl—express as “x bbl/d of oil equivalent” and specify as necessary the conditions at which the value was measured, the fluid type and composition, and the equivalency conversion factor Use of m ³ /d or m3/d for cubic meter per day preferred to bbl/d for barrel per day; 1 bbl/d = 42 galUS/d = 1.589873E–01 m ³ /d
BOPD (not permitted: see “bbl/d”)	barrel of oil per day	Use of m ³ /d or m3/d for cubic meter per day preferred to bbl/d for barrel per day; 1 bbl/d = 42 galUS/d = 1.589873E–01 m ³ /d Specify the conditions at which the value was measured Specify the fluid type as necessary
Bq	becquerel	Do not use Ci for curie; 1 Ci = 3.7E+10 Bq
Bscf/d (use restricted: see “Bcf/d”)	billion (10 ⁹) standard cubic feet per day	Use Bcf/d for billion cubic feet per day Use Bscf/d only if the conditions at which the value was measured are specified
Btu (see “J”)	British thermal unit (international)	Use of J for joule preferred; 1 Btu (international) = 1.055056E+03 J
BWPD (not permitted: see “bbl/d”)	barrel of water per day	Use of m ³ /d or m3/d for cubic meter per day preferred to bbl/d for barrel per day; 1 bbl/d = 42 galUS/d = 1.589873E–01 m ³ /d Specify the conditions at which the value was measured Specify the fluid type as necessary
C	coulomb	
c	cycle	
cc (not permitted: see “cm ³ or cm3”)	cubic centimeter	Use cm ³ or cm3 for cubic centimeter
cd	candela	
cf (not permitted: see “ft ³ or ft3”)	cubic foot	Except for gas, use ft ³ or ft3 for cubic foot Use for gas in the unit symbols Bcf, Mcf, MMcf, and Tcf and express the value without scientific notation
Ci (not permitted: see “Bq”)	curie	Use Bq for becquerel; 1 Ci = 3.7E+10 Bq
C/kg	coulomb per kilogram	Do not use R for roentgen; 1 R = 2.58E–04 C/kg

Unit Symbol	Unit Name	Comment
cm	centimeter	Use of this unit symbol or m for meter preferred to ft for foot; 1 ft = 3.048E-01 m = 3.048E+01 cm Use of this unit symbol or mm for millimeter preferred to in for inch; 1 in = 2.54E+01 mm = 2.54E+00 cm
cm ² or cm2	square centimeter	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use of this unit symbol or m ² or m2 for square meter preferred to ft ² or ft2 for square foot; 1 ft ² = 9.290304E-02 m ² = 9.290304E+02 cm ² Use of this symbol or mm ² or mm2 for square millimeter preferred to in ² or in2 for square inch; 1 in ² = 6.5416E+02 mm ² = 6.5416E+00 cm ²
cm ³ or cm3	cubic centimeter	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Do not use cc for cubic centimeter Do not use fl ozUK for fluid ounce (UK); 1 fl ozUK = 2.841308E+01 cm ³ Do not use fl ozUS for fluid ounce (US); 1 fl ozUS = 2.957353E+01 cm ³ Use of this unit symbol or mm ³ or mm3 for cubic millimeter preferred to in ³ or in3 for cubic inch; 1 in ³ = 1.638706E+04 mm ³ = 1.638706E+01 cm ³
c/min	cycle per min	Use preferred to allowable customary unit rpm for revolution per minute; 1.0 rpm = 1.0 c/min
cm/s	centimeter per second	Use of this unit symbol or m/s for meter per second preferred to ft/min for foot per minute; 1 ft/min = 5.08E-03 m/s = 5.08E-01 cm/s Use of this unit symbol or mm/s for millimeter per second preferred to in/s for inch per second; 1 in/s = 2.54E+01 mm/s = 2.54E+00 cm/s Use of this unit symbol, mm/s for millimeter per second, m/h for meter per hour, or min/m for minute per meter preferred to ft/h for foot per hour; 1 ft/h = 8.466667E-03 cm/s = 8.466667E-02 mm/s = 3.048E-01 m/h = 1.968504E+02 min/m
cm/s ² or cm/s2	centimeter per second squared	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use preferred to Gal for galileo; 1 Gal = 1.0E+00 cm/s ²
counts/s	counts per second	Do not use cps for counts per second

Unit Symbol	Unit Name	Comment
cP (see "Pa.s")	centipoise	Use of Pa.s for pascal-second preferred; 1 cP = 1.0E-03 Pa.s
cps (not permitted: see "counts/s")	count per second	Use counts/s for counts per second
cSt (not permitted: see "mm ² /s or mm ² /s")	centistokes	Use mm ² /s or mm ² /s for square millimeter per second; 1 cSt = 1.0E+00 mm ² /s
cu	capture unit	Use of 1/m for per meter preferred; 1 cu = 1.0E-01 1/m
D	darcy	
d	day	Use spelled-out unit name when not used with a measured value or depending on context
dAPI (not permitted: see "API")	API (American Petroleum Institute) gravity	Use API for API gravity
dB	decibel	The unit of sound intensity. decibel (dB) preferred to bel (B), which can be confused with "byte," the basic unit of information in computer storage and processing.
dega (see "°")	degree (angle)	Use of ° for degree (angle) preferred; one format must be used consistently in a document
degAPI (not permitted: see "API")	degree API (American Petroleum Institute) gravity	Use API for API gravity
degB	degree baumé	Use degB only as necessary as a historical secondary unit for fluid density and specify the conversion from conventional relative density (S) as Light liquids: $d = (144.3/S) - 144.3$ Heavy liquids: $d = 144.3 - (144.3/S)$, where d = relative density in degree baumé
degC	degree Celsius	Do not use °C for degree Celsius (degF - 32)/1.8 = degC For one-degree increments: 1 degC = 1.8 degF
degF	degree Fahrenheit	Do not use °F for degree Fahrenheit 1.8 degC + 32 = degF For one-degree increments: 1 degF = 0.56 degC
degR	degree Rankine	Do not use °R for degree Rankine

Unit Symbol	Unit Name	Comment
dm ³ or dm3	cubic decimeter	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Except for gas, use of this unit symbol or m³ or m3 for cubic meter preferred to ft³ or ft3 for cubic foot; 1 ft³ = 2.831685E–02 m³ = 2.831685E+01 dm³ For gas, use Bcf for billion cubic feet, Mcf for thousand cubic feet, MMcf for million cubic feet, and Tcf for trillion cubic feet and express the value without scientific notation
dm ³ /kg or dm3/kg	cubic decimeter per kilogram	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use preferred to ft³/lbm or ft3/lbm for cubic foot per pound (mass); 1 ft³/lbm = 6.242796E+01 dm³/kg
dm ³ /s or dm3/s	cubic decimeter per second	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use of this unit symbol or L/s for liter per second preferred to - ft³/min or ft3/min for cubic foot per minute; 1 ft³/min = 4.719474E–01 dm³/s = 4.719474E–01 L/s - ft³/s or ft3/s for cubic foot per second; 1 ft³/s = 2.831685E+01 dm³/s = 2.831685E+01 L/s
ehp (see “kW”)	horsepower (electric)	Use of kW for kilowatt preferred; 1 ehp = 7.46E–01 kW
emf (not permitted: see “V”)	electromotive force	Use V for volt (the unit symbol for electromotive force)
eq (use restricted: see “mmol”)	equivalent	Use mmol for millimole; can be used as a secondary unit with the “equivalent” defined 1 meq = [1 mmol/(valence of the substance)] [†]
eq/L (not permitted: see “mol/m ³ or mol/m3”)	equivalent per liter	Use mol/m³ or mol/m3 to express the normality as a whole-number multiple of the molarity dependent on the valence of the solution; (mol/m³)/valence[†] Do not use N for normality
erg	erg	The symbol for erg is the name of the unit Use of J for joule preferred; 1 erg = 1E–07 J
eV	electronvolt	
F	farad	
fl ozUK (not permitted: see “cm ³ or cm3”)	fluid ounce (UK)	Use cm³ or cm3 for cubic centimeter 1 fl ozUK = 2.841308E+01 cm³

Unit Symbol	Unit Name	Comment
fl ozUS (not permitted: see “cm ³ or cm3”)	fluid ounce (US)	Use cm ³ or cm3 for cubic centimeter 1 fl ozUS = 2.957353E+01 cm ³
fm ² or fm2	square femtometer	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use preferred to b for barn; 1 b = 1.0E+02 fm ^{2†}
fph (not permitted: see “min/m,” “m/h,” mm/s,” or “cm/s”)	foot per hour	Use of min/m for minute per meter, m/h for meter per hour, mm/s for millimeter per second, or cm/s for centimeter per second preferred to ft/h for foot per hour; 1 ft/h = 1.968504E+02 min/m = 3.048E–01 m/h = 8.466667E–02 mm/s = 8.466667E–03 cm/s
ft (see “m” or “cm”)	foot	Use of m for meter or cm for centimeter preferred; 1 ft = 3.048E–01 m = 3.048E+01 cm Do not use □ for foot
ft ² or ft2 (see “m ² or m2” or “cm ² or cm2”)	square foot	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use of m ² or m2 for square meter or cm ² or cm2 for square centimeter preferred; 1 ft ² = 9.290304E–02 m ² = 9.290304E+02 cm ²
ft ³ or ft3 (see “m ³ or m3” or “dm ³ or dm3”)	cubic foot	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Do not modify the unit symbol ft ³ or ft3 (e.g., standard ft ³)—specify the conditions at which the value was measured Except for gas, use of m ³ or m3 for cubic meter or dm ³ or dm3 for cubic decimeter preferred; 1 ft ³ = 2.831685E–02 m ³ = 2.831685E+01 dm ³ For gas, use Bcf for billion cubic feet, Mcf for thousand cubic feet, MMcf for million cubic feet, and Tcf for trillion cubic feet (do not use cf by itself for cubic foot) and express the value without scientific notation

Unit Symbol	Unit Name	Comment
ft ³ or ft3 (standard) (not permitted: see “ft ³ or ft3”)	cubic foot at standard conditions	Do not modify the unit symbol ft³ or ft3; instead, specify the conditions at which the value was measured Except for gas, use of m³ or m3 for cubic meter or dm³ or dm3 for cubic decimeter preferred; 1 ft³ = 2.831685E–02 m³ = 2.831685E+01 dm³ Do not use scf for standard cubic foot except in Bscf for billion standard cubic feet, Mscf for thousand standard cubic feet, MMscf for million cubic feet, and Tscf for trillion cubic feet with the conditions at which the value was measured specified and the value expressed without scientific notation
ft ³ /bbl or ft3/bbl (see “L/m ³ or L/m3” or “m ³ /m ³ or m3/m3”)	cubic foot per barrel	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use of L/m³ or L/m3 for liter per cubic meter or m³/m³ or m3/m3 for cubic meter per cubic meter preferred; 1 ft³/bbl = 1.78073E+02 L/m³ = 1.78073E–01 m³/m³
ft ³ /d or ft3/d (see “m ³ /h or m3/h” or “m ³ /d or m3/d”)	cubic foot per day	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Except for gas, use of m³/h or m3/h for cubic meter per hour or m³/d or m3/d for cubic meter per day preferred; 1 ft³/d = 1.179869E–03 m³/h = 2.831685E–02 m³/d For gas, use Bcf/d for billion cubic feet per day, Mcf/d for thousand cubic feet per day, MMcf/d for million cubic feet per day, and Tcf/d for trillion cubic feet per day (do not use cf/d for cubic feet per day without a prefix) and express the value without scientific notation
ft ³ /d or ft3/d (standard) (not permitted: see “ft ³ /d or ft3/d”)	cubic foot per day at standard conditions	Do not modify the unit symbol—specify the standard conditions at which the value was measured Except for gas, use of m³/h or m3/h for cubic meter per hour or m³/d or m3/d for cubic meter per day preferred; 1 ft³/d = 1.179869E–03 m³/h = 2.831685E–02 m³/d For gas, use Bscf/d for billion standard cubic feet per day, Mscf/d for thousand standard cubic feet per day, MMscf for million cubic feet per day, and Tscf for trillion cubic feet per day (do not use scf/d for standard cubic foot per day without a prefix) with the conditions at which the value was measured specified and the value expressed without scientific notation

Unit Symbol	Unit Name	Comment
ft/h (see “min/m,” “m/h,” “mm/s,” or “cm/s”)	foot per hour	Use of min/m for minute per meter, m/h for meter per hour, mm/s for millimeter per second, or cm/s for centimeter per second preferred to ft/h for foot per hour; 1 ft/h = 1.968504E+02 min/m = 3.048E–01 m/h = 8.466667E–02 mm/s = 8.466667E–03 cm/s Do not use fph for foot per hour
ft.lbf (see “kJ”) (use for work or energy)	foot-pound (force)	Use of kJ for kilojoule preferred for work or energy unit values; 1 ft.lbf = 1.355818E–03 kJ Do not use for torque unit values, for which use of N.m for newton-meter preferred to lbf.ft for pound (force)-foot; 1 lbf.ft = 1.355818E+00 N.m
ft ³ /lbm or ft3/lbm (see “dm ³ /kg or dm3/kg”)	cubic foot per pound (mass)	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use of dm ³ /kg or dm3/kg for cubic decimeter per kilogram preferred; 1 ft ³ /lbm = 6.242796E+01 dm ³ /kg
ft/min (see “m/s” or “cm/s”)	foot per minute	Use of m/s for meter per second or cm/s for centimeter per second preferred; 1 ft/min = 5.08E–03 m/s = 5.08E–01 cm/s
ft ³ /min or ft3/min (see “dm ³ /s or dm3/s” or “L/s”)	cubic foot per minute	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use of dm ³ /s or dm3/s for cubic decimeter per second or L/s for liter per second preferred; 1 ft ³ /min = 4.719474E–01 dm ³ /s = 4.719474E–01 L/s
ft/s (see “m/s”)	foot per second	Use of m/s for meter per second preferred; 1 ft/s = 3.048E–01 m/s
ft ² /s or ft2/s (see “m ² /s or m2/s”)	square foot per second	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use of m ² /s or m2/s for square meter per second preferred; 1 ft ² /s = 9.290304E–02 m ² /s
ft ³ /s or ft3/s (see “dm ³ /s or dm3/s” or “L/s”)	cubic foot per second	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use of dm ³ /s or dm3/s for cubic decimeter per second or L/s for liter per second preferred; 1 ft ³ /s = 2.831685E+01 dm ³ /s = 2.831685E+01 L/s
G (not permitted: see “g _n ”)	acceleration due to gravity	Use g _n for acceleration due to gravity (SPE- and IEEE-accepted unit symbol) 1 g _n = 9.806650E+00 m/s ²
G (not permitted: see “T”)	gauss	Use T for tesla; 1 G = 1 Gs = 1.0E–04 T

Unit Symbol	Unit Name	Comment
g	gram	Use preferred to ozm for ounce (mass); 1 ozm = 2.834952E+01 g
Gal (see "cm/s ² or cm/s ² ")	galileo	Use of cm/s ² or cm/s ² for centimeter per second squared preferred; 1 Gal = 1.0E+00 cm/s ²
galUK (see "m ³ or m ³ ")	gallon (UK)	Use of m ³ or m ³ for cubic meter preferred 1 galUK = 4.546092E-03 m ³
galUS (see "m ³ or m ³ ")	gallon (US)	Use of m ³ or m ³ for cubic meter preferred 1 galUS = 3.785412E-03 m ³
galUS/d (see "m ³ /d or m ³ /d")	gallon per day	Use of m ³ /d or m ³ /d for cubic meter per day preferred 1 galUS/d = 3.785412E-03 m ³ /d
galUS/min (see "m ³ /h or m ³ /h")	gallon per minute	Use of m ³ /h or m ³ /h for cubic meter per hour preferred 1 galUS/min = 2.271247E-01 m ³ /h Do not use gpm for gallon per minute
galUS/sack (use restricted: see "L/m ³ or L/m ³ " or "L/metric ton")	gallon (US) per sack	Use of L/m ³ or L/m ³ for liter per cubic meter preferred to galUS/sack for gallon per sack to express the volume of fluid per volume of solid (concentration). Use of L/metric ton for liter per metric ton preferred to galUS/sack for gallon (US) per sack to express the volume of fluid per mass of solid (specific volume); for portland cement: 1 galUS/sack = ~88.78 L/t To use sack as a unit, specify the material and its volume in m ³ or m ³ for cubic meter (preferable to characterizing it by its mass) Do not use sk for sack
gAPI	API gamma ray	
GB	gigabyte	One gigabyte equals one billion bytes.
g/cc (not permitted: see "g/cm ³ or g/cm ³ ")	gram per cubic centimeter	Use g/cm ³ or g/cm ³ for gram per cubic centimeter to specify mass density
g/cm ³ or g/cm ³	gram per cubic centimeter	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Do not use g/cc for gram per cubic centimeter
GHz	gigahertz	
g/kg	gram per kilogram	Use preferred to allowable customary unit wt % for weight percent or use kg/kg for kilogram/kilogram or express as a mass fraction of x %; 1 wt % = 1.0E-02E kg/kg = 1.0E+01 g/kg

Unit Symbol	Unit Name	Comment
g/m ³ or g/m3	gram per cubic meter	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use of this unit symbol or kg/m³ or kg/m3 for kilogram per cubic meter preferred to lbm/ft³ or lbm/ft3 for pound (mass) per cubic foot; 1 lbm/ft³ = 1.601846E+01 kg/m³ = 1.601846E+04 g/m³ Use of this unit symbol preferred to lbm/1,000 galUS or lbm/1,000 galUK for pound (mass) per thousand gallons; 1 lbm/1,000 galUS = 1.198264E+02 g/m³ or 1 lbm/1,000 gal UK = 9.977633E+01 g/m³ Use of this unit symbol preferred to PTB for pound per thousand barrels (salt content); 1 PTB = 2.853010E+00 g/m³ = 2.853010E+00 mg/dm³ Do not use ppt for pound per thousand (gallons)
g _n	acceleration due to gravity	Use this SPE- and IEEE-accepted unit symbol instead of G for acceleration due to gravity 1 g_n = 9.806650E+00 m/s²
GPa	gigapascal	
gpm (not permitted: see "galUS/min")	gallon per minute	Use of m³/h or m3/h for cubic meter per hour preferred to galUS/min for gallon per minute; 1 galUS/min = 2.271247E-01 m³/h
Gs (not permitted: see "T")	gauss	Use T for tesla; 1 Gs = 1 G = 1.0E-04 T
GW	gigawatt	
Gy	gray	Do not use rad for radiation absorbed dose to avoid confusion with SI unit symbol rad for radian 1 rad = 1.0E-02 Gy
H	henry	
h	hour	Use spelled-out unit name when not used with a measured value or depending on context Do not use hr for hour
ha (see "m ² or m2")	hectare	Use of m ² or m2 preferred; 1 ha = 1.0E+04 m ²
ha.m	hectare-meter	Use of this unit symbol or m³ or m3 for cubic meter preferred to acre.ft for acre-foot; 1 acre.ft = 1.233489E+03 m³ = 1.233489E-01 ha.m
hhp (see "kW")	hydraulic horsepower	Use of kW for kilowatt preferred; 1 hhp = 7.46043E-01 kW

Unit Symbol	Unit Name	Comment
hp (see “W” or “kW”)	horsepower (550 ft.lbf/s) (also called “mechanical” or “standard” horsepower)	Use of W for watt or kW for kilowatt preferred; 1 hp = 7.456999E+02 W = 7.456999E–01 kW
hp.h (see “MJ”)	horsepower-hour	Use of MJ for megajoule preferred; 1 hp.h = 2.684520E+00 MJ
hr (not permitted: see “h”)	hour	Use h for hour
Hz	hertz	
in (see “mm” or “cm”)	inch	The unit symbol in is not followed by a period unless it is at the end of a sentence Do not use □□ for inch Use of mm for millimeter or cm for centimeter preferred; 1 in = 2.54E+01 mm = 2.54E+00 cm
in ² or in2 (see “mm ² or mm2” or “cm ² or cm2”)	square inch	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use of mm ² or mm2 for square millimeter or cm ² or cm2 for square centimeter preferred; 1 in ² = 6.5416E+02 mm ² = 6.5416E+00 cm ²
in ³ or in3 (see “mm ³ or mm3” or “cm ³ or cm3”)	cubic inch	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use of mm ³ or mm3 for cubic millimeter or cm ³ or cm3 for cubic centimeter preferred; 1 in ³ = 1.638706E+04 mm ³ = 1.638706E+01 cm ³
in/s	inch per second	Use of mm/s for millimeter per second or cm/s for centimeter per second preferred; 1 in/s = 2.54E+01 mm/s = 2.54E+00 cm/s
J	joule	Do not use kgf.m for kilogram (force)–meter; 1 kgf.m = 9.806650E+00 J Use preferred to Btu for British thermal unit (international); 1 Btu (international) = 1.055056E+03 J Use preferred to erg; 1 erg = 1E–07 J
K	kelvin	
kC	kilocoulomb	Use preferred to A.h for ampere-hour; 1 A.h = 3.6E+00 kC

Unit Symbol	Unit Name	Comment
kg	kilogram	Use preferred to lbm for pound (mass); 1 lbm = 4.535924E-01 kg Exemption: Only M-I SWACO collateral uses lb for pound (mass) and in combined unit symbols Use preferred to sack, for which the material must be specified and its volume in m ³ or m3 for cubic meter (preferable to characterizing it by its mass); for portland cement, 1 sack = ~42.64 kg Do not use sk for sack
kgf (not permitted: see "N")	kilogram (force)	Use N for Newton; 1 kgf = 9.80665E+00 N Do not modify the unit symbol kg Do not use kp for kilopond
kgf.m (not permitted: see "J")	kilogram (force)–meter	Use J for joule; 1 kgf.m = 9.806650E+00 J
kg/kg	kilogram per kilogram	Use preferred to allowable customary unit wt % for weight percent or use g/kg for gram/kilogram or express as a mass fraction of x %; 1 wt % = 1.0E-02E kg/kg = 1.0E+01 g/kg
kg/m ³ or kg/m3	kilogram per cubic meter	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use of this unit symbol or g/m ³ or g/m3 for gram per cubic meter preferred to lbm/ft ³ or lbm/ft3 for pound (mass) per cubic foot; 1 lbm/ft ³ = 1.601846E+01 kg/m ³ = 1.601846E+04 g/m ³ Use preferred to lbm/galUS for pound (mass) per gallon (US); 1 lbm/galUS = 1.198264E+02 kg/m ³
kHz	kilohertz	
kip (not permitted: see "N")	kip (= 1,000 lbf)	Use N for newton; 1 kip = 4.448222E+03 N Do not use klbf for 1,000 lbf
kJ	kilojoule	Use preferred to kW.h for kilowatt-hour; 1 kW.h = 3.6E+03 kJ Use preferred to ft.lbf for foot-pound (force) for energy or work values; 1 ft.lbf = 1.355818E-03 kJ
klbf (not permitted: see "lbf")	kilopound (force) (= 1,000 lbf)	Use of N for Newton preferred; 1 klbf = 4.448222E+03 N Use lbf instead of klbf (SI prefixes cannot be used with non-SI unit symbols); 1E+03 lbf = 1 klbf Do not use kip for 1,000 lbf
km	kilometer	

Unit Symbol	Unit Name	Comment
km ² or km2	square kilometer	Power can be superscript trademark symbol or in line; one format must be used consistently in a document
km ³ or km3	cubic kilometer	Power can be superscript trademark symbol or in line; one format must be used consistently in a document
km/h	kilometer per hour	Use preferred to allowable customary unit knot for knot to express nautical speed values; 1 knot = 1.852E+00 km/h Use preferred to mi/h for mile per hour to express non-nautical speed values; 1 mi/h = 1.609344E+00 km/h
knot (see “km/h”)	knot	Allowable customary unit for nautical speed values or preferably use km/h for kilometer per hour; 1 knot = 1.852E+00 km/h
kp (not permitted: see “N”)	kilopond	Use N for Newton; 1 kp = 9.80665E+00 N Do not use kgf for kilogram (force)
kPa	kilopascal	Use of this unit symbol or MPa for megapascal preferred to - atm for atmosphere (standard); 1 atm = 1.01325E–01 MPa = 1.01325E+02 kPa at standard conditions of 760-mm Hg at 0 degC or 14.696 lbf/in² - bar for bar; 1 bar = 1.0E–01 MPa = 1.0E+02 kPa - psi for pound (force) per square inch; 1 psi = 6.894757E+00 kPa = 6.894757E–03 MPa Do not use kpsi for 1,000 pounds (force) per square inch or ksi for kip (1,000 lbf) per square inch; 1 ksi = 1 kpsi = 6.894757E+03 kPa
kpsi (not permitted: see “psi”)	thousand pounds (force) per square inch	Use psi for pound (force) per square inch (SI prefixes cannot be used with non-SI unit symbols); 1.0E+03 psi = 1 kpsi Use of kPa for kilopascal or MPa for megapascal preferred to psi; 1 psi = 6.894757E–03 MPa = 6.894757E+00 kPa
ksi (not permitted: see “psi”)	kip per square inch (1 kip = 1,000 lbf)	Use of kPa for kilopascal preferred to psi for pound (force) per square inch; 1 ksi = 6.895E+03 kPa
kV	kilovolt	
kW	kilowatt	Use preferred to ehp for horsepower (electric); 1 ehp = 7.46E–01 kW Use preferred to hhp for hydraulic horsepower; 1 hhp = 7.46043E–01 kW Use preferred to hp for horsepower (550 ft.lbf/s) (also called “mechanical” or “standard” horsepower); 1 hp = 7.456999E–01 kW

Unit Symbol	Unit Name	Comment
kW.h (see “kJ”)	kilowatt-hour	Use of kJ for kilojoule preferred; 1 kW.h = 3.6E+03 kJ
L	liter	Use restricted to the measurement of liquids and gases Do not use lowercase or script letter l for the unit symbol Do not spell the unit name “litre” (SLB style uses American spelling; <i>Merriam Webster’s Collegiate Dictionary</i>, 11th ed.) 1 L = 1.0E–03 m³ or m³ for cubic meter = 1.0E+00 dm³ or dm³ for cubic decimeter; 1 kL = 1.0E+00 m³ Do not use pt for pint; 1 pt = 4.731765E–01 L Do not use qt for quart; 1 qt = 9.463529E–01 L Use preferred to sack, for which the material must be specified and its volume in m³ or m³ for cubic meter (preferable to characterizing it by its mass); for portland cement, 1 sk = ~1 ft³ = ~28.3 L Do not use sk for sack
lbf (see “N”)	pound (force)	Use of N for Newton preferred; 1 lbf = 4.448222E+00 N Do not use klbf for kilopound (force); 1 klbf = 1.0E+03 lbf
lbf/100 ft ² or lbf/100 ft ² (not permitted: see “Pa”)	pound (force) per 100 square feet (unit for yield point)	Use Pa for pascal; 1 lbf/100 ft ² = 4.788026E–01 Pa
lbf.ft (see “N.m”) (use for torque values)	pound (force)-foot (unit for torque)	Use of N.m for newton-meter preferred; 1 lbf.ft = 1.355818E+00 N.m Do not use for energy or work unit values, for which use of kJ for kilojoule preferred to ft.lbf for foot-pound (force); 1 ft.lbf = 1.355818E–03 kJ
lbm (see “kg”)	pound (mass)	Use of kg for kilogram preferred; 1 lbm = 4.535924E–01 kg Use of lbm preferred to allowable customary unit symbol ppa for “pound of proppant added per barrel of clean fluid” with expression as “x lbm of proppant added per barrel of clean fluid” Do not use # for pound (mass) Exemption: Only M-I SWACO collateral uses lb for pound (mass) and in combined unit symbols

Unit Symbol	Unit Name	Comment
lbm/ft ³ or lbm/ft3 (see “kg/m ³ or kg/m3” or “g/m ³ or g/m3”)	pound (mass) per cubic foot	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use of kg/m ³ or kg/m3 for kilogram per cubic meter or g/m ³ or g/m3 for gram per cubic meter preferred; 1 lbm/ft ³ = 1.601846E+01 kg/m ³ = 1.601846E+04 g/m ³
lbm/galUS (see “kg/m ³ or kg/m3”)	pound (mass) per gallon	Use of kg/m ³ or kg/m3 for kilogram per cubic meter preferred; 1 lbm/galUS = 1.198264E+02 kg/m ³
lbm/1,000 galUK (see “g/m ³ or g/m3”)	pound (mass) per thousand gallons	Use of g/m ³ or g/m3 for gram per cubic meter preferred; 1 lbm/1,000 galUK = 0.977633E+01 g/m ³ Do not use ppt for pound per thousand (gallons)
lbm/1,000 galUS (see “g/m ³ or g/m3”)	pound (mass) per thousand gallons	Use of g/m ³ or g/m3 for gram per cubic meter preferred; 1 lbm/1,000 galUS = 1.198264E+02 g/m ³ Do not use ppt for pound per thousand (gallons)
lm	lumen	
L/m ³ or L/m3	liter per cubic meter	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use of this unit symbol or m ³ /m ³ or m3/m3 for cubic meter per cubic meter preferred to ft ³ /bbl or ft3/bbl for cubic foot per barrel; 1 ft ³ /bbl = 1.78073E+02 L/m ³ = 1.78073E–01 m ³ /m ³ Use of L/m ³ or L/m3 for liter per cubic meter preferred to galUS/sack for gallon per sack to express the volume of fluid per volume of solid (concentration)
L/min	liter per minute	Do not use Lpm
Lpm (not permitted: see “L/min”)	liter per minute	Use L/min for liter per minute
L/s	liter per second	Use of this unit symbol or dm ³ /s or dm3/s for cubic decimeter per second preferred to - ft³/min or ft3/min for cubic foot per minute; 1 ft³/min = 4.719474E–01 dm³/s = 4.719474E–01 L/s - ft³/s or ft3/s for cubic foot per second; 1 ft³/s = 2.831685E+01 dm³/s = 2.831685E+01 L/s

Unit Symbol	Unit Name	Comment
L/metric ton	liter per metric ton	Use of L/metric ton for liter per metric ton preferred to galUS/sack for gallon (US) per sack to express the volume of fluid per mass of solid (specific volume); for portland cement: 1 galUS/sack = ~88.78 L/ metric ton Do not use sk for sack
lx	lux	
M (not permitted: see "mol/m ³ or mol/m ³ ")	molarity	Use mol/m ³ or mol/m ³ for mole per cubic meter to express as the amount-of-substance concentration (volume based); 1 mol/L = 1.0E+03 mol/m ³ [†] Do not use mol/L for mole per liter
m	meter	Do not spell the unit name "metre" (SLB style uses American spelling: <i>Merriam Webster's Collegiate Dictionary</i> , 11th ed.) Use of this unit symbol or cm for centimeter preferred to ft for foot; 1 ft = 3.048E-01 m = 3.048E+01 cm Use preferred to allowable customary unit yd for yard; 1 yd = 3 ft = 9.144E-01 m
m ² or m2	square meter	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use preferred to - acre for acre; 1 acre = 4.046856E+03 m² and 1 acre (US survey) = 4.046873E+03 m² - ha for hectare; 1 ha = 1.0E+04 m² Use of this unit symbol or cm ² or cm2 for square centimeter preferred to ft ² or ft2 for square foot; 1 ft ² = 9.290304E-02 m ² = 9.290304E+02 cm ²

Unit Symbol	Unit Name	Comment
m ³ or m3	cubic meter	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use preferred to - bbl for barrel; 1 bbl = 42 galUS = 1.589873E−01 m³ - galUK or galUS; 1 galUK = 4.546092E−03 m³ and 1 galUS = 3.785412E−03 m³ Except for gas, use of this unit symbol or dm³ or dm3 for cubic decimeter preferred to ft³ or ft3 for cubic foot; 1 ft³ = 2.831685E−02 m³ = 2.831685E+01 dm³ For gas, use Bcf for billion cubic feet, Mcf for thousand cubic feet, MMcf for million cubic feet, and Tcf for trillion cubic feet per day and express the value expressed without scientific notation Use of this unit symbol or ha.m for hectare-meter preferred to acre.ft for acre-foot; 1 acre.ft = 1.233489E+03 m³ = 1.233489E−01 ha.m Use preferred to allowable customary unit yd³ for cubic yard; 1 yd³ = 7.645549E−01 m³ 1 L = 1.0E−03 m³
	megampere	
Ma	million years	Do not use MY for million years
mA	milliampere	
Mcf	thousand cubic feet	Use for gas preferred to m³ or m3 for cubic meter or dm³ or dm3 for cubic decimeter; values are not expressed in scientific notation 1 Mcf = 1,000 ft³ = 2.831685E+01 m³ = 2.831685E+04 dm³
Mcf/d	thousand cubic feet per day	Use for gas preferred to m³/h or m3/h for cubic meter per hour or m³/d or m3/d for cubic meter per day; values are not expressed in scientific notation 1 Mcf/d = 1,000 ft³/d = 1.179869E m³/h = 2.831685E+01 m³/d Mscf/d for thousand standard cubic feet per day is used only if the conditions at which the value was measured are specified
mCi	millicurie	
mD	millidarcy	

Unit Symbol	Unit Name	Comment
m ³ /d or m3/d	cubic meter per day	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use preferred to galUS/d for gallon per day; 1 galUS/d = 3.785412E-03 m³/d Except for gas, use of this unit or m³/h or m3/h for cubic meter per hour preferred to ft³/d or ft3/d for cubic foot per day; 1 ft³/d = 1.179869E-03 m³/h = 2.831685E-02 m³/d For gas, use Bcf/d for billion cubic feet per day, Mcf/d for thousand cubic feet per day, MMcf/d for million cubic feet per day, and Tcf/d for trillion cubic feet per day and express the value without scientific notation
meq (use restricted: see “mmol”)	milliequivalent	Use mmol for millimole; eq for equivalent or meq for milliequivalent can be used as a secondary unit with the “equivalent” defined 1 meq = [1 mmol/(valence of the substance)][†]
metric ton	metric ton; long-ton; UK ton; tonUK (1,000 kg; 2,204 lbm)	Do not use tonne, tonUK, dead-weight ton, or long-ton for unit name. Transition Technologies and EPRs prefer “metric ton,” which follows the U.N.’s Sustainable Development Goal terminology. 1 metric ton = 1,000 kg = 2,204 lbm
MeV	megaelectronvolt	
Mg (see “metric ton”)	megagram	Can be confused with the chemical symbol for magnesium. Convert to lbm, kg, metric ton, or US ton. 1 Mg = 1,000 kg = 1 metric ton 1 US ton = 0.902 Mg = 902 kg
mg	milligram	
mGal (see “cm/s ² or cm/s2”)	milligal	Use of cm/s² or cm/s2 for centimeter per second squared preferred; 1 mGal = 1.0E-03 cm/s²
mg/dm ³ or mg/dm3	milligram per cubic decimeter	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use of this unit symbol preferred to PTB for pound per thousand barrels (salt content); 1 PTB = 2.853010E+00 g/m³ = 2.853010E+00 mg/dm³
m/h	meter per hour	Use of this unit or min/m for minute per meter, mm/s for millimeter per second, or cm/s for centimeter per second preferred to ft/h for foot per hour; 1 ft/h = 3.048E-01 m/h = 1.968504E+02 min/m = 8.466667E-02 mm/s = 8.466667E-03 cm/s

Unit Symbol	Unit Name	Comment
m ³ /h or m3/h	cubic meter per hour	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use preferred to galUS/min for gallon per minute; 1 galUS/min = 2.271247E-01 m ³ /h Do not use gpm for gallon per minute Except for gas, use of this unit or m ³ /d or m3/d for cubic meter per day preferred to ft ³ /d or ft3/d for cubic foot per day; 1 ft ³ /d = 1.179869E-03 m ³ /h = 2.831685E-02 m ³ /d For gas, use Bcf/d for billion cubic feet of gas per day, Mcf/d for thousand cubic feet per day, MMcf/d for million cubic feet per day, and Tcf/d for trillion cubic feet per day and express the value without scientific notation
mho (not permitted: see "S")	mho	Use S for siemens for mho; 1 mho = 1.0E+00 S Do not use Ŧ (inverted Greek letter Omega) for mho
mho/m (not permitted: see "S/m")	mho per meter	Use S/m for siemens per meter; 1.0 mho/m = 1.0E+00 S/m Do not use Ŧ (inverted Greek letter Omega) for mho
MHz	megahertz	
mi	mile	
mi/h (use restricted: see "km/h")	mile per hour	Use of km/h for kilometer per hour preferred for non-nautical speed values; 1 mi/h = 1.609344E+00 km/h Do not use mph for mile per hour
mil/yr (not permitted: see "mm/a")	mil (0.001 in) per year	Use mm/a for millimeter per year; 1 mil per year = 2.54E-02 mm/a Do not use mpy for mil per year
min	minute	Use spelled-out unit name when not used with a measured value or depending on context Do not use m or ms for minute
mina	minute (angle)	Do not use ¢ (prime) for minute (angle)
min/m	minute per meter	Use of this unit or m/h for meter per hour, mm/s for millimeter per second, or cm/s for centimeter per second preferred to ft/h for foot per hour; 1 ft/h = 1.968504E+02 min/m = 3.048E-01 m/h = 8.466667E-02 mm/s = 8.466667E-03 cm/s
MJ	megajoule	Use preferred to hp.h for horsepower-hour; 1 hp.h = 2.684520E+00 MJ
mL	milliliter	
mm	millimeter	Use of this unit symbol or cm for centimeter preferred to in for inch; 1 in = 2.54E+01 mm = 2.54E+00 cm

Unit Symbol	Unit Name	Comment
mm ² or mm2	square millimeter	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use of this symbol or cm ² or cm2 for square centimeter preferred to in ² or in2 for square inch; 1 in ² = 6.5416E+02 mm ² = 6.5416E+00 cm ²
mm ³ or mm3	cubic millimeter	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use of this unit symbol or cm ³ or cm3 for cubic centimeter preferred to in ³ or in3 for cubic inch; 1 in ³ = 1.638706E+04 mm ³ = 1.638706E+01 cm ³
m ³ /m ³ or m3/m3	cubic meter per cubic meter	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use preferred to customary unit symbol vol % for volume percent; 1 vol % = 1.0E–02 m ³ /m ³ or express as a volume fraction of x % Do not use vol/vol for volume per volume Use of this unit symbol or L/m ³ or L/m3 for liter per cubic meter preferred to ft ³ /bbl or ft3/bbl for cubic foot per barrel; 1 ft ³ /bbl = 1.78073E+02 L/m ³ = 1.78073E–01 m ³ /m ³
mm/a	millimeter per year	Do not use mil/yr or mpy for mil (0.001 in) per year; 1 mil per year = 1 mpy = 2.54E–02 mm/a
MMcf	million (10 ⁶) cubic feet	Use for gas preferred to m ³ or m3 for cubic meter or dm ³ or dm3 for cubic decimeter; values are expressed without scientific notation 1 MMcf = 1,000,000 ft ³ = 2.831685E+04 m ³ = 2.831685E+07 dm ³
MMcf/d	million (10 ⁶) cubic feet per day	Use for gas preferred to m ³ /h or m3/h for cubic meter per hour or m ³ /d or m3/d for cubic meter per day; values are expressed without scientific notation 1 MMcf/d = 1,000,000 ft ³ /d = 1.179869E+03 m ³ /h = 2.831685E+04 m ³ /d MMscf/d for million standard cubic feet per day used only if the conditions at which the value was measured are specified
mmho (not permitted: see “mS”)	millimho	Use mS for millisiemens; 1 mmho = 1.0E+00 mS Do not use Ŧ (inverted Greek letter Omega) for mho

Unit Symbol	Unit Name	Comment
mmol	millimole	Use preferred to eq for equivalent or meq for milliequivalent, which can be used as necessary as a secondary unit with the “equivalent” defined; 1 meq = [1 mmol/(valence of the substance)] [†]
mm/s	millimeter per second	Use of this unit symbol or cm/s for centimeter per second preferred to in/s for inch per second; 1 in/s = 2.54E+01 mm/s = 2.54E+00 cm/s Use of this unit symbol, cm/s for centimeter per second, m/h for meter per hour, or min/m for minute per meter preferred to ft/h for foot per hour; 1 ft/h = 8.466667E–02 mm/s = 8.466667E–03 cm/s = 3.048E–01 m/h = 1.968504E+02 min/m
mm ² /s or mm2/s	square millimeter per second	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Do not use cSt for centistokes; 1 cSt = 1.0E+00 mm ² /s
MMscf/d (use restricted: see “MMcf/d”)	million (10 ⁶) standard cubic feet per day	Use MMcf/d for million cubic feet per day; express the value without scientific notation Use MMscf/d only if the conditions at which the value was measured are specified
mol	mole	
mol/L (not permitted: see “mol/m ³ or mol/m3”)	mole per liter	Use mol/m ³ or mol/m3 for mole per cubic meter to express as the amount-of-substance concentration (volume based); 1 mol/L = 1.0E+03 mol/m ³ Do not use M for molarity
mol/m ³ or mol/m3	mole per cubic meter	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Do not use M for molarity or mol/L for mole per liter—express as the amount-of-substance concentration (volume based); 1 mol/L = 1.0E+03 mol/m ³ [†] Do not use N for normality or eq/L for equivalent per liter—express the normality as a whole-number multiple of the molarity dependent on the solution valence; (mol/m ³)/valence [†]
mon (not permitted: see “month”)	month	Always spelled out
month	month	Always spelled out Do not use mon for month

Unit Symbol	Unit Name	Comment
MPa	megapascal	Use of this unit symbol or kPa for kilopascal preferred to - atm for atmosphere (standard); 1 atm = 1.01325E+01 MPa = 1.01325E+02 kPa at standard conditions of 760-mm Hg at 0 degC or 14.696 lbf/in² - bar for bar; 1 bar = 1.0E+01 MPa = 1.0E+02 kPa - psi for pound (force) per square inch; 1 psi = 6.894757E-03 MPa = 6.89477E+00 kPa Do not modify the unit symbol (e.g., MPa (absolute))—specify the conditions at which the value was measured or use an expression such as “gauge pressure = 15 MPa”
MPa (absolute) (not permitted: see “MPa”)	megapascal (absolute)	Do not modify the unit symbol—specify the conditions at which the value was measured
MPa (gauge) (not permitted: see “MPa”)	megapascal (gauge)	Do not modify the unit symbol—specify the conditions at which the value was measured
mPa	millipascal	
mph (not permitted: see “km/h”)	mile per hour	Use of km/h for kilometer per hour preferred to mi/h for mile per hour for non-nautical speed values; 1 mi/h = 1.609344E+00 km/h
mpy (not permitted: see “mm/a”)	mil (0.001 in) per year	Use mm/a for millimeter per year; 1 mpy = 2.54E-02 mm/a Do not use mil/yr for mil per year
Mrayl	megarayl	Use for acoustic impedance unit
mS	millisiemens	Do not use mmho for millimho; 1 mmho = 1.0E+00 mS Do not use mΩ for millimho
ms	millisecond	Do not use msec for millisecond
ms (not permitted: see “min”)	minute	Use min for minute
m/s	meter per second	Use of this unit symbol or cm/s for centimeter per second preferred to ft/min for foot per minute; 1 ft/min = 5.08E-03 m/s = 5.08E-01 cm/s Use preferred to ft/s for foot per second; 1 ft/s = 3.048E-01 m/s
m ² /s or m2/s	square meter per second	Power can be superscript trademark symbol or in line; one format must be used consistently in a document Use of m²/s or m2/s for square meter per second preferred to ft²/s or ft2/s for square foot per second; 1 ft²/s = 9.290304E-02 m²/s Do not use St for Stokes; 1 St = 1.0E-04 m²/s

Unit Symbol	Unit Name	Comment
Mscf/d (use restricted: see "Mcf/d")	thousand standard cubic feet per day	Use Mcf/d for thousand cubic feet per day; express the value without scientific notation Use Mscf/d only if the conditions at which the value was measured are specified
msec (not permitted: see "ms")	millisecond	Use ms for millisecond
mT	millitesla	
mV	millivolt	
MY (not permitted: see "Ma")	million years	Use Ma for million years
m $\bar{\Omega}$ (not permitted: see "mS")	millimho	Use mS for millisiemens; 1 m $\bar{\Omega}$ = 1.0E+00 mS Do not use mmho for millimho Do not use $\bar{\Omega}$ (inverted Greek letter Omega) for mho
N	newton	Use preferred to lbf for pound (force); 1 lbf = 4.448222E+00 N Do not use kgf for kilogram (force) or kp for kilopond; 1 kgf = 1 kp = 9.80665E+00 N Do not use kip or klpf (= 1,000 lbf); 1 kip = 1 klpf = 4.448222E+03 N Use preferred to ozf for ounce (force); 1 ozf = 2.780139E-01 N
N (not permitted: see "mol/m ³ or mol/m3")	normality	Use mol/m ³ or mol/m3 to express the normality as a whole-number multiple of the molarity dependent on the solution valence; (mol/m ³)/valence [†] Do not use eq/L for equivalent per liter
nano Ω (not permitted: see "nohm")	nanohm	Use nohm for nanohm
nAPI	API neutron	
nm	nanometer	Do not use Å for angstrom; 1 Å = 1.0E-01 nm
N.m	newton-meter	Use preferred to lbf.ft for pound (force)-foot for torque values; 1 lbf.ft = 1.355818E+00 N.m Do not use for energy or work unit values, for which use of kJ for kilojoule preferred to ft.lbf for foot-pound (force); 1 ft.lbf = 1.355818E-03 kJ
nohm	nanohm	Do not use nano Ω
ns	nanosecond	
nT	nanotesla	
Oe (not permitted: see "A/m")	oersted	Use A/m for ampere per meter; 1 Oe = 7.957747E+01 A/m
ohm	ohm	Do not use Ω (Greek letter Omega) for ohm
ohm.cm	ohm-centimeter	Do not use Ω .cm

Unit Symbol	Unit Name	Comment
ohm.m	ohm-meter	Do not use Ω .m
ozf (see “N”)	ounce (force)	Use of N for newton preferred; 1 ozf = 2.780139E-01 N
ozm (see “g”)	ounce (mass)	Use of g for gram preferred; 1 ozm = 2.834952E+01 g
P (see “Pa.s”)	poise	Use of Pa.s for pascal-second preferred; 1 P = 1.0E-01 Pa.s
Pa	pascal	Use preferred to lbf/100 ft ² for pound (force) per 100 square feet; 1 lbf/100 ft ² = 4.788026E-01 Pa
Pa.s	pascal-second	Use preferred to P for poise and cP for centipoise; 1 cP = 1.0E-03 Pa.s and 1 P = 1.0E-01 Pa.s
pF	picofarad	
ppa (see “lbm”)	pound of proppant added per barrel of clean fluid	Allowable customary unit or use simple unit symbol lbm for pound (mass)—express as “x lbm of proppant added per barrel of clean fluid” Use of kg for kilogram preferred; 1 lbm = 4.535924E-01 kg
ppb	part per billion	Allowable customary unit or use scientific notation ($\times 10^{-9}$) or a ratio of unit symbols (e.g., ug/kg)
ppk	part per thousand	Allowable customary unit or use scientific notation ($\times 10^{-3}$) or a ratio of unit symbols (e.g., g/kg) See also ppt for part per thousand; if ppk or ppt is used, use one consistently
ppm	part per million	Allowable customary unit or use scientific notation ($\times 10^{-6}$) or a ratio of unit symbols (e.g., ug/g) 1 ppm = 1 ug/g
ppmv	part per million by volume	Allowable customary unit
ppt	part per thousand	Allowable customary unit or use scientific notation ($\times 10^{-3}$) or a ratio of unit symbols (e.g., g/kg) See also ppk for part per thousand; if ppk or ppt is used, use one consistently
ppt (not permitted: see “lbm/1,000 galUS” or “lbm/1,000 galUK”)	pound per thousand (gallons)	Use lbm/1,000 galUS pound (mass) per 1,000 galUS or lbm/1,000 gal UK for pound (mass) per 1,000 galUK Use of g/m ³ or g/m3 for gram per cubic meter preferred; 1 lbm/1,000 galUS = 1.198264E+02 g/m ³ or 1 lbm/1,000 gal UK = 9.977633E+01 g/m ³

Unit Symbol	Unit Name	Comment
psi (see “kPa” or “MPa”)	pound (force) per square inch	Use of kPa for kilopascal or MPa for megapascal preferred; 1 psi = 6.894757E–03 MPa = 6.894757E+00 kPa Do not modify the psi unit symbol (e.g., psi [absolute], psig)—specify the conditions at which the value was measured or use an expression such as “gauge pressure = 230 kPa” Do not use kpsi for 1,000 pounds (force) per square inch or ksi for kip (1,000 lbf) per square inch; 1 ksi = 1 kpsi = 6.894757E+03 kPa
psi (absolute) (not permitted: see “psi”)	pound (force) per square inch (absolute)	Do not modify the unit symbol—specify the conditions at which the value was measured Use of kPa for kilopascal or MPa for megapascal preferred to psi; 1 psi = 6.894757E–03 MPa = 6.894757E+00 kPa
psi (gauge) (not permitted: see “psi”)	pound (force) per square inch (gauge)	Do not modify the unit symbol—specify the conditions at which the value was measured—specify the conditions at which the value was measured or use an expression such as “gauge pressure = 230 kPa” Use of kPa for kilopascal or MPa for megapascal preferred to psi; 1 psi = 6.894757E–03 MPa = 6.894757E+00 kPa
psia (not permitted: see “psi”)	pound (force) per square inch (absolute)	Do not modify the unit symbol—specify the conditions at which the value was measured Use of kPa for kilopascal or MPa for megapascal preferred to psi; 1 psi = 6.894757E–03 MPa = 6.894757E+00 kPa
psig (not permitted: see “psi”)	pound (force) per square inch (gauge)	Do not modify the unit symbol—specify the conditions at which the value was measured—specify the conditions at which the value was measured or use an expression such as “gauge pressure = 230 kPa” Use of kPa for kilopascal or MPa for megapascal preferred to psi; 1 psi = 6.894757E–03 MPa = 6.894757E+00 kPa
pt (not permitted: see “L”)	pint	Use L for liter; 1 pt = 4.731765E–01 L
PTB (see g/m ³ or mg/dm ³)	pound per thousand barrels (salt content)	Use of g/m ³ or g/m ³ for gram per cubic meter or mg/dm ³ or mg/dm ³ for milligram per cubic decimeter preferred; 1 PTB = 2.853010E+00 g/m ³ = 2.853010E+00 mg/dm ³
pu (see “%”)	porosity unit	Allowable customary unit or use % for percent and should be specified that this is not the “true” porosity (e.g., use terms such as uncorrected porosity, limestone porosity, or sandstone matrix porosity)
qt (not permitted: see “L”)	quart	Use L for liter; 1 qt = 9.463529E–01 L

Unit Symbol	Unit Name	Comment
R (not permitted: see “C/kg”)	roentgen	Use C/kg for coulomb per kilogram; 1 R = 2.58E–04 C/kg
rad	radian	
rad (not permitted: see “Gy”)	radiation absorbed dose	Use Gy for gray; 1 rad = 1.0E–02 Gy
rad/m	radian per meter	Use preferred to °/m or dega/m at 0.01 or 1/30 scale for degree (angle) per meter; 1 °/m = 1.745329E–02 rad/m
RB/d (not permitted: see “bbl/d”)	reservoir barrel per day	Do not modify the unit symbol bbl/d—specify the reservoir conditions at which the value was measured Use of m ³ /d or m3/d for cubic meter per day preferred to bbl/d for barrel per day; 1 bbl/d = 42 galUS/d = 1.589873E–01 m ³ /d Do not use bbl/d (reservoir) for reservoir barrel per day
rem	rem	Use Sv for sievert; 1 rem = 1.0E–02 Sv†
res bbl (not permitted: see “bbl”)	reservoir barrel	Do not modify the unit symbol bbl—specify the reservoir conditions at which the value was measured Use of m ³ or m3 for cubic meter preferred to bbl for barrel; 1 bbl = 42 galUS = 1.589873E–01 m ³ Do not use bbl (reservoir) for reservoir barrel
rpm (see “c/min”)	revolution per minute	Allowable customary unit or preferably use c/min for cycle per minute; 1.0 rpm = 1.0 c/min
S	siemens	Do not use mho for mho; 1 mho = 1.0E+00 S Do not use Ū (inverted Greek letter Omega) for mho
s	second	Use spelled-out unit name when not used with a measured value or depending on context Do not use sec for second
sack (use restricted: see “kg” or “L”)	sack	Use of kg for kilogram and L for liter preferred to sack; for portland cement, 1 sack = ~42.64 kg and 1 sack = ~1 ft ³ = ~28.3 L The absolute volume of a 94-lbm sack of portland cement is 3.59 galUS. To use sack as a unit, specify the material and its volume in m ³ or m3 for cubic meter (preferable to characterizing it by its mass). Use of L/metric ton for liter per metric ton preferred to galUS/sack for gallon per sack to express the volume of fluid per mass of solid (specific volume). Use of L/m ³ or L/m3 for liter per cubic meter preferred to galUS/sack for gallon per sack to express the volume of fluid per volume of solid (concentration). Do not use sk for sack

Unit Symbol	Unit Name	Comment
scf (use restricted: see "ft ³ or ft3")	standard cubic foot	Use ft³ or ft3 and specify the standard conditions at which the value was measured Except for gas, use of m³ or m3 for cubic meter or dm³ or dm3 for cubic decimeter preferred; 1 ft³ = 2.831685E–02 m³ = 2.831685E+01 dm³ For gas, use Bscf for billion standard cubic feet, Mscf for thousand standard cubic feet, MMscf for million standard cubic feet, and Tscf for trillion standard cubic feet only if the standard conditions at which the value was measured are specified; express the value without scientific notation If standard conditions are not specified for gas, use Bcf for billion cubic feet, Mcf for thousand cubic feet, MMcf for million cubic feet, and Tcf for trillion cubic feet
scf/d (use restricted: see "ft ³ /d or ft3/d")	standard cubic feet per day	Use ft³/d or ft3/d and specify the standard conditions at which the value was measured Except for gas, use of m³/h or m3/h for cubic meter per hour or m³/d or m3/d for cubic meter per day preferred; 1 ft³/d = 1.179869E–03 m³/h = 2.831685E–02 m³/d For gas, use Bscf/d for billion standard cubic feet per day, Mscf/d for thousand standard cubic feet per day, MMscf/d for million standard cubic feet per day, and Tscf/d for trillion standard cubic feet per day only if the standard conditions at which the value was measured are specified; the value is expressed without scientific notation If standard conditions are not specified for gas, use Bcf/d for billion cubic feet per day, Mcf/d for thousand cubic feet per day, MMcf/d for million cubic feet per day, and Tcf/d for trillion cubic feet per day
sec (not permitted: see "s")	second	Use s for second
seca	second (angle)	Do not use " (double prime) for second
sg	specific gravity	Specific gravity uses water as the reference density (1 g/cm³ = 1.0E+03 kg/m³); other materials used for the reference density must be specified "Relative density" is preferred to "specific gravity" for the unitless ratio of the density of a material divided by a reference density

Unit Symbol	Unit Name	Comment
sk (not permitted: see “sack,” “kg,” or “L”)	sack	Use of kg for kilogram and L for liter preferred to sack To use the unit sack, specify the material and its volume in m ³ or m3 for cubic meter (preferable to characterizing it by its mass); for portland cement, 1 sack = ~42.64 kg and 1 sack = ~1 ft ³ = ~28.3 L
S/m	siemens per meter	Do not use mho/m for mho per meter; 1 mho/m = 1.0E+00 S/m Do not use Ū/m for mho/m
sr	steradian	
St (not permitted: see “m ² /s or m2/s”)	Stokes	Use m ² /s or m2/s for square meter per second 1.0E–E04 St = 1 m ² /s
STB (not permitted: see “bbl”)	stock-tank barrel	Do not modify the unit symbol bbl—specify the stock-tank conditions at which the value was measured Use of m ³ or m3 for cubic meter preferred to bbl for barrel; 1 bbl = 42 galUS = 1.589873E–01 m ³ Do not use bbl (stock tank) for stock-tank barrel
STB/d (not permitted: see “bbl/d”)	stock-tank barrel per day	Do not modify the unit symbol bbl/d—specify the stock-tank conditions at which the value was measured Use of m ³ /d or m3/d for cubic meter per day preferred to bbl/d for barrel per day; 1 bbl/d = 42 galUS/d = 1.589873E–01 m ³ /d Do not use STB/d for stock-tank barrel per day
STOOIP (not permitted: see “bbl”)	stock-tank original oil in place	Do not modify the implied unit symbol bbl—specify the stock-tank conditions at which the value was measured Use of m ³ or m3 for cubic meter preferred to bbl for barrel; 1 bbl = 42 galUS = 1.589873E–01 m ³ Do not use bbl (stock tank) or STB for stock-tank barrel
su (not permitted: see “%”)	saturation unit	Use % for percent to express the percentage of fluid in the pore space (e.g., “a fluid fraction of x %”)
Sv	sievert	Do not use rem for rem; 1 rem = 1.0E–02 S†
T	tesla	Use instead of G or Gs for gauss; 1 G = 1 Gs = 1.0E–04 T
t	ton (in space-constrained copy)	For space constraints, you can use t for metric ton or US ton as long as, further into the document, you specify “metric ton” or “US ton”. Transition Technologies and EPRs prefer “metric ton,” which follows the U.N.’s Sustainable Development Goal terminology.

Unit Symbol	Unit Name	Comment
t/h	metric ton per hour	For geothermal applications only. Customary for Europe and Asia. Use galUS/min for US audience.
TB	terabyte	One terabyte equals 1,000,000,000,000 bytes.
Tcf	trillion cubic feet (trillion = 10^{12})	Use for gas preferred to m^3 or m^3 for cubic meter or dm^3 or dm^3 for cubic decimeter; values are expressed without scientific notation 1 Tcf = 1,000,000,000,000 ft^3 = 2.831685E+10 m^3 = 2.831685E+13 dm^3
Tcf/d	trillion cubic feet per day	Use for gas preferred to m^3/h or m^3/h for cubic meter per hour or m^3/d or m^3/d for cubic meter per day; values are expressed without scientific notation 1 Tcf/d = 1,000,000,000,000 ft^3/d = 1.179869E+09 m^3/h = 2.831685E+10 m^3/d
US ton	US ton	Do not use standard ton, short-ton, or tonUS; 1 US ton = 2,000 lbm = 907 kg Transition Technologies and EPRs prefer “metric ton,” which follows the United Nation’s Sustainable Development Goal terminology.
u	unified atomic mass unit	Do not use amu for atomic mass unit, which is equal to 1/12 of the mass of an atom of the nuclide ^{12}C 1 u = $1.6605402 \times 10^{-27}$ kg with a combined standard uncertainty of $0.0000010 \times 10^{-27}$ kg†
ue	microstrain	Do not use $\mu\epsilon$ or $\mu strain$ for microstrain
uF	microfarad	Do not use μF for microfarad
ug	microgram	Do not use μg for microgram
uL	microliter	Do not use μL for microliter
um	micrometer	Do not use μ or μm for micrometer Do not use micron for the unit name
us	microsecond	Do not use μs for microsecond
uV	microvolt	Do not use μV for microvolt
V	Volt	Do not use the abbreviation emf as a unit symbol for electromotive force
vol % (see “%” or “ m^3/m^3 or m^3/m^3 ”)	volume percent	Allowable customary unit or preferably use % for percent (e.g., “a volume fraction of x %”) or use the ratio m^3/m^3 or m^3/m^3 ; 1 vol % = $1.0E-02 m^3/m^3$
V AC	volts alternating current	Do not use VAC
V DC	volts direct current	Do not use VDC
vol/vol (not permitted: see “%” or “ m^3/m^3 or m^3/m^3 ”)	volume per volume	Use % for percent (e.g., “a volume fraction of x %”) or Use the ratio m^3/m^3 or m^3/m^3 ; 1 % vol/vol = $1.0E-02 m^3/m^3$

Unit Symbol	Unit Name	Comment
W	watt	Use of this unit symbol or kW for kilowatt preferred to hp for horsepower (550 ft.lbf/s) (also called “mechanical” or “standard” horsepower); 1 hp = 7.456999E+02 W = 7.456999E–01 kW
w (not permitted: see “week”)	week	Always spelled out Do not use wk for week
Wb	weber	
week	week	Always spelled out Do not use w or wk for week
wk (not permitted: see “week”)	week	Always spelled out Do not use w for week
wt % (see “%,” “kg/kg,” or “g/kg”)	weight percent	Allowable customary unit or preferably use % for percent (e.g., “a mass fraction of x %”) or Use ratio of mass unit symbols: kg/kg for kilogram per kilogram or g/kg for gram/kilogram 1 wt % = 1.0E–02E kg/kg = 1.0E+01 g/kg
yd (see “m”)	yard	Allowable customary unit or preferably use m for meter; 1 yd = 3 ft = 9.144E–01 m
yd ³ (see “m ³ or m3”)	cubic yard	Allowable customary unit or preferably use m ³ or m3 for cubic meter; 1 yd ³ = 7.645549E–01 m ³
yr (see “a”)	year	Restrict use of yr to combined units (e.g., m ³ /yr) Spell out “year” for values used in a descriptive or literary manner Use a for annum only in combined units for measured time values (e.g., m/a)
μ (not permitted: see “um”)	micron	Do not use prefix alone for um for micrometer
μe (not permitted: see “ue”)	microstrain	Use ue for microstrain
μF (not permitted: see “uF”)	microfarad	Use uF for microfarad
μg (not permitted: see “ug”)	microgram	Use ug for microgram
μL (not permitted: see “uL”)	microliter	Use uL for microliter
μm (not permitted: see “um”)	micrometer	Use um for micrometer Do not use micron for the unit name
μs (not permitted: see “us”)	microsecond	Use us for microsecond
μstrain (not permitted: see “ue”)	microstrain	Use ue for microstrain

Unit Symbol	Unit Name	Comment
μV (not permitted: see “uV”)	microvolt	Use uV for microvolt
Ω (not permitted: see “ohm”)	ohm	Use ohm, not the Greek letter Omega
$\Omega.\text{cm}$ (not permitted: see “ohm.cm”)	ohm-centimeter	Use ohm.cm for ohm-centimeter
$\Omega.\text{m}$ (not permitted: see “ohm.m”)	ohm-meter	Use ohm.m for ohm-meter
S (not permitted: see “S”)	mho	Use S for siemens; $1 \text{ S} = 1.0\text{E}+00 \text{ S}$ Do not use mho for mho
S/m (not permitted: see “S/m”)	mho per meter	Use S/m for siemens per meter; $1.0 \text{ S/m} = 1.0 \text{ mho/m} = 1.0\text{E}+00 \text{ S/m}$ Do not use S (inverted Greek letter Omega) for mho
sq (not permitted: use the power ² or 2)	square	Use the power ² or 2 instead of sq for square; one format (superscript trademark symbol or in line) must be used consistently in a document

Unit Prefixes

SI Unit Prefixes

The SI unit prefixes are used to form decimal multiples and submultiples of SI units. They are joined to the unit names (e.g., kilometer, not kilo-meter). Their use with customary unit abbreviations and unit names should be avoided.

SI Unit Prefixes		
Prefix	Symbol	Factor
yotta	Y	10 ²⁴
zetta	Z	10 ²¹
exa	E	10 ¹⁸
peta	P	10 ¹⁵
tera	T	10 ¹²
giga	G	10 ⁹
mega	M	10 ⁶
kilo	k	10 ³
hecto	h	10 ²
deka	da	10 ¹
deci	d	10 ⁻¹
centi	c	10 ⁻²
milli	m	10 ⁻³
micro	μ	10 ⁻⁶
nano	n	10 ⁻⁹
pico	p	10 ⁻¹²
femto	f	10 ⁻¹⁵
atto	a	10 ⁻¹⁸
zepto	z	10 ⁻²¹
yocto	y	10 ⁻²⁴

Customary Unit Prefixes

These prefixes are used strictly with specified customary unit symbols.

SI Unit Prefixes		
Prefix	Unit Symbols	Factor
B	Bcf, Bcf/d, Bscf/d	billion (10 ⁹)
M	Mcf, Mcf/d, Mscf/d	thousand (10 ³)
MM	MMcf, MMcf/d, MMscf/d	million (10 ⁶)
T	Tcf, Tcf/d	trillion (10 ¹²)

Source:

This list of unit symbols is based on the *Petrotechnical Open Software Corporation (POSC) Recommended Practice (RP) 66 V2* specification, part 5
(http://www.posc.org/technical/data_exchange/RP66V2/rp66v2_sec5.htm)

Notes:

Except as noted, conversion factors are from *The SI Metric System of Units and SPE Metric Standard* (Society of Petroleum Engineers (June 1984, 2nd printing) and are to exact values or seven-digit accuracy.

†Taylor, B. N.: *Guide for the Use of the International System of Units (SI)*: NIST Special Publication 811, (1995).

‡Recommended Practices for Exploration & Production Data Digital Interchange: Part 5: API-SI Unit Symbols (1st ed): API Recommended Practice 66, V2 (May 1, 1996).

“Units” previously on the *Communications Handbook* “Unit Abbreviations” list that were moved to the “List of spellings and terminology”:

- spf: shots per foot
- std: standard
- spm: strokes per minute
- computer terms (dpi, etc.)